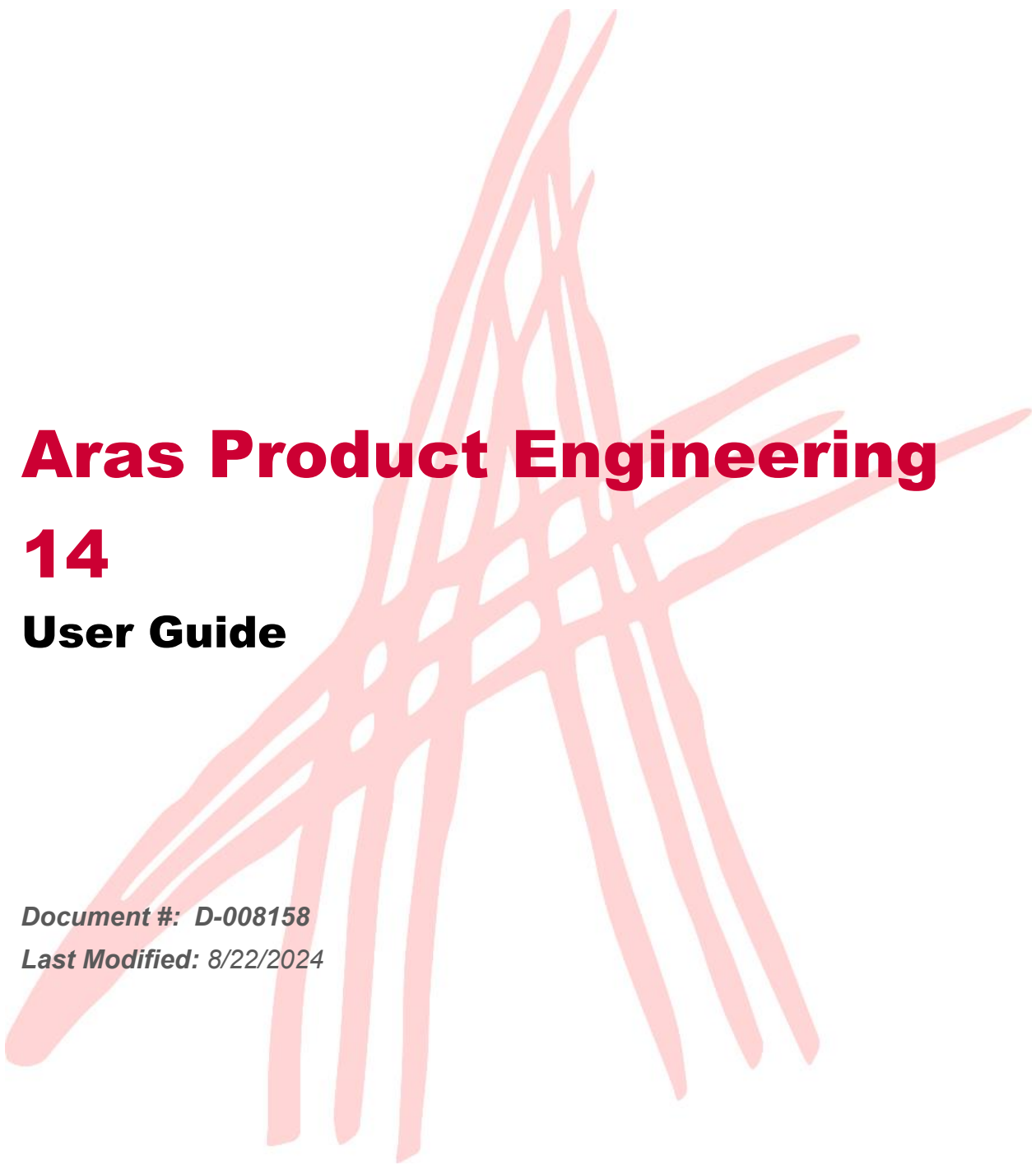




# **Aras Product Engineering**

## **14**

### **User Guide**

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# 1 Application Overview

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The Aras Product Engineering application is a business-ready solution for product definition and engineering change management across the enterprise. Product Engineering includes core PLM data and process functionality such as Parts, Bill of Materials (BOM), Documents, CAD Documents, Approved Manufacturer Lists/Approved Vendor Lists (AML/AVL) and Change items.

Benefits of the Aras Product Engineering application include:

- Visibility into key development indicators increases engineering effectiveness and provides a basis for continuous improvement.
- Secure online management of product definition processes fosters the development of high-quality products.
- Engineering change management reduces cycle times and streamlines business operations.
- Controlled product information release between departments, divisions, suppliers, and customers provides effective coordination.
- Reuse of product and process information increases productivity and controls costs.
- Industry best practices for configuration management and engineering change are incorporated using CMII principles and processes.
- Unparalleled flexibility enables application modification to address proprietary engineering processes and specific competitive practices.

## 2 User Interface (UI) Overview

The Aras Innovator user interface (UI) is dynamic; UI components, such as dialogs, toolbars, and buttons are shown or hidden depending on the context. Additionally, when a UI component is displayed in a lighter color, it indicates that this component is not available in the current context. This section describes essential UI interactions and components to navigate in the Product Engineering application.

### 2.1 Navigation Panel



Clicking the **Navigation** button displays the **Navigation Panel**. This panel displays the Aras Innovator Table of Contents. A User's permissions determine the categories that appear in the Navigation

Panel. The Navigation Panel is auto hidden when you click outside the TOC. Click the **Pin** button  in the Navigation Panel if you wish to pin it to the UI.

The Product Engineering application has several categories on the Navigation panel:

- **Change Management** – Contains Engineering Change Items, such as Problem Reports (PRs), Engineering Change Requests (ECRs) and Engineering Notices (ECNs). This list may vary depending on how an administrator configures the system.
- **Design** – Contains Part and Product items.
- **Documents** – Contains Document and CAD Document items.
- **Sourcing** – Contains Manufacturers, Manufacturer Parts, and Vendor items.

### 2.2 Search Page

Hover or click on an item in the Navigation Panel to access the search page.

#### 2.2.1 Search Toolbar

The **Search** toolbar enables you to search for items and view the search results.

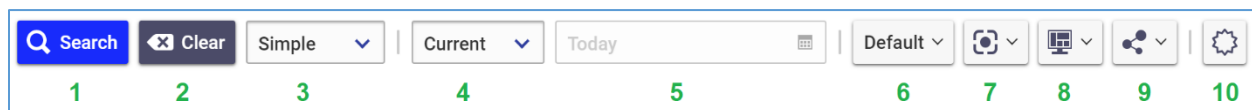


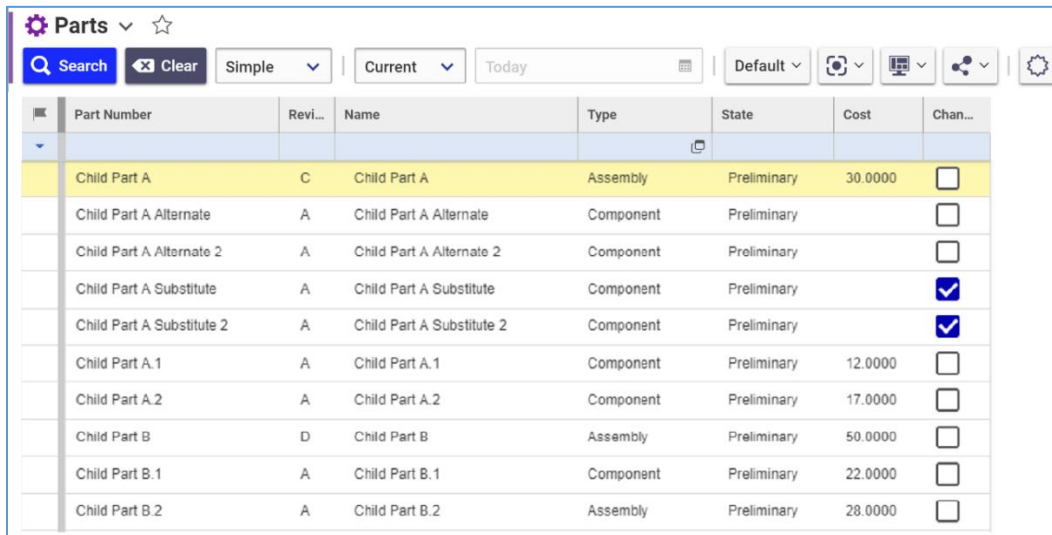
Figure 1.

1. Click the **Search** button to run a search.
2. Click the **Clear** button to clear search criteria.
3. **Search Modes** dropdown menu:
  - Use **Simple** search when the columns that appear in the search grid are sufficient for the desired search criteria.
  - Use **Advanced** search when entering complex search criteria where each criterion can use one of these operators: =, ≠, like, not like, null, not null.

- Use **AML** search to create a complex search query.
  - Use **Hidden** to hide the search criteria from the search grid.
- Query Type dropdown menu:
    - **Current** returns the current version of items.
    - **Latest** returns the most recent version of items based on a specified date.
    - **Released** returns the most recent version in the released state on a specified date.
    - **Effective** returns the effective version on a specified date.
  - The **Query Date** dialog box enables you to select a date in conjunction with the **Latest**, **Released**, and **Effective** Query Types. By default, it is set to Today.
  - The **Layouts** menu presents a list of layouts to select from. This menu also provides the ability to save a new layout, update a default layout, and delete a layout.
  - The **Refine** menu presents a list of item Properties and Extended Properties to select that appear in the Search Grid.
  - The **Display** menu allows you to select the level of details to display when an item is selected from the search results.
    - Choose **Form** to display the selected item's form at the bottom pane of the search results page.
    - Choose **Properties** to display details in a pane to the left of the search results page.
    - Choose **Hide** to show no details.
  - The **Share** menu presents options to export search results.
  - Click the **New** button to create a new Item.

## 2.2.2 Search Grid

The Search Grid shows search results for items of one ItemType. Double-click on an item in the search grid to open it for viewing, or right-click it to access the context menu. If you select multiple items, the context menu displays multi-item actions applicable for that ItemType.



| Part Number               | Revi... | Name                      | Type      | State       | Cost    | Chan...                             |
|---------------------------|---------|---------------------------|-----------|-------------|---------|-------------------------------------|
| Child Part A              | C       | Child Part A              | Assembly  | Preliminary | 30.0000 | <input type="checkbox"/>            |
| Child Part A Alternate    | A       | Child Part A Alternate    | Component | Preliminary |         | <input type="checkbox"/>            |
| Child Part A Alternate 2  | A       | Child Part A Alternate 2  | Component | Preliminary |         | <input type="checkbox"/>            |
| Child Part A Substitute   | A       | Child Part A Substitute   | Component | Preliminary |         | <input checked="" type="checkbox"/> |
| Child Part A Substitute 2 | A       | Child Part A Substitute 2 | Component | Preliminary |         | <input checked="" type="checkbox"/> |
| Child Part A.1            | A       | Child Part A.1            | Component | Preliminary | 12.0000 | <input type="checkbox"/>            |
| Child Part A.2            | A       | Child Part A.2            | Component | Preliminary | 17.0000 | <input type="checkbox"/>            |
| Child Part B              | D       | Child Part B              | Assembly  | Preliminary | 50.0000 | <input type="checkbox"/>            |
| Child Part B.1            | A       | Child Part B.1            | Component | Preliminary | 22.0000 | <input type="checkbox"/>            |
| Child Part B.2            | A       | Child Part B.2            | Assembly  | Preliminary | 28.0000 | <input type="checkbox"/>            |

Figure 2.

In Simple Search mode, you can enter simple search criteria in the blue filter row of the Search Grid to filter the results against these criteria.

All searches are case insensitive. Use \* or % as a wildcard search. Use \* to search for all non-blank values. Use 0 or 1 to filter on properties with check boxes: 0 corresponds to not checked, 1 corresponds to checked values.

For example, you can type **Parent\*** or **Parent%** in the **Name** blue cell of the Parts Search Grid to get all the Parts that have names that start with the word Parent.

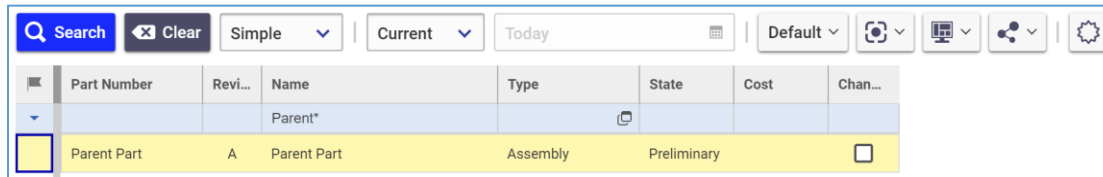


Figure 3.

The first column shows the claim status of Items. Use this column to search for items claimed by you or someone else. For details, refer to section [2.4 Item Claiming](#).

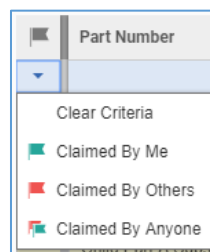


Figure 4.

## 2.2.3 Paging Controls

The **Paging Controls** toolbar enables you to move from page to page in the Search grid, specify the total number of pages/results, and specify the maximum number of results that display on a page.



Figure 5.

1. Click the **Prev** button to go to the previous page.
2. Click the **Next** button to go to the next page.
3. Displays the **Current page**.
4. Click the **Show Total Pages and Results** button to view the total number of pages and search results.
5. Use the **Results per Page** box to specify the number of results that appear on a page.
6. Enter a number in the **Maximum Results** box to specify the maximum number of results returned from a search.



## 2.2.4 Context Menu

The context menu displays the available actions for items appearing in the Search grid. Right-click an item in the grid to access the context menu. The context menu is an alternative access point to the functionality provided in the item window toolbar when you open an item. Each context menu contains commands specific to the ItemType and state of the item and is subject to permissions. As such, context menus are dynamic and variable. The commands common to most context menus are explained here.

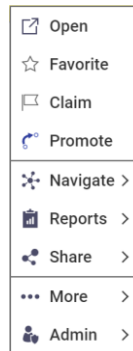


Figure 6.

- **Open** displays an Item in a new tab in view mode.
- **Favorite** allows you to favorite an item.
- **Claim** explicitly locks an item to prevent editing by other users and keeps the item locked/claimed between sessions. For details, refer to section [2.4 Item Claiming](#).
- **Unclaim** releases the claim from the item.
- **Promote** takes the item to its next life cycle state if any is available. For details, refer to section [4.3.1 Manual State Promotion](#).
- **Navigate** contains exploration options for the selected item.
  - **Structure Browser** is used for exploration and comparison of item's relationships. For details, refer to section [11.2 Comparing Items](#).
  - **Where Used** is used for determining where an Item is used by other Items. For details, refer to section [11.1 Finding Where an Item is Used](#).
- **Reports** displays the Reports available for the selected item.
- **Share** displays options for sharing the selected item.
- **More** displays additional actions available for the selected item.
  - **Properties** displays the item's properties in the **Properties** dialog.
  - **Save As** starts the copy process by opening a form page tab to create a new item.
  - **Delete** displays options for deleting the current version or all versions of an item.

## 2.3 Item View

A view of an item appears as a new **Page** tab with:

1. The Page Title bar

2. The Item toolbar
3. The Form accordion
4. The Relationships accordion
5. The Accordion Arrow button, which folds and unfolds an accordion

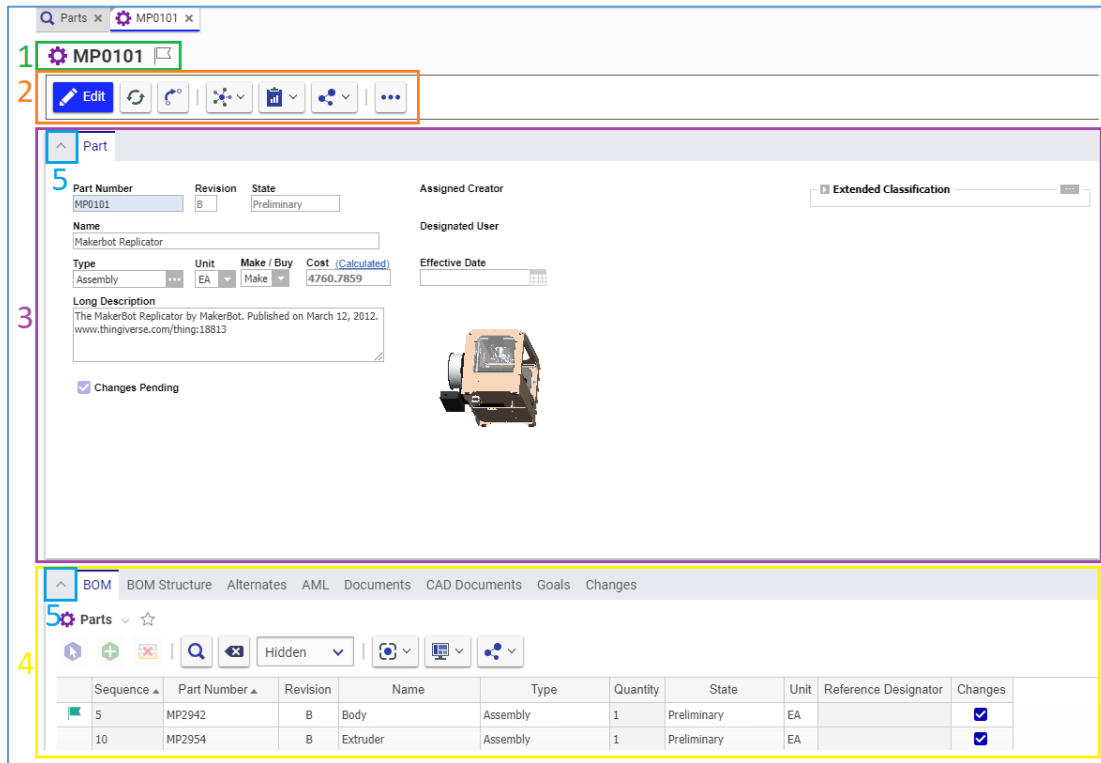


Figure 7.

### 2.3.1 Item Toolbar

The contents of the **Item** toolbar depend on the ItemType, state of the item, and permissions. When you create a new item, the **Item** toolbar displays the following buttons:

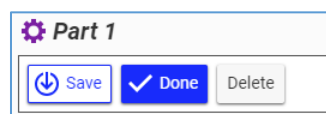


Figure 8.

- **Save** enables you to save changes to the Item and continue editing it.
- **Done** enables you to save changes to the Item and unclaim it.
- **Delete** enables you to delete the Item.

When editing a saved/existing item, the **Item** toolbar displays the following buttons:

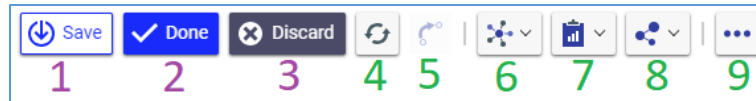


Figure 9.

1. **Save** saves changes to the item and enables you to continue editing it.
2. **Done** saves changes to the item, unclaims it, and quits edit mode.
3. **Discard** reverts all changes to the item, unclaims it, and quits edit mode.
4. **Refresh** updates the data in the View with the latest information.
5. **Promote** takes the item to its next life cycle state if any is available. For details, refer to section [4.3.1 Manual State Promotion](#).
6. **Navigate** displays a menu with options to explore the item's history, usage, and relationships.
7. **Reports** displays a menu with reports available for this ItemType.
8. **Share** displays a menu with export options available for this ItemType.
9. **More** displays a menu with additional capabilities available for this ItemType.

When viewing an item, the **Item** toolbar displays the same buttons, except the **Edit** button, which replaces the **Save**, **Done**, and **Discard** buttons:



Figure 10.

10. **Edit** claims the item and launches it in edit mode.

## 2.3.2 Relationships Accordion

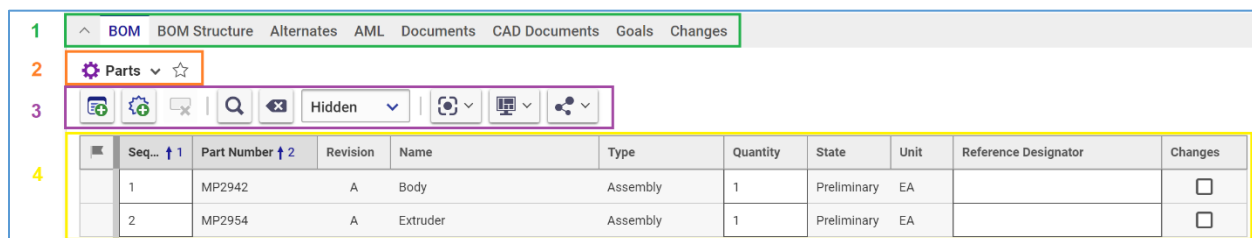


Figure 11.

The **Relationships** accordion hosts:

1. The **Relationships** accordion tabs
2. The **Relationships** title bars
3. The **Relationships** toolbars
4. The **Relationships** content area

### 2.3.3 Relationships Toolbar

Each **Relationships** accordion tab has a **Relationships** toolbar to provide functionality specific to its relationship type and **Relationships** content area. The **Relationships** toolbars are dynamic and variable. The contents of the **Relationships** toolbar depend on the relationship type, state of the item, and permissions. The buttons common to most **Relationships** toolbars are explained here.

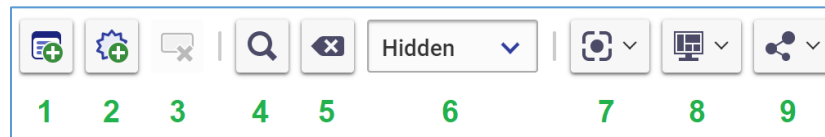


Figure 12.

1. The **Add Items** button launches a **Search dialog** to search for and add existing items to the **Relationships** grid.
2. The **New Item** button creates a new row in the **Relationships** grid. When the item is saved, the related item is created.
3. The **Delete Row** button removes the selected row (relationship) from the table without deleting the related item.
4. The **Run Search** button runs the search criteria on the related items.
5. The **Clear Search Criteria** button clears the search criteria on the related items.
6. The **Search Modes** menu allows selecting the mode for the search criteria. For details on the search modes, refer to section [2.2.1 Search Toolbar](#).
7. The **Refine** menu presents a list of item Properties and Extended Properties to select which will be displayed in the **Relationships** grid.
8. The **Display** menu provides:
  - a. **Redline View** to compare each item to its previous Released revision
  - b. **Save Layout** to allow you to save the layout of the relationship grid
9. The **Share** menu displays options to export search results available for an item.

### 2.3.4 Relationships Content Area

A **Relationships** content area can display a **Relationships grid** or a **Relationships structure**, such as a multiple level Tree Grid View.

**Note:** For further information regarding Tree Grid View usage, refer to the *Aras Innovator – Tree Grid View Administrator Guide*.

In a **Relationships** content area, single clicking on an item highlights the **item** for further actions. Right-clicking highlights the **item** and opens its context menu.

In a **Relationships grid**, double-clicking opens the Item in a new tab. Standard Aras Innovator search procedure are applicable to the **Relationships** grids.

### 2.3.5 Relationship Context Menu

A **Relationship** context menu contains commands for a relationship and related items. This menu can be accessed by right clicking on an item.

The **Relationship** context menus are dynamic and variable. The list of available actions depends on the type and life cycle state of the related item, the relationship type, and permissions. Actions that are common to most relationship context menus are explained here.

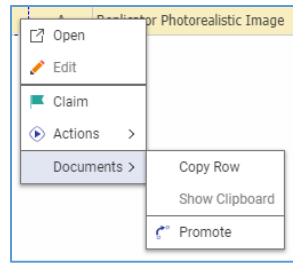


Figure 13.

- **Open** displays an Item in a new tab in view mode.
- **Edit** claims the item and launches it in edit-in-row mode.
- **Done Editing** unclaims the item, saves changes made to it in its table row, and stops edit-in-row mode.
- **Discard Changes** unclaims the item, undoes changes made to it in its table row, and stops edit-in-row mode.
- **Replace** replaces the Item in its table row with another Item while retaining any existing relationship information.
- **Remove** deletes the Item from the table.
- **Claim** explicitly locks an item to prevent editing by other users and keeps the item locked/claimed between sessions. For details, refer to section [2.4 Item Claiming](#).
- **Unclaim** releases the claim from the item.
- **Actions** → **Add Item(s) to Change** opens the **Choose Change Item** dialog to select the change type with an option to create a new change or add to an existing change. For details, refer to section [13 Change Management](#).
- **Copy Row** copies the Item to the clipboard.
- **Show Clipboard** opens the Clipboard Manager dialog.
- **Promote** takes the item to its next life cycle state if any is available. For details, refer to section [4.3 Manual Promotion and Revise](#).

## 2.4 Item Claiming

Claiming an item explicitly locks the item to prevent editing by other users. Users can edit only items claimed by themselves. An unclaimed item automatically becomes claimed when a user puts it in Edit mode, and it becomes unclaimed when the user exits Edit mode.

An item can be claimed by selecting the **Claim** flag  on the item page or by right-clicking on the item from a search grid or relationship grid and selecting **Claim**.

When you claim an item, the item will display a green claim flag . When an item is claimed by another user, the item will display a red claim flag .

## 3 Permissions

### 3.1 Viewing Permissions

To view the current Permissions for an Item:

1. Open the Item.
2. On the Item toolbar and click **More** → **Permissions**. The **Permissions** submenu appears. The current Item Permission is checked.

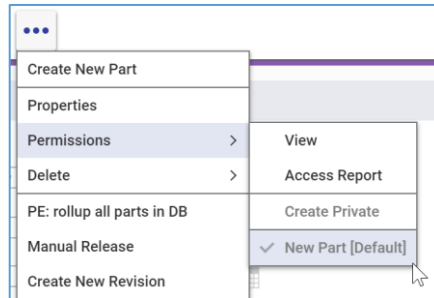


Figure 14.

3. Click **View**. The view of the current Item Permission appears showing the **Access** accordion with identities and their rights.

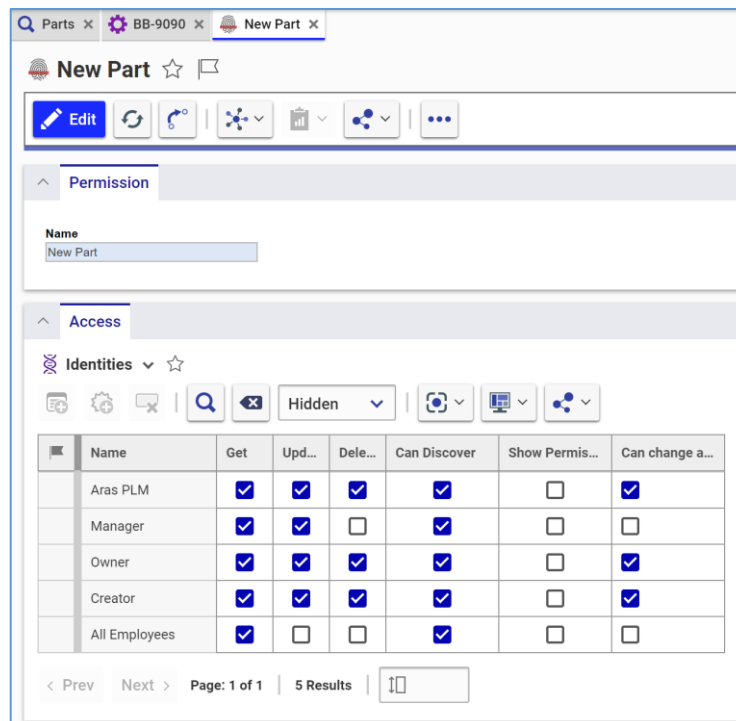


Figure 15.

## 3.2 Understanding Permissions

A Permission defines what access rights a user or group has to an Item. Permissions are assigned to an ItemType to define security for all Items of that type. Permissions can also be assigned to Life Cycle states to change the security of an Item over its lifetime. Permissions define which Identities may:

- **Get** to retrieve Items in viewing mode.
- **Update** to claim and edit Items.
- **Delete** to delete Items.
- **Can Discover** to search for Items.
- **Show Permission Warning** to show a warning message if access is restricted to the identity.
- **Can Change Access** to change Item's Permissions.

Some standard Aras Innovator Identities that can be tied to Permissions are explained here:

- **Manager** is responsible for the management, usage, and review of the Item and is defined as the **Designated User** property on the Item. This property is optional.
- **Owner** is the Item owner and is defined as the **Assigned Creator** property on the Item. This property is optional.
- **Creator** is the creator of the Item and is defined as the **Created By** property on the Item. The Creator is assigned automatically and cannot be changed.

**Note:** The Aras PLM system Identity represents the change management process and does not contain any individuals. Its permissions should not be changed.

## 3.3 Defining Private Permissions

A Private Permission is a Permission set specifically for a given Item, overriding its current Item Permission. The configured Permission is unique to the given Item and is not shared with any other Items. Private Permissions can be set for confidential Items to prevent viewing by other Identities. Once set, the Private Permission controls access to the Item by imposing a new set of access rights.

It is possible to roll back to the default Item Permission. Once the default permission is reapplied, the Private Permission is deleted.

The **Can Change Access** grant in an Item Permission defines who can create Private Permissions. By default, the Item Creator and Assigned Creator have this grant. Aras Administrators can define who can change access rights for Items.

The following subsections assume that the user has change access rights.

### 3.3.1 Creating Private Permissions

To create a private permission for an Item, follow this procedure:

1. Open the Item that requires a private permission.
2. On the Item toolbar, click **Edit**.
3. Click **More → Permissions → Create Private**. A new Private Permission appears, pre-populated with the grants defined by the default Permission.

4. Update Identities as needed. The following sections of this document can be referenced for additional information on updating Identities in Private Permissions:
  - a. [3.3.1.1 Adding Identities to Private Permission](#)
  - b. [3.3.1.2 Creating Identities for Private Permission](#)
  - c. [3.3.1.3 Replacing Identities in Private Permissions](#)
  - d. [3.3.1.4 Removing Identities from Private Permissions](#).
5. Select the appropriate access for each Identity in the **Access** grid.
6. Click **Done** to create the Private Permission.

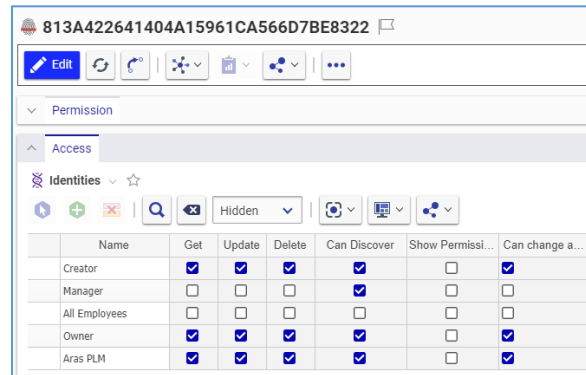


Figure 16.

7. On the **Item** toolbar, click **Done**. The new Private Permission is assigned to the Item and takes control of the access rights to the Item.

### 3.3.1.1 Adding Identities to a Private Permission

Use the following procedure to add identities to a Private Permission:

1. Click **Add Identities**  on the **Access** accordion toolbar on the Private Permission view. The search dialog appears.
2. Search for and select the necessary Identity.
3. Select OK to add the Identity to the Private Permission.

### 3.3.1.2 Creating Identities for Private Permission

Use the following procedure to create identities and add them to a Private Permission:

1. Click **New Identity**  on the **Access** accordion toolbar on the Private Permission view. A **new row** appears at the bottom of the grid.



|  | Name          | Get                                 | Upd...                              | Dele...                             | Can Discover                        | Show Permis...           | Can change a...                     |
|--|---------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------------------|-------------------------------------|
|  | Aras PLM      | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> |
|  | Manager       | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            |
|  | Owner         | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> |
|  | Creator       | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> |
|  | All Employees | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            |
|  |               | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/>            |

Figure 17.

2. In the **Name** cell of the **new row**, enter a name for the new Identity.
3. Click **Save** to save the new Identity.
4. (Optional) To add Members to an Identity, right-click the new identity and click **Open**. This opens the new Identity.

The screenshot shows the 'Test Group' configuration window. The 'Identity' tab is selected, displaying fields for Name (Test Group), Is Alias (checkbox), Description, and Users Passwords (Maximum Password Age, Password History Length). The 'Member' tab is also visible, showing a list of identities and a search bar.

Figure 18.

5. Click **Edit** to configure the new Identity and add Members.
6. Click **Done** to save your changes.

### 3.3.1.3 Replacing Identities in Private Permissions

Use the following procedure to replace identities in a Private Permission with an existing Identity:

1. Right-click the Identity to be replaced and then click **Replace**. The search dialog appears.

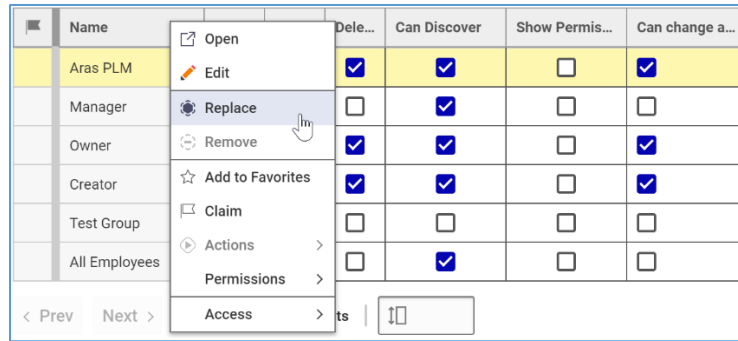


Figure 19.

2. Search for and select the necessary Identity, which replaces the previous Identity. The grants are retained for the selected Identity.

### 3.3.1.4 Removing Identities from Private Permissions

Use the following procedure to remove identities from a Private Permission:

1. Click on the Identity to be removed to select it.
2. Click **Delete Access**  on the **Access** accordion toolbar.
3. Click **Save** to save the Permission and remove the Identity.

### 3.3.2 Rolling Back to Default Permissions

Use the following procedure to restore the default Permission on an Item:

1. Open the Item.
2. Click **Edit** on the Item toolbar.
3. On the **Item** toolbar, click **More** → **Permissions** → **Current Item Permission [Default]**. A blue checkmark appears next to the default Permission, which takes control of the access rights to the Item. The Private Permission is deleted.

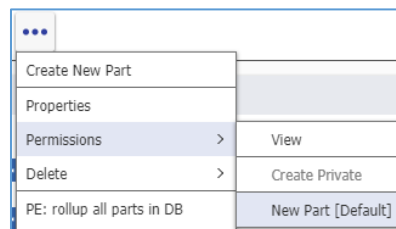


Figure 20.

4. Click **Done**.

**Note:** Once a Private Permission is assigned to an Item, the Item cannot be directly deleted through the Item toolbar using standard delete procedures. Use the procedure outlined above to roll back to the default permissions and avoid errors.

## 4 Versioning and Promoting

ItemTypes in the Product Engineering application are versionable. Versionable Items keep a history of changes by storing a new generation of the item each time a change is made.

Every Item in Aras Innovator can be promoted through a defined Life Cycle, which is a series of states that an Item traverses during its existence. Most ItemTypes in the Product Engineering application have their own unique default Life Cycle.

In this section, the term *Item* refers to a versionable Item with a Life Cycle.

### 4.1 Item Revisions and Versions

The sequence of Claim, Save, and Unclaim actions results in the creation of a new generation of the Item. When an Item reaches the Released state in its Life Cycle, no further changes can be made to the Released revision. Any further changes to the Item require it to be revised to create a new revision in the Preliminary state.

#### 4.1.1 Version Exploration

The **Item versions** dialog displays all versions of an Item. To access this dialog, either:

- In the Item Search Grid, right-click the Item and click **Navigate → Versions**.
- Open the Item, on the **Item** toolbar, click **Navigate**  **→ Versions**.

|  | Part Number | Revision | Name    | Type      | State       | Cost    | Changes                  | generation |
|--|-------------|----------|---------|-----------|-------------|---------|--------------------------|------------|
|  | Child A     | A        | Child A | Component | Preliminary | 10.0000 | <input type="checkbox"/> | 1          |
|  | Child A     | A        | Child A | Component | Preliminary | 15.0000 | <input type="checkbox"/> | 2          |
|  | Child A     | A        | Child A | Component | Preliminary | 17.0000 | <input type="checkbox"/> | 3          |
|  | Child A     | A        | Child A | Assembly  | Preliminary | 29.0000 | <input type="checkbox"/> | 4          |
|  | Child A     | A        | Child A | Assembly  | Preliminary | 30.0000 | <input type="checkbox"/> | 5          |
|  | Child A     | A        | Child A | Assembly  | Preliminary | 35.0000 | <input type="checkbox"/> | 6          |
|  | Child A     | A        | Child A | Assembly  | Preliminary | 30.0000 | <input type="checkbox"/> | 7          |

Figure 21.

To examine details specific to an old version of an Item, double-click on the corresponding row in the **Item versions** dialog.

**Note:** Previous versions of Items are read-only and cannot be edited.

To view where an old version of an Item is used, right-click the old version in the **Item versions** dialog and click **Where Used**.

|  | Part Number | Revisi... | Name    | Type      | State       | Cost    | Changes                  | generation |
|--|-------------|-----------|---------|-----------|-------------|---------|--------------------------|------------|
|  | Child A     | A         | Child A | Component | Preliminary | 10.0000 | <input type="checkbox"/> | 1          |
|  | Child A     | A         | Child A | Component | Preliminary | 15.0000 | <input type="checkbox"/> | 2          |
|  | Child A     | A         | Child A | Component | Preliminary | 17.0000 | <input type="checkbox"/> | 3          |

Figure 22.

## 4.1.2 Version Deletion

To delete a version of an Item, a user must have **Delete** permission rights granted for the current Life Cycle state of the Item. For details on access rights, refer to section [3. Permissions](#). This subsection assumes the user has **Delete** rights.

**Note:** Claimed Items cannot be deleted.

To delete an old version of an Item:

1. Go to the **Item versions** dialog for the Item.
2. Double-click on the version of the Item to be deleted to open it.
3. On the Item toolbar, click **More**  → **Delete** → **Purge This Version**. A confirmation dialog appears.
4. In the confirmation dialog, click **Purge** to delete the specific version of the Item.

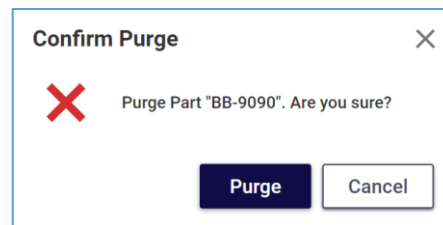


Figure 23.

It is also possible to delete the current version of an Item using the **Purge This Version** action, which can be accessed from the Item view or through the context menu in the Item Search Grid.

**Note:** If the current version of the Item is the only version existing in Aras Innovator, its purging deletes the Item completely. To avoid unwanted deletion, check Item versions before purging the current version.

## 4.2 Item History

The Product Engineering application can track changes made to an Item at both the item and property levels, including who made the change and when. By default, the following actions result in the creation of a history record: Add, Update, Promote, Reset or Set Default Lifecycle State, and Delete.

To view the history of an Item, access the **Item History** dialog:

1. Open the Item.
2. On the Item toolbar, click **Navigate**  → **History**.

The **Item History** dialog shows a grid of changes including time stamps, Identities, Actions, Comments, Item states, Revisions, and Generations.

| When [...]             | Who [...]                   | Action  | Comment [...] | Item State  | Rev | Gen | Created_on_tick ▲  |
|------------------------|-----------------------------|---------|---------------|-------------|-----|-----|--------------------|
| 10/16/2017 10:08:00 AM | <a href="#">Mike Miller</a> | Add     |               | Preliminary | A   | 1   | 63683189283760700  |
| 11/7/2017 1:06:00 PM   | <a href="#">Terry Adams</a> | Update  |               | Preliminary | A   | 1   | 636831896180851700 |
| 11/7/2017 1:37:00 PM   | <a href="#">Mike Miller</a> | Promote |               | Released    | A   | 1   | 636831896190018000 |

Figure 24.

## 4.3 Manual Promotion and Revise

Items can be released and/or revised either via change processes or manually. The change management processes can automatically release the Items where Life Cycles are automatically controlled by Workflows if these Items are in the affected items list. If an item has never been released or revised via a change process, it is possible to release and revise it manually. This section describes the manual process. For details on the release and revise features with change management, refer to [Section 13. Change Management](#).

Manual release and revise features are available only when all the following criteria are true:

- No generations of the Item have been previously released or revised via a change process.
- The Item is not claimed.
- The user performing the action is the Assigned Creator of the Item.

### 4.3.1 Manual State Promotion

In the default Item Life Cycle, the Assigned Creator can manually promote an Item to its next life cycle state if the requirements for manual release and revision are met, and if there is a state available.

An Item can be promoted from either the Item view or its context menu in the Items Search Grid.

To promote an Item using its view:

1. Open the Item.
2. On the Item toolbar, click **Promote** .
3. In the promotion dialog, select the desired state to promote the Item to by either:
  - a. Double-clicking the state to which the item should be promoted.

- b. Clicking the state to promote to and then clicking **Promote to Selected State**.

To promote an Item using its context menu:

1. Go to the Items Search Grid.
2. Right-click the **Item** and then click **Promote**.

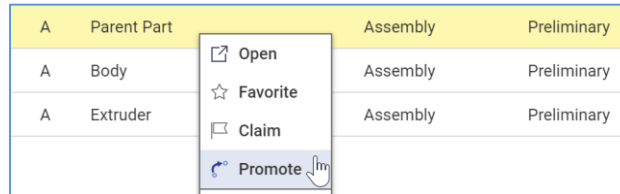


Figure 25.

3. In the promotion dialog, select the desired state to promote the Item to by either:
  - a. Double-clicking the state to which the item should be promoted.
  - b. Clicking the state to promote to and then clicking **Promote to Selected State**.

### 4.3.2 Manual Release

In addition to the manual promotion feature, you can release an Item in one action. In the default Item Life Cycle, the Assigned Creator can manually promote an Item to the Released state if the requirements for manual release are met by using either the Item view or its context menu in the Items Search Grid.

To release an Item using its view:

1. Open the Item.
2. On the Item toolbar, click **More**  **→ Manual Release**.

The promoted Item gets the **Released** value in the **State** field. The **plus** button also appears in the **Revision** field for manual revision.

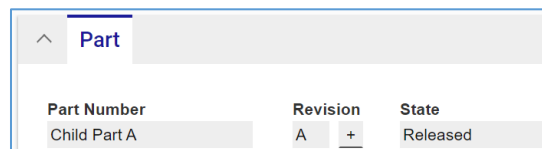


Figure 26.

To release an Item using its context menu:

1. Go to the Items Search Grid.
2. Right-click the **Item** and then click **More → Manual Release**.

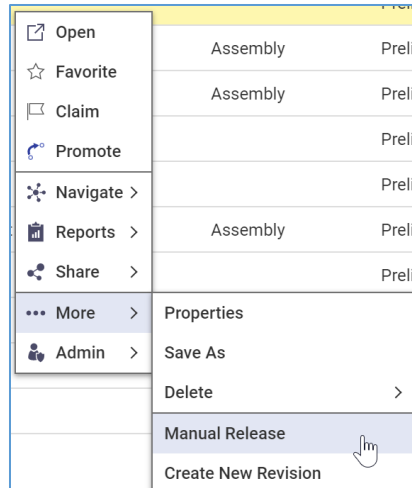


Figure 27.

### 4.3.3 Manual Revise

The Assigned Creator can manually create a new major revision of an Item if the above-mentioned requirements for manual revision are met and if the Item is in the Released state either from the Item view or from the Item's context menu in the Items Search Grid.

To revise an Item manually using its view:

1. Open the Item.
2. Use one of the following options:
  - a. On the Item toolbar, click **More**  → **Create New Revision**.
  - b. Click the **+** button next to the **Revision** field and select **OK** in the confirmation prompt.

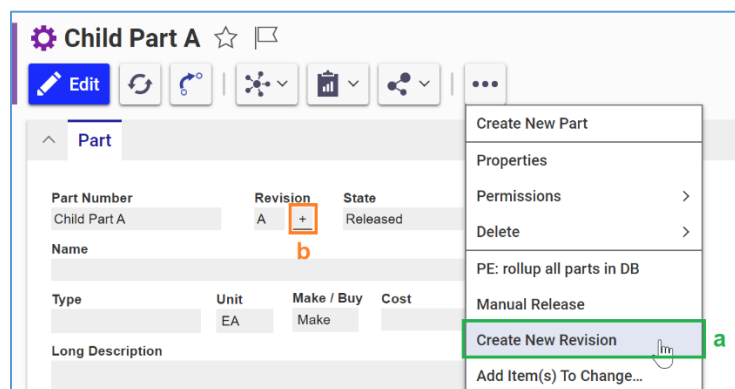


Figure 28.

The revised Item gets the new major value in the **Revision** field and the Preliminary state. Under **Revision**, the **plus** button disappears.

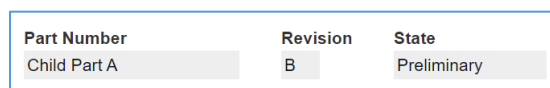


Figure 29.

Use the following procedure to revise an Item using its context menu:

1. Go to the Items Search Grid.
2. Right-click the **Item**, and then click **More → Create New Revision**.

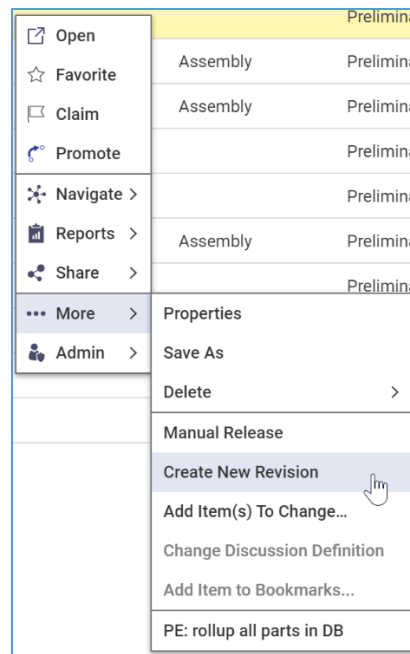


Figure 30.

**Note:** If you cannot see the latest Item generation in a Search Grid, click **Search** to reload the Grid. Another option is to use the Version exploration procedure.



## 5 Parts

Parts are the basic component of any Bill of Materials (BOM) management application. Parts can have a classification, such as Assembly, Component, Material or Software. A part can be bought or made in-house. It can be associated with other items. It can have alternate or substitute parts. It can have a list of approved manufacturers (AML) for making the part, and a list of approved vendors (AVL) for purchasing the part. It can have associated documentation, such as drawings and specifications. It can have a Bill of Material (BOM) and/or it can be part of another part's BOM. It can be the top-level assembly for a product model.

The following figure shows the underlying Data Model for a Part. Each of the relationships shown in the figure is described in detail in the following sections.

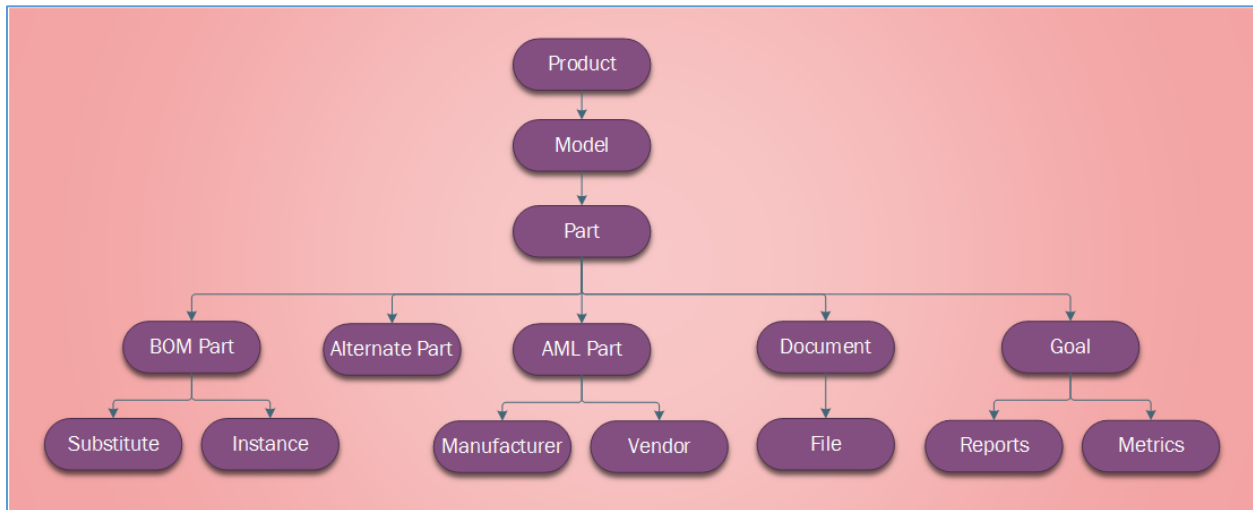


Figure 31.

Each part has a life cycle associated with it. Refer to 5.3 Part Life Cycle, [4.3. Manual Promotion and Revise](#) and [13. Change Management](#) for details.

Part reports, such as a Multilevel BOM report or a BOM Costing Report, are available to output information.

### 5.1 Creating a Part

This section outlines creating a Part in a standard instance of Aras Innovator. Your administrator may configure Parts as part of a different category in the Navigation panel, along with different or additional fields on the Part Form.

To create a Part:

1. Go to **Contents → Design → Parts**.
2. Click **Create New Part**. A New Part Form will appear. Enter the necessary information.

The screenshot shows the 'Part 1' form in the Aras INNOVATOR application. The form is divided into several sections. At the top, there are buttons for 'Save', 'Done', and 'Delete'. Below these, the 'Part' section contains fields for 'Part Number', 'Revision', 'State', 'Assigned Creator', 'Name', 'Designated User', 'Type', 'Unit', 'Make / Buy', 'Cost', 'Effective Date', 'Long Description', and 'Control Type'. The 'Part Number' field is highlighted in yellow. The 'Long Description' field has a 'Select an image' link. Below the form, there is a 'BOM' section with tabs for 'BOM Structure', 'Alternates', 'AML', 'Documents', 'CAD Documents', 'Goals', and 'Changes'. At the bottom, there is a toolbar with icons for 'Parts', 'Hidden', and other functions. A table at the very bottom has columns for 'Sequence', 'Part Number', 'Revision', 'Name', 'Type', 'Quantity', 'State', 'Unit', 'Reference Designator', and 'Changes'.

Figure 32.

- **Part Number** – (Required) This is the part number. The Part Number must be unique.
  - **Name** – This is the name for the Part.
  - **Type** – This is the classification type for the Part. The classification dialog displays the Part classification tree. The Part classification tree can be adjusted and extended as needed by an administrator.
  - **Unit** – This is the unit of measure for the Part:
  - **Make/Buy** – This specifies whether the Part is made in-house or whether it is bought from a manufacturer.
  - **Long Description** – This is where you can provide additional information about the Part.
  - **Assigned Creator** – This is the user responsible for the design or the content of the Part. The Assigned Creator is often identified as the Owner of the item, who may be different from the user creating the Part in the system.
  - **Designated User** – This is the user responsible for the management, usage, and review of the Part.
  - **Effective Date** – This is the date when a Part actually goes into production or the process that signifies its availability. This can be different from the Release date, which is when the Part reaches the Released state in its life cycle.
  - **Select an image** – This allows you to select an image using the Image Browser to generate a thumbnail for the Part.
  - **Extended Classification** – This allows you to assign values to Items based on the Item's extended classification.
3. Once you have entered in the necessary information, click **Save** or **Done** to save your changes and create the Part.

## 5.2 Part Relationships

A Part can have relationships to other Items. The Relationships accordion keeps these relationships grouped in separate Relationships tabs. Each Relationships tab represents a single relationship type.

The screenshot displays the Aras Part Relationships interface for Part BB-9090. The interface is divided into two main sections. The top section, titled 'Part', contains various fields for part information: Part Number (BB-9090), Revision (A), State (Preliminary), Assigned Creator, Name (Non-Threaded 63mm Bottom Bracket), Designated User, Type (Component), Unit (EA), Make / Buy (Buy), Cost, Effective Date (7/15/2021 12:00:00 AM), Long Description, Changes Pending (checkbox), and Control Type. The bottom section features a red-bordered accordion menu with tabs for different relationship types: BOM, BOM Structure, Alternates, AML, Documents, CAD Documents, Goals, and Changes. The 'BOM' tab is currently selected and highlighted.

Figure 33.

- **BOM** – A single-level BOM editor that enables you to view and manage the direct children of a Part. For more details, refer to section [6.1 Part BOM Relationships](#).
- **BOM Structure** – A multi-level, read-only view of the Part's BOM. For more details, refer to section [6.2 BOM Structures](#).
- **Alternates** – A list of alternate Parts for a Part. For more details, refer to section [6.7 Alternates](#).
- **AML** – An Approved Manufacturing List for a Part. For more details, refer to section [7.7 Approved Manufacturer List \(AML\)](#).
- **Documents** – A list of documents for a Part. For more details, refer to section [8.4 Managing Part Documents](#).
- **CAD Documents** – A list of CAD documents, such as 2D/3D models and drawings for a Part. For more details, refer to section [9.6 Managing CAD Documents in Parts](#).
- **Goals** – The Goals set for this Part. For more details, refer to section [6.11 Design Goals](#).
- **Changes** – Displays a list of change items, such as Problem Report, ECR or ECN, where the Part is an affected item. For more details, refer to section [13. Change Management](#).

## 5.3 Part Life Cycle

Each Part has a Life Cycle associated with it. The following figure shows the default Life Cycle Map of a Part. A newly created Part is assigned the Preliminary state by default. It can be released manually or via a change process. Refer to the [4.3 Manual Promotion and Revise](#) section for the details on manual release. The Product Engineering application provides change management processes, which are used to release Parts after appropriate reviews and approvals. During the change processes, a Part goes through different life cycle states depending on the step in the workflow. For more details, refer to section [13. Change Management](#).

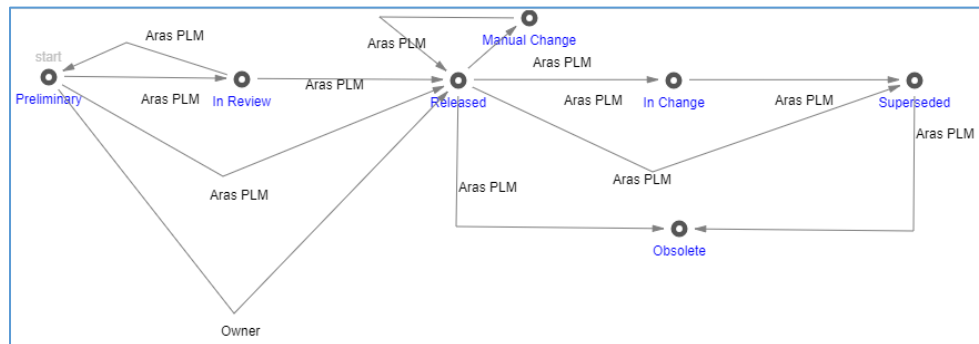


Figure 34.

The current State of a Part can be viewed in the following places:

- The **State** column of the Parts Search Grid.

| Part Number | Revi... | Name     | Type     | State       | Cost | Chan...                  |
|-------------|---------|----------|----------|-------------|------|--------------------------|
| MP2942      | A       | Body     | Assembly | Preliminary |      | <input type="checkbox"/> |
| MP2954      | A       | Extruder | Assembly | Preliminary |      | <input type="checkbox"/> |

Figure 35.

- The **State** read-only property box of the Part form.

MP2942

Part

Part Number: MP2942

Revision: A

State: Preliminary

Figure 36.

- The **Life Cycle** dialog, where the current State is highlighted.

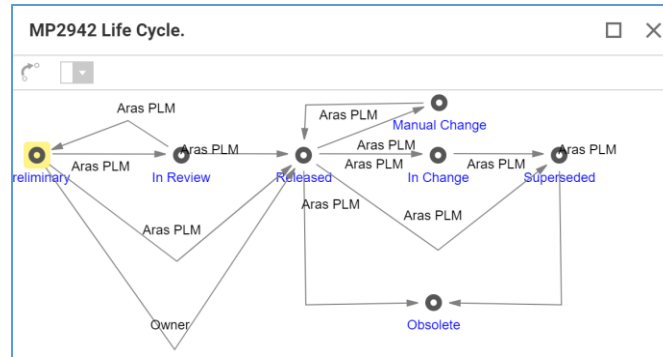


Figure 37.

To access the **Life Cycle** dialog for a Part:

1. Open the Part
2. On the Part toolbar, click **Navigate**  → **Life Cycle**.

## 5.4 Part Deletion

It is possible to delete all or some versions of a Part. In the former case, the Part is deleted from Aras Innovator completely. For details on the latter case, refer to section [4.1.2 Version Deletion](#).

To delete a Part completely, a user must have the **Delete** rights granted in Permissions for all Life Cycle states the Part has including the current one. For example, if the Part is currently in the Released state and was in Preliminary before, the user must have the **Delete** rights granted in Permissions for both states. An error message will be provided if the user does not have Delete rights. For details on access rights, refer to section [3. Permissions](#). This subsection assumes the user has these rights.

You can delete a Part either directly from the Part form, or by using the context menu in the Parts Search Grid.

To delete a Part completely through the Part Form:

1. Open the Part.
2. On the Part toolbar, click **More**  → **Delete** → **Delete All Versions**. A notification dialog appears.
3. Click **OK** to delete the Part completely. Click **Cancel** to cancel.

To delete a Part using the context menu:

1. Go to the Part Search Grid.
2. Right-click the **Part** → **Delete All Versions**. A notification dialog appears.
3. Click **OK** to delete the Part completely. Click **Cancel** to cancel.

## 6 Bill of Materials

A Part may be an assembly of other Parts keeping them in a Bill of Materials (BOM). A BOM is a list of components, parts, materials, other assemblies, and the quantities of each that are needed to put together this Part. A BOM is sometimes called a parts list, product structure, or associated list. In process industries, it is also known as a formula, recipe, or ingredients list.

A Part with a BOM is a Parent that has Parts as Children. In technical terms, it is a Part that has *Part BOM Relationships* to other Parts. A Child can be a single Part or another assembly (Parent) creating a multi-level BOM structure.

A Part may have Alternates. Alternates are Parts that can replace this Part globally in the scope of a company. For more details on this topic, refer to section [6.7 Alternates](#). A Part can have Substitutes, which are Parts that can replace this Part in a specific BOM only. For more details on this topic, refer to section [6.8 Substitutes](#).

A Child Part can be used in a Parent multiple times. Each of the child instances may need to be identified individually to provide its specific X, Y and Z coordinates or Reference Designation information. Use the **Instances** tab of a BOM relationship to specify instance information.

The Goals of a Parent Part are automatically rolled up from all its Children. For more details on Goals, refer to section [6.11 Design Goals](#).

### 6.1 Part BOM Relationships

A Part BOM relationship is a relationship between a Parent Part and its Child Part defined in the **BOM** relationships accordion tab of the Parent Part.

To view a Part BOM relationship:

1. Open the Parent Part item.
2. Go to the **BOM** tab.
3. In the **BOM** grid, right-click a Child → **BOM** → **Open**. The view of the Part BOM relationship between the Parent and the Child appears.

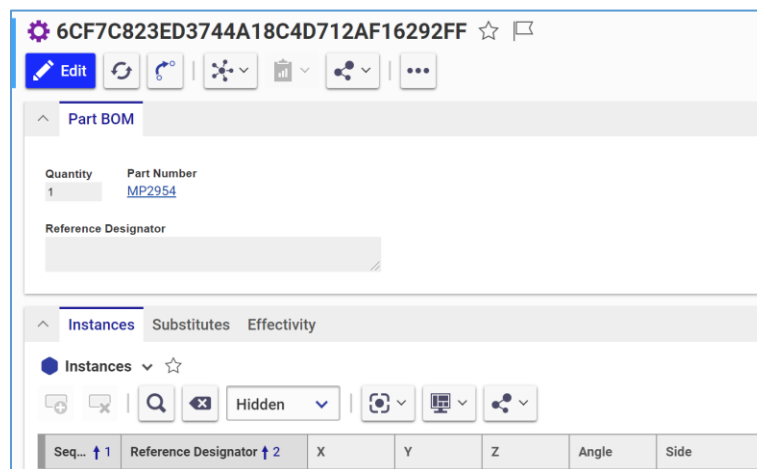


Figure 38.

- **Quantity** – The number of Parts used in the assembly.
- **Part Number** – The part number.

- **Reference Designator** – Used primarily in electronic assemblies to list labels of each Instance.

The relationships accordion of the Part BOM contains three tabs:

- **Instances** – Refer to section [6.6 Instances](#) for details.
- **Substitutes** – Refer to section [6.8 Substitutes](#) for details.
- **Effectivity** – Refer to section [6.10 Effectivity Management](#) for details.

## 6.2 BOM Structures

A multi-level BOM is a structure of Parts and is often designed to be configurable. This approach enables you to have a single product structure for numerous product configurations.

Aras Innovator provides functionality to resolve a BOM structure using:

- Structure Resolution, which identifies Part versions depending on a Resolution Mode. Refer to section [6.9 Applying Structure Resolution Modes](#) for details.
- Effectivity, which identifies valid Parts to be used under different conditions. Refer to section [6.10 Effectivity Management](#) for details.

To view the BOM Structure for a Parent Part:

1. Open the Parent Part.
2. Go to the **BOM Structure** accordion tab of a Parent Part.

| Part Number | Revision | State       | Sequence | Quantity | Claimed By | Name     | Effectivity | Reference Designator |
|-------------|----------|-------------|----------|----------|------------|----------|-------------|----------------------|
| MP2942      | A        | Preliminary | 1        | 1        |            | Body     |             |                      |
| MP2954      | A        | Preliminary | 2        | 1        |            | Extruder |             |                      |

Figure 39.

**Note:** The BOM Structure tab leverages a Tree Grid View to display the multi-level BOM structure for the Part. You can refer to the *Aras Innovator - Tree Grid View Administrator Guide* for information on the available toolbar options, along with instructions on configuring the items displayed in the BOM Structure.

## 6.3 Setting up a BOM

There are two ways to set up a BOM Relationship:

1. Create all the necessary Children and then create a Parent and populate the **BOM** tab with its Children. This approach builds a new assembly using existing Parts and is outlined in [6.3.1 Creating a BOM from Existing Parts](#).
2. Create a Parent and then create its Children using the BOM tab. This approach is convenient for populating nested BOMs or creating BOM structures quickly without getting into the details of each Child. However, it is still necessary to complete each Child later. This approach is outlined in [6.3.2 Creating a BOM from the BOM tab](#).

**Note:** An Administrator can restrict Aras Innovator to one approach.

### 6.3.1 Creating a BOM from Existing Parts

This approach is used for building a new assembly using existing Parts. With this approach, all necessary Children are created prior to creating the Parent Part and populating the BOM tab of the Parent Part with Children. This section assumes that a Parent Part and all necessary Children exist.

To set up a BOM:

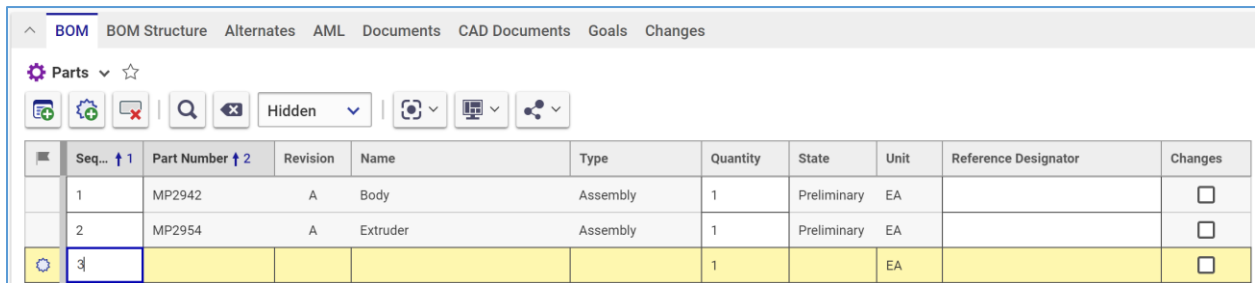
1. Open the Parent Part.
2. On the Parent toolbar, click **Edit**.
3. Go to the **BOM** accordion tab.
4. Click **Add Parts**. The Search dialog appears.
5. Search for and select the necessary Children to be added to the **BOM** table.
6. Enter relationship information, such as Quantity and Reference Designator, as needed. Double-click the **Quantity** cells of each Child and enter the required quantity.
7. Click **Save** or **Done**. The BOM is created.

### 6.3.2 Creating a BOM from the BOM tab

This approach is convenient for populating nested BOMs or creating BOM structures quickly without getting into the details of each Child. With this approach, all Children are created directly from within the BOM tab. Note that you still need to complete the details for each Child later. This section assumes that a Parent Part is created.

To set up a BOM:

1. Open the Parent Part.
2. On the Parent toolbar, click **Edit**.
3. Go to the **BOM** accordion tab.
4. Click **New Part**. A new row appears in the **BOM** table.



| Seq... | Part Number | Revision | Name     | Type     | Quantity | State       | Unit | Reference Designator | Changes                  |
|--------|-------------|----------|----------|----------|----------|-------------|------|----------------------|--------------------------|
| 1      | MP2942      | A        | Body     | Assembly | 1        | Preliminary | EA   |                      | <input type="checkbox"/> |
| 2      | MP2954      | A        | Extruder | Assembly | 1        | Preliminary | EA   |                      | <input type="checkbox"/> |
| 3      |             |          |          |          | 1        |             | EA   |                      | <input type="checkbox"/> |

Figure 40.

5. Click the **Part Number** cell and enter a unique part number for the Child.

**Note:** The Part Number must be unique. It is the only required Part property provided by the user.

6. Enter any additional data by clicking on the corresponding cell in the BOM table. Refer to section [5.1 Creating a Part](#) for a description of the Part fields.
7. Click **Save** or **Done**. The BOM and the Children are created.



## 6.4 Editing a BOM

You can update an existing BOM to add, replace, or remove Children from the BOM.

### 6.4.1 Adding New Children

To add new Children to an existing BOM, refer to section [6.3 Setting up a BOM](#).

### 6.4.2 Replacing Children

To replace Children in a BOM:

1. Open the Parent Part.
2. On the Parent toolbar, click **Edit**.
3. Go to the **BOM** accordion tab.
4. Right-click the Child to be replaced and then click **Replace**. The Search dialog appears.
5. Search for and select a replacement Part. The Child in the BOM is replaced with the Part selected. The BOM relationship data, such as Quantity and Reference Designator, is retained.
6. Click **Save** or **Done**. The Children are replaced.

### 6.4.3 Removing Children

To remove Children from a BOM:

1. Open the Parent Part.
2. Click **Edit** on the Parent toolbar.
3. Go to the **BOM** accordion tab.
4. Click on the Child or Children to be deleted and then click **Delete Part BOM**.
5. Click **Save** or **Done**. The selected Children are removed from the BOM.

## 6.5 Editing Parts within a BOM

To edit Children (related Parts) from the Parent BOM:

1. Open the Parent Part.
2. Click **Edit** on the Parent toolbar.
3. Go to the **BOM** accordion tab.
4. Right-click a Child and click **Edit**. The Child becomes claimed, and its row becomes editable. Enter any edits by clicking on the corresponding cell in the BOM table.
5. Right-clicking on the Child and clicking **Discard Changes** will discard any changes. The Child becomes unclaimed, its properties are rolled back, and its row becomes read-only.

**Note:** After you save the Child's Parent, the Child remains claimed with all new data saved. It is possible to continue editing the Child in its row, but it is not possible to discard this data.

6. Right-click the Child Part and then click **Done Editing**. The Child becomes unclaimed, **its row** becomes read-only, and its new property values are saved.
7. Click **Done** on the toolbar of Child's Parent to save your changes.

**Note:** To edit properties that are not present in the BOM table, right-click the Child Part and then click **Open**. The Child Part form appears allowing you to edit those properties.

8. Click **Save** or **Done** to save your changes.

## 6.6 Instances

A Parent Part can have multiple Instances of the same Child in its BOM. It may be necessary to identify information about each instance. For example, an assembly (Parent) can be fastened using several of the same bolt. It may be needed to specify the location of each instance.

The Part BOM form contains the **Instances** relationship tab. This relationship tab displays the properties of each Child Part Instance for its identification in a BOM:

- **Reference Designator** – This is a unique Instance label. The Reference Designator data is used primarily in electronic diagrams. For example, a circuit board consists of four Instances of the same resistor. Each Instance has its own Reference Designator to identify it unambiguously within the electrical schematic or on the printed circuit board.
- **X** – This is an X coordinate of Instance location in the assembly.
- **Y** – This is a Y coordinate of Instance location in the assembly.
- **Z** – This is a Z coordinate of Instance location in the assembly.
- **Angle** – This is an angle of Instance location in the assembly.
- **Side** – This is from where (Top or Bottom) the Instance location is measured.

**Note:** The coordinates and angle do not have a unit of measure. Their values must be entered in the same units.

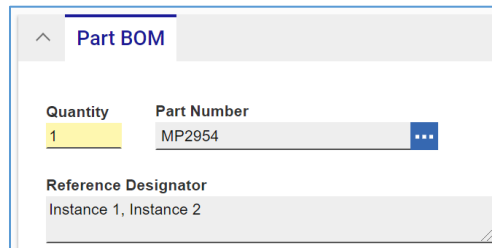
### 6.6.1 Specifying Instances

To specify Instances of a Child Part in a BOM:

1. Claim the Parent Part.
2. Open the Part BOM relationship as described in [6.1 Part BOM Relationships](#).
3. Click **Edit** on the Part BOM toolbar.
4. Click **New BOM Instance** on the **Instances** relationship accordion toolbar. A new row appears in the Instances table.
5. Click on the corresponding cell in the row to enter the appropriate information.
6. If it is necessary to remove an Instance:
  - a. Click the Instance to select it.
  - b. Click **Delete BOM Instance** on the **Instances** relationship accordion toolbar.
  - c. On the **Part BOM** toolbar, click **Save**. The **Instance** disappears.

**Note:** The **Instance** disappears immediately if you do not save the Part BOM after it was added.

- On the Part BOM view, in the Reference Designator text box, enter a comma-separated list of Reference Designators for all the Instances.



The screenshot shows a 'Part BOM' form. It has a 'Quantity' field with the value '1' and a 'Part Number' field with the value 'MP2954'. Below these is a 'Reference Designator' field containing the text 'Instance 1, Instance 2'.

Figure 41.

- On the Part BOM toolbar, click **Done**.
- Go to the **Parent** Part form. The list of Instances is displayed in the corresponding Child's row.

|  | Seq... | Part Number | Revision | Name     | Type     | Quantity | State       | Unit | Reference Designator   | Changes                  |
|--|--------|-------------|----------|----------|----------|----------|-------------|------|------------------------|--------------------------|
|  | 1      | MP2942      | A        | Body     | Assembly | 1        | Preliminary | EA   |                        | <input type="checkbox"/> |
|  | 2      | MP2954      | A        | Extruder | Assembly | 1        | Preliminary | EA   | Instance 1, Instance 2 | <input type="checkbox"/> |

Figure 42.

- Unclaim the Parent Part.

## 6.7 Alternates

A Part may have an Alternate Part that can replace it globally. A Part is interchangeable with its Alternate regardless of where the Part is used in BOM structures.

The presence of an Alternate in a Part is for informational purposes only. To replace a Part with its Alternate in a BOM, you must follow the Child Part replacement procedure outlined in section [6.4.2. Replacing Children](#).

### 6.7.1 Assigning Alternates

To add an existing Part to another Part as an Alternate:

- Open the Part where the Alternate will be added.
- Click **Edit** on the **Part** toolbar.
- Go to the Alternates relationship tab.
- Click **Add Parts** on the Alternates accordion toolbar. The Search dialog appears.
- Search for and select a Part to add as an Alternate for the current Part.
- Click **Save** or **Done** on the Part toolbar.

### 6.7.2 Creating Alternates

To create a new Part and add it to another Part as an Alternate:

- Open the Part where the Alternate will be added.

2. Click **Edit** on the **Part** toolbar.
3. Go to the Alternates relationship tab.
4. Click **New Part** on the Alternates accordion toolbar. A blank row appears in the Alternates grid.

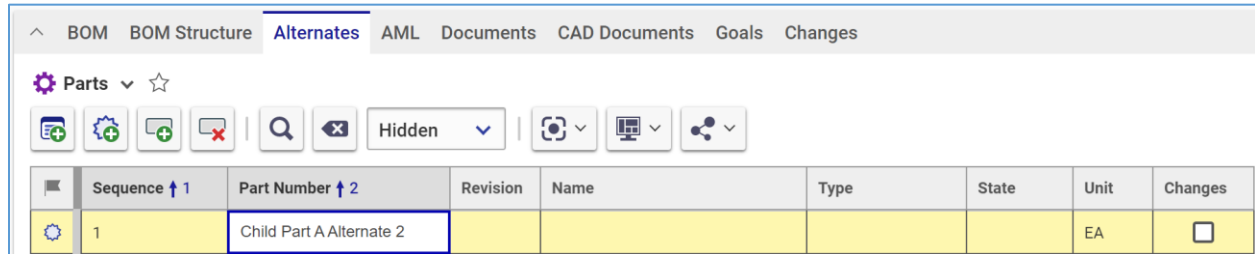


Figure 43.

5. Click the **Part Number** cell to specify a unique part number for the new Part.
6. If it is necessary to provide additional data, click the corresponding cell and enter the data.
7. Click **Save** or **Done** on the Part toolbar.

### 6.7.3 Replacing Alternates

To replace an Alternate in a Part:

1. Open the Part where the Alternate will be added.
2. Click **Edit** on the **Part** toolbar.
3. Go to the Alternates relationship tab.
4. Right-click the Alternate and then click **Replace**. The Search dialog appears.
5. Search for and select a replacement Alternate Part.
6. Click **Save** or **Done** on the **Part** toolbar.

### 6.7.4 Removing Alternates

To remove an Alternate from a Part:

1. Open the Part where the Alternate will be added.
2. Click **Edit** on the **Part** toolbar.
3. Go to the Alternates relationship tab.
4. Click the Alternate and then click **Delete Part Alternate** on the Alternates relationship tab toolbar.
5. Click **Save** or **Done** on the **Part** toolbar.

## 6.8 Substitutes

A Part may have a Substitute Part that can replace it only in specific BOM structures.

The presence of a Substitute in a BOM is for informational purposes only. To replace a Part with its Substitute in a BOM, you must follow the Child Part replacement procedure outlined in section [6.4.2 Replacing Children](#).

## 6.8.1 Assigning Substitutes

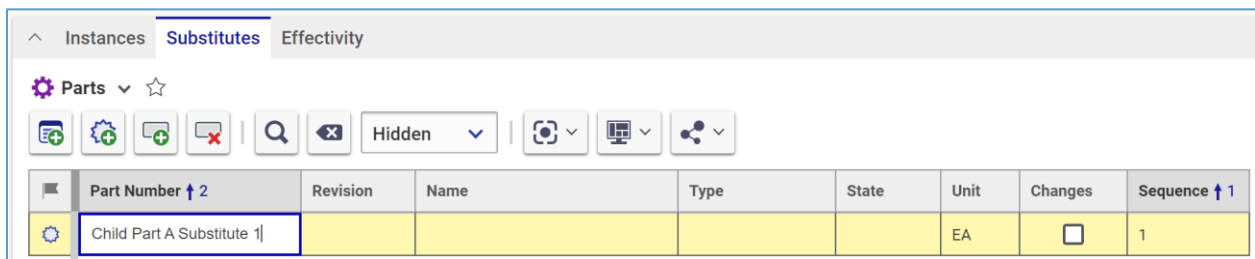
To add a Substitute Part for a Child Part in a BOM:

1. Claim the Parent Part.
2. Open the Part BOM relationship as described in [6.1 Part BOM Relationships](#).
3. Click **Edit**.
4. In the Substitutes tab, click **Add Parts** on the accordion toolbar. The Search dialog appears.
5. Search for and select a Part to be added as a Substitute for the current Part.
6. Click **Done** on the **Part BOM** toolbar.
7. Unclaim the Parent.

## 6.8.2 Creating Substitutes

To create a new Part and add it as a Substitute Part for a Child Part in a BOM:

1. Claim the Parent Part.
2. Open the Part BOM relationship as described in [6.1 Part BOM Relationships](#).
3. Click **Edit**.
4. In the Substitutes tab, click **New Part** on the Substitutes accordion toolbar. A blank row appears in the Substitutes grid.



| Part Number ↑ 2           | Revision | Name | Type | State | Unit | Changes                  | Sequence ↑ 1 |
|---------------------------|----------|------|------|-------|------|--------------------------|--------------|
| Child Part A Substitute 1 |          |      |      |       | EA   | <input type="checkbox"/> | 1            |

Figure 44.

5. Click the **Part Number** cell to specify a unique part number for the new Part.
6. If it is necessary to provide additional data, click the corresponding cell and enter the data.
7. Click **Save** or **Done** on the Part toolbar.

### 6.8.3 Replacing Substitutes

To replace a Substitute in a Part:

1. Claim the Parent Part.
2. Open the Part BOM relationship as described in [6.1 Part BOM Relationships](#).
3. Click **Edit**.
4. In the Substitutes tab, right-click the Substitute and then click **Replace**. The Search dialog appears.
5. Search for and select a replacement Substitute Part.
6. Click **Done** on the **Part BOM** toolbar.
7. Unclaim the Parent.

### 6.8.4 Removing Substitutes

To remove a Substitute from a Part:

1. Claim the Parent Part.
2. Open the Part BOM relationship as described in [6.1 Part BOM Relationships](#).
3. Click **Edit**.
4. In the Substitutes tab, click the Substitute and then click **Delete BOM Substitute** on the Substitutes accordion toolbar.
5. Click **Save** or **Done** on the Part toolbar.
6. Unclaim the Parent.

## 6.9 Applying Structure Resolution Modes

The following Structure Resolution Modes are available to dynamically resolve a multi-level Part BOM structure using item versions as specified in the Resolution Mode:

- **Default** – The version of the related Part as stored on the Part BOM Relationship is used. This is the standard resolution.
- **Latest** – The latest version of the related Part is used in the resolution.
- **Latest Released** – The latest Released version of the related Part is used in the resolution.
- **Latest Released or Latest** – The latest Released version of the related Part is used in the resolution. If there are no Released versions of the related Part, then the latest version is used.

To apply a Structure Resolution Mode to a BOM structure:

1. Open the Part.
2. In the **BOM Structure** accordion tab, click **Parameters** .
3. Select the desired Structure Resolution Mode in the Value drop-down list of the StructureResolutionMode row of the Parameters dialog.

The image shows a 'Parameters' dialog box with a close button (X) in the top right corner. Inside the dialog, there are two columns: 'Property' and 'Value'. The 'Property' column has a single entry 'StructureResolutionMode' highlighted in yellow. The 'Value' column shows a dropdown menu with the following options: 'Default' (selected), 'Default', 'Latest', 'Latest Released', and 'Latest Released Or Latest'. There are green and red checkmark icons in the top left of the dialog.

| Property                | Value   |
|-------------------------|---------|
| StructureResolutionMode | Default |

Figure 45.

4. Click **Apply**. The BOM is resolved to a Part structure according to the selected Mode.

To explain how different Resolution Modes work, this section uses a set of Parts that have multiple versions as follows:

- **Child A:** A – Released; B – Preliminary
  - **Child A.1:** A – Preliminary
  - **Child A.2:** A, B, C – Released
  - **Child A.3:** A, B – Released; C – Preliminary
- **Child B:** A, B – Released
  - **Child B.1:** A – Released
  - **Child B.2:** A, B, C – Released; D – Preliminary
- **Child C:** A – Preliminary
  - **Child C.1:** A – Preliminary
  - **Child C.2:** A, B – Released

From these Parts, a simple BOM structure example is created as described in the following table.

Table 1: A simple BOM example

| BOM Level |   |   | Part      | Revision | State       |
|-----------|---|---|-----------|----------|-------------|
| 0         |   |   | Parent    | A        | Preliminary |
|           | 1 |   | Child A   | B        | Preliminary |
|           |   | 2 | Child A.1 | A        | Preliminary |
|           |   | 2 | Child A.2 | C        | Released    |
|           |   | 2 | Child A.3 | C        | Preliminary |
|           | 1 |   | Child B   | B        | Released    |
|           |   | 2 | Child B.1 | A        | Released    |
|           |   | 2 | Child B.2 | B        | Released    |
|           | 1 |   | Child C   | A        | Preliminary |
|           |   | 2 | Child C.1 | A        | Preliminary |
|           |   | 2 | Child C.2 | B        | Released    |

### 6.9.1 Default Resolution Mode

The Default Resolution Mode uses the versions of related child Parts as stored on the BOM Relationships. This is the standard resolution in Aras Innovator.

The following image shows the resolved structure when the Default resolution mode is selected:

| Part Number | Revision | State       |
|-------------|----------|-------------|
| Child A     | B        | Preliminary |
| Child A.1   | A        | Preliminary |
| Child A.2   | C        | Released    |
| Child A.3   | C        | Preliminary |
| Child B     | B        | Released    |
| Child B.1   | A        | Released    |
| Child B.2   | B        | Released    |
| Child C     | A        | Preliminary |
| Child C.1   | A        | Preliminary |
| Child C.2   | B        | Released    |

Figure 46.

### 6.9.2 Latest Resolution Mode

When the Latest resolution mode is selected, the structure is resolved starting from the top using the latest versions of child items. When a child item is resolved to its latest version, then its child items are resolved using this resolution mode. This is done recursively.

The Latest resolution mode does not consider the release state of the child Parts.

The following image shows the resolved structure when Latest resolution mode is selected:



| Part Number | Revision | State       |
|-------------|----------|-------------|
| Child A     | B        | Preliminary |
| Child A.1   | A        | Preliminary |
| Child A.2   | C        | Released    |
| Child A.3   | C        | Preliminary |
| Child B     | B        | Released    |
| Child B.1   | A        | Released    |
| Child B.2   | D        | Preliminary |
| Child C     | A        | Preliminary |
| Child C.1   | A        | Preliminary |
| Child C.2   | B        | Released    |

Figure 47.

The latest revision D of Child B.2 is shown where this Part version is in the Preliminary state.

### 6.9.3 Latest Released Resolution Mode

When the Latest Released resolution mode is selected, the structure is resolved starting from the top using the latest versions of child items that are in Released state. When the child item is resolved to its latest Released version, then its child items are resolved using this resolution mode. This is done recursively.

If the related child item has no versions that are in Released state, then a blank row is displayed for that item. There is no further processing on that item.

The following image shows the resolved structure when Latest Released resolution mode is selected:

| Part Number | Revision | State    |
|-------------|----------|----------|
| Child A     | A        | Released |
| Child A.2   | C        | Released |
| Child A.3   | B        | Released |
| Child B     | B        | Released |
| Child B.1   | A        | Released |
| Child B.2   | C        | Released |
|             |          |          |

Figure 48.

The latest Released versions of child Parts are resolved. A blank row is displayed for Child C because it has no version that is in the Released state.

### 6.9.4 Latest Released Or Latest Resolution Mode

When the Latest Released or Latest resolution mode is selected, the structure is resolved starting from the top using the latest versions of child items that are in Released state. If there are no versions that are in Released state, then the latest version is used. When the child item is resolved to a version, then its child items are resolved using this resolution mode. This is done recursively.

The following image shows the resolved structure when Latest Released or Latest resolution mode is selected:

| Part Number | Revision | State       |
|-------------|----------|-------------|
| Child A     | A        | Released    |
| Child A.2   | C        | Released    |
| Child A.3   | B        | Released    |
| Child B     | B        | Released    |
| Child B.1   | A        | Released    |
| Child B.2   | C        | Released    |
| Child C     | A        | Preliminary |
| Child C.1   | A        | Preliminary |
| Child C.2   | B        | Released    |

Figure 49.

The latest Released versions of child Parts are resolved. Child C does not have any version that is in the Released state. Therefore, its latest Preliminary version is resolved.

## 6.10 Effectivity Management

Effectivity Management is incorporated into Part BOM Structures. Using effectivity, conditions for which to include or exclude items in specific configurations can be expressed.

The PE application provides commonly used effectivity variables out-of-the-box along with the ability to easily add additional effectivity variables.

Effectivity scopes organize lists of relevant effectivity variables. For example, if one product uses Model and Unit effectivity and another uses Model, Lot, and Date effectivity, two different effectivity scopes can be created. The PE application provides an Aras Part BOM Scope with Model, Unit and Date variables. Additional effectivity scopes and variables can be created by an Administrator.

The following table defines terms, acronyms, and abbreviations used for Effectivity in the PE application.

**Table 2:** Effectivity Terminology

| Term                   | Definition                                                                                                                                                         |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Effectivity            | Identification of valid uses of an item in a structure if this item's use is conditional.                                                                          |
| Effectivity Variable   | A variable that influences effectivity decisions, such as a date, model, unit, batch, lot, plant, etc.                                                             |
| Effectivity Scope      | A built-in ItemType that represents a list of relevant Effectivity Variables.<br>For example, the Scope may contain a Model, Unit, and Date Effectivity Variables. |
| Effectivity Expression | A representation of an effectivity condition in the Boolean Expression Language.<br>For example:<br>Model = "Model X" and (Unit >= 10 and Unit <=20)               |

| Term                 | Definition                                                                                                                                                                                                                                                                                                                                |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Effectivity Criteria | <p>The criteria to resolve a structure by effectivity.</p> <p>Effectivity conditions set on relationships are evaluated against the input criteria to determine the inclusion or exclusion of conditional items in the resolved structure. For example: resolve the structure for criteria:</p> <p>(Model = "Model X" and Unit = 15).</p> |

To access the Effectivity Services, go to **Contents → Administration → Effectivity Services**.

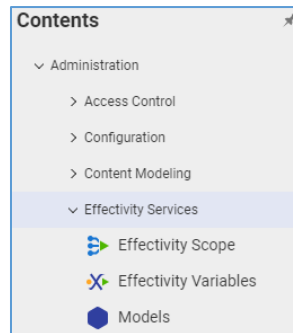


Figure 50.

### 6.10.1 Viewing Effectivity on BOM Structure tab

Effectivity, which is set on Part BOM relationships, can be viewed from the **BOM Structure** tab.

To view an Effectivity on a multi-level BOM:

1. Open the **BOM Structure** tab of the Part.
2. BOM Effectivities are displayed in the Effectivity column.

| BOM BOM Structure Alternates AML Documents CAD Documents Goals Changes |          |             |          |          |           |                                  |                                                               |
|------------------------------------------------------------------------|----------|-------------|----------|----------|-----------|----------------------------------|---------------------------------------------------------------|
| Part Number                                                            | Revision | State       | Sequence | Quantity | Locked By | Name                             | Effectivity                                                   |
| MP2942                                                                 | B        | Preliminary | 5        | 1        |           | Body                             |                                                               |
| MP2954                                                                 | B        | Preliminary | 10       | 1        |           | Extruder                         |                                                               |
| MP2961                                                                 | A        | Released    | 15       | 1        |           | Shipping                         |                                                               |
| MP2956                                                                 | A        | Released    | 20       | 1        |           | Gantry                           |                                                               |
| MP2968                                                                 | A        | Released    | 25       | 1        |           | XY Stage                         |                                                               |
| MP2943                                                                 | B        | Preliminary | 30       | 1        |           | Build Platform                   |                                                               |
| MP2952                                                                 | B        | Preliminary | 35       | 1        |           | Electronics                      |                                                               |
| MP2967                                                                 | A        | Released    | 40       | 1        |           | Wiring                           |                                                               |
| MP2938                                                                 | B        | Preliminary | 45       | 1        |           | Additional Parts                 |                                                               |
| MP2935                                                                 | A        | Released    | 5        | 2        |           | Filament Spool Holder            | Model = Replicator OR (Model = [Replicator 2] AND Unit <= 99) |
| MP2361                                                                 | A        | Preliminary | 6        | 1        |           | Filament Heavy Duty Spool Holder | Model = [Replicator 2] AND Unit >= 100                        |
| MP2962                                                                 | B        | Released    | 10       | 8        |           | Side Clip Short                  |                                                               |
| MP2937                                                                 | A        | Released    | 15       | 2        |           | Acrylic Side Cover               |                                                               |
| MP4000                                                                 | A        | Preliminary | 16       | 1        |           | Filament                         | Date >= [12/19/2018]                                          |

Figure 51.

- To view full details, right-click a BOM row → **View Effectivity**. The Manage Effectivity Expressions split pane appears on the right side of the **BOM Structure** tab with a list of effectivities set on the selected BOM.

Figure 52.

## 6.10.2 Managing Effectivity on a BOM

### 6.10.2.1 Setting Effectivity on a BOM

Multiple Effectivities can be set on a single Part BOM Relationship. These Effectivities can use the same or a different Effectivity Scope depending on how your business uses multiple Effectivities.

To set Effectivity on a BOM using the Part Item view:

- In the Manage Effectivity Expressions split pane, click **Create effectivity** . The Effectivity Expression dialog appears.

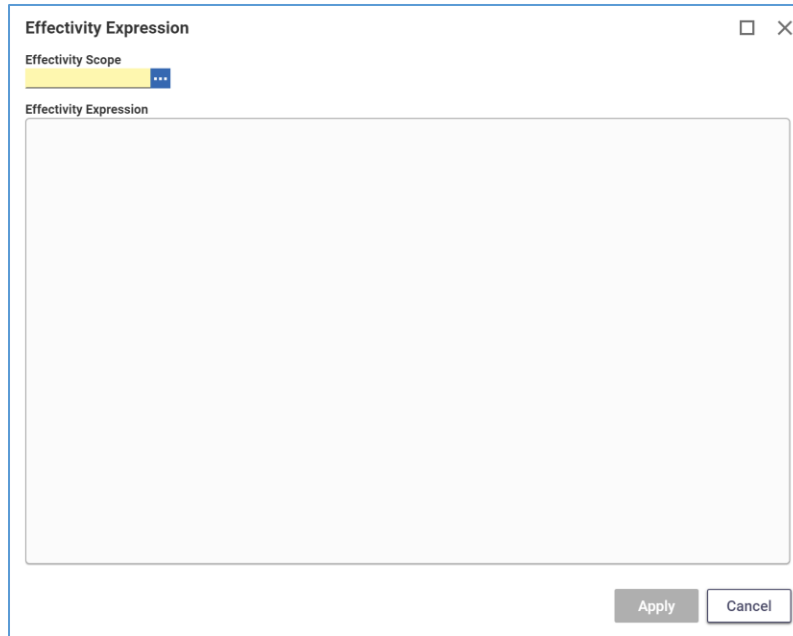


Figure 53.

2. Set the Effectivity Scope field by either typing or using the ellipsis to search for an Effectivity Scope.
3. Enter your Effectivity Expression in the Effectivity Expression text field.

**Note:** Aras Innovator supports both simple and complex Effectivity Expressions using the free-form Boolean expression language. Effectivity Expressions can be created using Boolean operators, like `equal`, `greater than and equal`, `less than and equal`, `not`, `and`, and `or`. The Effectivity Expression editor provides guided editing user interface as an Effectivity Expression is entered.

4. Click **Apply**.

#### 6.10.2.2 Updating Effectivity on a BOM

To update Effectivity on a BOM:

1. In the Manage Effectivity Expressions split pane, right-click on an Effectivity → **Edit Effectivity**.
2. Update the Effectivity Expression as required.
3. Click **Apply**. The updated Effectivity Expression appears in the Manage Effectivity Expressions split pane. The updated Effectivity Expression is set on the BOM.

#### 6.10.2.3 Removing Effectivity from a BOM

To remove an Effectivity from a BOM with the help of a Part Item view:

1. In the Manage Effectivity Expressions split pane, click on the Effectivity to be deleted.
2. Click **Delete effectivity** .
3. In the confirmation dialog, click **OK** to delete the effectivity.

### 6.10.3 Resolving a BOM Structure by Effectivity

When no Effectivity Criteria is yet specified, the unfiltered BOM structure is displayed. The unfiltered BOM structure can contain extra Parts. In the example below, Parts for two different models are both included.

|  |        |   |             |    |   |                                           |                        |
|--|--------|---|-------------|----|---|-------------------------------------------|------------------------|
|  | MP2954 | B | Preliminary | 10 | 1 | Extruder                                  |                        |
|  | MP1705 | B | Released    | 5  | 2 | MK7 Thermal Core                          |                        |
|  | MP2607 | D | Released    | 10 | 2 | Cartridge Heater 40W 24V Right Angle Exit |                        |
|  | MP2966 | C | Released    | 20 | 1 | Wire-Filament Extruder Guide              |                        |
|  | MP2959 | C | Released    | 25 | 1 | Nylon Spacer with Flange                  |                        |
|  | MP2506 | B | Released    | 30 | 2 | MK8 Nozzle 0.4mm                          | Model = [Replicator 2] |
|  | MP2505 | A | Preliminary | 30 | 1 | Nozzle 0.3mm                              | Model = Replicator     |
|  | MP1708 | B | Released    | 35 | 2 | MK7 Heatsink                              |                        |
|  | MP2685 | C | Released    | 40 | 4 | Spacer Black 16.5mm length 3.2mm ID 7...  |                        |
|  | MP2675 | A | Released    | 45 | 2 | Tube Fitting .45in Tube                   |                        |
|  | MP2350 | C | Released    | 50 | 2 | Fan 24V 40x40x10 Hi Performance           |                        |
|  | MP2953 | D | Released    | 55 | 1 | Extruder Hardware                         |                        |
|  | MP2589 | A | Preliminary | 60 | 1 | Bar Mount                                 | Model = Replicator     |
|  | MP2590 | A | Released    | 60 | 1 | MK8 Bar Mount                             | Model = [Replicator 2] |
|  | MP1872 | C | Released    | 65 | 2 | Stepper Motor NEMA17                      |                        |

Figure 54.

To resolve a BOM by Effectivity:

1. On the **BOM Structure** toolbar, click **Set Effectivity Criteria**. The Effectivity Criteria Filter dialog appears.
2. Click the ellipsis in the Scope field.
3. Search for and select a proper Effectivity Scope. The selected Effectivity Scope populates the Scope field and Effectivity Variables of this Scope appear as editable rows.

Effectivity Criteria Filter

Scope

Aras Part BOM Scope

...

| Variable | Value |
|----------|-------|
| Unit     |       |
| Date     |       |
| Model    |       |

Figure 55.

4. Provide the Effectivity Criteria you need to resolve the BOM structure in the Value cells of the Effectivity Variables.

**Note:** It is possible to resolve a structure either fully or partially. To fully-resolve a structure, a value must be entered for every effectivity variable on the **Effectivity Criteria Filter** dialog for the selected scope. To resolve a structure partially, one or more effectivity variable values can be left blank. When left blank, those effectivity variables are ignored while evaluating if the Part BOM relationship is effective.

5. Click **Apply**.

**Note:** If the effectivities set within the BOM structure use more than one effectivity scope, then during structure resolution, the effectivities are evaluated for the scope selected on the Effectivity Criteria Filter dialog.

If there are multiple effectivities set on a single Part BOM relationship, they are evaluated using OR logic during structure resolution. This means that if at least one of the effectivities set on the relationship meet the effectivity criteria, the child item is evaluated as effective for the Parent and displayed in the resolved structure. IF your business requires evaluating such cases using AND logic, an administrator can change the configuration.

To return to the unfiltered structure, remove the Effectivity Criteria:

1. On the **BOM Structure** toolbar, click **Set Effectivity Criteria**.
2. Click **Clear**.
3. Click **Apply**.

## 6.11 Design Goals

### 6.11.1 Adding a Design Goal

You can add design goals to a Part item using the Goals tab in the relationship accordion. A design goal is either defined by cost or weight. The supplied values for each Child Part can then be used in a part structure to calculate the cost of a Parent Part. To add a design goal for a Part:

1. Open the Part.
2. Click **Edit**.
3. Go to the **Goals** tab.
4. Click **New Part Goal** on the Goals relationships toolbar. A new row appears in the Goals table.

| Goal ▲ | Target | Estimated | Calculated | Actual | Comments |
|--------|--------|-----------|------------|--------|----------|
| ▼      |        |           |            |        |          |
| Cost   |        |           |            |        |          |
| Weight |        |           |            |        |          |

Figure 56.

5. Enter the appropriate values for each cell of the new row:
  - **Goal** – (Required) Cost or Weight.
  - **Target** – The design target value for the cost or weight.
  - **Estimate** – The estimated value for the cost or weight.
  - **Calculated** – The roll-up calculated value for the cost or weight based on the child component values in the BOM. This value is not editable.
  - **Actual** – The actual value for the cost or weight. For cost, this may be the value that was negotiated with the vendor. For weight, this may be the value resulting from a measurement.
  - **Comments** – Remarks about the design goal.

**Note:** Goals and their calculations do not have a unit of measure. Their values must be entered in the same units.

6. Click **Save** to save your changes.

**Note:** A Part can have at most one Cost and one Weight Goal row. If there are several rows for Goals of the same type, you will receive an error upon saving the Part. Delete the extra row to resolve the error.

### 6.11.2 Rolling Up Costs

Costs are entered as either Target, Estimate, Calculated, or Actual. Based on the values supplied in the design goal, a calculated cost/weight rollup will occur automatically for a BOM. To initiate a roll up, you must specify at least a Target on the parent. The calculated value is determined using the values supplied in the design goal based on the following hierarchy from lowest to highest accuracy: Target, Estimated, Calculated, Actual.

In other words:

- If the Target and Estimated values are supplied, then the Estimated value is used.
- If the Target, Estimated and Calculated values are supplied, the Calculated value of the part is used.
- If Actual value is specified, it overrides any of the other values defined.

Specifying only Target values on the child parts will prevent a roll-up calculation. As

long as the children cost goal values are Target, they do not alter the parent's goal value. The Target goal for the Parent will remain as long as there is at least one child part that has only a Target value for its cost goal. However, as soon as all children have an Estimated or Actual goal value, the parent's value changes to Calculated. The parent's Goal value is automatically re-calculated whenever any Child has a Goal changed.

In the following example, Parent Part contains two immediate children: Child A and Child B.



Part Number: Parent, Revision: A, State: Preliminary

Name: Parent Part

Type: Assembly, Unit: EA, Make / Buy: Make, Cost (Calculated): 31.0000

| Part Number ↑ 2 | Revision | Name    | Quantity | Cost    |
|-----------------|----------|---------|----------|---------|
| Child A         | A        | Child A | 1        | 17.0000 |
| Child B         | A        | Child B | 1        | 14.0000 |

| Goal ↑ | Target  | Estimated | Calculated | Actual  |
|--------|---------|-----------|------------|---------|
| Cost   | 10.0000 | 15.0000   |            | 17.0000 |

| Goal ↑ | Target | Estimated | Calculated | Actual |
|--------|--------|-----------|------------|--------|
| Cost   |        | 14.0000   |            |        |

Figure 57.

Child A has a Target cost goal of 10.00, and an Estimated goal of 15.00, but the Actual cost is 17.00. Actual cost is used for the rollup calculation.

Child B has an Estimated cost goal of 14.00 so that is used for the rollup calculation. If a user supplies an Actual cost goal to Child B, then this value will be used for the rollup calculation.

To initiate a roll up, you must specify at least a Target on the parent. A user then assigns a cost goal on Parent Part with an Estimated goal of 30.00.

Now the calculated cost of Parent Part appears automatically and is 31.00.

**Note:** Goals and their calculations are unitless. Their values must be entered in the same units. Ensure that you enter Cost in the same currency and Weight in the same unit of mass.

## 6.12 BOM Reports

To present complex multi-level BOMs in a well-organized layout, Aras Innovator provides four standard types of Reports:

- **Multilevel BOM Report** – Shows all Parts of a BOM with their position in the hierarchy and their quantity.
- **BOM Quantity Rollup Report** – Shows total quantities of all Parts of a BOM.
- **BOM Costing Report** – Calculates the cost of a Parent from Actual or Estimated Costs of its Children.
- **Part Circular Reference Report** – Detects Circular References in a BOM.

To access these standard Reports, either:

- Go to **Contents** → **Parts** → **Search Parts**, right-click on a Part → **Reports**.
- Open a Part, and on the Part toolbar, click **Reports** , and then select the **Report** you need.

To print a Report, use the Print button  available in the top left corner of each Report tab. The print button uses standard browser print function. Depending on the browser and your specific configuration, various print choices, such as printing to a printer or saving to PDF, are available.

### 6.12.1 Multilevel BOM Report

The Multilevel BOM Report shows a BOM hierarchy where you can see the properties for each Part: BOM level, Quantity, AML Status, Manufacturer, and Manufacturer Part.

| Bill of Materials Report |             |                                               |          |            |                              | Generated on: 2/25/2019, 4:43:53 PM |
|--------------------------|-------------|-----------------------------------------------|----------|------------|------------------------------|-------------------------------------|
| BOM Level                | Part Number | Name                                          | Quantity | AML Status | Manufacturer                 | Manufacturer Part                   |
| 0                        | MP0101      | Makerbot Replicator                           | 1        |            |                              |                                     |
| 1                        | MP2942      | Body                                          | 1        |            |                              |                                     |
| 2                        | MP2667      | Spacer Black 5-16 in length .14in ID .25in OD | 1        |            |                              |                                     |
| 2                        | MP2322      | RGB LED Strip Common Anode                    | 1        |            |                              |                                     |
| 2                        | MP2660      | Spacer Black 1-2 in length .14in ID .25in OD  | 2        |            |                              |                                     |
| 2                        | MP2939      | Body Fan Assembly                             | 1        |            |                              |                                     |
| 3                        | MP2347      | Fan 24V 40x40x10 1m leads                     | 1        |            |                              |                                     |
| 3                        | MP2685      | Spacer Black 16.5mm length 3.2mm ID 7mm OD    | 2        |            |                              |                                     |
| 2                        | MP2940      | Body Hardware                                 | 1        |            |                              |                                     |
| 3                        | MP2295      | M3 Square Nut                                 | 44       |            |                              |                                     |
| 3                        | MP0370-004  | M3x16 Socket Cap Bolt                         | 38       | Approved   | McMaster-Carr Supply Company | 91290A120                           |
| 3                        | MP0370-006  | M3x25 Socket Cap Bolt                         | 4        | Approved   | McMaster-Carr Supply Company | 91290A125                           |
| 3                        | MP2728      | Support Blind Hole Mount                      | 5        |            |                              |                                     |
| 3                        | MP0370-008  | M3x35 Socket Cap Bolt                         | 2        | Approved   | McMaster-Carr Supply Company | 91290A135                           |

Figure 58.

### 6.12.2 BOM Quantity Rollup Report

The BOM Quantity Rollup Report displays a list of BOM Parts where for each Part displays the total quantity required for this BOM.

| Quantity Rollup Report for MP0101 |                                               |                     | Generated on: 2/25/2019, 4:46:58 PM |
|-----------------------------------|-----------------------------------------------|---------------------|-------------------------------------|
| Part Number                       | Name                                          | Calculated Quantity |                                     |
| MP2942                            | Body                                          | 1                   |                                     |
| MP2954                            | Extruder                                      | 1                   |                                     |
| MP2961                            | Shipping                                      | 1                   |                                     |
| MP2956                            | Gantry                                        | 1                   |                                     |
| MP2968                            | XY Stage                                      | 1                   |                                     |
| MP2943                            | Build Platform                                | 1                   |                                     |
| MP2952                            | Electronics                                   | 1                   |                                     |
| MP2967                            | Wiring                                        | 1                   |                                     |
| MP2938                            | Additional Parts                              | 1                   |                                     |
| MP1705                            | MK7 Thermal Core                              | 2                   |                                     |
| MP2667                            | Spacer Black 5-16 in length .14in ID .25in OD | 1                   |                                     |
| MP2935                            | Filament Spool Holder                         | 2                   |                                     |
| MP2459                            | M3 Thumb Nuts                                 | 4                   |                                     |
| MP1531                            | ABS Black 1kg Spool 1.75mm                    | 1                   |                                     |
| MP2295                            | M3 Square Nut                                 | 84                  |                                     |
| MP2584                            | PTFE Tubing 14in OD 18in ID                   | 2                   |                                     |
| MP2753                            | Shaft 8mm x 424mm                             | 2                   |                                     |
| MP2827                            | Nylon 440 screw                               | 6                   |                                     |

Figure 59.

### 6.12.3 BOM Costing Report

The BOM Costing Report shows the Cost Goal Rollup for a given BOM where each Part displays its level in the BOM and its properties at that level: Quantity, Cost value, and Cost Basis.

| Bill of Materials Costing Report |             |                                               |          |           |            | Generated on: 2/25/2019, 4:50:03 PM |
|----------------------------------|-------------|-----------------------------------------------|----------|-----------|------------|-------------------------------------|
| BOM Level                        | Part Number | Name                                          | Quantity | Cost      | Cost Basis |                                     |
| 0                                | MP0101      | Makerbot Replicator                           | 1        | 4760.7859 | Calculated |                                     |
| 1                                | MP2942      | Body                                          | 1        | 443.5369  | Calculated |                                     |
| 2                                | MP2667      | Spacer Black 5-16 in length .14in ID .25in OD | 1        | 0.0600    | Actual     |                                     |
| 2                                | MP2322      | RGB LED Strip Common Anode                    | 1        | 3.8000    | Actual     |                                     |
| 2                                | MP2660      | Spacer Black 1-2 in length .14in ID .25in OD  | 2        | 0.6500    | Actual     |                                     |
| 2                                | MP2939      | Body Fan Assembly                             | 1        | 9.9000    | Calculated |                                     |
| 3                                | MP2347      | Fan 24V 40x40x10 1m leads                     | 1        | 9.0000    | Actual     |                                     |
| 3                                | MP2685      | Spacer Black 16.5mm length 3.2mm ID 7mm OD    | 2        | 0.4500    | Actual     |                                     |
| 2                                | MP2940      | Body Hardware                                 | 1        | 5.6800    | Calculated |                                     |

Figure 60.

### 6.12.4 Circular Reference Report

A circular reference occurs in a series of parts or assemblies where a Child Part adds its Parent Part to its own BOM, causing an “infinite” series of references.

For example, a circular reference would occur if there is a Parent Part that has two Children A and B and a user were to edit Child A to add the Parent Part to its BOM. The BOM Relationship from Child A to the Parent Part creates a Circular Reference in this case.

When the Circular Reference Report is run, it detects these cases in a BOM structure, indicating where the circular references occur.

| List of circular dependencies between Parts |                                                         | Generated on: 2/25/2019, 5:07:55 PM                     |
|---------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| Loop #                                      | Source Part                                             | Referenced Part                                         |
| 1                                           | Child Part B.2 gen.5 (1AADE136F54B4D74A78F71B06EA71E52) | Parent Part gen.14 (73678DA3F16042E4A77729723FB71E2A)   |
|                                             | Parent Part gen.14 (73678DA3F16042E4A77729723FB71E2A)   | Child Part B gen.14 (C120442B993344A89B7C8654AD73C84E)  |
|                                             | Child Part B gen.14 (C120442B993344A89B7C8654AD73C84E)  | Child Part B.2 gen.5 (1AADE136F54B4D74A78F71B06EA71E52) |

Figure 61.

If there are no Circular References in a BOM, the Circular Reference Report displays an appropriate message.

| List of circular dependencies between Parts       |  | Generated on: 2/25/2019, 5:09:45 PM |
|---------------------------------------------------|--|-------------------------------------|
| No circular dependencies between Parts are found. |  |                                     |

Figure 62.

## 6.13 Comparing BOMs

To compare complex multi-level BOMs side-by-side, Aras Innovator provides the Structure Browser. Refer to section [11.2 Comparing Items](#) for information about comparing BOMs. All examples in that section are applicable to BOMs and Parts.

## 6.14 Finding Where a Part is Used

Aras Innovator offers the Where Used feature to determine where an Item is used by other Items. Refer to section [11.1 Finding Where an Item is Used](#) for information about how to find out where a Part is used. All examples in that section are applicable to BOMs and Parts.

## 6.15 Sorting BOM Structures

Columns in the **BOM Structure** accordion tab can be sorted in ascending or descending order by clicking on the column headers.

Sorting is applied to each BOM level recursively. When a level of Parts is sorted, then children of each Part are sorted within this Part.

| Part Number | Sequence | Revision | State       |
|-------------|----------|----------|-------------|
| Child A     | 1        | B        | Preliminary |
| Child A.1   | 1        | A        | Preliminary |
| Child A.3   | 3        | C        | Preliminary |
| Child A.2   | 2        | C        | Released    |
| Child B     | 2        | B        | Released    |
| Child B.2   | 2        | D        | Preliminary |
| Child B.1   | 1        | A        | Released    |

Figure 63.

To sort across multiple columns, press and hold the **Ctrl** key while clicking on the column headers.

| Part Number | Sequence | Revi...<br>↓ 2 | State<br>↑ 1 |
|-------------|----------|----------------|--------------|
| Child A     | 1        | B              | Preliminary  |
| Child A.3   | 3        | C              | Preliminary  |
| Child A.1   | 1        | A              | Preliminary  |
| Child A.2   | 2        | C              | Released     |
| Child B     | 2        | B              | Released     |
| Child B.2   | 2        | B              | Released     |
| Child B.1   | 1        | A              | Released     |

Figure 64.

## 7 Sourcing Items

To simplify the sourcing process, the Product Engineering (PE) application uses a data model based on the Part as the central item. A Part can be made in-house or purchased. In the latter case, the Part can have an AML (Approved Manufacturers List) to maintain one or more Manufacturer Parts associated with it. Each Manufacturer Part can be produced by only one Manufacturer and be sold by many Vendors. A Vendor may sell the same Part from more than one Manufacturer and can include all their available Parts with their Manufacturer information.

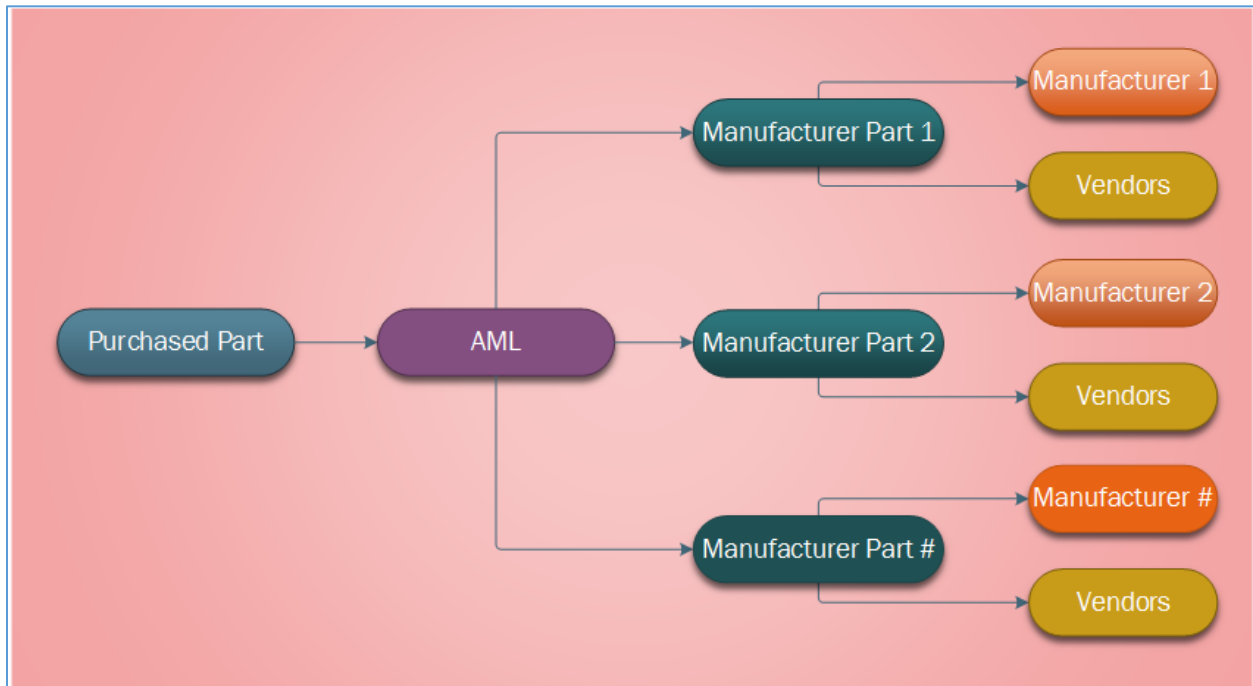


Figure 65.

Each Manufacturer Part has the Manufacturer property to denote who makes the Part. You can assign only one Manufacturer to a Manufacturer Part. If there are several Manufacturers making similar, equivalent, or identical Parts, you must create the corresponding number of Manufacturer Parts.

**Note:** By default, only members of the Component Engineering Identity can create and maintain Sourcing Items. An administrator can change this accessibility as needed.

The recommended workflow for Sourcing is the following sequence of steps:

1. Create Manufacturers.
2. Create Manufacturer Parts.
3. Create Vendors.
4. Approve Manufacturers, Manufacturer Parts, and Vendors.
5. Add Manufacturer Parts to the AML of the corresponding Part.

## 7.1 Creating a Manufacturer

A Manufacturer produces Manufacturer Parts. To create a Manufacturer:

1. Go to **Contents → Sourcing → Manufacturers**.
2. Click **Create New Manufacturer**. The view of a new Manufacturer appears. Enter the necessary information.

The screenshot shows a web interface for creating a new manufacturer. At the top, there's a title bar 'Manufacturer 1' with 'Save', 'Done', and 'Delete' buttons. Below this is a tabbed interface with 'Manufacturer' and 'Manufacturer Parts' tabs. The 'Manufacturer' tab is active and contains several input fields: 'Name' (highlighted in yellow), 'State', 'Main Phone', 'Contact Name', and 'Responsible' (which has a dropdown arrow). The 'Manufacturer Parts' tab is also visible and contains a table with the following columns: 'Part Number', 'Name', 'Description', 'State', 'Claimed By [...]', 'Unit Price', 'Unit', and 'Files'.

Figure 66.

- **Name** – (Required) The name of the Manufacturer. The name of the Manufacturer must be unique.
  - **Main Phone** – The number used to call the Manufacturer.
  - **Contact Name** – The name of the contact person for the Manufacturer.
  - **Responsible** – The Identity responsible for the Manufacturer. This role is often identified as the Owner of the Manufacturer item and may be different from the user creating the Manufacturer in the system.
3. Once you have entered the necessary information, click **Done** to save your changes.

## 7.2 Creating a Manufacturer Part

Manufacturer Parts contain a Manufacturer property to specify the Manufacturer that makes the Part. Only one Manufacturer can be assigned to a Manufacturer Part. If there are several Manufacturers making similar, equivalent, or identical Parts, a Manufacturer Part must be created for each of them.

To create a new Manufacturer Part:

1. Go to **Contents → Sourcing → Manufacturer Parts**.
2. Click **Create New Manufacturer Part**. The view of a new Manufacturer Part appears. Enter the necessary information.

Figure 67.

- **Part Number** – (Required) The Manufacturer Part's part number.
  - **Responsible** – The Identity responsible for the Manufacturer Part. This role is often identified as the Owner of the Manufacturer Part item and may be different from the user creating the Manufacturer in the system.
  - **Name** – The name of the Manufacturer Part.
  - **Unit** – The unit of measure for the Manufacturer Part.
  - **Unit Price** – The price per Manufacturer Part unit.
  - **Description** – Remarks about the Manufacturer Part.
  - **Manufacturer** – The Manufacturer of the Manufacturer Part.
3. If it is necessary to add a file, in the Files tab, select New File  to select a file for upload, or drag and drop the desired file into the Files area.
  4. Once you have entered the necessary information, click **Done** to save your changes.

## 7.3 Creating a Vendor

A Vendor sells Manufacturer Parts. You can assign multiple Vendors to a Manufacturer Part. To create a new Vendor:

1. Go to **Contents** → **Sourcing** → **Vendors**.
2. Click **Create New Vendor**. The view of a new Vendor appears.



**Vendor 1**

Save Done Delete

**Vendor**

Name

Main Phone Main Fax

Contact Name

Address

City

State Country Zip Code

Web Site

Description

**Parts**

Manufacturer Parts

Part Number ↑ 2 Name Manufacturer [...] State Unit Price Unit Files Catalog Number ↑ 1

Figure 68.

- **Name** – (Required) The name of the Vendor. The vendor name must be unique.
  - **Main Phone** – The phone number for the Vendor.
  - **Main Fax** – The fax number for the Vendor.
  - **Contact Name** – The name of the contact person for the Vendor.
  - **Address** – The vendor's street address.
  - **City** – The Vendor's city.
  - **State** – The Vendor's US State / Canadian Province.
  - **Country** – The Vendor's country.
  - **Zip Code** – The Vendor's postal code.
  - **Web Site** – The Vendor's website address.
  - **Description** – Remarks about the Vendor.
3. Assign Manufacturer Parts to the Vendor as described in section [7.3.1. Assigning Manufacturer Parts to a Vendor](#).
  4. If it is necessary, create new Manufacturer Parts and assign them to the Vendor as described in section [7.3.2. Creating Manufacturer Parts from a Vendor](#).
  5. Once you have entered the necessary information, click **Done** to save your changes.

### 7.3.1 Assigning Manufacturer Parts to a Vendor

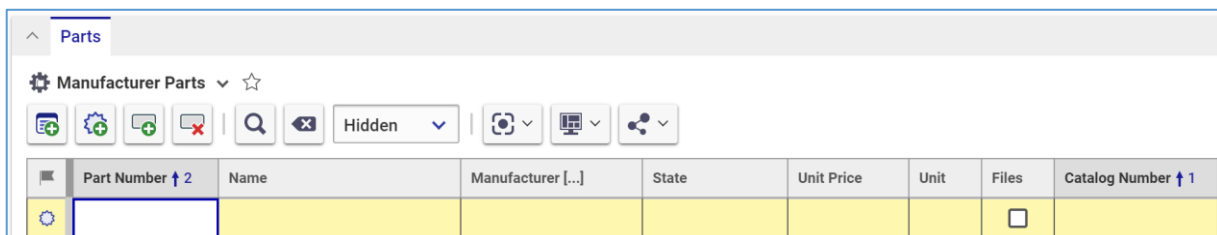
To add an existing Manufacturer Part to a Vendor:

1. Open the Vendor Item.
2. Click **Edit**.
3. Go to the **Parts** tab.
4. Click **Add Manufacturer Parts** . The Search dialog appears.
5. Search for and select a Manufacturer Part that the Vendor supplies.
6. Click **Save** or **Done** to save your changes.

### 7.3.2 Creating Manufacturer Parts from a Vendor

To create a new Manufacturer Part and add it to a Vendor:

1. Open the Vendor Item.
2. Click **Edit**.
3. Go to the **Parts** tab.
4. Click **New Manufacturer Part** . A blank row appears in the **Manufacturer Parts** grid.



| Part Number ↑ 2 | Name | Manufacturer [...] | State | Unit Price | Unit | Files                    | Catalog Number ↑ 1 |
|-----------------|------|--------------------|-------|------------|------|--------------------------|--------------------|
|                 |      |                    |       |            |      | <input type="checkbox"/> |                    |

Figure 69.

5. Enter the appropriate values for each cell of the new row.
6. Click **Save** or **Done** to save your changes.

### 7.3.3 Replacing Manufacturer Parts in a Vendor

To replace a Manufacturer Part in a Vendor:

1. Open the Vendor Item.
2. Click **Edit**.
3. Go to the **Parts** tab.
4. Right-click the **Manufacturer Part** to be replaced and then click **Replace**. The Search dialog appears.
5. Search for and select a replacement Manufacturer Part. The selected Manufacturer Part replaces the old one.
6. Click **Save** or **Done** to save your changes.

### 7.3.4 Removing Manufacturer Parts from a Vendor

To remove a Manufacturer Part from a Vendor:

1. Open the Vendor Item.
2. Click Edit.
3. Go to the **Parts** tab.
4. Click the **Manufacturer Part** to be deleted and then click **Delete Vendor Part** .
5. Click **Save** or **Done** to save your changes.

## 7.4 Life Cycle of Sourcing Items

The PE application has Sourcing Life Cycles for Manufacturer, Part AML, and Vendor. Each Sourcing Life Cycle is based on the same Life Cycle Map and can be configured by an Administrator as needed.

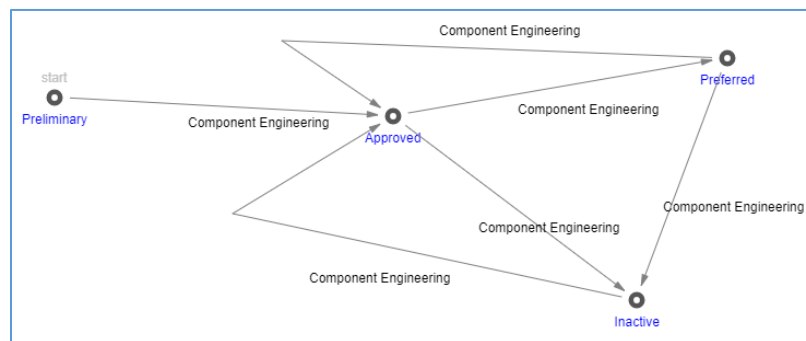


Figure 70.

When created, each Sourcing Item starts in the Preliminary state. To indicate their approval status, Items can be promoted to the Approved state. From the Approved state, the Item can further be promoted to the Preferred state. To deactivate an approved item that is no longer used, the item can be promoted to the Inactive state.

## 7.5 Promoting Sourcing Items

You can promote a Sourcing Item either from its view or from its context menu in the Search Grid. For details, refer to section [4.3 Manual Promotion and Revise](#).

**Note:** By default, only members of the Component Engineering identity have access rights to promote Sourcing Items. An Aras administrator can change this access if it is necessary.

## 7.6 Approved Vendors Report

The Approved Vendors Report provided in Aras Innovator displays a list of all approved Vendors. You can access the Approved Vendors Report by either:

- Going to **Contents** → **Sourcing** → **Vendors**: Right-click on a Vendor → **Reports** → **Approved Vendors Report**.

- Opening the Vendor: On the **Vendor** toolbar, click **Reports**  → **Approved Vendors Report**.

To print the Approved Vendors Report, use the Print button  available in the top left corner of the Report. The print button uses standard browser print function. Depending on the browser and your specific configuration, various print choices, such as printing to a printer or saving to PDF, are available.

## 7.7 Approved Manufacturer List (AML)

The Approved Manufacturer List (AML) is a list of Manufacturer Parts associated with a given Part.

### 7.7.1 Assigning Existing Manufacturer Parts to a Part

To assign a Manufacturer Part to a Part:

1. Open the Part.
2. Click **Edit**.
3. Go to the **AML** tab.
4. Click **Add Manufacturer Parts**  on the **AML** accordion toolbar. The Search dialog appears.
5. Search for and select the necessary Manufacturer Parts to add to the AML.
6. Click **Save** or **Done** to save your changes.

### 7.7.2 Creating Manufacturer Parts from an AML

To create a new Manufacturer Part and add it to a Part:

1. Open the Part.
2. Click **Edit**.
3. Go to the **AML** tab.
4. Click **New Manufacturer Part** . A blank row appears in the AML grid.

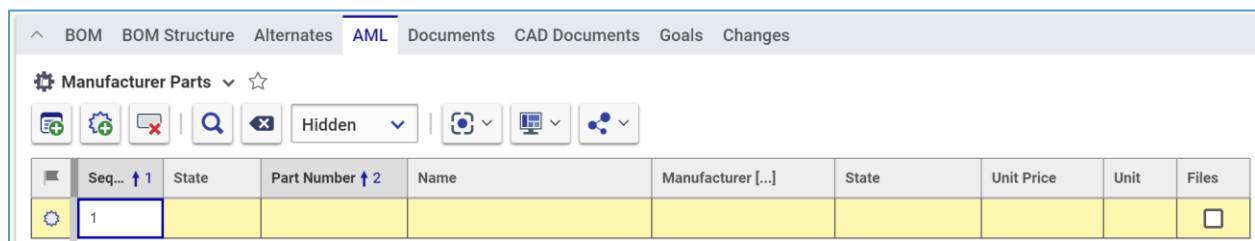


Figure 71.

5. Enter the appropriate values for each cell of the new row.
6. Click **Save** or **Done** to save your changes.

### 7.7.3 Replacing Manufacturer Parts in an AML

Use the following procedure to replace a Manufacturer Part in a Part:

1. Open the Part.
2. Click **Edit**.
3. Go to the AML tab.
4. Right-click the **Manufacturer Part** to be replaced and then click **Replace**. The Search dialog appears.
5. Search for and select a replacement Manufacturer Part. The selected Manufacturer Part replaces the Manufacturer Part.
6. Click **Save** or **Done** to save your changes.

### 7.7.4 Removing Manufacturer Parts from an AML

Use the following procedure to remove a Manufacturer Part from a Part:

1. Open the Part.
2. Click **Edit**.
3. Go to the **AML** tab.
4. Click the **Manufacturer Part** to be removed and then click **Delete Approved Manufacturer** .
5. Click **Save** or **Done** to save your changes.

### 7.7.5 Promoting Manufacturer Parts for a Part

For a given Part, you can promote its Manufacturer Parts to one of the Approved, Preferred, or Inactive states from the Part's AML. For details on AML life cycle states, refer to section [7.4 Life Cycle of Sourcing Items](#).

**Note:** By default, only members of the Component Engineering identity have access rights to promote Sourcing Items and AMLs. An Aras administrator can change this access if it is necessary. To be available for promotion, a Sourcing Item must be unclaimed and in the Viewing mode.

To promote a Manufacturer Part from an AML:

1. Open the Part associated with the Manufacturer Part.
2. Go to the **AML** tab.

3. Right-click on the Manufacturer Part → **AML** → **Promote**.

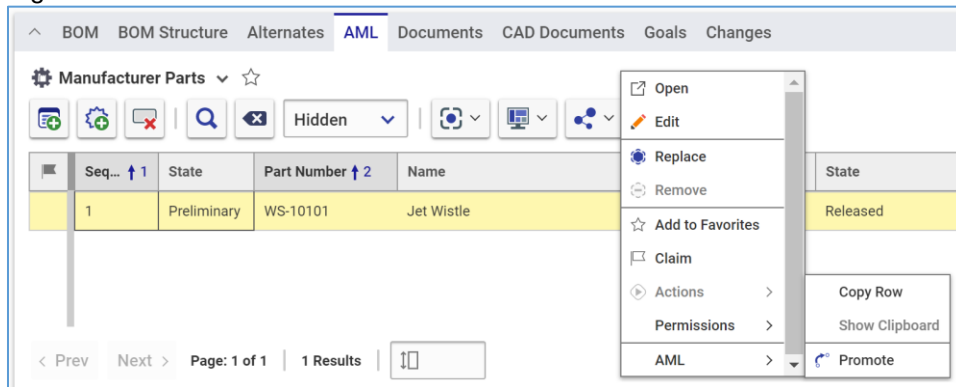


Figure 72.

4. In the promotion dialog, select the desired state to promote the Item to by either:
  - a. Double-clicking the state to which the item should be promoted.
  - b. Clicking the state to promote to and then clicking **Promote to Selected State**.

## 7.8 Finding Where Used and Comparing Sourcing Items

Aras Innovator offers the Where Used feature to determine where an Item is used by other Items. Refer to section [11.1 Finding Where an Item is Used](#) for further information. All examples in that section are applicable to Manufacturers, Manufacturer Parts, and Vendors.

To compare Items side-by-side, Aras Innovator offers the Structure Browser. Refer to section [11.2 Comparing Items](#) for further information. All examples in that section are applicable to Manufacturers, Manufacturer Parts, and Vendors.

## 8 Documents

A Part can have one or more Documents associated with it. A Document can be a drawing, a specification, quality control, etc. Documents can have associated files. Documents are an important component of any Part since they can describe how to make the specific Part or its expected performance and function. Documents also play an important role in Change Management.

### 8.1 Documents Life Cycle

By default, the Document Life Cycle has the following states: Preliminary, In Review, Released, Manual Change, In Change, Superseded, and Obsolete. An Aras administrator can change the Life Cycle Map as needed to meet specific business needs.

The following figure shows the default Document Life Cycle. A new Document starts in the **Preliminary** state and can be released manually in its early stages, or via a change process as it matures. Refer to sections [4.3. Manual Promotion and Revise](#) and [13. Change Management](#) for details.

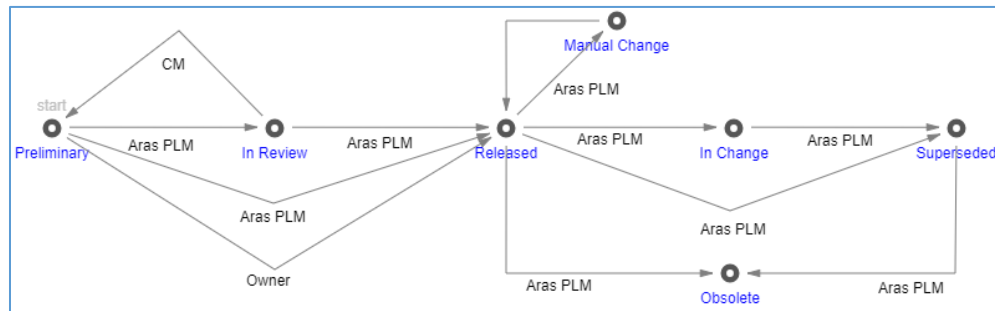


Figure 73.

### 8.2 Creating a Document

To create a new Document:

1. Go to **Contents** → **Documents** → **Documents**.
2. Click **Create New Document**. The new Document appears. Enter the necessary information.

**Document 1**

Save Done Delete

Document

Document Number Revision State Assigned Creator [Select an image...](#)

Name Designated User

Type Authoring Tool Version Effective Date

Description From Template

☐ Files Attached  
☐ Changes Pending  
☐ Template

Files Changes

Files ☆

Hidden

Related File [...] File Type [...] Comments Sequencing Is Shared

Figure 74.

- **Document Number** – (Required) The number for the Document. The Document Number must be unique.
- **Name** – The name of the Document.
- **Type** – This is the classification type for the Document. The classification dialog displays the Document classification tree. The Document classification tree can be adjusted and extended as needed by an administrator.
- **Authoring Tool** – The application used to create the Document.
- **Version** – The version of the authoring tool.
- **Description** – Remarks about the Document.
- **Assigned Creator** – This is the user responsible for the design or the content of the Document. The Assigned Creator is often identified as the Owner of the item, who may be different from the user creating the Document in the system.
- **Designated User** – This is the user responsible for the management, usage, and review of the Document.
- **Effective Date** – This is the date when a Document actually goes into publishing or the process that signifies its availability. This can be different from the release date, which is when the Document reaches the Released state in its life cycle.
- **Select an image** – This allows you to select an image using the Image Browser to generate a thumbnail for the Document. For more details, refer to section [2.4.6 Item Thumbnail](#).
- **Extended Classification** – This allows you to assign values to Items based on the Item's extended classification.
- **Template** – Identifies whether a Document is a template.

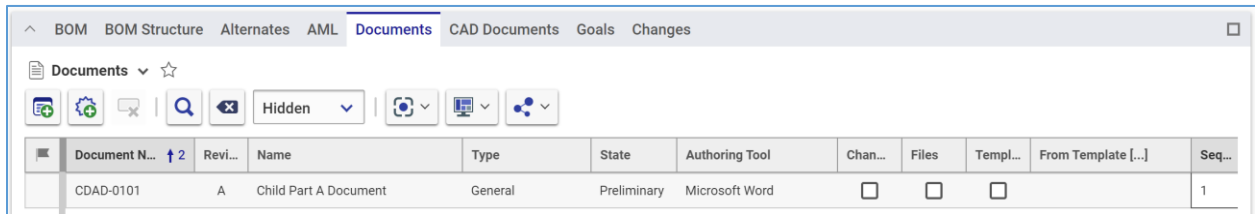


3. If it is necessary to add a file, in the Files tab, select New File  to select a file for upload, or drag and drop the desired file into the Files area.
4. Once you have entered the necessary information, click **Save** or **Done** to save your changes.

## 8.3 Creating a Document from a Part

To create a new Document and add it to a Part:

1. Open the **Part** where the Document will be added.
2. Click **Edit**.
3. Go to the **Documents** tab.
4. Click **New Document** . A new Document view appears. A row also appears in the Documents grid on the Part.
5. If it is necessary to add a file to the Document, in the Files tab, select New File  to select a file for upload, or drag and drop the desired file into the Files area.
6. Once you have entered the necessary information, click **Done** to save your changes.
7. Close the new Document. The new Document appears in the Documents tab on the Part.



| Document N... | Revi... | Name                  | Type    | State       | Authoring Tool | Chan...                  | Files                    | Templ...                 | From Template [...] | Seq... |
|---------------|---------|-----------------------|---------|-------------|----------------|--------------------------|--------------------------|--------------------------|---------------------|--------|
| CDAD-0101     | A       | Child Part A Document | General | Preliminary | Microsoft Word | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                     | 1      |

Figure 75.

8. Click **Save** or **Done** on the Part to save your changes.

## 8.4 Managing Part Documents

For a single Part, you can add, replace, and remove Documents.

### 8.4.1 Adding Documents to a Part

To add an existing document to a Part:

1. Open the **Part** where the Document will be added.
2. Click **Edit**.
3. Go to the **Documents** tab.
4. Click **Add Documents** . The Search dialog appears.

5. Search for and select a Document to add.
6. Click **Save** or **Done** to save your changes.

## 8.4.2 Replacing Documents in a Part

To replace a document in a Part:

1. Open the **Part** where the Document will be added.
2. Click **Edit**.
3. Go to the **Documents** tab.
4. Right-click on the Document to be replaced → **Replace**. The Search dialog appears.
5. Search for and select a Document. The selected Document replaces the selected Document.
6. Click **Save** or **Done** to save your changes.

## 8.4.3 Removing Documents from a Part

To remove a Document from a Part:

1. Open the **Part** where the Document will be added.
2. Click **Edit**.
3. Go to the **Documents** tab.
4. Click the **Document** to be deleted and then click **Delete Part Document** .
5. Click **Save** or **Done** to save your changes.

# 8.5 Managing Document Files

For a single Document, you can add, download, replace, and remove Files. All Files are kept in the Aras Innovator Vault.

Files are not versionable, but Documents are. When you make an edit to Document Files, a new version of the Document is created. Old versions of the Document reference old Files. For example, if you replaced File A with File B, the previous version of the Document has File A, and the current one has File B. To access old Files, you can view the corresponding Document version. For details about how to access Document versions, refer to section [4.1.1 Versions Exploration](#).

## 8.5.1 Adding Files to a Document

To add a File to a Document:

1. Open the **Document**.
2. Click **Edit**.
3. Go to the **Files** tab.
4. Click **New File** . The standard operating system dialog for file browsing and selection appears.

5. Search for and select a File. A new row appears in the **Files** grid with the file attached.

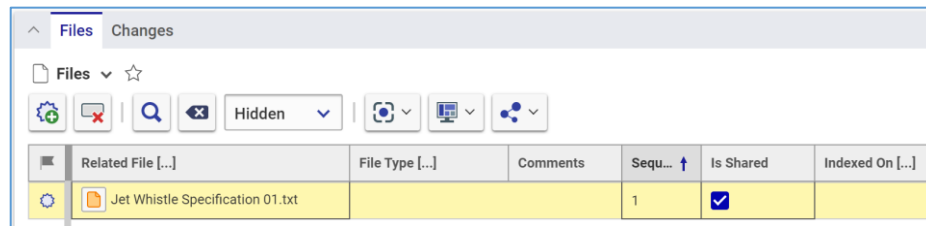


Figure 76.

6. Enter any necessary information:
  - **File Type** – The file type. This property is an optional property, and if left blank, Aras Innovator will automatically assign a file type depending on the File's extension once the Document is saved.
  - **Comments** – Remarks about the File.
  - **Is Shared** – Determines whether the File is shared with other users. If it is necessary to hide the File from other users, uncheck the Is Shared check box.
7. Click **Save** or **Done** to save your changes.

## 8.5.2 Working with Files

In this section, it is assumed that you are working with the current Document version.

To work with Files attached to a Document, you can download them from the Aras Innovator Vault, where they are kept.

Files can be downloaded by clicking on the hyperlink in the Files tab on the Document:

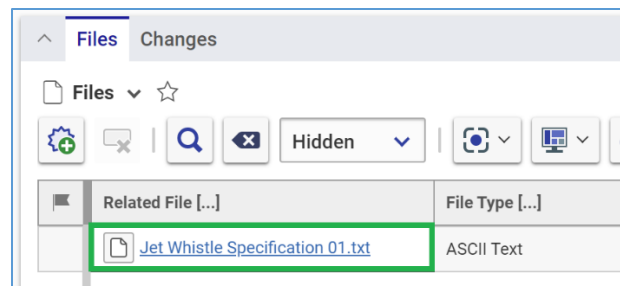


Figure 77.

Files can also be downloaded through the **Manage File** dialog:

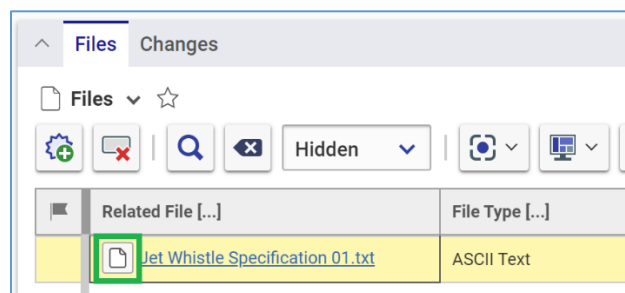


Figure 78.

You can either select the hyperlink in the **Manage File** dialog or use the **Download file** button  to download the File.

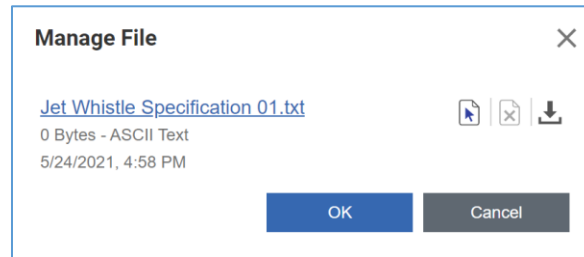


Figure 79.

### 8.5.3 Replacing Files in a Document

To replace a File, you can remove the old File from a Document and then attach the replacing File while keeping the Document in the **Edit** mode. Refer to sections [8.5.1 Adding Files to a Document](#) and [8.5.4 Removing Files from a Document](#) to use this approach.

Files can also be replaced using the **Manage file** dialog:

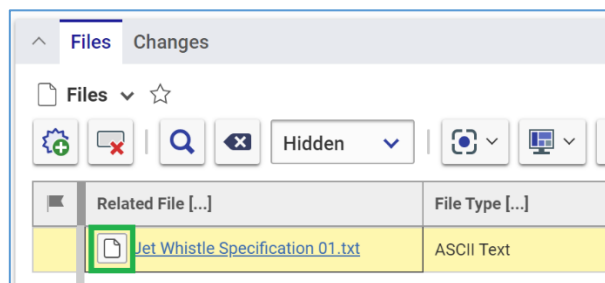


Figure 80.

You can click the **Select new file** button  to replace the existing file.

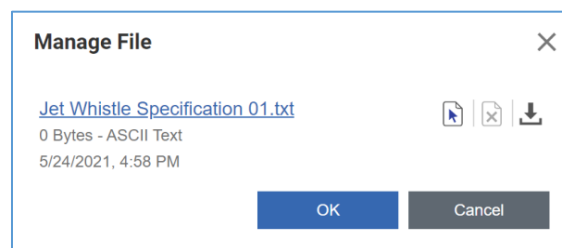


Figure 81.

### 8.5.4 Removing Files from a Document

To remove a File from a Document:

1. Open the **Document**.
2. Click **Edit**.
3. Go to the **Files** tab.

4. Click the file to be removed and click **Delete Document File** .
5. Click **Save** or **Done** to save your changes.

**Note:** The **File** disappears immediately if the Document was not saved after it was added.

## 8.6 Deleting Documents

It is possible to delete all or some versions of a Document. This subsection outlines the former case, in which the Document is deleted from Aras Innovator completely. For details on the latter case, refer to section [4.1.2 Version Deletion](#).

To delete a Document completely, you must have **Delete** rights granted in Permissions for all Life Cycle states of the Document, including the current one. For example, if the latest Document revision is currently in the Preliminary state, but has Released revisions, you must have the **Delete** rights granted in Permissions for both life cycle states. For details on access rights, refer to section [3 Permissions](#). This subsection assumes that you have all necessary rights.

**Note:** Documents that are claimed by any user or are used on other items cannot be deleted.

You can delete a Document using either its view or the context menu in the Documents Search Page.

To delete a Document using its view:

1. Open the Document.
2. On the Document toolbar, click **More → Delete → Delete All Versions**.

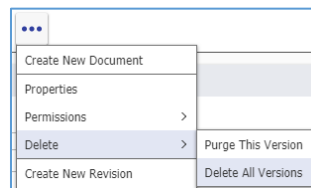


Figure 82.

3. Click **OK** in the confirmation dialog to delete the Document.

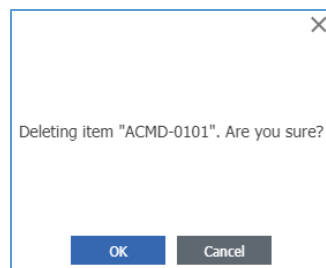


Figure 83.

To delete a Document using its context menu:

1. Go to the Documents Search Page.
2. Right-click the Document to be deleted and click **More → Delete → Delete All Versions**.
3. Click **OK** in the confirmation dialog to delete the Document.

If a Document has only one version, you can delete it using the **Purge This Version** feature. For details about this feature, refer to section [4.1.2 Version Deletion](#). For details about how to find out if the current version is the only one, refer to section [4.1.1 Versions Exploration](#).

## 8.7 Finding Where Used and Comparing Documents

Aras Innovator offers the Where Used feature to determine where an Item is used by other Items. Refer to section [11.1 Finding Where an Item is Used](#) for further information. All examples in that section are applicable to Documents.

To compare Items side-by-side, Aras Innovator offers the Structure Browser. Refer to section [11.2 Comparing Items](#) for further information. All examples in that section are applicable to Documents.

## 9 CAD Documents

Along with Documents, a Part can have one or more Computer Aided Design (CAD) Documents associated with it. A CAD Document can be a drawing, a specification, 3D model, etc. created in a CAD authoring tool.

A CAD Document can have Files associated with it that are kept in the Aras Innovator Vault. If you have purchased a third-party integration to a CAD tool, you can attach a native File to a Part using your CAD Authoring tool and a viewable File. This subscription enables you to view and work with native Files using Aras Innovator.

**Note:** Some features of CAD Documents, such as transformation of native files into other viewable formats, require an Aras subscription.

### 9.1 Creating a CAD Document

To create a CAD Document:

1. Go to **Contents** → **Documents** → **CAD Documents**.
2. Select **Create New CAD Document**. The new CAD Document appears. Enter the necessary information.

The screenshot shows the 'CAD Document 1' form. At the top, there are three buttons: 'Save' (with a download icon), 'Done' (with a checkmark icon), and 'Delete'. Below these is a tabbed interface with 'CAD Document' selected. The form contains several input fields: 'Document Number' (highlighted in yellow), 'Revision', 'State', 'Name', 'Type', 'Authoring Tool' (with a dropdown arrow), 'Version', and a large 'Description' text area. To the right of these fields are 'Assigned Creator' and 'Designated User' (both with dropdown arrows and a 'Select an image...' link), and a 'From Template' section with checkboxes for 'Changes Pending', 'Standard', and 'Template'. At the bottom right, there are two sections: 'Native File' with a 'Select file...' button, and 'Viewable File' with a 'Select file...' button. At the very bottom, there are four tabs: 'Structure' (selected), 'Parents', 'Files', and 'Changes'.

Figure 84.

- **Document Number** – (Required) The CAD Document number.
- **Name** – The name of the CAD Document.
- **Type** – This is the classification type for the CAD Document. The classification dialog displays the CAD Document classification tree. The CAD Document classification tree can be adjusted and extended as needed by an administrator.
- **Authoring Tool** – The application used to create the CAD Document.

- **Version** – The version of the authoring tool.
- **Description** – Remarks about the CAD Document.
- **Assigned Creator** – This is the user responsible for the design or the content of the CAD Document. The Assigned Creator is often identified as the Owner of the item, who may be different from the user creating the CAD Document in the system.
- **Designated User** – This is the user responsible for the management, usage, and review of the CAD Document.
- **Select an image** – This allows you to select an image using the Image Browser to generate a thumbnail for the CAD Document.
- **Changes Pending** – The Changes Pending flag is automatically set when the CAD Document is associated with a change item.
- **Standard** – If the CAD Document follows a standard, select the Standard check box.
- **Template** – If the CAD Document is going to be used as a template, select the Template check box.
- **Extended Classification** – This allows you to assign values to Items based on the Item's extended classification.

**Note:** There are no extended classifications provided for CAD Documents out of the box.

3. If it is necessary to attach a native or viewable File to the CAD Document, use the procedure described in section [9.7.1 Attaching Native and Viewable CAD Files to a CAD Document](#).
4. If it is necessary to attach additional Files to the CAD Document, use the procedure described in section [9.7.2 Attaching Other Files to a CAD Document](#).
5. Once you have entered the necessary information, click **Save** or **Done** to save your changes.

## 9.2 Creating a CAD Document from a Part

To create a new CAD Document and attach it to a Part:

1. Open the **Part** where the CAD Document will be added.
2. Click **Edit**.
3. Go to the **CAD Documents** tab.
4. Click **New CAD Document** . A new CAD Document row appears in the CAD Documents grid on the Part.

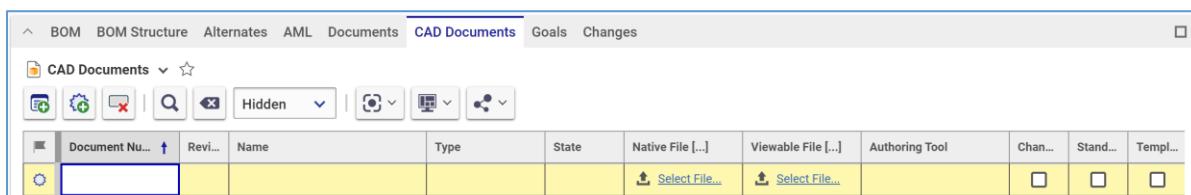


Figure 85.

5. Click the **Document Number** cell and enter a document number for the CAD Document.

**Note:** The Document Number is required and must be provided by the user.



6. Enter any additional data by clicking on the corresponding cell in the BOM table.
7. Click **Save** or **Done** to save your changes.

## 9.3 CAD Documents Life Cycle

By default, the CAD Document Life Cycle has the following states: Preliminary, In Review, Released, In Change, Superseded, and Obsolete. An administrator can change the Life Cycle Map if it is necessary to meet specific business requirements.

The figure below shows the default CAD Document Life Cycle. A new CAD Document starts in the **Preliminary** state and can be promoted either automatically or manually. Both the simplified and CMII change management processes can automatically promote the CAD Documents. The associated Life Cycles are automatically controlled by Workflows for those CAD Documents that are included in the affected items list. CAD Documents that are not included in the list because they do not require a full change process can be promoted using either manual release or revision if certain requirements are met. By default, a CAD Document Owner can release a CAD Document. For further details on this feature, refer to section [4.3. Manual Promotion and Revise](#).

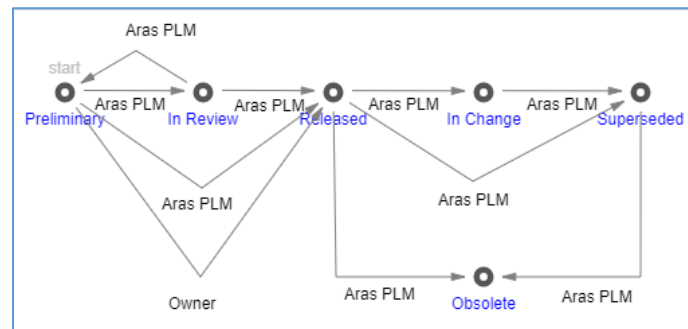


Figure 86.

## 9.4 CAD Document Relationships

A CAD Document can have relationships with other items. The Relationships accordion keeps these relationships grouped in separate Relationships tabs. Each Relationships tab represents a single relationship type.

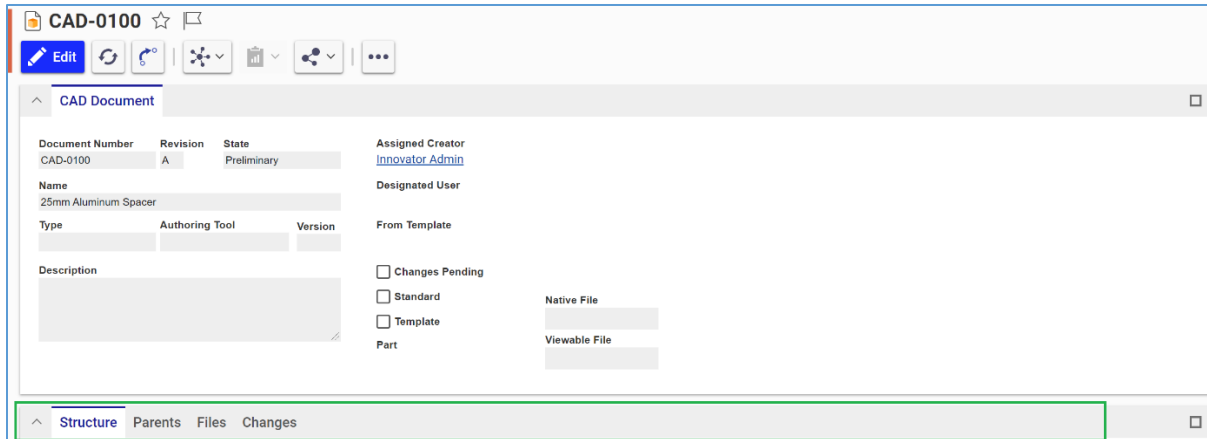


Figure 87.

- **Structure** – A single-level CAD hierarchy editor that enables you to view and manage the direct Children of this CAD Document. For more details, refer to section [9.5 Hierarchy of CAD Documents](#).
- **Parents** – A read-only list of the CAD Document's direct Parents.
- **Parts** – A read-only list of the Parts attached to the CAD Document. This tab appears only when a CAD Document is attached to more than one Part. For more details, refer to section [9.10.2 Parts Associated with a CAD Document](#).
- **Files** – A list of Files attached to the CAD Document. Typically, this tab hosts additional CAD Document Files. The **Native File** property of the CAD Document is a link to a native CAD File. The **Viewable File** property of the CAD Document is a link to a viewable representation of the native CAD File. For more details, refer to section [9.7 Managing CAD Files](#).
- **Changes** – Displays a list of change items, such as DCOs, Express ECOs and EDRs, where this CAD Document is an affected item. For more details, refer to sections [13. Change Management](#).

## 9.5 Hierarchy of CAD Documents

If you have purchased a third-party integration to a CAD authoring application, you can import Part families that were created with this CAD application into Aras Innovator. The Product Engineering (PE) application's capabilities enable you to manage the imported Part families as well as to create new ones.

A Part family is a multi-level hierarchy of CAD Documents that consist of single-level CAD Documents Relationships. The PE application supports the management of multi-level CAD Documents hierarchies through creating, replacing, and removing single-level CAD Documents Relationships. A single-level CAD Documents Relationship is a Parent-Child relationship between two CAD Documents. If a Parent has a Child that is also a Parent, it is a multi-level CAD Documents hierarchy.

If you add a CAD Document to another CAD Document, you create a Parent-Child relationship between them. If the Parent had no Children before, then a new hierarchy of CAD Documents is created and vice versa. If you remove a CAD Document from another CAD Document, you delete a Parent-Child relationship between them. If the removed Child was the last one, then the hierarchy of CAD Documents is deleted.

A multi-level hierarchy of CAD Documents can represent a proper BOM hierarchy.

There are two ways to set up a multi-level hierarchy of CAD Documents:

- Add existing Children to a Parent. This approach is used when building a new assembly using existing CAD Documents. For more details on this approach, refer to section [9.5.1 Creating a CAD Document Relationship](#).
- Create new Children while adding them to a Parent. This approach is convenient for populating nested CAD Documents or creating hierarchies quickly without getting into the details of each Child. However, it is still necessary to complete each Child later. For more details on this approach, refer to section [9.5.2 Creating a CAD Document from another CAD Document](#).

### 9.5.1 Creating a CAD Document Relationship

This approach is used for building a new assembly using existing CAD Documents.

To add an existing CAD Document to another CAD Document:

1. Open the Parent CAD Document.
2. On the Parent toolbar, click **Edit**.
3. Go to the **Structure** accordion tab.
4. On the **Structure** accordion toolbar, click **Add CAD Documents**.
5. Search for and select the necessary Children to be added.

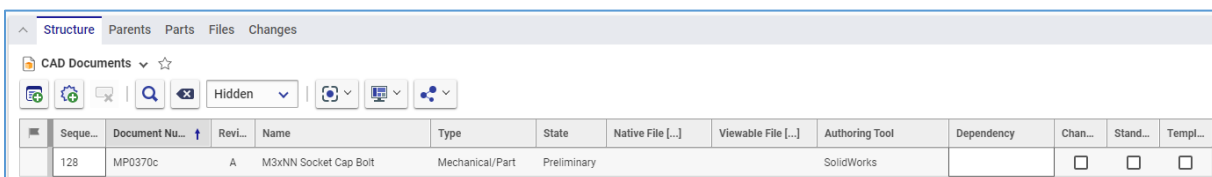


Figure 88.

6. Specify the Dependency type, if necessary, by selecting the ellipsis in the Dependency cell, or by typing in the dependency.
7. Click **Save** or **Done** to save your changes and create the Parent-Child relationship between the CAD Documents.

### 9.5.2 Creating a CAD Document from Another CAD Document

To create a new CAD Document and add it to another CAD Document:

1. Open the Parent CAD Document.
2. On the Parent toolbar, click **Edit**.
3. Go to the **Structure** accordion tab.
4. Click **New CAD Document**. A new row appears at the bottom of the **CAD Documents** grid.

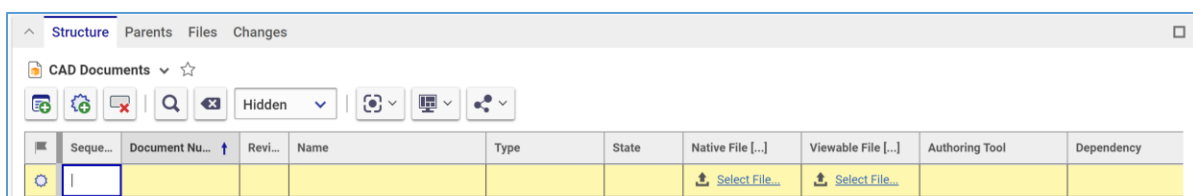


Figure 89.

5. Click the **Document Number** cell and enter a document number for the Child.

**Note:** The Document Number is required and should be provided by the user.

6. Enter any additional data by clicking on the corresponding cell or hyperlink in the CAD Documents grid. Refer to section [9.1 Creating a CAD Document](#) for a description of the CAD Document fields.
7. Click **Save** or **Done**. The new CAD Document is created and attached to the Parent CAD Document.

### 9.5.3 Replacing a CAD Documents Relationship

To replace a Child CAD Document in a CAD Document Relationship:

1. Open the Parent CAD Document.
2. On the Parent toolbar, click **Edit**.
3. Go to the **Structure** accordion tab.
4. Right-click the Child CAD Document to be replaced and then click **Replace**. The Search dialog appears.
5. Search for and select a replacement CAD Document. The selected CAD Document replaces the existing CAD Document.
6. Click **Save** or **Done**. The Children are replaced.

### 9.5.4 Removing a CAD Documents Relationship

To remove a Child CAD Document from a Parent CAD Document:

1. Open the Parent CAD Document.
2. On the Parent toolbar, click **Edit**.
3. Go to the **Structure** accordion tab.
4. Click on the CAD Document(s) to be deleted and then click **Delete CAD Document**.
5. Click **Save** or **Done**. The selected Children are removed.

## 9.6 Managing CAD Documents in Parts

For a single Part, you can attach, replace, and remove CAD Documents. If you have purchased a third-party CAD integration tool, you can use it to attach native CAD and viewable files to a Part using the CAD authoring tool. This subscription enables you to view and work with CAD File representations in Aras Innovator using viewers for Parts and CAD Documents.

**Note:** You cannot attach Parts to a CAD Document.

### 9.6.1 Adding CAD Documents to a Part

To add an existing CAD Document to a Part:

1. Open the **Part** where the CAD Document will be added.
2. Click **Edit**.
3. Go to the **CAD Documents** tab.

4. Click **Add CAD Documents** . The Search dialog appears.
5. Search for and select a CAD Document to add.
6. Click **Save** or **Done** to save your changes.

## 9.6.2 Replacing CAD Documents in a Part

To replace a CAD Document in a Part:

1. Open the **Part** where the CAD Document will be added.
2. Click **Edit**.
3. Go to the **CAD Documents** tab.
4. Right-click on the CAD Document to be replaced → **Replace**. The Search dialog appears.
5. Search for and select a CAD Document. The selected CAD Document replaces the selected CAD Document.
6. Click **Save** or **Done** to save your changes.

## 9.6.3 Removing CAD Documents from a Part

To remove a CAD Document from a Part:

1. Open the **Part** where the CAD Document will be added.
2. Click **Edit**.
3. Go to the **CAD Documents** tab.
4. Click the **CAD Document** to be deleted and then click **Delete Part CAD** .
5. Click **Save** or **Done** to save your changes.

## 9.6.4 Working with CAD Documents from a Part

If you have purchased a third-party CAD integration tool, you can use it to attach a native CAD File to a Part, using the CAD authoring tool, for viewing and working in Aras Innovator.

Once the integration license is used, and files are imported into Aras Innovator, the viewers will appear in a sidebar on the left-hand side of the Part view, allowing you to view and work with the CAD Files and images attached to the Part.

| Document Number | Revisi... | Name                        | Type                | State       |
|-----------------|-----------|-----------------------------|---------------------|-------------|
| MP0101-D        | B         | MakerBot Replicator Drawing | Mechanical/Drawing  | Preliminary |
| MP0101          | B         | MakerBot Replicator         | Mechanical/Assembly | Preliminary |

Figure 90.

- **Form** shows the Part view.

Figure 91.

- **Monolithic Viewer** is a simple tool for viewing and working with 3D CAD Model representations. For more details, refer to section [9.8 Working with CAD Files](#).

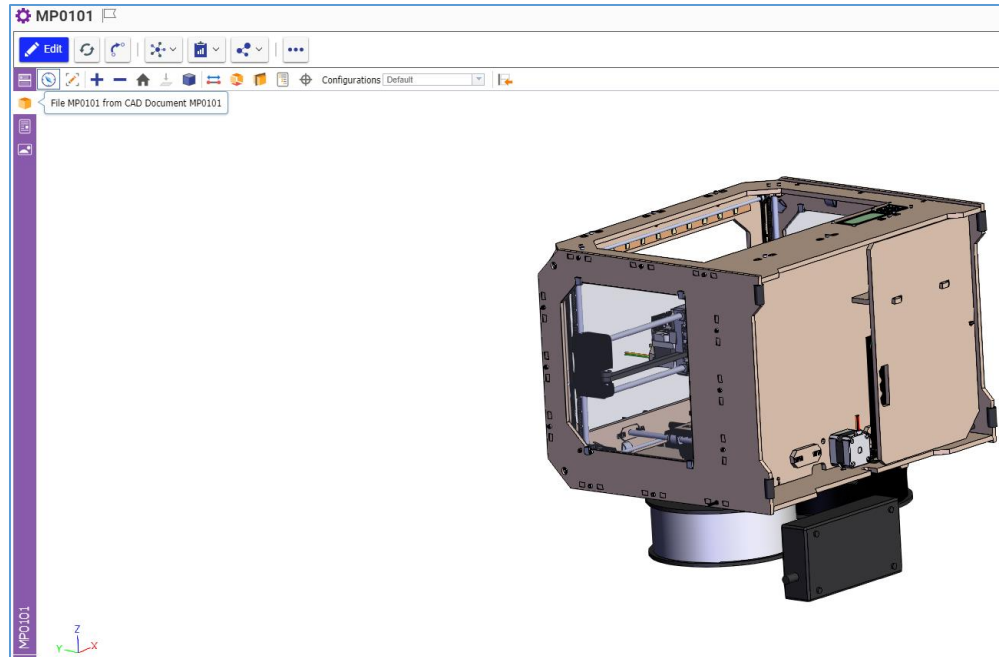


Figure 92.

- **PDF Viewer** displays the PDF file.

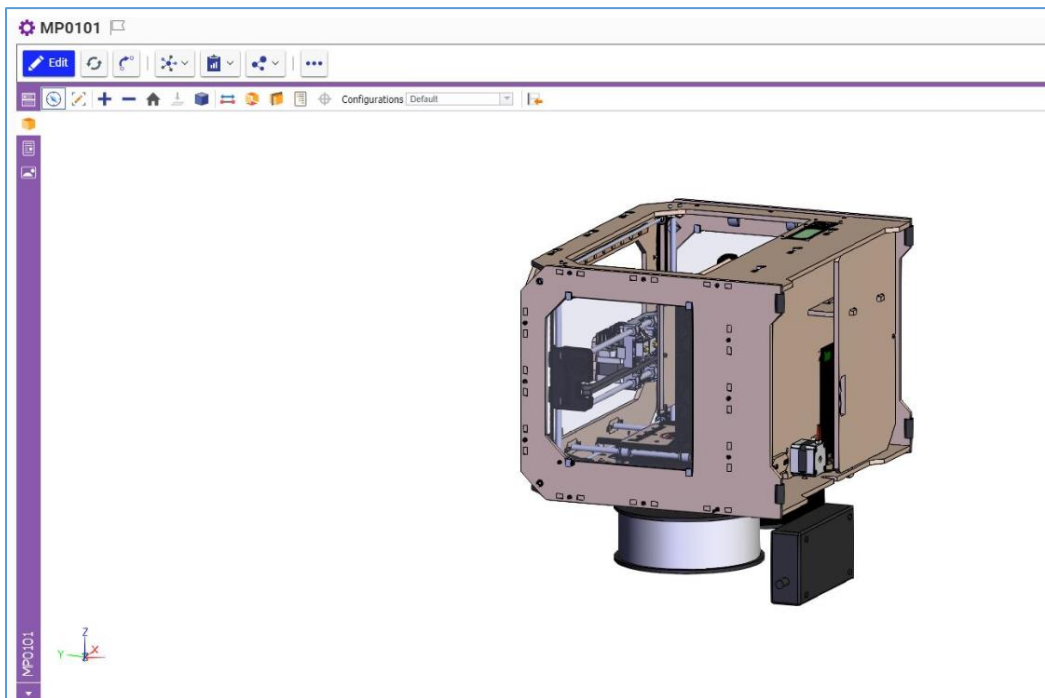


Figure 93.

- Image Viewer displays the image file.

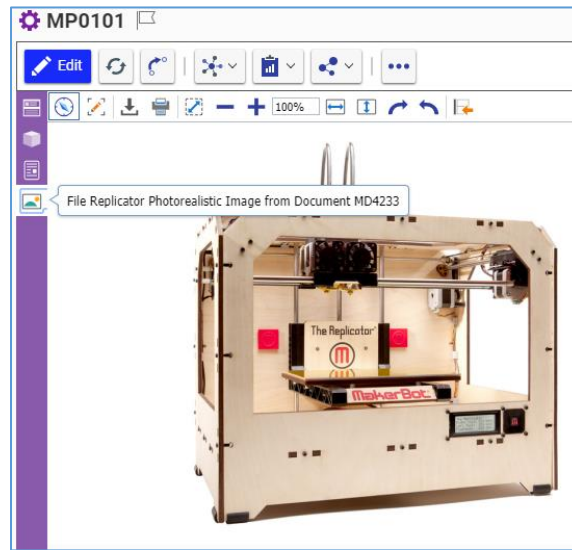


Figure 94.

If there are several Files of the same type, you are prompted to choose one.

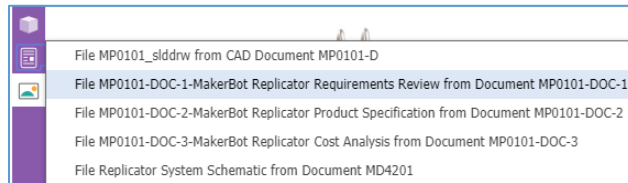


Figure 95.

Alternatively, you can view the list of all possible navigation options within the sidebar by selecting the arrow at the bottom of the sidebar and making your selection for the viewer you wish to view. This list includes the Part Form view. It is possible to view Files attached to the Part. The blue circle marks the currently selected viewer.

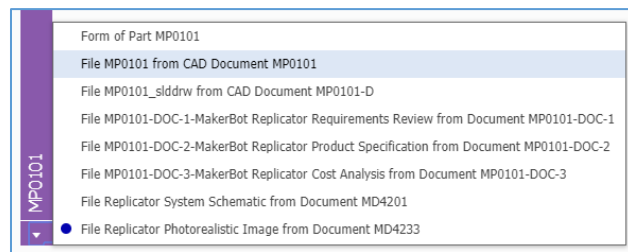


Figure 96.



## 9.7 Managing CAD Files

A CAD Document can have Files associated with it that are kept in the Aras Innovator Vault. If you have purchased a third-party CAD integration tool, you can use it to attach, download, replace, and remove native CAD and viewable files to a Part using the CAD authoring tool. This subscription also enables you to view and work with CAD File representations in Aras Innovator directly from the corresponding CAD Document.

In Aras Innovator, CAD Files and Files are not versionable, but CAD Documents are. When a CAD Document is versioned, the references to its Files remain the same. In other words, old versions of the CAD Document keep old Files. For example, if Version 1 of a CAD Document contains File A, then gets versioned, after which, File A is replaced with File B, the previous version of the CAD Document (Version 1) has File A, and the current version has File B. To access the old Files, you can view the appropriate version of the CAD Document. For details about how to access CAD Document versions, refer to section [4.1.1 Versions Exploration](#).

### 9.7.1 Attaching Native and Viewable CAD Files to a CAD Document

For a single CAD Document, you can attach only one native and one viewable CAD File using the following procedure:

1. Open the CAD Document.
2. Click **Edit**.
3. Click the **Select file** link in the Native file field. The Open file dialog appears.

Figure 97.

4. Select a Native CAD File. The **Select file** link is replaced with the name of the selected Native CAD File.

Figure 98.

5. Click the **Select file** link in the Viewable file field. The Open file dialog appears.

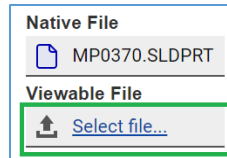


Figure 99.

6. Select a viewable representation of the Native CAD File. The **Select file** link is replaced with the name of the selected Viewable File.
7. Click **Save** or **Done** to save your changes. The names of the selected Files become links and the Files are attached to the CAD Document.

## 9.7.2 Attaching Other Files to a CAD Document

To attach additional Files to a CAD Document:

1. Open the CAD Document.
2. Click **Edit**.
3. Go to the **Files** accordion tab.
4. Click **New CADFiles** on the Files accordion toolbar. A new row appears at the bottom of the Files grid.

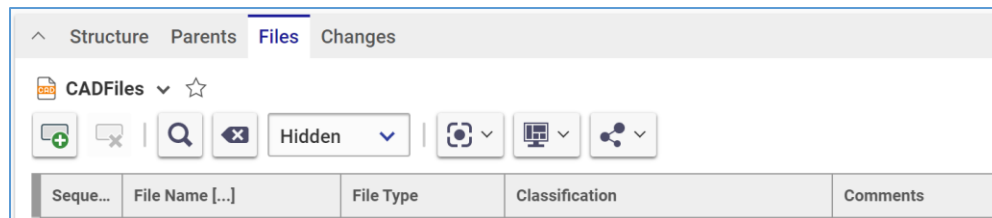


Figure 100.

5. Click the **Select File** link in the File Name cell. The Open file dialog appears.
6. Select the File to be uploaded. The **Select file** link is replaced with the name of the selected File.

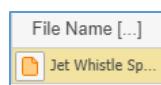


Figure 101.

**Note:** Aras Innovator will automatically provide a file type to the **File Type** cell depending on the File's extension when you save the CAD Document.

7. Specify the Classification of the File by clicking the **Classification** cell, selecting the **ellipsis** button, and selecting the appropriate classification.
8. Provide any additional information about the File in the **Comments** cell as needed.
9. Click **Save** or **Done** to save your changes and attach the additional File to the CAD Document.

## 9.7.3 Downloading Native and Viewable CAD Files from a CAD Document

You can download native and viewable CAD Files using either of the following methods:

- The file links on the CAD Document
- The Manage file property dialog

To download native and viewable CAD Files using the file links on the CAD Document, click the corresponding link on the CAD Document:

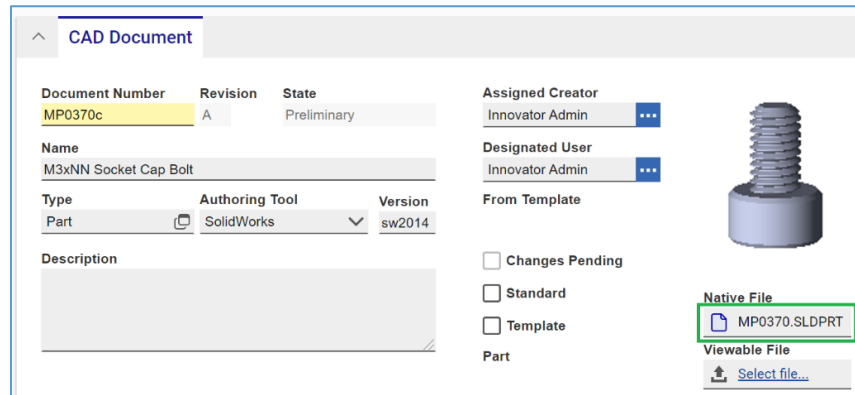


Figure 102.

When selecting the link of the Viewable File, a new browser window appears showing the viewable file and providing functionality for viewing and downloading.

**Note:** The view of a viewable file and functionality available depend on your browser.

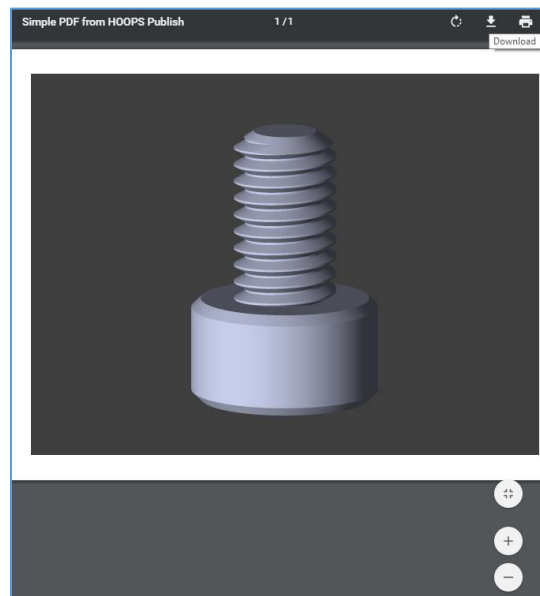


Figure 103.

To download files using the Manage file property dialog, select the **Manage file property** icon for the corresponding file:

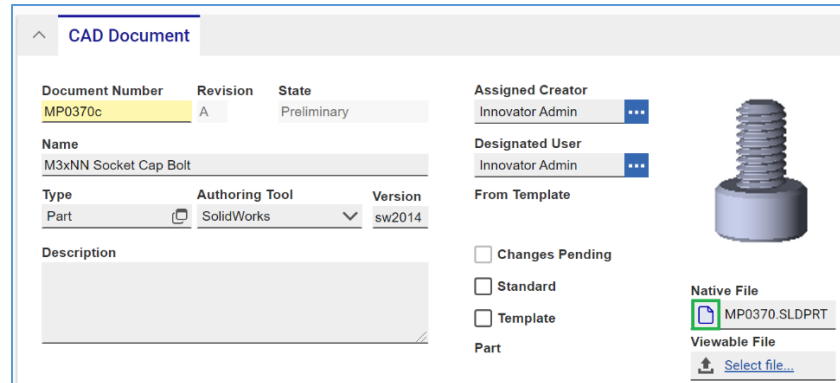


Figure 104.

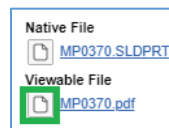


Figure 105.

You can either select the hyperlink in the Manage file property dialog or use the **Download file** button  to download the File.

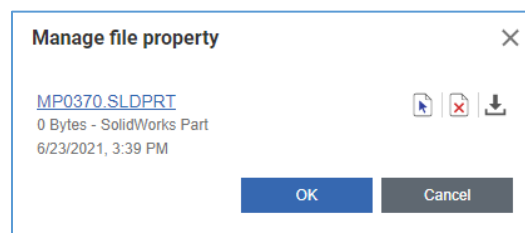


Figure 106.

## 9.7.4 Downloading Other Files from a CAD Document

You can download additional Files from a CAD Document using either of the following methods:

1. The File Name link in the Files grid.
2. The Manage file property dialog.

To download files from the CAD Document, click on the corresponding link in the Files tab on the CAD Document:

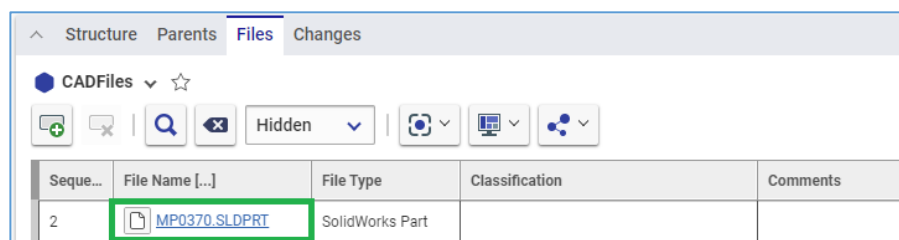


Figure 107.

To download files from the CAD Document using the Manage file property dialog, select the Manage file property button for the corresponding file:

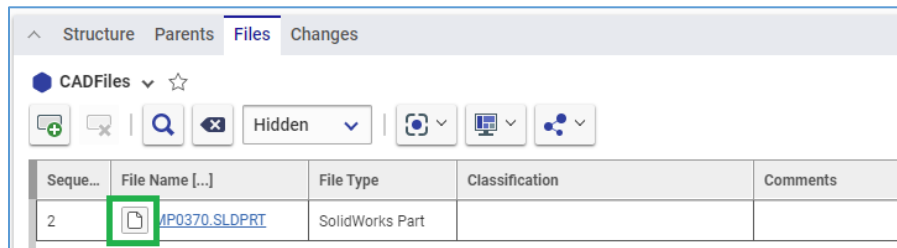


Figure 108.

You can either select the hyperlink in the Manage file property dialog or use the **Download file** button



to download the File.

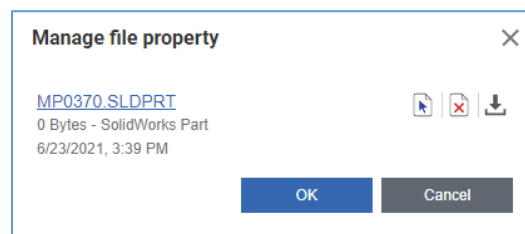


Figure 109.

If you are downloading a viewable file, such as a PDF, instead of starting the downloading process of your operating system, a new browser window appears showing the viewable file and providing functionality for viewing and downloading. Use the file download procedure of your browser to download the file and close the browser window with the viewable file when completed.

## 9.7.5 Replacing Native and Viewable CAD Files in a CAD Document

To replace the native and viewable files on a CAD Document:

1. Open the **CAD Document**.
2. Click **Edit**.
3. Click the **Manage file property** button under Native File.

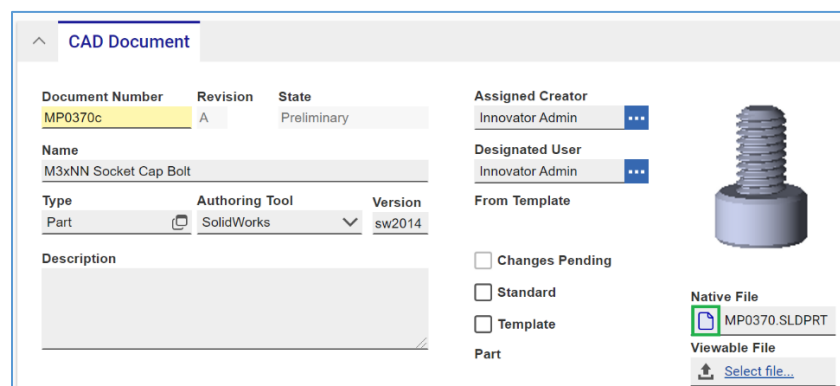


Figure 110.

The **Manage file property** dialog appears.

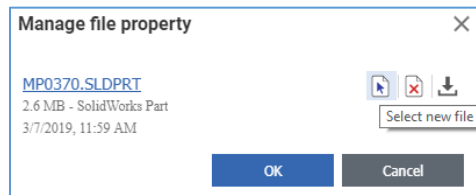


Figure 111.

4. Click the **Select new file** button.
5. Search for and select the replacement Native File. In the **Manage file property** dialog, the link with the name of the old native File is replaced with the name of the new Native File. There is no information on the new Native File.

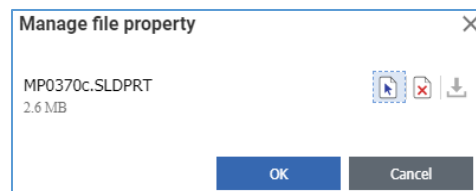


Figure 112.

6. Click **OK**. In the CAD Document view, a link with the name of the old native File is replaced with the name of the new native File.
7. Repeat steps 3-6 using the **Manage file property** button under Viewable File.
8. Click **Save** or **Done** to save your changes. The names of the replacement Files become links. These Files are attached to the CAD Document.



Figure 113.

## 9.7.6 Replacing Other Files in a CAD Document

To replace additional Files in a CAD Document:

1. Open the CAD Document.
2. Go to the Files accordion tab.
3. Click the **Manage file property** button in the File Name cell.

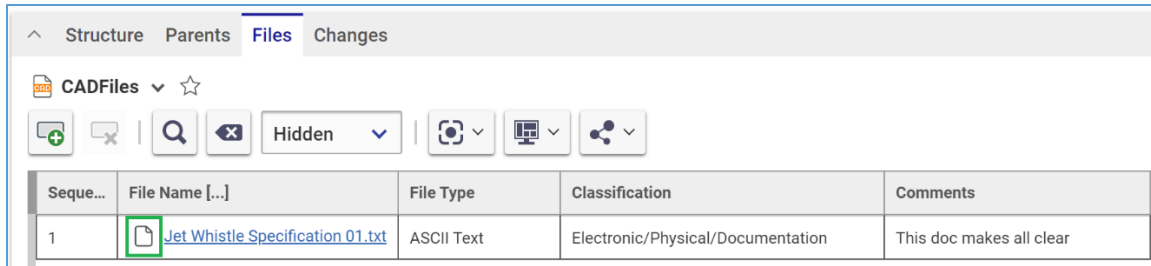


Figure 114.

The **Manage File** dialog appears.

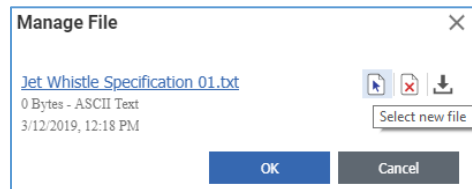


Figure 115.

- Click the **Select new file** button.
- Search for and select the replacing File. In the Manage file dialog, the link with the name of the old File is replaced with the name of the new File.

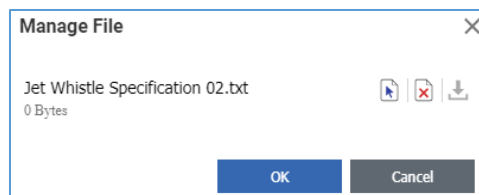


Figure 116.

- Click **OK**. In the File Name cell, the link with the name of the old File is replaced with the name of the new File. The rest of the information on the old File is kept.
- Click **Save** or **Done** to save your changes. In the **File Name** cell, the name of the new File becomes a link to it. The Files are replaced, and the File Type is updated

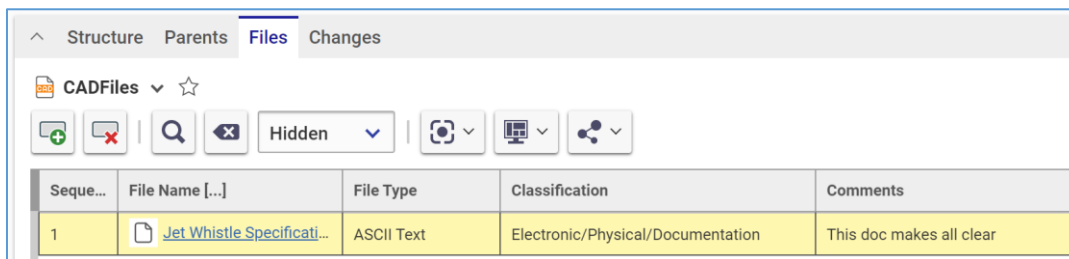


Figure 117.

## 9.7.7 Removing Native and Viewable CAD Files from a CAD Document

To remove native and viewable CAD Files from a CAD Document:

- Open the CAD Document.

- Click **Edit**.
- Click the **Manage file property** button under Native file.

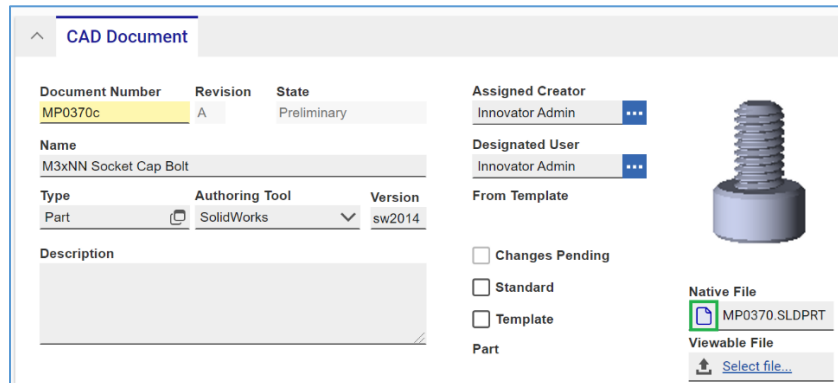


Figure 118.

The **Manage file property** dialog appears.

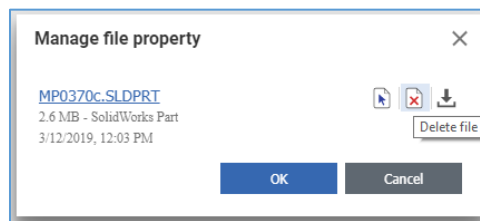


Figure 119.

- Click the **Delete file** button. A notification dialog appears.

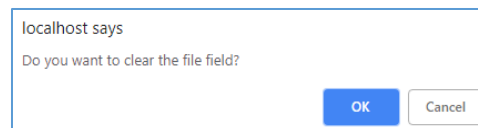


Figure 120.

- Click **OK** in the notification dialog. The Select File link replaces the link to the native File.

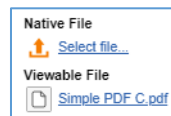


Figure 121.

- Click the **Manage file property** button under Viewable file. The Manage file property dialog appears.

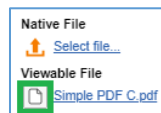


Figure 122.

- Click the **Delete file** button. A notification dialog appears.
- Click **OK** in the notification dialog. The Select File link replaces the link to the viewable File.



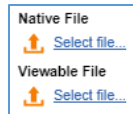


Figure 123.

- Click **Save** or **Done**. The native and viewable Files are removed from the CAD Document.

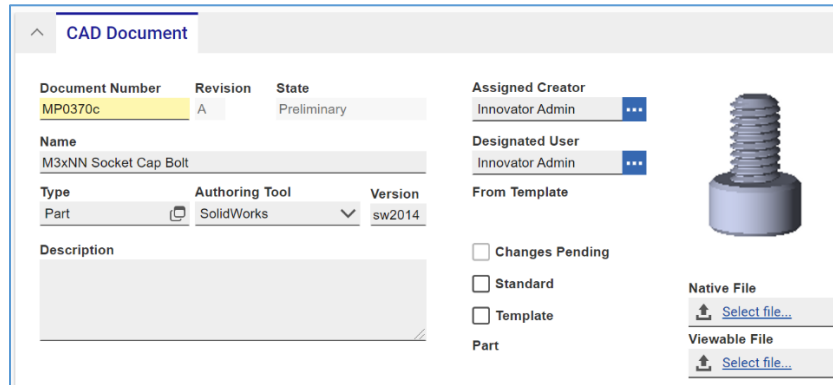


Figure 124.

## 9.7.8 Removing Other Files from a CAD Document

To remove a File from a Document:

- Open the CAD Document.
- Click **Edit**.
- Go to the **Files** accordion tab.
- Click the **File** row and then click **Delete Row** on the **Files** accordion toolbar. The **File** becomes unavailable.

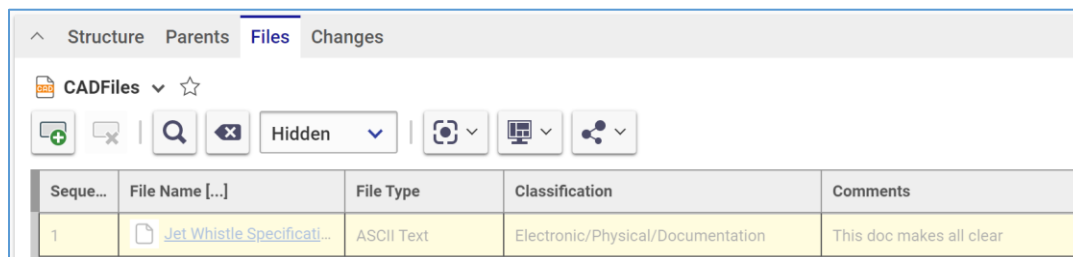


Figure 125.

- Click **Save** or **Done** on the **CAD Document** toolbar. The File disappears from the grid.

**Note:** The File disappears immediately if the CAD Document was not saved after it was added.

## 9.8 Working with CAD Files

If you have purchased a third-party CAD integration tool, you can use it to attach native CAD and viewable files to a Part using the CAD authoring tool. This subscription also enables you to view and work with CAD File representations in Aras Innovator directly from the corresponding CAD Document.

## 9.8.1 Accessing CAD Models from CAD Documents

Once the integration license is obtained, and files are imported into Aras Innovator, the CAD Document displays the Sidebar viewer icon on its left side.

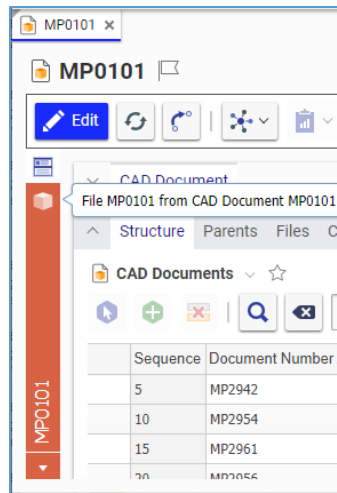


Figure 126.

The Sidebar hosts tabs that provide access to tools for viewing and working with CAD Models, PDF files and images attached to the CAD Document. The Sidebar contents depend on your license, CAD authoring tool, CAD and other Files attached to a CAD Document. The most used tabs are as follows:

- **Form** shows the CAD Document view.

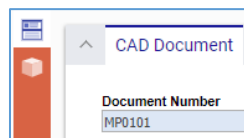


Figure 127.

- **Monolithic Viewer** is a simple tool for viewing and working with 3D CAD models.

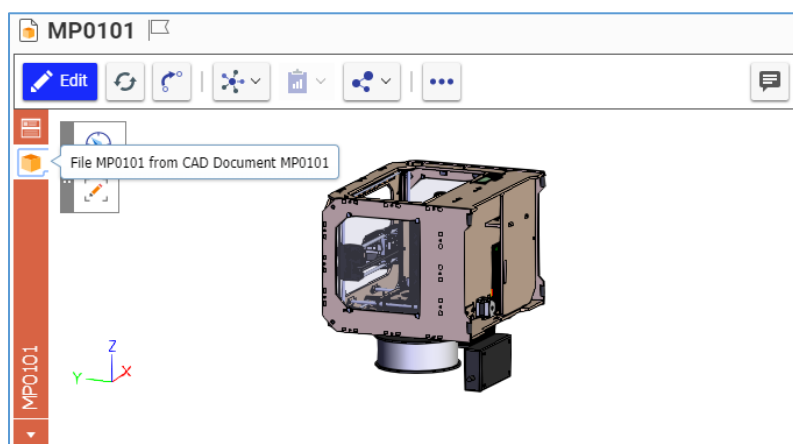


Figure 128.

- **PDF Viewer** displays the PDF file.

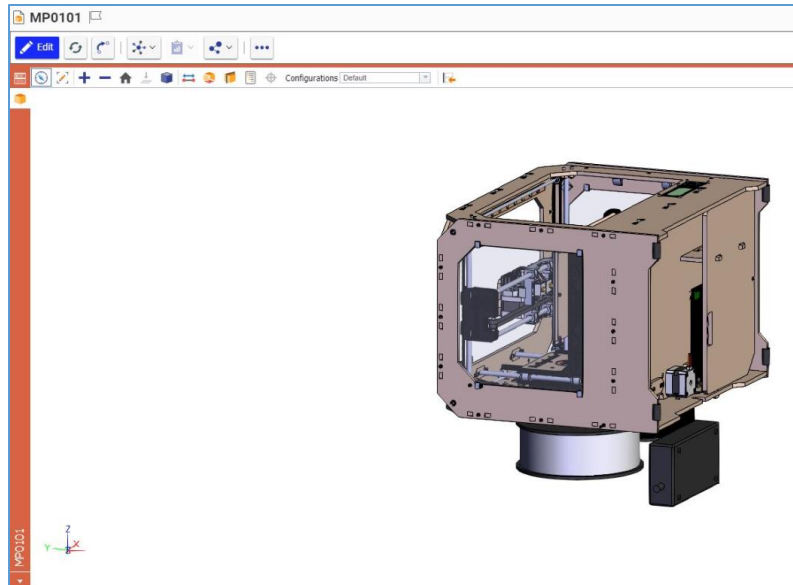


Figure 129.

Alternatively, you can view the list of all possible navigation options within the sidebar by selecting the arrow at the bottom of the sidebar and making your selection for the viewer you wish to view. This list includes the CAD Document Form view. It is possible to view Files attached to the CAD Document. The blue circle marks the currently selected viewer.

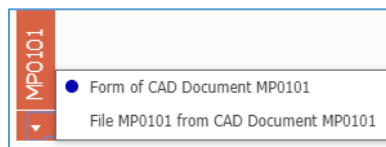


Figure 130.

## 9.8.2 Viewing CAD Models

When the viewer is selected, the CAD Model appears as follows:

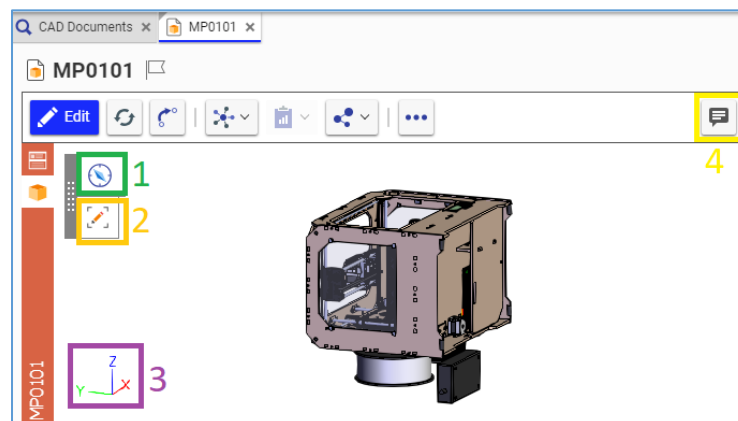


Figure 131.

The view includes the following functionality:

1. The **View** toolbar button
2. The **Markup** toolbar button
3. The **Model Orientation** indicator
4. The **Discussion** button

**Note:** As the **Discussion** button is on the CAD Document toolbar, it can be accessed from anywhere in the CAD Document. However, it is most convenient to use it from the CAD Files view because you can add snapshots of the CAD Model.

You can view a CAD Model from different sides by clicking and dragging at any place on the Model canvas. The **Model Orientation** indicator tells you which side is toward you and where the other sides are.

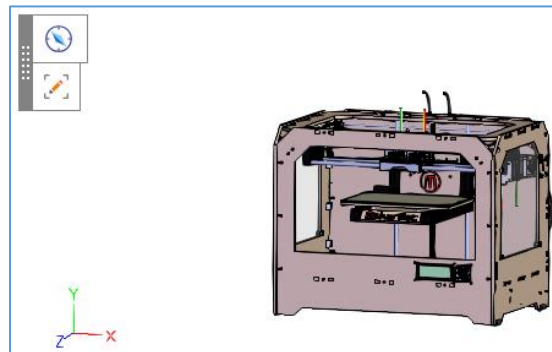


Figure 132.

Use the Viewing toolbar and Model context menu for advanced viewing.

#### 9.8.2.1 Using the Viewing Toolbars

Click the **View** toolbar button to launch the viewing mode and to access the Basic Viewing toolbar:

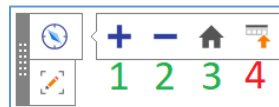


Figure 133.

1. **Zoom Up** increases the CAD Model size on the screen.
2. **Zoom Down** decreases the CAD Model size on the screen.
3. **Reset View** sets the default CAD Model orientation and size on the screen.
4. **Switch To Standard Toolbar** shows the **Standard Viewing** toolbar.

The **Standard Viewing** toolbar includes the same buttons with the exception of the **Switch To Standard Toolbar** button. The **Switch To Basic Toolbar** button appears instead. There are additional features on the **Standard Viewing** toolbar:

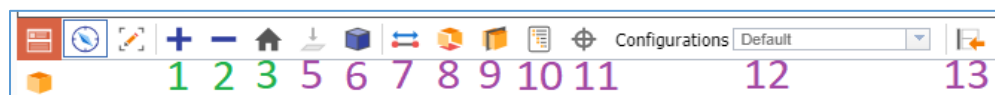


Figure 134.

5. **Orient to Face** shows the CAD Model oriented to you with the surface (part) selected.

6. **Set Display Style** shows/hides a toolbar with options on how to display the CAD Model.
7. **Measure** shows/hides a toolbar with options for measuring the CAD Model.
8. **Exploded View** shows/hides the toolbar with the exploding slider.
9. **Cross Section** shows/hides a toolbar with options for cross-section viewing of the CAD Model.
10. **Model Browser** shows/hides viewing options as trees.
11. **Product Manufacturer Information** displays annotations and attributes contained in a 3D solid model that complement the geometric definition and complete the digital data set.
12. **Configurations** is the list of configurations available within the CAD Model.
13. **Switch to Basic Toolbar** shows the **Basic Viewing** toolbar.

### 9.8.2.2 Using the CAD Model Context menus

When viewing a CAD Model, **right-click** the CAD Model to access its context menu:

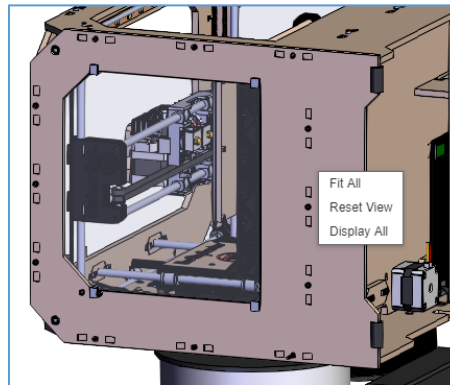


Figure 135.

- **Fit All** sets the CAD Model size so that all parts are shown on the screen.
- **Reset View** sets the default CAD Model orientation and size on the screen.
- **Display All** shows all the Model parts.

Click an individual part to select it and then right-click to access the part context menu:

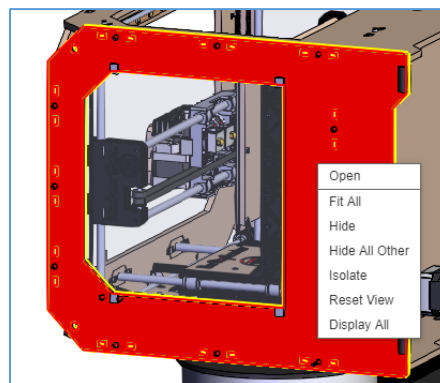


Figure 136.

- **Open** opens the CAD Document view of the selected part.

- **Fit All** sets the CAD Model size so that all parts are shown on the screen.
- **Hide** hides the selected part from showing on the screen.
- **Hide All Other** hides everything appearing on the screen except the selected part.
- **Isolate** makes everything transparent except the selected part.
- **Reset View** sets the default CAD Model orientation and size on the screen.
- **Display All** shows all the Model parts.

### 9.8.2.3 Setting the CAD Model Display Style

The **Display Style** toolbar provides the following display options for the CAD Model. Refer to section [9.8.2.1 Using the Viewing Toolbars](#) to access the button.

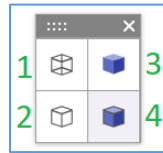


Figure 137.

1. **Wireframe** displays wireframes of the CAD Model assembly and all its parts. Surfaces are transparent.

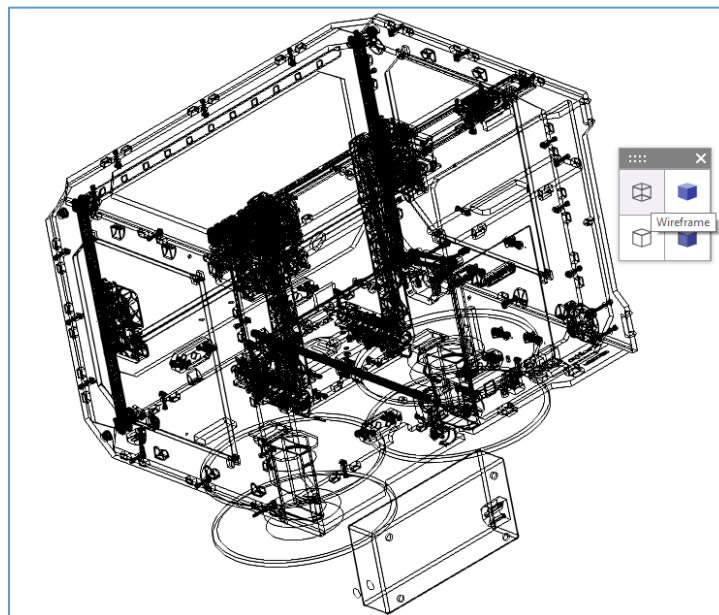


Figure 138.

2. **Hidden Line** displays the white surfaces and black edges of the CAD Model assembly.

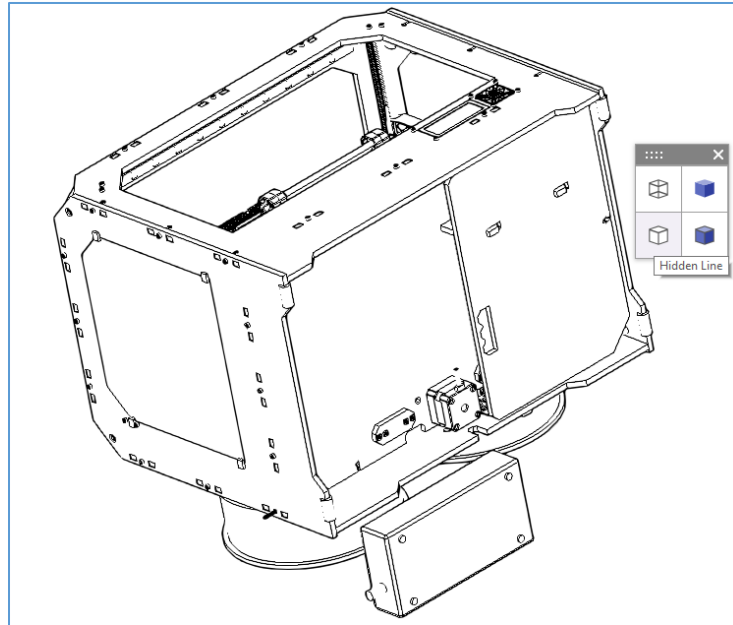


Figure 139.

3. **Shaded** displays the CAD Model assembly surfaces in color.

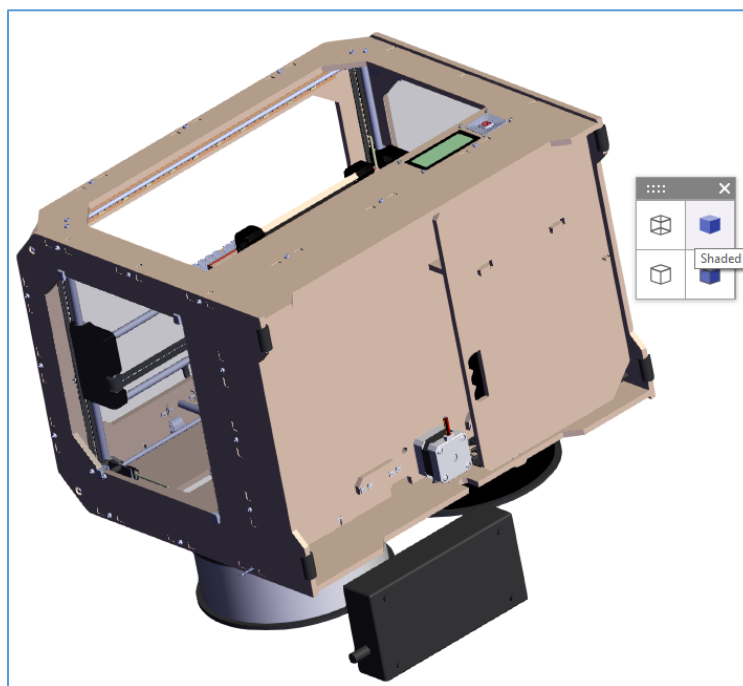


Figure 140.

4. **Wireframe on Shaded** displays the CAD Model assembly with both wireframes and colored surfaces. This is the default display style.

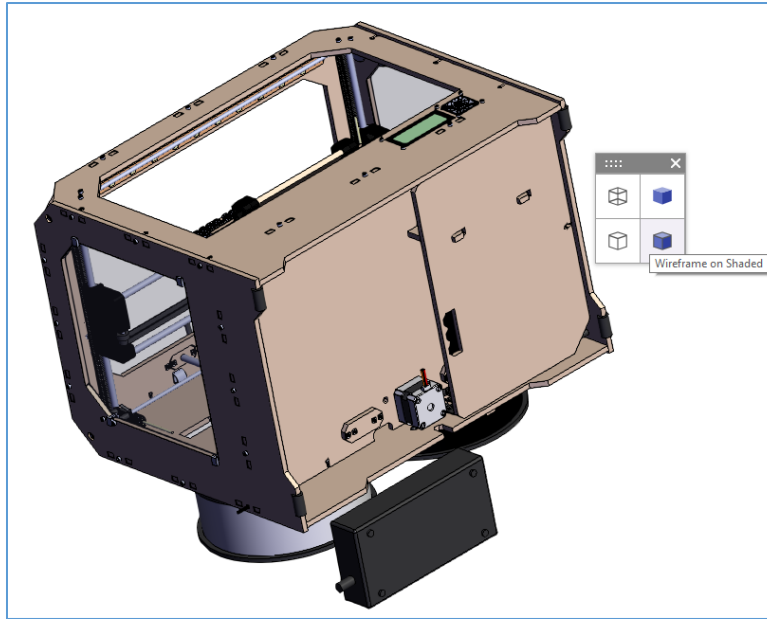


Figure 141.

#### 9.8.2.4 Measuring CAD Models

The **Measure** toolbar enables you to specify the type of measurement to use for the CAD Model. Measurements are displayed in the CAD Model authored units. Refer to section [9.8.2.1 Using the Viewing Toolbars](#) to access the button.

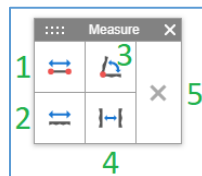


Figure 142.

1. **Measure Point to Point** shows the distance between two arbitrary points. Click this button and then click the CAD Model to set these points and measure the distance between them.



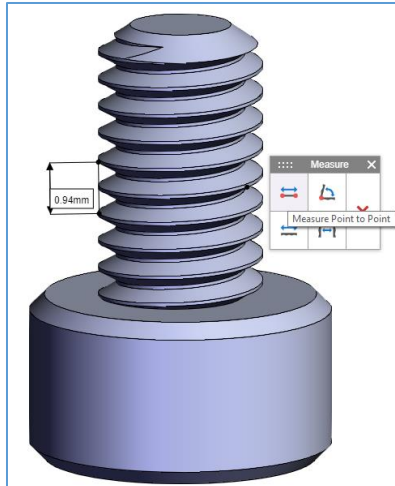


Figure 143.

2. **Measure Edges** shows the length or radius of an arbitrary edge depending on the edge type. Click this button and then click an edge to measure it. The measured edges are bolded.

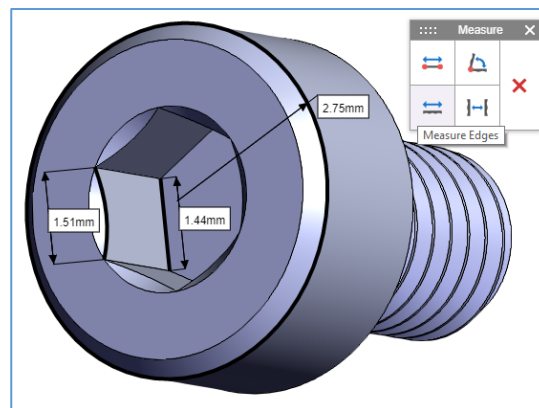


Figure 144.

3. **Measure Angle Between Faces** shows the angle in grades between two arbitrary surfaces of the CAD Model assembly. Click this button and then click the appropriate surfaces to measure them.

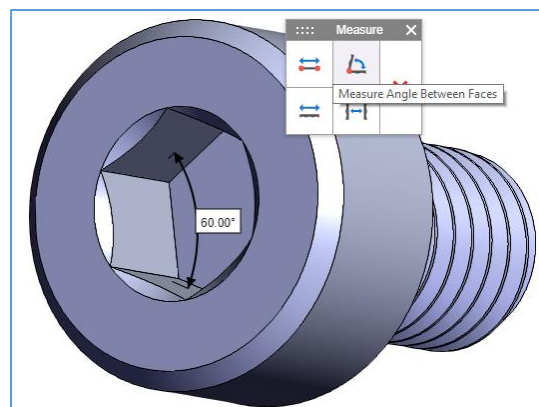


Figure 145.

4. **Measure Distance Between Faces** shows the distance between two arbitrary surfaces of the CAD Model assembly. Click this button and then click the appropriate surfaces to measure the distance between them.

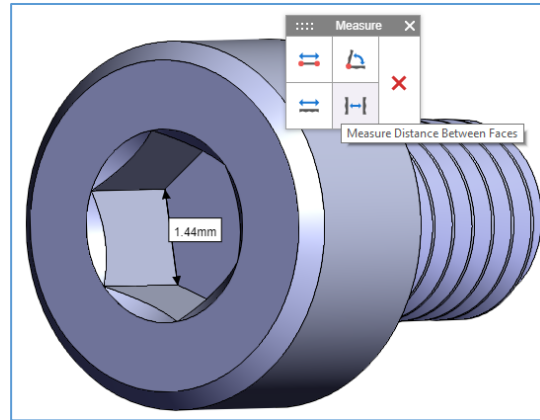


Figure 146.

5. Click **Delete All** to clear all measurements from the CAD Model.

#### 9.8.2.5 Exploding CAD Models

The **Exploded View** toolbar enables you to view assemblies as a set of separated parts to be fitted. Drag the **slider**. As you move it to the right, the more separated the parts become. Refer to section [9.8.2.1 Using the Viewing Toolbars](#) to access the button.

You can use other viewing and markup features, like measure or labels, in the exploded view.

To quit the exploded view, you can click the **Reset View** button on the viewing toolbar or context menu.

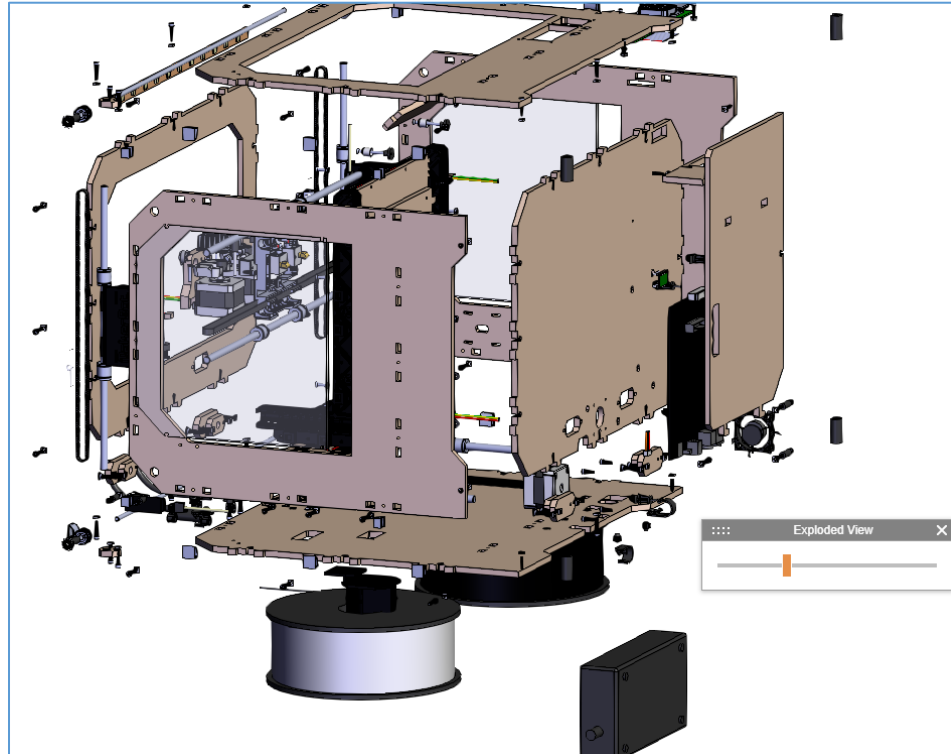


Figure 147.

### 9.8.2.6 Cross Section Viewing of the CAD Model

The **Cross Section** toolbar enables you to view the CAD Model in sections providing the following features. Refer to section [9.8.2.1 Using the Viewing Toolbars](#) to access the button.

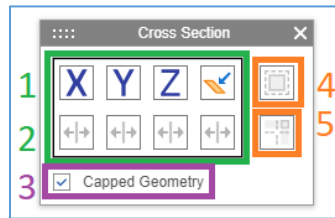


Figure 148.

1. Use the **set of plane buttons** to specify cutting planes for the CAD Model. The **X**, **Y**, and **Z** buttons set the cutting planes according to the Model coordinate system. The **Pick Plane** button sets an arbitrary cutting plane: click a surface and then click this button. To cut the CAD Model in sections, drag the cutting planes.

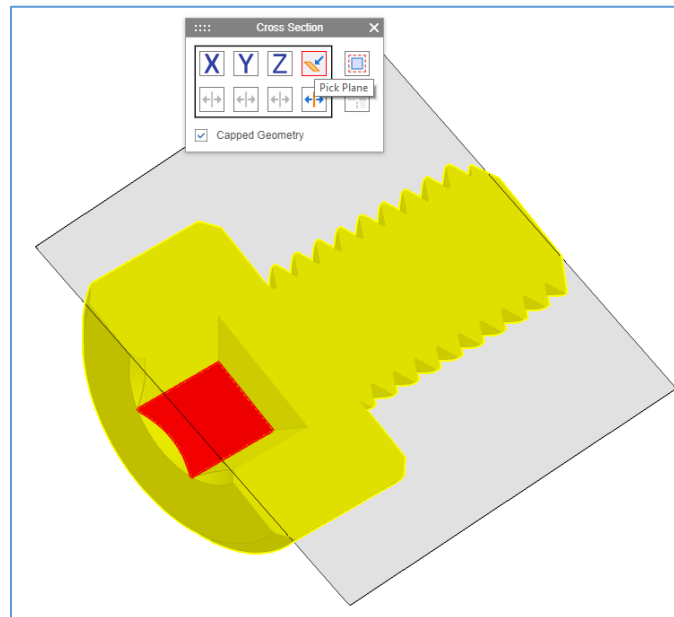


Figure 149.

2. Each **plane** button includes an **Invert Section** button. Click this button to determine which of the cut sections is visible.

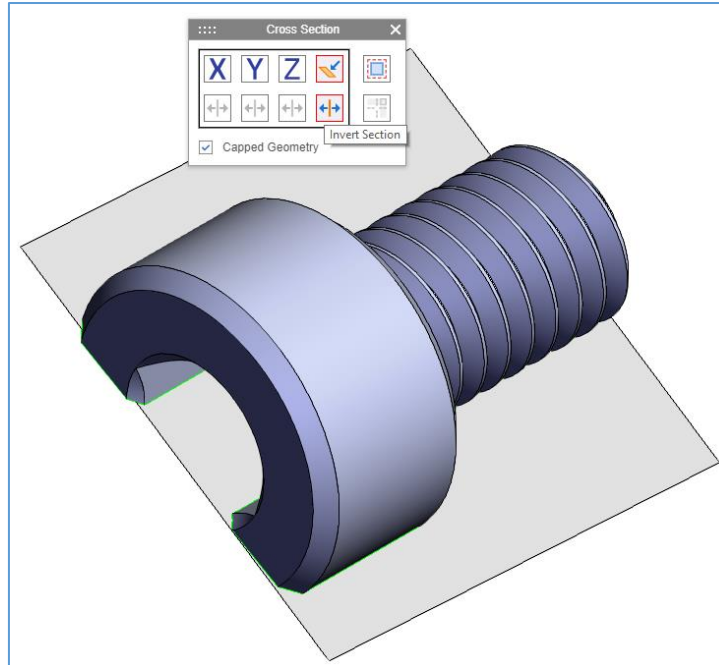


Figure 150.

3. Click the **Capped Geometry** check box to display the cut section in green.

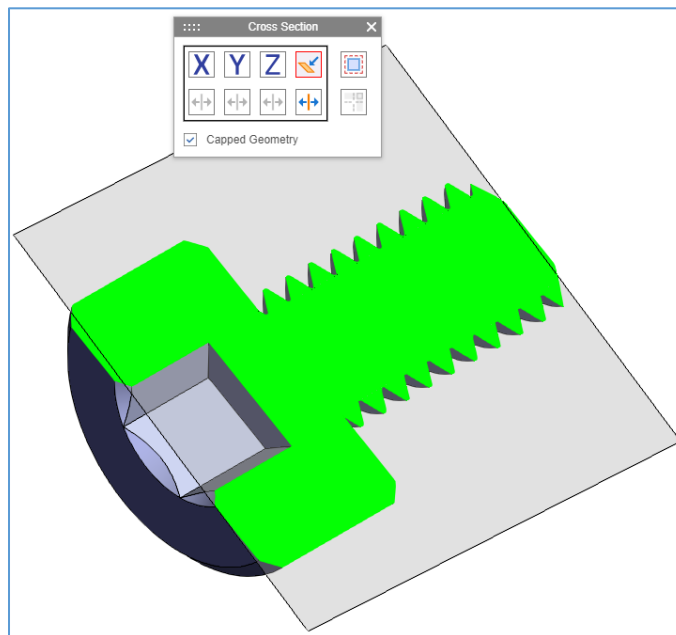


Figure 151.

If you do not select the checkbox, the cut section is hollow.

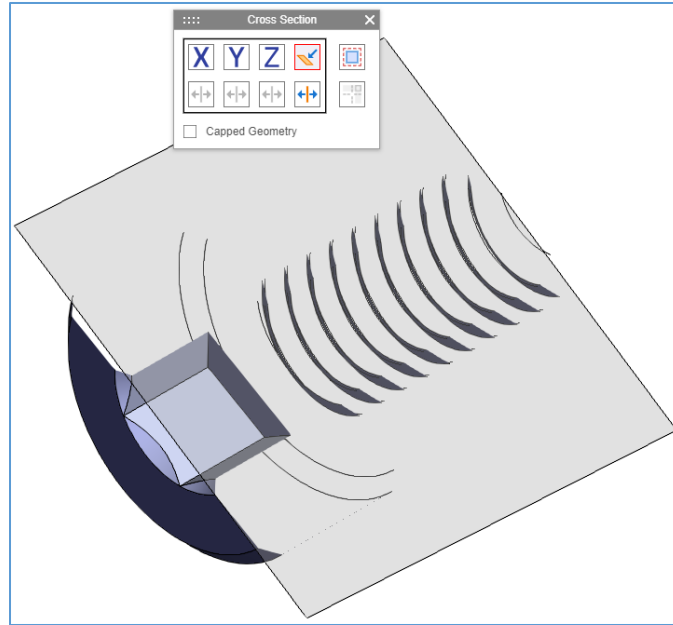


Figure 152.

4. Click the **Toggle Cutting Plane Display** button to hide/show the cutting planes.

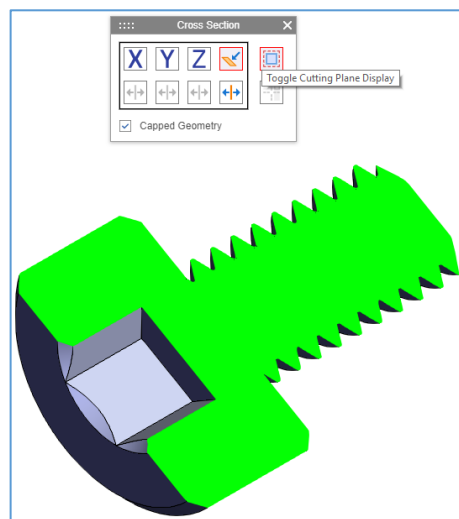


Figure 153.

5. When using several cutting planes, the **Toggle Cross Section** button becomes available, enabling you to select which sides of the planes can cut.

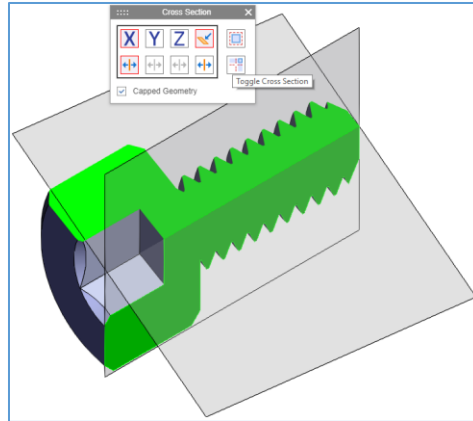


Figure 154.

6. Click the **Toggle Cross Section** button to toggle the cutting sides of the planes. You can combine this button with the Invert Section buttons of these planes.

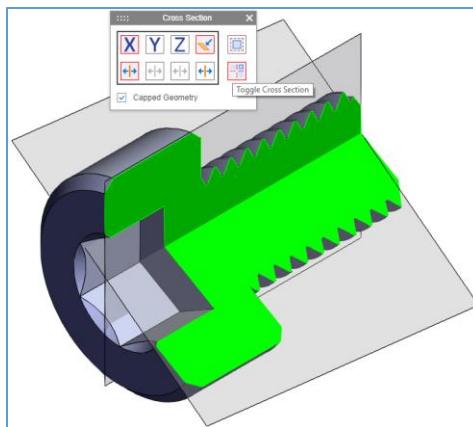


Figure 155.

To enable other viewing and markup functionality in the exploded view, hide the cutting planes.

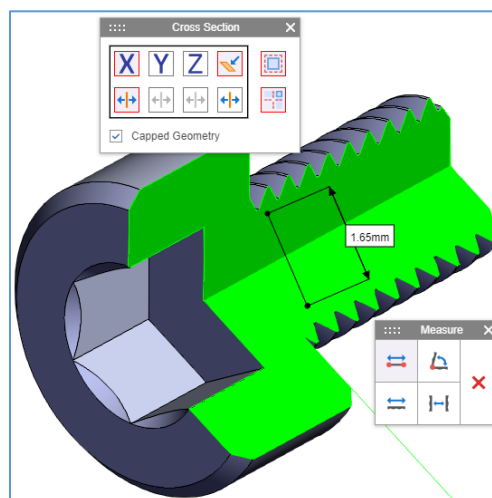


Figure 156.

To quit the cross-section view of a Model, close the **Cross Section** toolbar.

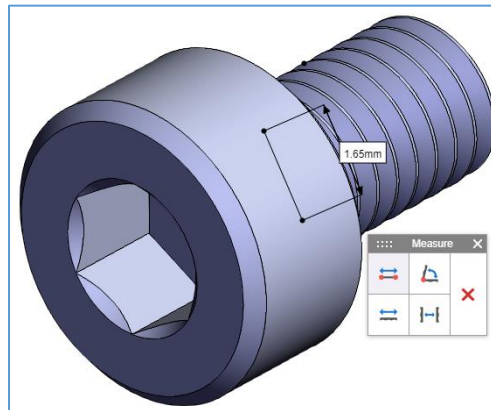


Figure 157.

### 9.8.2.7 Browsing the CAD Model

You can use the **Model Browser** to switch quickly between CAD Model viewing positions. The **Model Browser** appears at the left side with the **Views** tab. It displays viewing position options specific to the CAD Model. Click an option to apply its view. Refer to section [9.8.2.1 Using the Viewing Toolbars](#) to access the button.

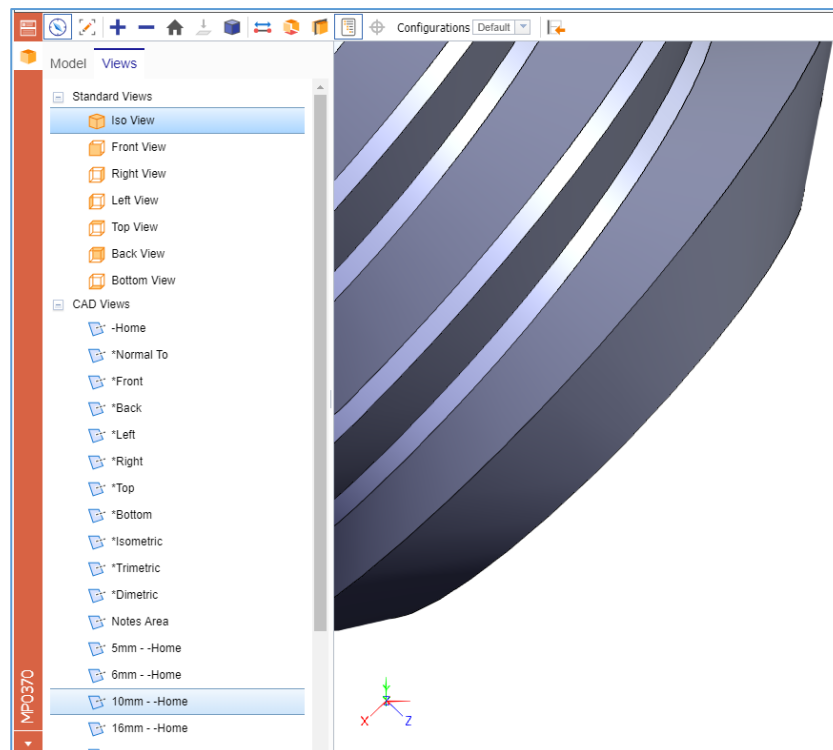


Figure 158.

For assemblies, the **Model Browser** provides the **Model** tab for quick navigation through the assembly structure. It appears as a tree of CAD Documents. Click a part on the CAD Model, and its CAD Document is highlighted in the tree. You can also click a CAD Document in the tree, and its part is highlighted on the CAD Model.

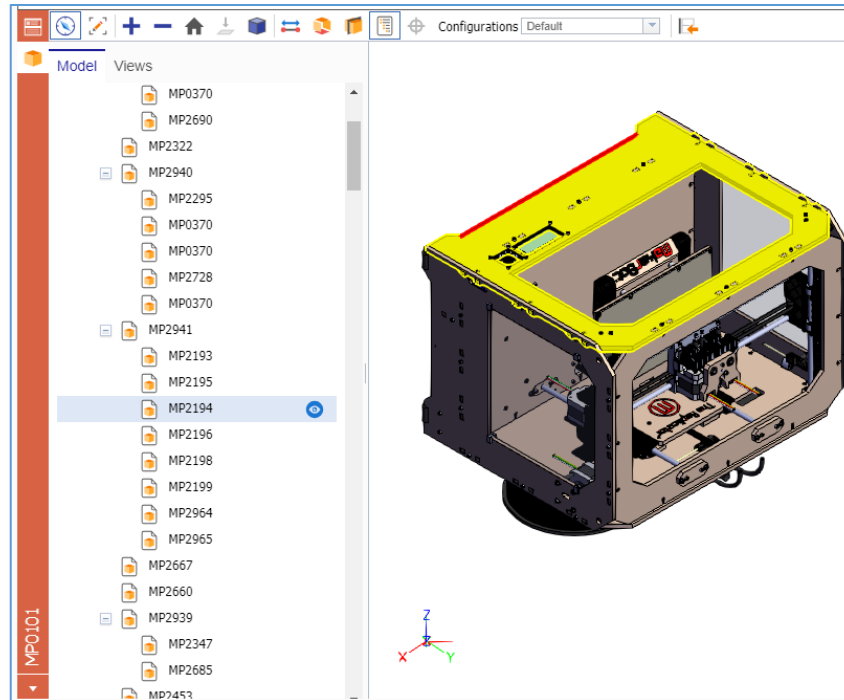


Figure 159.

Each CAD Document has the **Toggle Part Display** button that enables you to hide/show this part on the CAD Model. Hidden parts in the CAD Documents appear light grey in the tree.

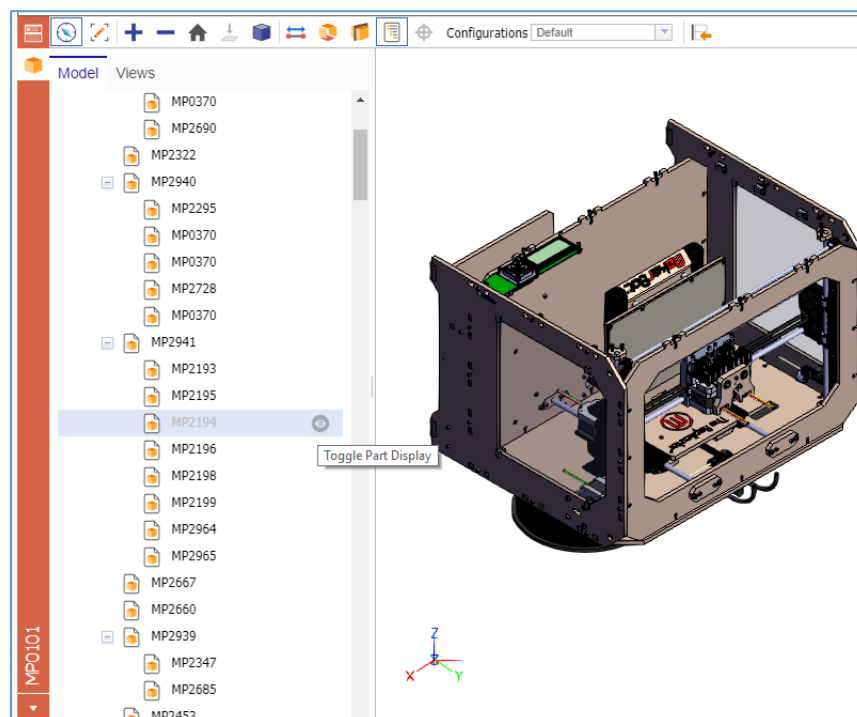


Figure 160.

The context menu is available for CAD Document in the tree.



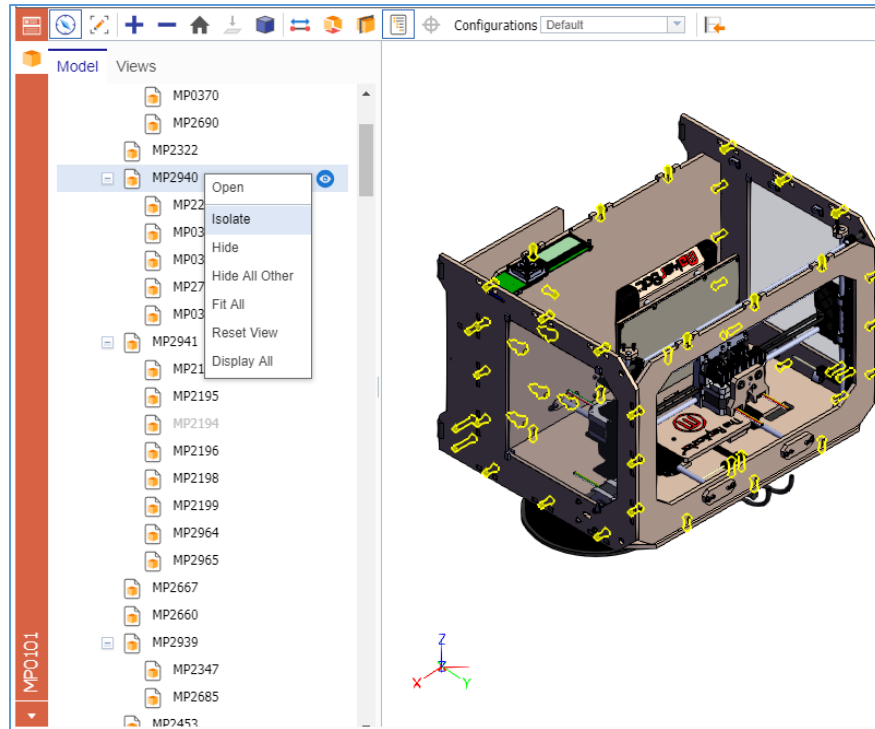


Figure 161.

Click the **Model Browser** button on the **Standard Viewing** toolbar to quit the Model Browser. To show hidden parts, click **Display All** in the context menu.

### 9.8.3 Marking up CAD Models

The CAD integration subscription enables you to mark areas of interest on CAD Models right in Aras Innovator. To do so, position the CAD Model appropriately and then use a **Markup** toolbar as described in the following subsections. The Model context menu and viewing functionality are not available in markup mode. To use them, launch viewing mode.

#### 9.8.3.1 Using the Markup Toolbars

Click the **Markup** toolbar button to launch markup mode and access the **Basic Markup** toolbar. The following buttons appear:



Figure 162.

1. **Scribble** shows/hides a toolbar that enables you to draw arbitrary lines on the CAD Model.
2. **Highlight** shows/hides a toolbar that enables you to draw rectangular areas on the CAD Model.
3. **Label** shows/hides a toolbar that enables you to add labels to the CAD Model.
4. **Delete** removes selected markup items from the CAD Model.
5. **Switch To Standard Toolbar** displays the **Standard Markup** toolbar.

The **Standard Markup** toolbar includes the same buttons with the exception of the **Switch To Standard Toolbar** button. The **Switch To Basic Toolbar** button appears instead. There are additional buttons on the **Standard Markup** toolbar:

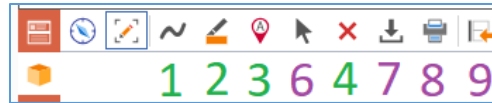


Figure 163.

1. **Select** selects markups on the CAD Model.
2. **Download File** downloads a .jpg image of the CAD Model.
3. **Print** uses the standard browser print function for CAD Model images. Depending on the browser and your specific configuration, various print choices, such as printing to a printer or saving to PDF, are available.
4. **Switch to Basic Toolbar** shows the **Basic Markup** toolbar.

Unlike the **Basic Markup** toolbar, the **Standard Markup** toolbar is dynamic. The **Standard Markup** toolbar does not show/hide an additional toolbar containing functionality for a selected markup feature (buttons 1-3), it displays functionality for the selected markup feature. The selected feature and functionality options are enclosed by blue boxes. To hide the functionality options, click the selected feature.

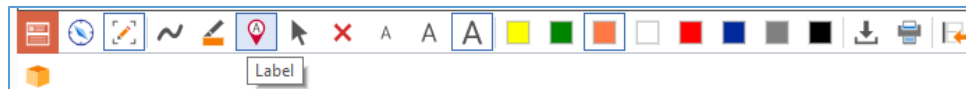


Figure 164.

Markup features discussed in the following subsections assume that you have accessed the corresponding feature.

### 9.8.3.2 Drawing on CAD Models

You can draw arbitrary lines on a CAD Model using the scribble feature. Refer to section [9.8.3.1 Using the Markup Toolbars](#) to access the button.

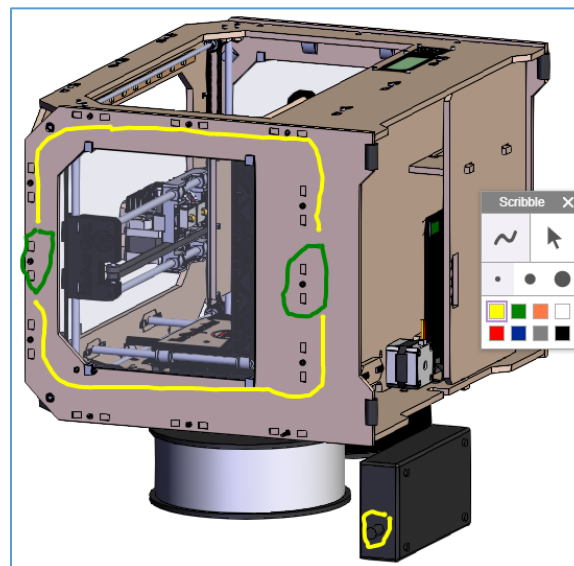


Figure 165.

When accessed from the **Basic Markup** toolbar, the scribble feature appears as the **Scribble** toolbar.

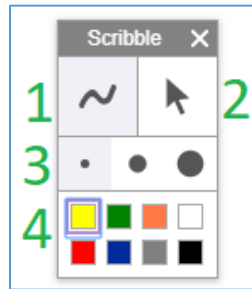


Figure 166.

When accessed from the **Standard Markup** toolbar, the scribble feature is hosted on this toolbar.

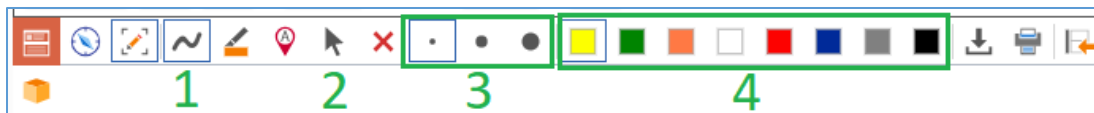


Figure 167.

The scribble feature provides the following functionality:

1. **Scribble** draws arbitrary lines on the CAD Model.
2. **Select** selects markups on the CAD Model.
3. You can select one of the following line widths: **Small**, **Medium**, and **Large**.
4. You can select one of the colors displayed on the toolbar.

Click a feature or option to use it.

Click **Select** and then click a line to select it. When selected, the line has **blue edges with circles**. You cannot reposition selected lines.

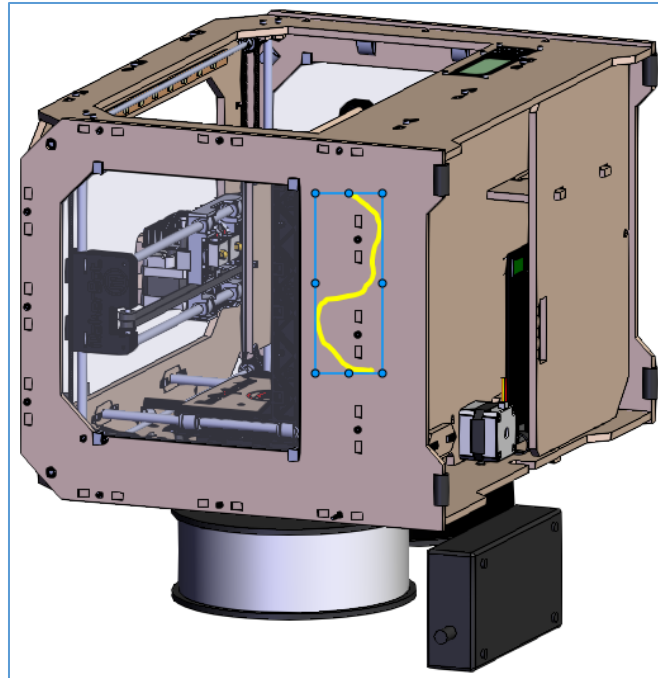


Figure 168.

### 9.8.3.3 Highlighting Areas on CAD Models

You can specify arbitrary rectangular areas on a CAD Model using the highlighting feature. Refer to section [9.8.3.1 Using the Markup Toolbars](#) to access the button.

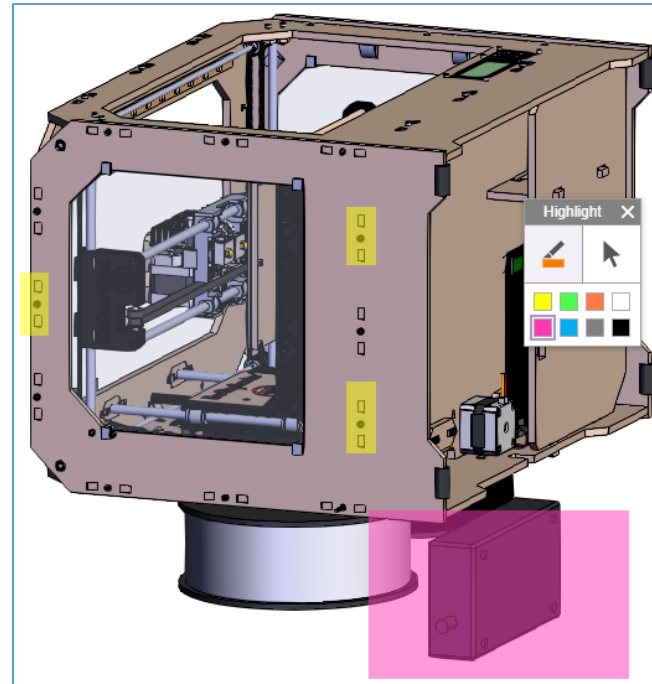


Figure 169.

When accessed from the **Basic Markup** toolbar, the highlighting feature appears as the **Highlight** toolbar.

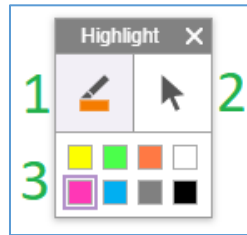


Figure 170.

When accessed from the **Standard Markup** toolbar, the highlighting feature is hosted on this toolbar.



Figure 171.

The highlighting feature provides the following functionality:

1. **Highlight** draws arbitrary rectangles on the CAD Model.
2. **Select** selects markups on the CAD Model.
3. A set of rectangle colors.

Click a feature or option to use it.

Click **Select** and then click a rectangle area to select this area. When selected, the rectangle area has **blue edges with circles**.

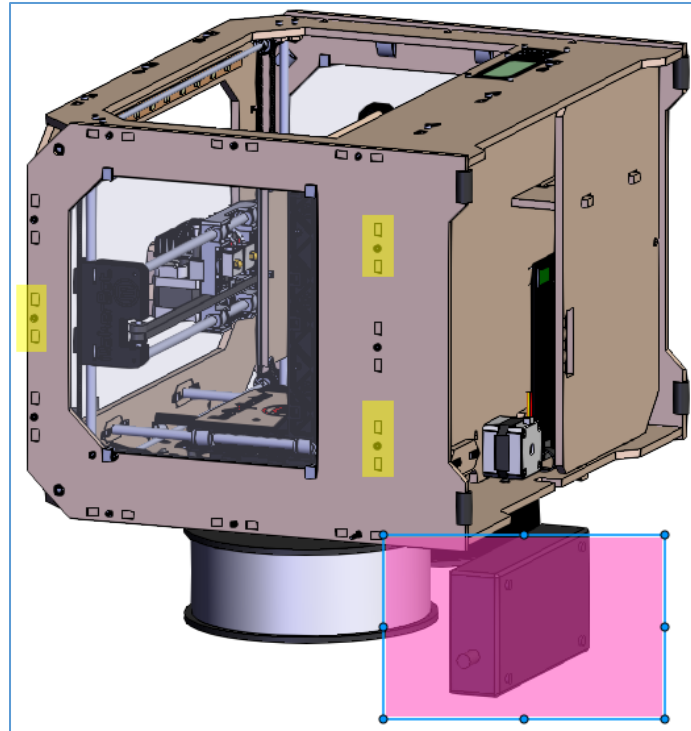


Figure 172.

Drag the **blue circles** on the selected rectangle **edges** to reposition the rectangle.

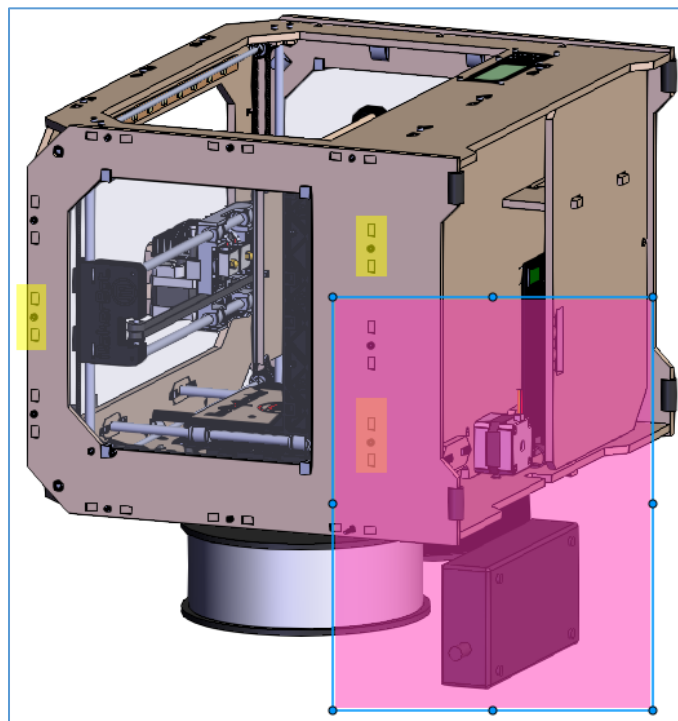


Figure 173.

#### 9.8.3.4 Putting Labels on CAD Models

You can label parts of a CAD Model using the label feature. Refer to section [9.8.3.1 Using the Markup Toolbars](#) to access the button.

When accessed from the **Basic Markup** toolbar, the label feature appears as the **Label** toolbar.

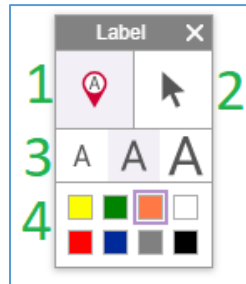


Figure 174.

When accessed from the **Standard Markup** toolbar, the label feature appears on this toolbar.



Figure 175.

The label feature provides the following functionality:

1. **Label** adds labels to the CAD Model.
2. **Select** selects markup items on the CAD Model.
3. A set of font sizes: **Small**, **Medium**, and **Large**.
4. A set of label frame colors.

Click a feature or option to use it.

To create a label:

1. Click the **Label** button.
2. Click on the area of the CAD Model that you want to label.
3. Enter the appropriate text inside the label.
4. Click **Select** and then click a label to select it. When selected, the label has **blue edges with a circle at the bottom**.

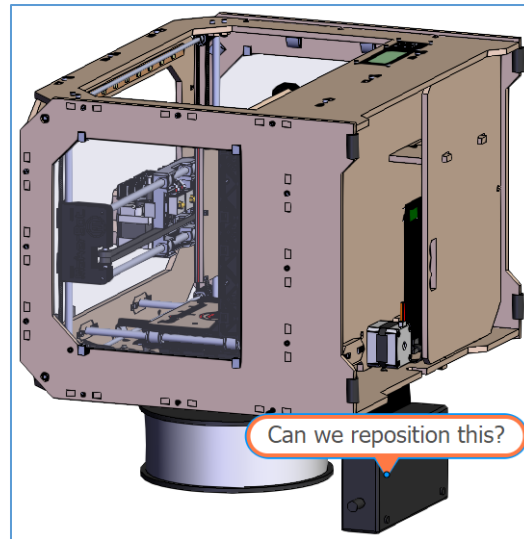


Figure 176.

To reposition a label, select and drag **its colored frame from right above the blue bottom circle** to the appropriate position on the CAD model.

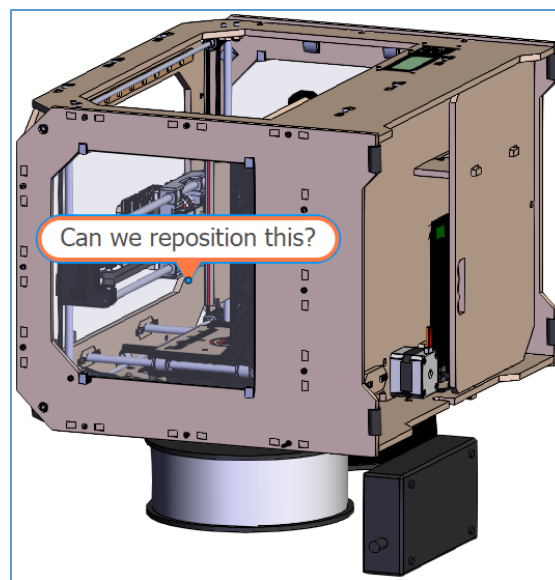


Figure 177.

To change where a label points to, select the label, and drag **the blue bottom circle**.



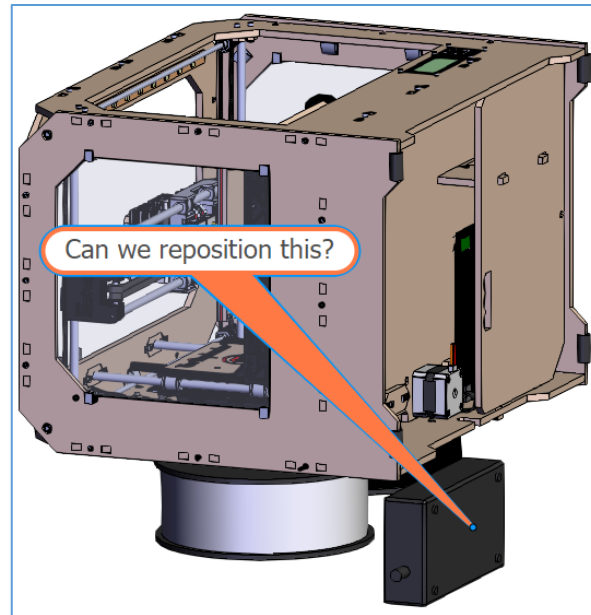


Figure 178.

To enter or change the label text, click inside the label, and edit the text.

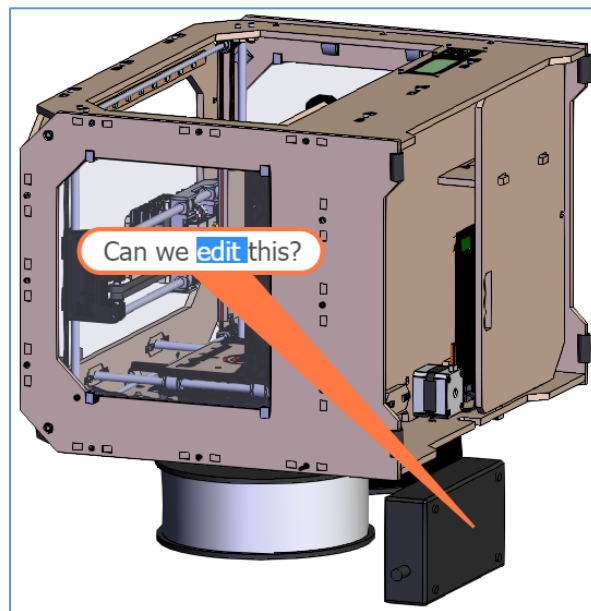


Figure 179.

### 9.8.3.5 Selecting Markups on CAD Models

You can select a single or several markup items for further actions. Selected items have **bold blue edges** with **circles**. Further actions, position and number of the **circles** depend on the markup type of each item.

To select a single markup item, click **Select** on a **Markup** toolbar and then click the **item**.

To select several markup items, click **Select** on a **Markup** toolbar and drag the cursor to specify the rectangular selecting area.

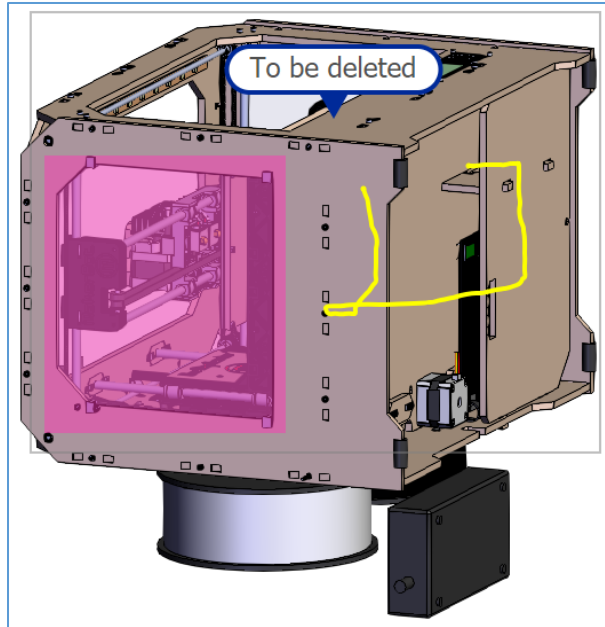


Figure 180.

All the markup items within the selected area become selected.

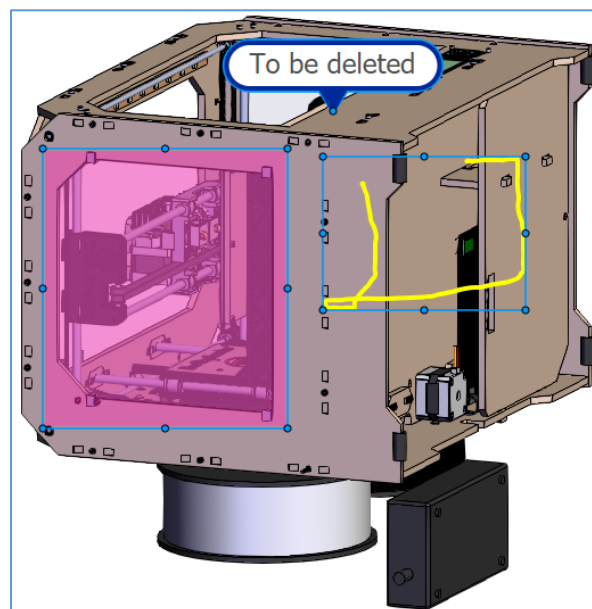


Figure 181.

#### 9.8.3.6 Deleting Markups on CAD Models

To delete markup items, select them and then use one of the following options:

- Click **Delete** on a **Markup** toolbar.
- Press **Delete** on your keyboard.

## 9.8.4 Downloading CAD Models

Click **Download File** on the **Standard Markup** toolbar. The standard downloading process of your browser saves a .jpg image of the CAD Model representation on your local machine. Refer to section [9.8.3.1 Using the Markup Toolbars](#) to access the button.

## 9.8.5 Printing CAD Models

Click **Print** on the **Standard Markup** toolbar. The standard print function of your browser starts. Depending on the browser and your specific configuration, various print choices, such as printing to a printer or saving to PDF, are available. Refer to section [9.8.3.1 Using the Markup Toolbars](#) to access the button.

# 9.9 Deleting CAD Documents

It is possible to delete all or some versions of a CAD Document. In the former case, the CAD Document is deleted from Aras Innovator completely. For details on the latter case, refer to section [4.1.2 Version Deletion](#).

To delete a CAD Document completely, you must have **Delete** rights granted in Permissions for all Life Cycle states associated with that CAD Document. For example, if the new CAD Document revision is currently in the Preliminary state, but has Released revisions, you must have the **Delete** rights granted in Permissions for both life cycle states. For details on access rights, refer to section [3. Permissions](#). This subsection assumes that you have all necessary rights.

You cannot delete a CAD Document, which is attached to a Part or another CAD Document.

**Note:** CAD Documents that are claimed by any user or are used on other items cannot be deleted.

CAD Documents can be deleted using either its view or the context menu in the CAD Documents Search Page.

To delete a CAD Document from its view:

1. Open the CAD Document.
2. On the CAD Document toolbar, click **More → Delete → Delete All Versions**.

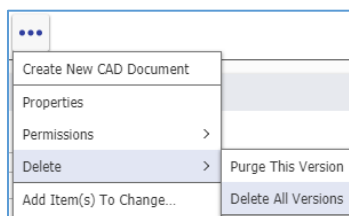


Figure 182.

3. Click **OK** in the confirmation dialog to delete the CAD Document.

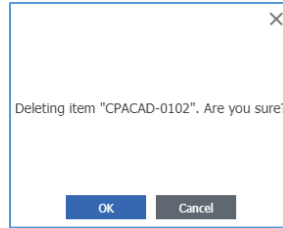


Figure 183.

To delete a CAD Document using its context menu:

1. Go to the Documents Search Page.
2. Right-click the Document to be deleted and click **More** → **Delete** → **Delete All Versions**.
3. Click **OK** in the confirmation dialog to delete the Document.

## 9.10 Finding Where a CAD Document is Used

There are several methods to determine where a CAD Document is used:

- The Parents accordion tab on the CAD Document
- Parts associated with a CAD Document
- The Where Used browser

### 9.10.1 CAD Document Parents

The **Parents** accordion tab of the CAD Document view shows a read-only list of the CAD Document's direct Parents.

| Document Number | Revision | Name                   | Type                | State       | Claimed By ... | Native File [...]             | Viewable File [...]        | Authoring Tool | Changes                  | Standard                 | Template                 | Gen |
|-----------------|----------|------------------------|---------------------|-------------|----------------|-------------------------------|----------------------------|----------------|--------------------------|--------------------------|--------------------------|-----|
| MP2940          | A        | Body Hardware          | Mechanical/Assembly | Released    |                | <a href="#">MP2940.SLDASM</a> | <a href="#">MP2940.pdf</a> | SolidWorks     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | 1   |
| MP2943          | A        | Build Platform         | Mechanical/Assembly | Released    |                | <a href="#">MP2943.SLDASM</a> | <a href="#">MP2943.pdf</a> | SolidWorks     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | 1   |
| MP2948          | A        | Cover Hardware         | Mechanical/Assembly | Preliminary |                | <a href="#">MP2948.SLDASM</a> | <a href="#">MP2948.pdf</a> | SolidWorks     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | 1   |
| MP2953          | A        | Extruder Hardware      | Mechanical/Assembly | Released    |                | <a href="#">MP2953.SLDASM</a> | <a href="#">MP2953.pdf</a> | SolidWorks     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | 1   |
| MP2956          | A        | Gantry                 | Mechanical/Assembly | Released    |                | <a href="#">MP2956.SLDASM</a> | <a href="#">MP2956.pdf</a> | SolidWorks     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | 1   |
| MP2963          | A        | Stepper Motor Assembly | Mechanical/Assembly | Released    |                | <a href="#">MP2963.SLDASM</a> | <a href="#">MP2963.pdf</a> | SolidWorks     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | 1   |
| MP2968          | A        | XY Stage               | Mechanical/Assembly | Released    |                | <a href="#">MP2968.SLDASM</a> | <a href="#">MP2968.pdf</a> | SolidWorks     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | 1   |

Figure 184.

1. Right-click a Parent CAD Document and click **View** open the selected Parent.
2. To download the native or viewable files, click on the link in the corresponding cell.

## 9.10.2 Parts Associated with a CAD Document

CAD Documents can be attached to a Part, but Parts cannot be attached to a CAD Document. You can use the Parts accordion tab on the CAD Document to explore which Parts include this CAD Document as an attachment.

**Note:** If the CAD Document is not attached anywhere, there are no Part links or Part accordion tabs.

If the CAD Document is attached to a single Part, there is a link to this Part but there is no Parts accordion tab.

**CAD Document**

Document Number: MP0370c Revision: A State: Preliminary

Name: M3xNN Socket Cap Bolt

Type: Part Authoring Tool: SolidWorks Version: sw2014

Description:

Assigned Creator: Innovator Admin

Designated User: Innovator Admin

From Template:

☐ Changes Pending

☐ Standard

☐ Template

[Part MP0190](#)

Figure 185.

If the CAD Document is attached to several Parts, the Part link and the Parts accordion tab appear. Clicking on the Part link opens the Parts accordion tab. Right-click a Part in the Parts grid and click View to open the selected Part.

**CAD Document**

Document Number: MP0370c Revision: A State: Preliminary

Name: M3xNN Socket Cap Bolt

Type: Part Authoring Tool: SolidWorks Version: sw2014

Description:

Assigned Creator: Innovator Admin

Designated User: Innovator Admin

From Template:

☐ Changes Pending

☐ Standard

☐ Template

[Part See tab](#)

Native File: [MP0370.SLDPR](#)

Viewable File: [MP0370.pdf](#)

**Parts** | Structure | Parents | Files | Changes

| Part Number | Re... | Name                  | Type      | State    | Locked By [...] | Cost   | Chan...                  | g... |
|-------------|-------|-----------------------|-----------|----------|-----------------|--------|--------------------------|------|
| MP0370-001  | A     | M3x5 Socket Cap Bolt  | Component | Released |                 | 0.0800 | <input type="checkbox"/> | 1    |
| MP0370-002  | A     | M3x6 Socket Cap Bolt  | Component | Released |                 | 0.0800 | <input type="checkbox"/> | 1    |
| MP0370-003  | A     | M3x10 Socket Cap Bolt | Component | Released |                 | 0.0800 | <input type="checkbox"/> | 1    |
| MP0370-004  | A     | M3x16 Socket Cap Bolt | Component | Released |                 | 0.0800 | <input type="checkbox"/> | 1    |

Figure 186.

If a previous (not the latest) revision of a CAD Document is attached to a Part, the **Part** link is displayed as broken.

Figure 187.

Figure 188.

### 9.10.3 Where Used CAD Document Browser

Aras Innovator offers the Where Used feature to determine where an Item is used by other Items. Refer to section [11.1 Finding Where an Item is Used](#) for further information. All examples in that section are applicable to CAD Documents.

## 9.11 Comparing CAD Documents

To compare complex multi-level hierarchies of CAD Documents side-by-side, Aras Innovator offers the Structure Browser. Refer to section [11.2 Comparing Items](#) for further information. All examples in that section are applicable to CAD Documents.

# 10 Products

A Product is an additional ItemType that represents something that a customer will buy from your company. A Product is related to one or more models, which are then related to a Parent Part item that contains the top-level BOM for the Product Model. Products may or may not be useful in your organization depending on your requirements and are not necessary for release or modification of Parts. As the Product is not directly under change management control, it has no Life Cycle.

## 10.1 Creating a Product

To create a Product:

1. Go to **Contents** → **Design** → **Products**.
2. Click **Create New Product**. The new Product view appears. Enter the necessary information.

Figure 189.

- **Product Number** – The Product number.
  - **Name** – The name of the Product.
  - **Description** – Remarks about the Product.
3. Add Models as described in section [10.2 Adding Models to a Product](#).
  4. Click **Save** or **Done** to save your changes and create the Product.

## 10.2 Adding Models to a Product

To add a Model to a Product:

1. Open the Product.

2. Click **Edit**.
3. Go to the **Models** accordion tab.
4. Click **New Model** on the Models accordion toolbar. A new row appears at the bottom of the Models grid.

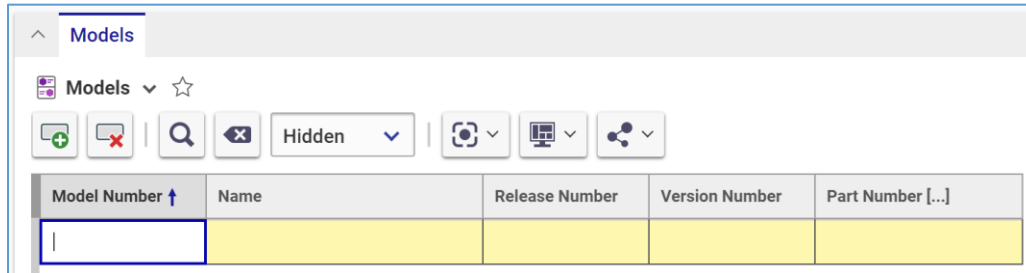


Figure 190.

5. Enter the necessary information by clicking on the corresponding cell in the Models grid.
  - **Model Number** – The Model number.
  - **Model Name** – The Model name.
  - **Release Number** – For software products, this is the release number.
  - **Version Number** – For software products, this is the version number.
  - **Part Number** – The Part that represents the Model.
6. Click **Save** or **Done** to save your changes. The Model is attached to the Part.

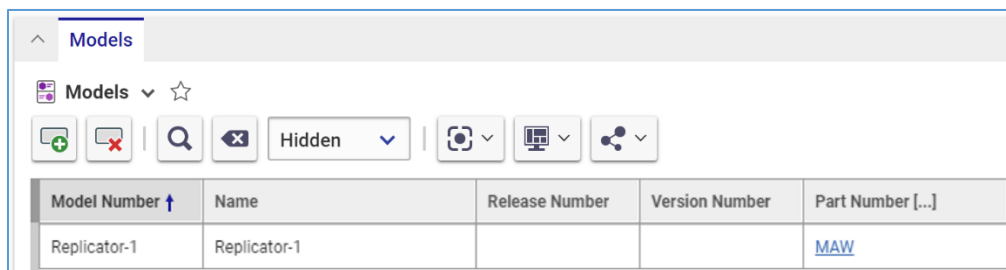


Figure 191.

## 10.3 Removing Models from a Product

To remove Models from a Product:

1. Open the Product.
2. Click **Edit**.
3. Go to the **Models** accordion tab.
4. Click the Model to be removed and then click **Delete Model** on the Models accordion toolbar. The Model becomes greyed out.



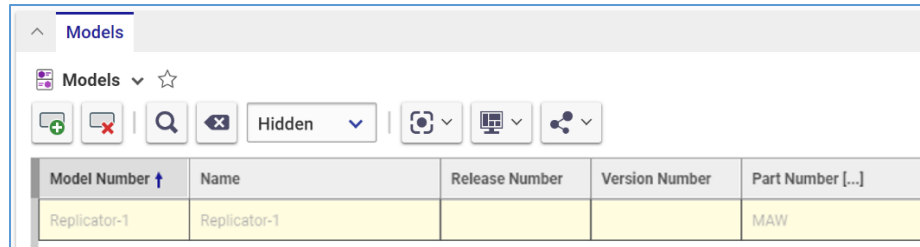


Figure 192.

- Click either **Save** or **Done** to save your changes. The **Model** disappears from the grid.

## 10.4 Deleting a Product

You can delete a Product either from its view or by using the context menu in the Product Search Page.

**Note:** A claimed Product cannot be deleted.

To delete a Product using its view:

- Open the **Product**.
- On the **Product** toolbar, click **More → Delete → Delete**.

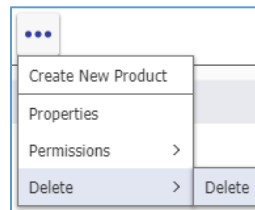


Figure 193.

- Click **OK** in the confirmation dialog to delete the Product.

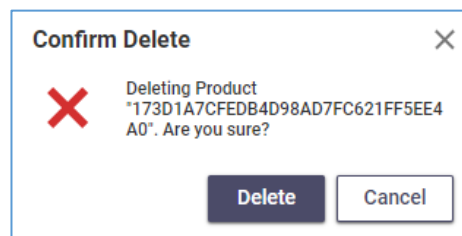


Figure 194.

To delete a Product using its context menu:

- Go to the Products Search Page.
- Right-click the Product to be deleted and then click **More → Delete**.
- Click **OK** in the confirmation dialog to delete the Product.

# 11 Exploring Usage and Comparing Items

The Product Engineering application features the Where Used and Structure browsers to explore and compare items and their structures.

In this section, the term *Item* means an item of explorable ItemTypes: Parts, Manufacturers, Manufacturer Parts, Vendors, Documents, and CAD Documents. Examples shown on Parts are generic and applicable to other Items. However, detailed information on an Item depends on its type. The appropriate subsections of ItemTypes provide additional information or features for explorations specific to these ItemTypes.

## 11.1 Finding Where an Item is Used

The **Where Used** browser shows upside-down trees of the Item and corresponding hierarchies where the given item is on the top with its Parents under, if any, and each branch represents a separate usage of the given Item including associated and related Items. There are also separate branches for single versions of each Item.

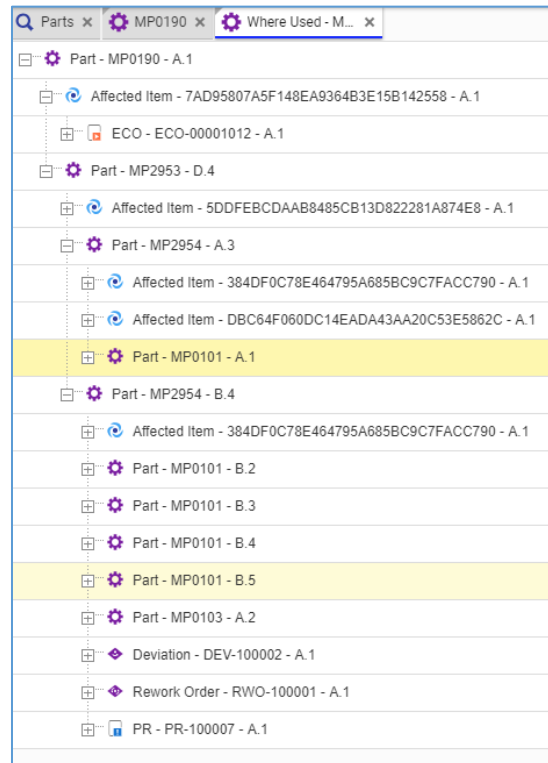


Figure 195.

To access the **Where Used** browser for an Item, use one of the following methods:

- Go to the Item Search Grid Page, right-click the Item, and then click **Navigate → Where Used**.

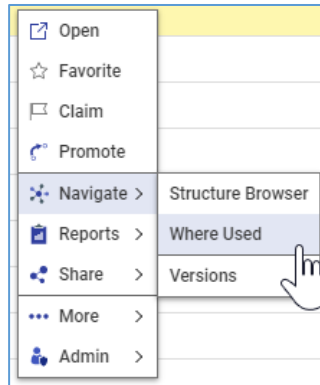


Figure 196.

- Open the Item, click **Navigate** on the **Item** toolbar, and then click **Where Used** in the drop-down menu.

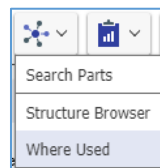


Figure 197.

When you access the Where Used browser, it appears as a new Aras Innovator view.  
To quit the Where Used Browser, close its view in Aras Innovator.

## 11.2 Comparing Items

To compare side-by-side complex multi-level hierarchies of Items, Aras Innovator offers the Structure Browser. You can also use it to explore the usage of a single Item.

There are several ways to access the Structure Browser:

- Go the Item Search Grid Page, right-click an Item to be compared, and then click **Navigate → Structure Browser**.

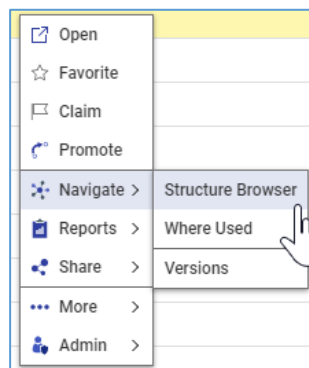


Figure 198.

- Open the view of an Item to be compared. Click **Navigate** on the **Item** toolbar, and then click **Structure Browser**.

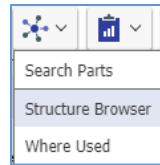


Figure 199.

When you access the Structure Browser, it appears as a new Aras Innovator view showing the Item hierarchy tree where you can see the Number and Revision for each Item.

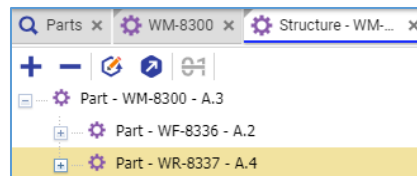


Figure 200.

In addition to the standard tree navigation features, the following buttons are available in the Structure Browser toolbar:

- **Expand All**, which opens the whole hierarchy tree.

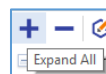


Figure 201.

- **Collapse All**, which closes the whole hierarchy tree.

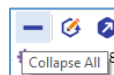


Figure 202.

To quit the Structure Browser, close its view.

### 11.2.1 Comparing Different Item Versions

Use the following procedure to compare the current version of an Item with an older one:

1. Click **Side-by-side with other Version** on the Structure Browser toolbar.



Figure 203.

The Item versions dialog appears.

| Item versions |          |                     |          |             |           |                                     |            |  |
|---------------|----------|---------------------|----------|-------------|-----------|-------------------------------------|------------|--|
| Part Number   | Revision | Name                | Type     | State       | Cost      | Changes                             | generation |  |
| MP0101        | A        | Makerbot Replicator | Assembly | In Change   | 4813.9059 | <input checked="" type="checkbox"/> | 1          |  |
| MP0101        | B        | Makerbot Replicator | Assembly | Preliminary | 4760.7859 | <input type="checkbox"/>            | 2          |  |
| MP0101        | B        | Makerbot Replicator | Assembly | Preliminary | 4760.7859 | <input type="checkbox"/>            | 3          |  |
| MP0101        | B        | Makerbot Replicator | Assembly | Preliminary | 4760.7859 | <input type="checkbox"/>            | 4          |  |
| MP0101        | B        | Makerbot Replicator | Assembly | Preliminary | 4760.7859 | <input checked="" type="checkbox"/> | 5          |  |

Figure 204.

- Double-click the **version** to be compared. The hierarchy tree of the selected Item version appears on the right side of the Structure Browser, and the Highlight button becomes available on the Structure Browser toolbar.

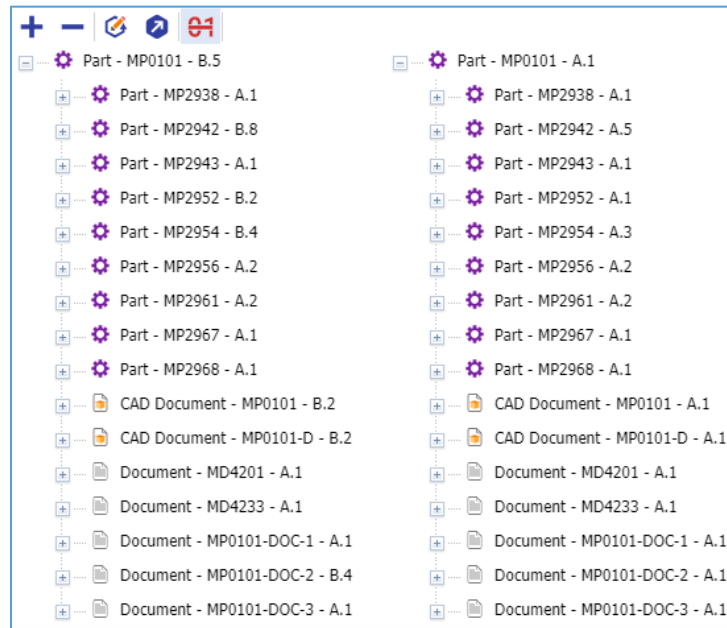


Figure 205.

- Click **Highlight** on the Structure Browser toolbar.

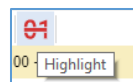


Figure 206.

The trees open. Differences are highlighted.

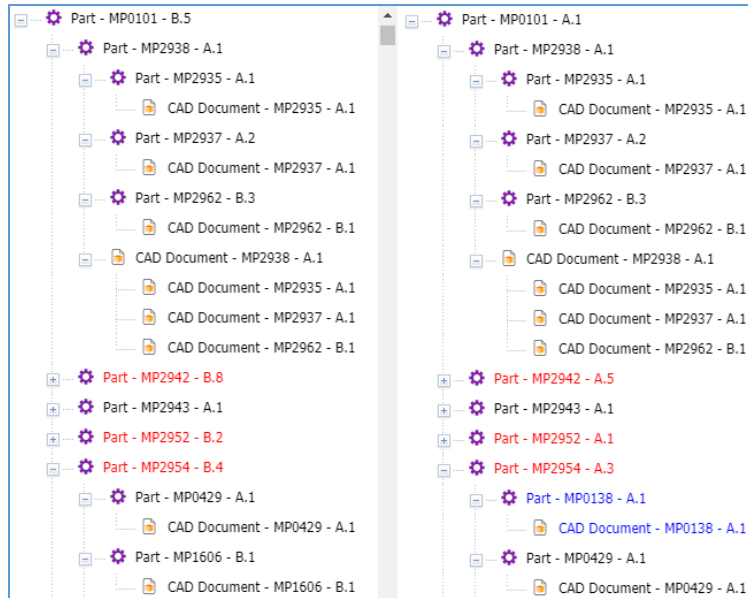


Figure 207.

- Black means that both Item versions use the same version of this Item.
  - Red means that both Item versions have different versions of this Item.
  - Blue means that this Item does not exist in the opposite Item version.
4. If it is necessary to view detailed information on an Item version, double-click the Item to open its view.

**Note:** You cannot compare more than two versions of an Item in one Structure Browser tab.

### 11.2.2 Comparing Different Items

To compare the hierarchies of two different CAD Documents:

1. Click **Side-by-Side with other ItemType** in the Structure Browser toolbar. The **Search dialog** appears.



Figure 208.

2. Using the standard search procedure, search for and select the Item to be compared. The tree of the selected Item hierarchy appears on the right side of the Structure Browser, and the Highlight button becomes available on the Structure Browser toolbar.

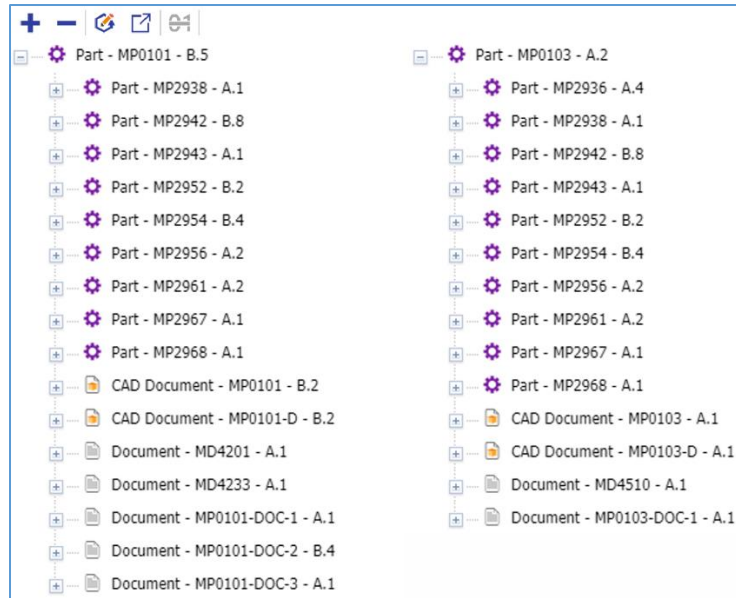


Figure 209.

- Click **Highlight** in the Structure Browser toolbar. The trees appear with differences highlighted.

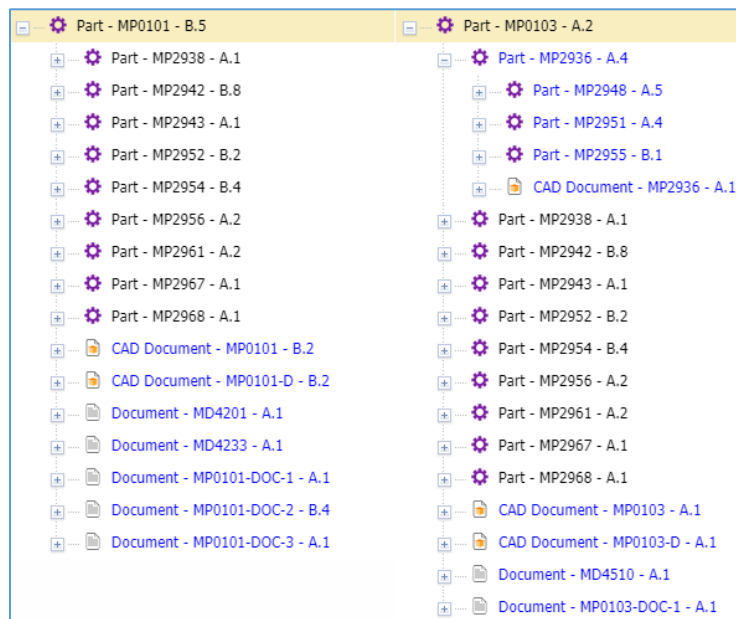


Figure 210.

- Black means that both Item hierarchies use the same version of this Item.
  - Red means that both Item hierarchies have different versions of this Item.
  - Blue means that this Item does not exist in the opposite Item hierarchy.
- If it is necessary to view detailed information on an Item, double-click the Item to open its view.

**Note:** You cannot compare more than two Item hierarchies in one Structure Browser tab.

## 12 Visual Collaboration for Discussing Items

You can discuss Parts, Documents, CAD Documents, CAD Models, etc. with your team right in Aras Innovator. To do so, use the discussion feature as described in the following subsections.

In this section, the term *Item* means an item of discussable ItemTypes: Parts, Documents, CAD Documents, CAD Models, etc. Examples shown on CAD Documents and CAD Models are generic and applicable to other Items.

To access the discussion feature, click the **Discussion** button on the right of the **Item** toolbar:

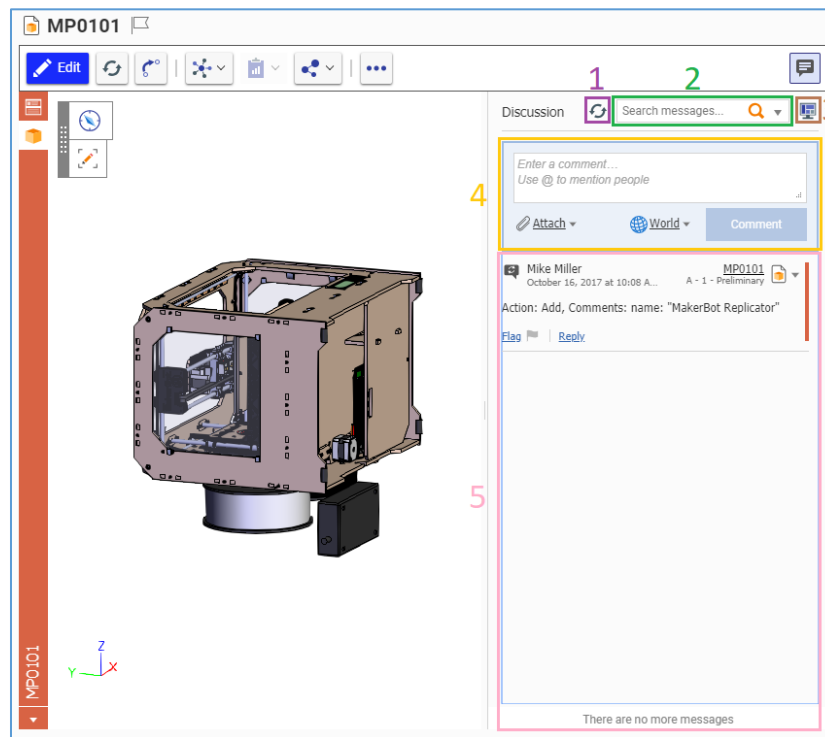


Figure 211.

1. Click the **Refresh** button to see the latest changes in the discussion.
2. Use the **Search** box to search for comment messages in the discussion.
3. Click the **Display Settings** button to change the discussion view.
4. Compose a message in the **Comment** field and click the **Comment** button to post it as part of a discussion.
5. The **Discussion** box displays all comment messages that are part of a discussion.

### 12.1 Searching Comments

You can use the **Search** box for simple and advanced searching of specific comment messages in a discussion. You can combine these two methods.

For a simple search, type your keywords in the **Search** box and press Enter on your keyboard.

For advanced search, click the **drop-down arrow** in the **Search** box and specify your search criteria in the following fields:



Discussion Search messages...

Type: All

Source: All

From: Anyone

Timeframe: All

Revision: All

State: All

Search Cancel

Figure 212.

- **Type** – Displays a drop-down list of comment message types: All, All Comments, Text Comments Only, Snapshot Comments Only, Audio Comments Only, Video Comments Only, Automated Messages Only.

Type: All

Source: All

From: All Comments

Timeframe: Text Comments Only

Revision: Snapshot Comments Only

Audio Comments Only

Video Comments Only

Automated Messages Only

Figure 213.

- **Source** – Displays a drop-down list of comment messages: All, This Item Only.

Source: All

From: This Item Only

Figure 214.

- **From** – Displays the author name associated with the comment messages.
- **Timeframe** – Displays a drop-down list to enables you to specify the timeframe during which comment messages were created: All, Within the Last, Date Range.

Timeframe: All

Revision: All

Within The Last

Date Range

Figure 215.

- **Revision** – Enables you to specify a Revision value of the item for search.
- **State** – Enables you to specify a life cycle state for search.

## 12.2 Setting up the Discussion Display

The **Display Settings** button enables you to configure the view of information that appears in a discussion.

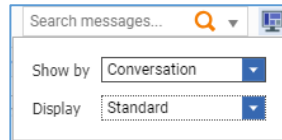


Figure 216.

- **Show by** – A drop-down list to specify the order of comment messages: **Conversation**, **Thread Date**, **Message Date**.

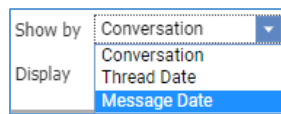


Figure 217.

- **Display** – A drop-down list to specify the display type: **Standard**, **Review**.

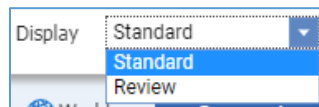


Figure 218.

## 12.3 Managing Your Comments

You can use the **Comment** box for composing messages and posting them in the discussion of an Item. To post one, enter your message in the text box and click **Comment**. You can also use the @ sign to send the message to specific people.

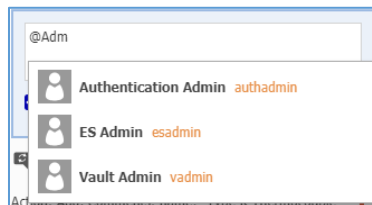


Figure 219.

By default, everyone can see a comment thread, including the message and all replies to it. An initial comment message has the **World** link when created. To change this accessibility, click the **World** link and then click the person you want to share the comment thread.

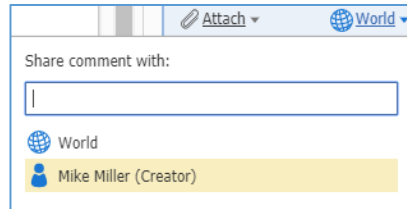


Figure 220.

The person's **name** replaces the **Word** link. Only you and this person can see this comment thread.



Figure 221.

If you want to add an image to your message, use the **Attach** link to attach the image file and click **Comment**. The **Attach** link displays the following options:

**Note:** You need to purchase a subscription for viewing and working with native CAD Models in Aras Innovator right from the corresponding Item and the discussion feature. Without this subscription, drawing and images are not available.

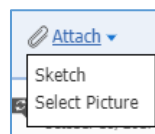


Figure 222.

- **Sketch** makes a snapshot of an Item view if you check the **Include snapshot** check box.

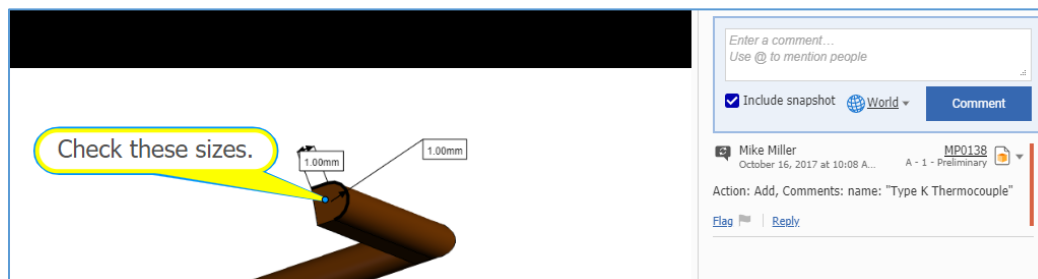


Figure 223.

Even if an Item has no 3D Model attached, you can use this feature to draw an arbitrary sketch using the markup tools.

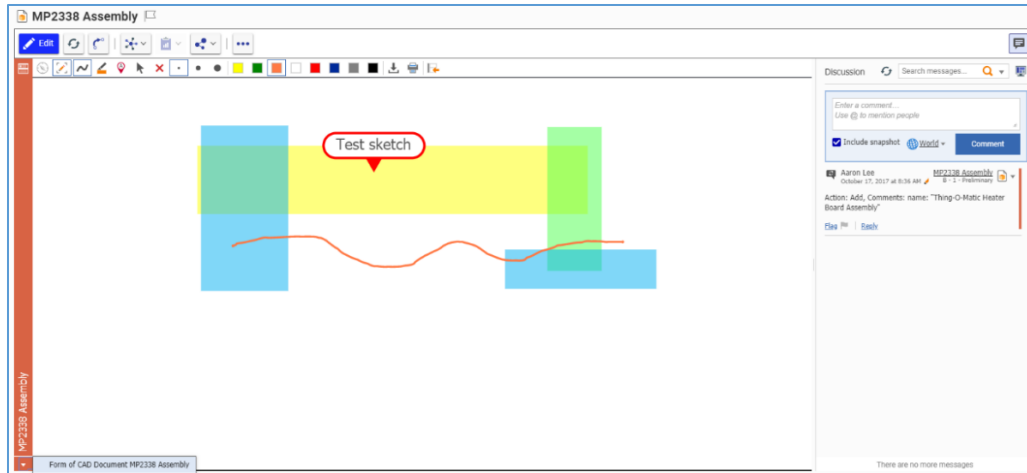


Figure 224.

- **Select Picture** opens the standard dialog of your operating system for file browsing. Search for and select an image. The uploaded image appears in the image view. You can draw on the uploaded image using the markup tools.

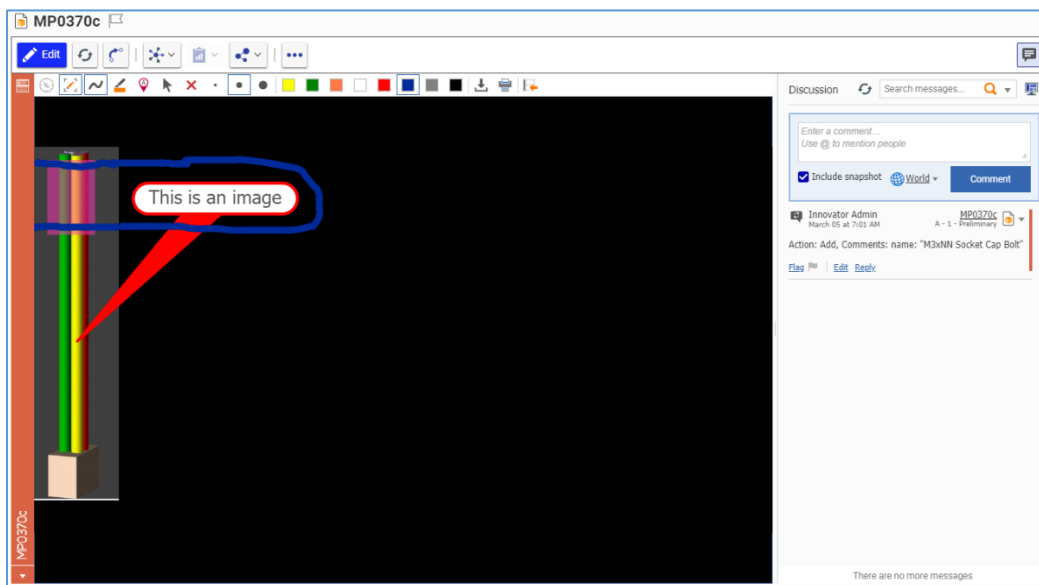


Figure 225.

When attached, the image appears as a thumbnail in a comment message.

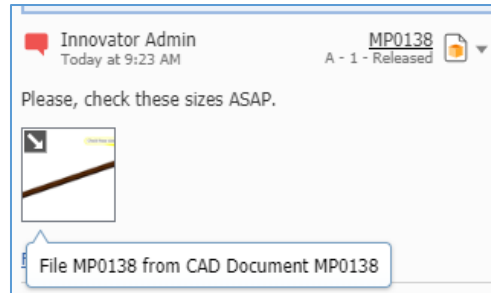


Figure 226.

Click the thumbnail, and the image appears in the Item view.

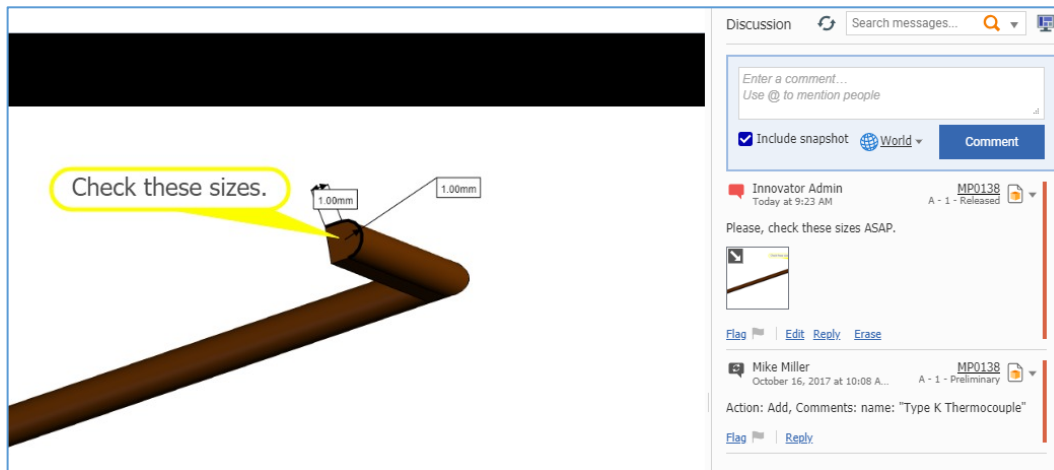


Figure 227.

If you need to edit your message, click the **Edit** link at the bottom of your comment, edit the message text, and click **Save**.

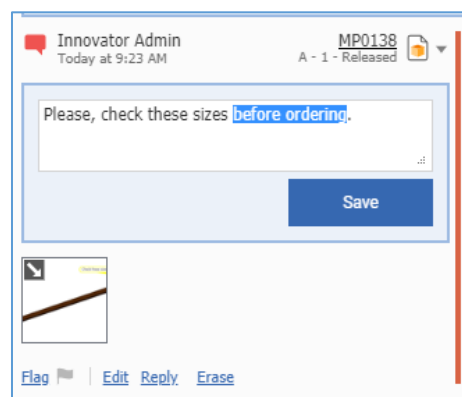


Figure 228.

You cannot edit or remove attached images. You can draw with the markup tools on the attached images. To save the changes, you need to attach the changed image as a snapshot to a new comment. To keep a history of changes, you can attach the changed snapshot as a reply to your own comment.

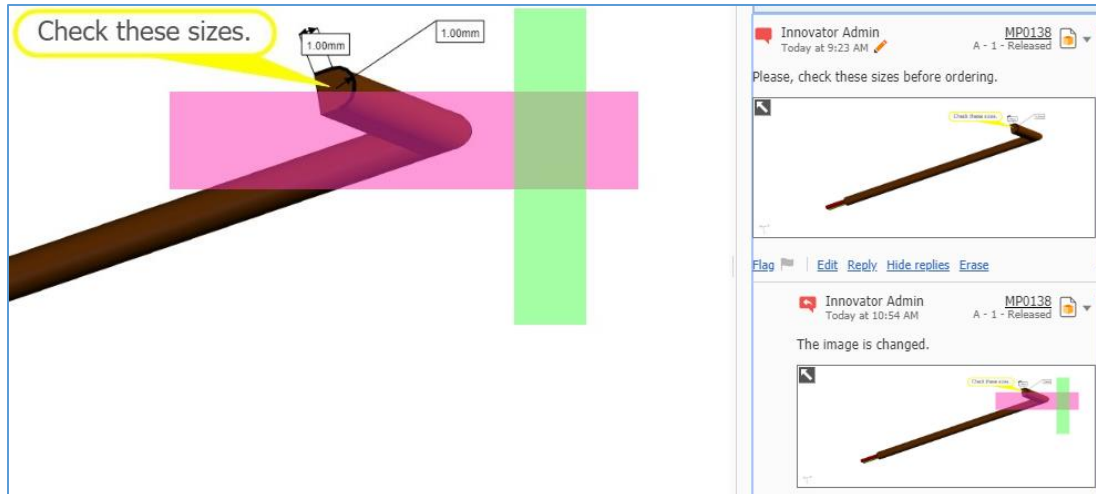


Figure 229.

To enlarge or reduce a thumbnail, click the **arrow** button at the top left corner of the thumbnail.

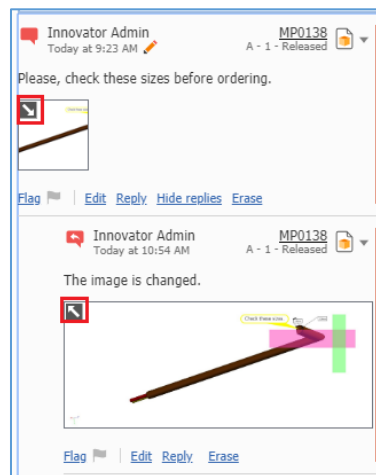


Figure 230.

You can mark important messages with the **Flag** link. For a comment with flags, click the **number** link after the flag icon to see who flagged the comment.

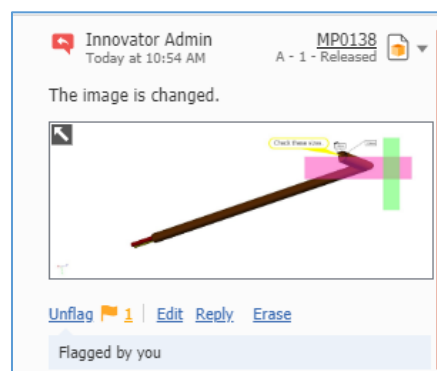


Figure 231.

Click the **Erase** link that appears at the bottom of the Comment box to delete your comment message in a discussion. You can only delete your own messages and attached files. You cannot delete other people's messages. The discussions keep a history of changes; metadata for erased messages is always available. This metadata provides the following information about a deleted message: author, when it was erased, related CAD Document, CAD Document version and state.

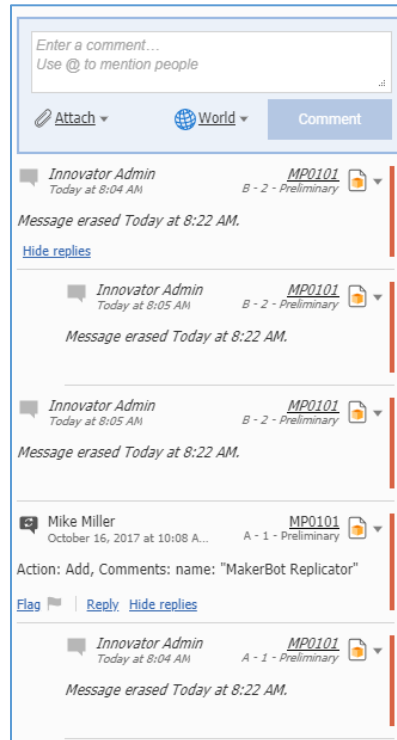


Figure 232.

You can quickly navigate through Item versions in discussions. Click the **drop-down arrow** after the displayed version that appears in the top right corner of a comment message. A drop-down list appears with the following options: **View This Version**, **View Current Version**, **Edit Current Version**, **Add Item to Bookmarks**. When selected, the view of a corresponding Item version appears.

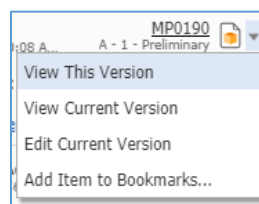


Figure 233.

## 12.4 Replying to Other's Comments

To reply to a comment:

1. Click the **Reply** link at the bottom of the **Comment** box. The **Reply** box appears.



Figure 234.

2. Enter your comment.
3. If necessary, attach an image as described in section [12.3 Managing Your Comments](#).
4. Click **Reply**.

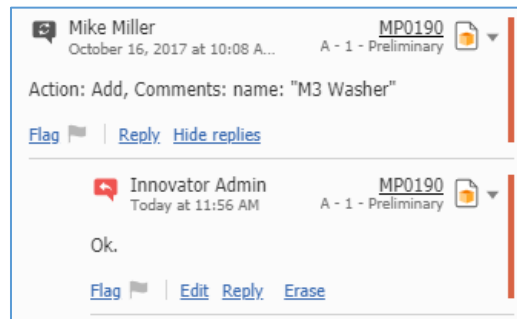


Figure 235.

You can edit and delete only your replies as described in section [12.3 Managing Your Comments](#).



## 13 Change Management

Change management is the process by which any required changes to Items are initiated, designed, reviewed, and implemented. In this section, the term *Item* means an item under direct change management control. In the Product Engineering (PE) solution, these Items are Parts, Documents, and CAD Documents.

The PE solution has three built-in processes to manage Item release, modification, and obsolescence:

- **CMII Process** — The Institute of Configuration Management developed the CMII change management process. It is a rigorous definition of how change management should be implemented throughout an organization to achieve better control over operations. You can use the CMII principles immediately out of the box or customize them to match your organization's specific requirements.
- **Simple Change Management** — This is a less rigorous process for organizations that do not require the entire CMII standard. Simple change management involves fewer steps and resources but still controls change in an organization based on a set of standard processes.
- **Express Change Management** — This is a streamlined process with a built-in Impact Matrix utility that quickly shows you how the change will affect other Items dependent on a changed Part or Document.

Aras Innovator can be configured to allow one or more of these processes to run in parallel depending on the complexity of the Parts and Documents involved and the requirements of the organization.

**Note:** You may be required to enter your password and e-signature to complete some of the CM assignment activities. This ensures that you are indeed the specified user required to perform this operation. The workflow administrator can set this feature on any workflow assignment.

### 13.1 CMII Change Management

The CMII Process uses three different ItemTypes to manage changes:

- The **Problem Report (PR)** is used for logging problems found with an Item. The PR does not force a resolution; it is a tool to track a problem, which may need to be addressed.
- The **Engineering Change Request (ECR)** is used for requesting a change to Items where a problem occurs or will occur if no action is taken. An ECR is usually created after one or more PRs have been filed and verified that a problem does exist. ECRs do not actually affect change on the Item(s).
- The **Engineering Change Notice (ECN)** is an authorization to make changes to an Item based on an approved ECR. It is used to describe modifications (real or proposed) to affected Items and is also used for notifying the appropriate users that the Item is changed. ECNs do affect change on the Item(s). Once an ECN has been approved and completed, a new Item is released, modified, or obsoleted depending on the actions taken.

#### 13.1.1 Problem Report (PR)

A Problem Report (PR) is a report about a problem that has been found. It does not force any resolution or initiate a corrective process. It is a simple notification that a problem with an Item has been discovered. The reported problem is the subject of further review, verification, and approval/rejection.

Anyone who has access to the Item in question can create a PR. An initiator can be a company employee or customer. For example, at a software company, a PR could be a new issue that has been reported by the helpdesk. At a services company, it could be a complaint call from a customer.

### 13.1.1.1 Understanding the PR Process

The PR workflow has the following roles:

- PR Creator – The user who initially created the PR in the system. This may or may not be the same person as other roles.
- PR Owner (**Assigned Creator**) – An engineer responsible for the research, identification, and verification of the problem. The Owner must verify the problem and provide detailed information about its origin as well as any possible corrective actions.
- Change Specialist I (CSI) – Coordinates PRs and ECRs through the analysis phase preparing for and conducting the Change Review Board meetings.
- The Configuration Management (CM) group.

A PR moves through various states during its life cycle as shown in the following figure.

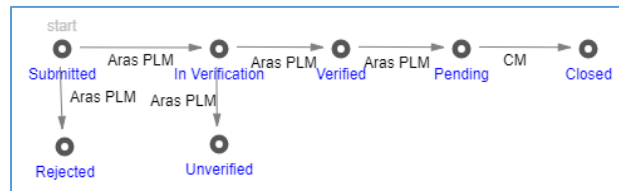


Figure 236.

The PR workflow together with the PR life cycle produces the following scenario:

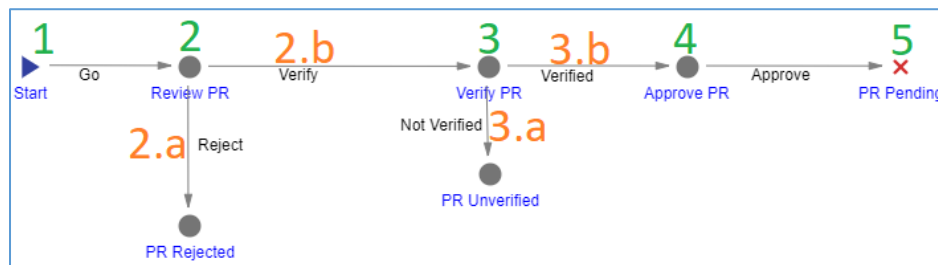


Figure 237.

1. A new PR is **Submitted** by a Creator.
2. A CSI **reviews the PR** to determine whether it should be:
  - a. In the **Rejected** state.
  - b. In the **In Verification** state. The CSI **identifies a PR Owner** and **sends the PR** to this identity for verification.
3. The PR Owner **verifies the PR** to determine whether it should be:
  - a. In the **Unverified** state. In this case, the PR Owner rejects the PR.
  - b. In the **Verified** state. Once the PR Owner has verified the PR, it goes to the CSI for approval.
4. The CSI checks the PR form. Once it is ensured that all the necessary information has been entered in the PR, **the CSI approves** the PR, promoting it to **Pending**.

5. A member of the CM group starts a further CMII process according to the PR results and closes the PR.

A CSI has access to and controls all PRs in the system regardless of their creator, activity, or state. For example, several PRs may be pending until they are all incorporated into one ECR. Therefore, when the final action does take place, the CSI needs to update the Final Action field with this information.

The following table shows the PR lifecycle states and workflow activities in terms of Roles and Permissions.

**Table 3:** The PR lifecycle and workflow activities with corresponding Roles and Permissions

| Life Cycle State | Workflow Activity | Can Add | Get                        | Update                     | Delete | Change access | Can be promoted   | Update responsible |
|------------------|-------------------|---------|----------------------------|----------------------------|--------|---------------|-------------------|--------------------|
| Submitted        | Review PR         | World   | All employees, CM, Creator | All employees, CM, Creator |        | CM            | Yes               | CSI                |
| Rejected         | PR Rejected       |         | All employees, CM          | CM                         |        | CM            | No, a final state |                    |
| In Verification  | Verify PR         |         | All employees, CM, Creator | All employees, CM, Creator |        | CM            | Yes               | Owner              |
| Unverified       | PR Unverified     |         | All employees, CM          | CM                         |        | CM            | No, a final state |                    |
| Verified PR      | Approve PR        |         | All employees, CM          | CM                         |        | CM            | Yes               | CSI                |
| Pending          | PR Pending        |         | All employees, CM, Creator | CM                         |        | CM            | Yes               | CM                 |
| Closed           |                   |         | All employees, Creator     |                            |        |               | No, a final state |                    |

The final PR states (Rejected, Unverified, Closed) are used to classify PRs in the system and in reports.

There are the following PR reports:

- PR Log Report – A list of all the PRs. For details, refer to section [13.1.1.6 PR Log Report](#).
- PR Report – A printable PR form. For details, refer to section [13.1.1.7 PR Report](#).

#### 13.1.1.2 Submitting a PR

A PR is a complex document that is filled in by different roles in the system. A PR creator is not expected to fill in the PR completely. This section defines the types of information that must be included in a newly created PR.

To create and submit a Problem Report:

1. Go to **Contents → Change Management → PRs**.
2. Click **Create New PR**. The view of a new PR appears. Enter the necessary information.

Figure 238.

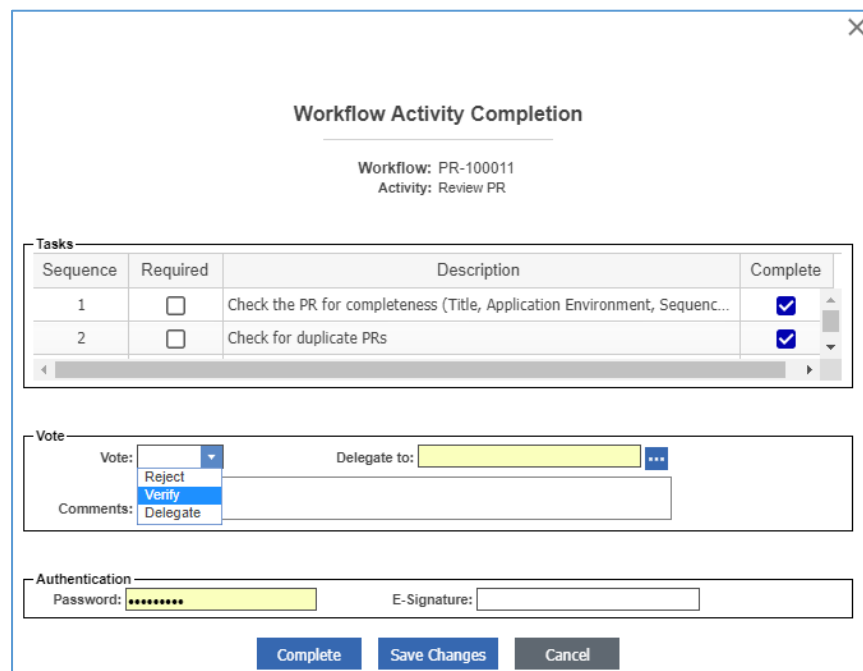
- **Title** – The title for the PR. The title should be as descriptive as possible.
  - **Application Environment** – The background information about the product and/or any related problems. For example, it could be the version or the server used for a software product, or it could be the temperature and the humidity level for an automotive part.
  - **Sequence Of Events Leading Up To The Problem** – The exact steps taken that led to the problem.
  - **Description Of The Problem** – A description of the problem. The description should describe the problem as precisely as possible.
  - **Affected Item** – The affected item.
  - **Basis** –
    - **Physical Hierarchy** – The Item (Part) itself and any Document associated with it. A bug in software would be classified as a physical hierarchy.
    - **Administrative Hierarchy** – Anything related to the process of running the business. An error in an expense report procedure would be classified as an administrative hierarchy.
3. Click **Done** on the **PR** toolbar. The PR is created and submitted for a CSI's decision.

### 13.1.1.3 Reviewing a PR

A CSI receives a **Submitted** PR to decide whether to reject or review it. This section defines which pieces of information the CSI must provide when the decision is made to review the PR.

Use the following procedure to review a Problem Report:

1. Open the **PR** view.
2. Click **Edit** on the **PR** toolbar.
3. Specify the priority number in the **Priority For Tech Review** field. The higher the priority, the lower the number selected.
4. Enter the name of the person who reported the problem in the **Reported By** text box. This Identity may not be the same person who created the PR in the system (the Creator).
5. Enter the name of the PR Owner in the **Assigned Creator** text box.
6. Click **Done** on the **PR** toolbar.
7. Go to **Contents → My Innovator → My InBasket → Search My InBasket**.
8. Go to the **Workflow Activity Completion** dialog for the PR.



**Workflow Activity Completion**

Workflow: PR-100011  
Activity: Review PR

| Sequence | Required                 | Description                                                               | Complete                            |
|----------|--------------------------|---------------------------------------------------------------------------|-------------------------------------|
| 1        | <input type="checkbox"/> | Check the PR for completeness (Title, Application Environment, Sequenc... | <input checked="" type="checkbox"/> |
| 2        | <input type="checkbox"/> | Check for duplicate PRs                                                   | <input checked="" type="checkbox"/> |

**Vote**

Vote: Reject Verify Delegate Delegate to:

Comments:

**Authentication**

Password:  E-Signature:

Figure 239.

9. Mark the tasks you completed in the **Tasks** field.
10. Select **Verify** in the **Vote** list.
11. Click **Complete**.

The PR awaits a decision of the PR Owner.

### 13.1.1.4 Verifying a PR

A PR Owner receives a PR **In Verification** to decide whether to reject or verify it. This section defines which pieces of information the PR Owner must provide when verifying the PR.

Use the following procedure to verify a Problem Report:

1. Open the **PR** view.
2. Click **Edit** on the **PR** toolbar.
3. Enter the steps that were taken to verify the problem as well as any special conditions or descriptions of the problem in the **Problem Verification** text box.
4. Enter an assessment of what would happen if this problem was not corrected in the **Ramifications If Not Resolved text** box
5. Assess the severity of the issue in the **Severity** box. The higher the severity; the lower the number selected.
6. Specify the product development phase where the problem occurred in the **Phase Caused** list.

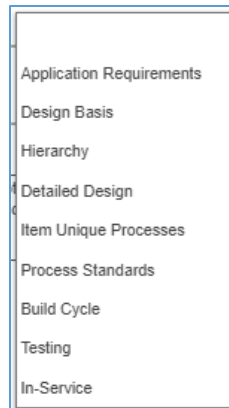
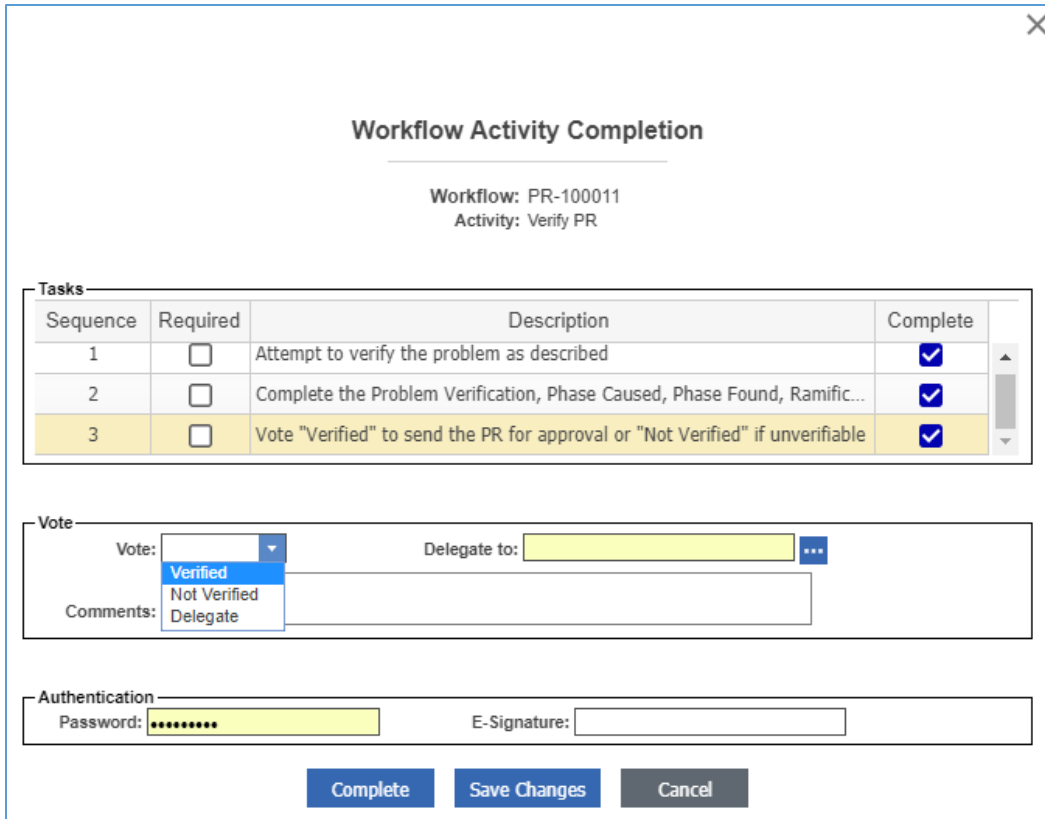


Figure 240.

7. Select the product development phase where the problem was found in the **Phase Found** list.
8. Click **Done** on the **PR** toolbar.
9. Go to **Contents → My Innovator → My InBasket --> Search My InBasket**.
10. Go to the **Workflow Activity Completion** dialog for the PR.



The dialog box is titled "Workflow Activity Completion" and shows details for Workflow: PR-100011 and Activity: Verify PR. It contains a "Tasks" table, a "Vote" section, and an "Authentication" section.

| Sequence | Required                 | Description                                                                   | Complete                            |
|----------|--------------------------|-------------------------------------------------------------------------------|-------------------------------------|
| 1        | <input type="checkbox"/> | Attempt to verify the problem as described                                    | <input checked="" type="checkbox"/> |
| 2        | <input type="checkbox"/> | Complete the Problem Verification, Phase Caused, Phase Found, Ramific...      | <input checked="" type="checkbox"/> |
| 3        | <input type="checkbox"/> | Vote "Verified" to send the PR for approval or "Not Verified" if unverifiable | <input checked="" type="checkbox"/> |

**Vote**

Vote: Verified Delegate to:

Comments:

**Authentication**

Password:  E-Signature:

**Complete** **Save Changes** **Cancel**

Figure 241.

11. Mark the tasks you completed in the **Tasks** field.
12. Select **Verified** in the **Vote** list.
13. Click **Complete**.

The Approve PR activity starts. The PR awaits a CSI's decision.

#### 13.1.1.5 Approving a PR

A CSI is responsible for the final approval of all PRs in the system regardless of their creator, activity, or state.

Use the following procedure to approve a Problem Report:

1. Open the **PR** view.
2. Check that all necessary information is present.
3. If it is necessary to correct the PR:
  - a. Click **Edit** on the **PR** toolbar.
  - b. If any final action takes place, enter information about it in the **Final Action** text box.
  - c. Provide any missing information.
  - d. Correct wrong information if any exists.
  - e. Attach auxiliary files if any exist.
  - f. Click **Done** on the **PR** toolbar.
4. Go to the **Workflow Activity Completion** dialog for the PR.

**Workflow Activity Completion**

Workflow: PR-100009  
Activity: Approve PR

| Sequence | Required                 | Description                                                    | Complete                            |
|----------|--------------------------|----------------------------------------------------------------|-------------------------------------|
| 1        | <input type="checkbox"/> | Check the PR for completeness                                  | <input checked="" type="checkbox"/> |
| 2        | <input type="checkbox"/> | Vote "Approve" to signify that the PR is complete and verified | <input checked="" type="checkbox"/> |

**Vote**

Vote:   Delegate to:  ...

Comments:

**Authentication**

Password:  E-Signature:

Figure 242.

5. Select the tasks you completed in the **Tasks** field.
6. Select **Approve** in the **Vote** list.

### 13.1.1.6 PR Log Report

You can get quick information about all the PRs submitted to Aras Innovator with the PR Log Report.

PR Log Report

Generated on: 3/19/2019, 7:09:37 PM

| PR Number | Title                                           | Priority | Date Reported          | Reported By    | Assigned Creator | Status          | ECR |
|-----------|-------------------------------------------------|----------|------------------------|----------------|------------------|-----------------|-----|
| PR-100001 | SD Card fit issues                              |          | 1/19/2018 10:30:00 AM  | Diane Prescott |                  | In Verification |     |
| PR-100002 | Cable routing within case                       |          | 11/13/2017 6:40:00 PM  | Dan Park       |                  | In Verification |     |
| PR-100003 | Screw pre-stress can break boards               |          | 11/13/2017 6:40:00 PM  | Dan Park       |                  | In Verification |     |
| PR-100004 | Manufacturing issues relating to case tolerance |          | 11/13/2017 6:40:00 PM  | Dan Park       |                  | In Verification |     |
| PR-100005 | Packaging not providing sufficient protection   |          | 11/15/2017 11:00:00 AM | Dan Park       |                  | Pending         |     |

Figure 243.

There are several ways to access the PR Log Report:

- Go to **Contents** → **Change Management** → **PRs**, right-click any **PR**, and then click **Reports** → **PR Log Report**.



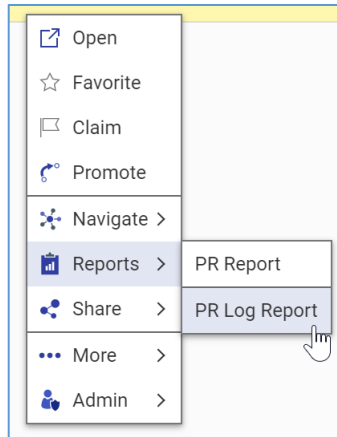


Figure 244.

- Open the view of any PR, on the **PR** toolbar, click **Reports**, and then click **PR Log Report**.

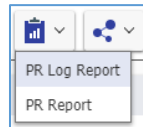


Figure 245.

To quit a Report, close its view in Aras Innovator.

At the top left corner of the PR Log Report view, there is a **Print** button, which uses the standard browser print function. Depending on the browser and your specific configuration, various print choices, such as printing to a printer or saving to PDF, are available.

### 13.1.1.7 PR Report

You can print or share a PR using the PR Report.

**PR Report** Generated on: 3/19/2019, 6:53:46 PM

|                                                                                                                                                                                                                |                                                                                                                                  |                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PR NUMBER<br>PR-100010                                                                                                                                                                                         | TITLE<br>SD Card fit issues                                                                                                      | STATUS<br>Closed                                                                                                                                            |
| APPLICATION ENVIRONMENT<br>Case and electronics                                                                                                                                                                | AFFECTED PART<br>MP0101                                                                                                          |                                                                                                                                                             |
| SEQUENCE OF EVENTS<br>Normal insertion of SD card                                                                                                                                                              | BASIS<br><input type="checkbox"/> Administrative Hierarchy<br><input checked="" type="checkbox"/> Physical Hierarchy             | PRIORITY FOR TECH REVIEW<br><input type="checkbox"/> 1<br><input type="checkbox"/> 2<br><input checked="" type="checkbox"/> 3<br><input type="checkbox"/> 4 |
| DESCRIPTION<br>The SD card slot is near a screw. Feedback from users notes that inserting the card properly is a challenge due to the proximity of the card and screw. Sometimes cards do not engage properly. | REPORTED BY<br>Innovator Admin                                                                                                   | ASSIGNED CREATOR<br>Innovator Admin                                                                                                                         |
| PROBLEM VERIFICATION<br>RAMIFICATIONS IF NOT RESOLVED<br><input type="checkbox"/> 1<br><input type="checkbox"/> 2<br><input type="checkbox"/> 3<br><input type="checkbox"/> 4                                  | SEVERITY<br><input type="checkbox"/> 1<br><input type="checkbox"/> 2<br><input type="checkbox"/> 3<br><input type="checkbox"/> 4 | PHASE CAUSED<br>PHASE FOUND<br>FINAL ACTION TAKEN                                                                                                           |

Figure 246.

There are several ways to access the PR Report:

- Go to **Contents → Change Management → PRs**, right-click a **PR** to be reported, and then click **Reports → PR Report**.

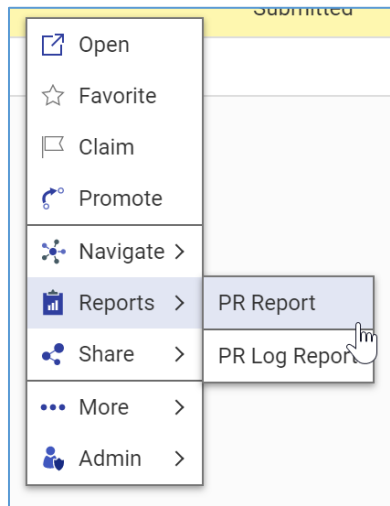


Figure 247.

- Open the view of a PR. On the **PR** toolbar, click **Reports**, and then click **PR Report**.

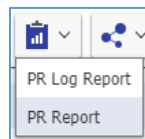


Figure 248.

To quit a Report, close its view in Aras Innovator.

At the top left corner of the PR Report view, there is a **Print** button, which uses the standard browser print function. Depending on the browser and your specific configuration, various print choices, such as printing to a printer or saving to PDF, are available.

### 13.1.2 Engineering Change Request (ECR)

An Engineering Change Request (ECR) is a request initiating a corrective process in response to some PRs or because of a potential problem identified through proactive thinking. An ECR is submitted to request a change to Items where a problem occurs or will occur if no action is taken.

An ECR may be quickly processed in the system using the Fast Track approach when the ECR is submitted to correct a small or local problem that does not have far-reaching consequences.

#### 13.1.2.1 Understanding the ECR Process

The ECR workflow has the following roles:

- An ECR Creator is a user who initially created the ECR in the system. This may or may not be the same person as other roles.
- An ECR Owner (**Assigned Creator**) is a person responsible for the technical review of the ECR. This person must be able to identify and verify the problem itself and to see the ramifications of both the problem and the solution.
- A Change Specialist I (CSI) coordinates PRs and ECRs through the analysis phase preparing for and conducting the Change Review Board meetings.

- The Change Review Board (CRB).
- The Configuration Management (CM) group.

An ECR moves between various states during its life cycle as shown in the following figure.

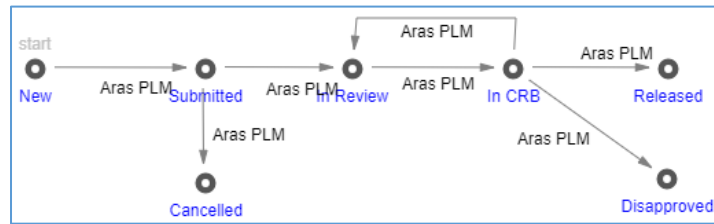


Figure 249.

The ECR workflow together with the ECR life cycle produces the following scenario:

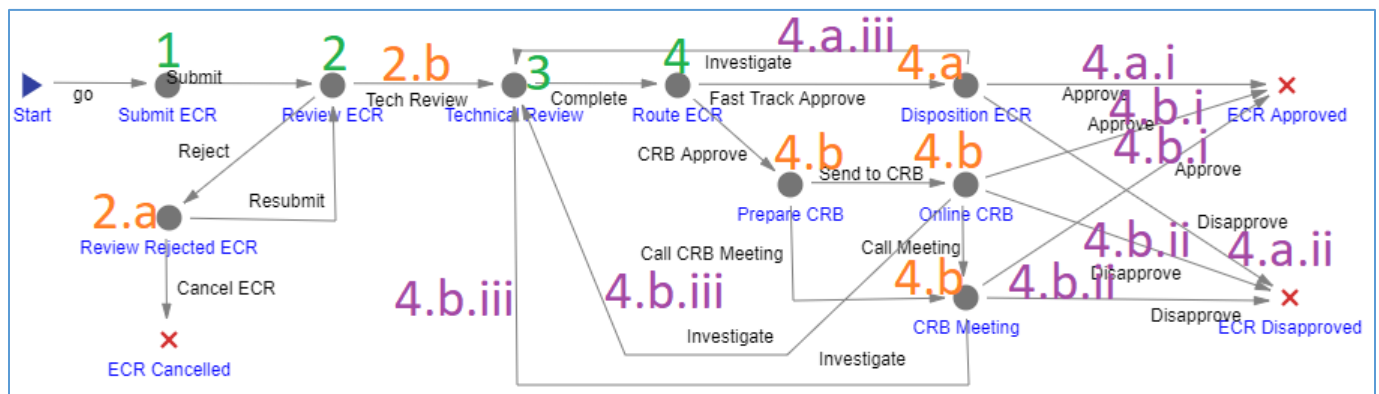


Figure 250.

- After a new ECR is created, an ECR Creator must declare that the ECR is ready for review voting **Submit the ECR**.
- The ECR enters the **Review ECR** activity **assigned to the CSI**. The CSI reviews the ECR thoroughly, and decides to either:
  - Reject the ECR**. The ECR goes **back to the creator**, who either resubmits or cancels the ECR. In the former case, the creator corrects the ECR according to CSI's comments for rejection.
  - Submit the ECR** for the technical review. The CSI assigns an Owner to the ECR.
- The ECR enters the **Technical Review** activity **assigned to the Owner**. The Owner conducts the technical review and provides data, comments, and recommendations. Once the Owner votes **Complete**, the ECR goes further.
- The ECR enters the **Route ECR** activity **assigned to the CSI**. The CSI must decide how to route the ECR. Different companies have different criteria for making this decision based on the cost estimates, time, and the nature of the problem itself. The CSI has the authority to vote either:
  - Fast Track Approve** for a fast-track ECR. The ECR enters the **Disposition ECR** activity **assigned to the Owner**. The Owner votes either:
    - ECR Approved** to approve the ECR.
    - ECR Disapproved** to reject the ECR.
    - Investigate** to send the ECR back for more investigation.

- b. **CRB Approve** for full CRB involvement. The ECR enters the **Prepare CRB** activity assigned to the **CSI**. The CSI defines the meeting type either through an in-person **CRB Meeting** or through an **Online CRB**. The CSI also prepares all the necessary information to present to the CRB. Regardless of the meeting type, the CRB votes either:
- i. **ECR Approved** to approve the ECR.
  - ii. **ECR Disapproved** to reject the ECR.
  - iii. **Investigate** to send the ECR back for more investigation.

The following table shows the ECR lifecycle states and workflow activities in terms of Roles and Permissions.

**Table 4:** The ECR lifecycle and workflow activities with corresponding Roles and Permissions

| Life Cycle State | Workflow Activity   | Can Add | Get                        | Update      | Delete | Change access | Can be promoted   | Update responsible |
|------------------|---------------------|---------|----------------------------|-------------|--------|---------------|-------------------|--------------------|
| New              |                     | World   | All employees, Creator, CM | Creator, CM |        | CM            |                   | Automatic          |
| Submitted        | Submit ECR          |         | All employees, Creator, CM | Creator, CM |        | CM            | Yes               | Creator            |
| In Review        | Review ECR          |         | All employees, CM          | CM          |        |               | Yes               | CSI                |
| In Review        | Review Rejected ECR |         | All employees, CM          | CM          |        |               | Yes               | Creator            |
| In Review        | Technical Review    |         | All employees, CM          | CM          |        |               | Yes               | Owner              |
| In Review        | Route ECR           |         | All employees, CM          | CM          |        |               | Yes               | CSI                |
| Cancelled        | ECR Cancelled       |         | All employees              |             |        |               | No, a final state |                    |
| In CRB           |                     |         | All employees, CM          | CM          |        |               |                   |                    |
| In CRB           | Disposition ECR     |         | All employees, CM          | CM          |        |               | Yes               | Owner              |
| In CRB           | Prepare CRB         |         | All employees, CM          | CM          |        |               | Yes               | CSI                |
| In CRB           | Online CRB          |         | All employees, CM          | CM          |        |               |                   | CRB                |
| In CRB           | CRB Meeting         |         | All employees, CM          | CM          |        |               | Yes               | CSI                |

| Life Cycle State | Workflow Activity | Can Add | Get           | Update | Delete | Change access | Can be promoted   | Update responsible |
|------------------|-------------------|---------|---------------|--------|--------|---------------|-------------------|--------------------|
| Disapproved      | ECR Disapproved   |         | All employees |        |        |               | No, a final state |                    |
| Life Cycle State | Workflow Activity | Can Add | Get           | Update | Delete | Change access | Can be promoted   | Update responsible |
| Released         | ECR Approved      |         | All employees |        |        |               | No, a final state |                    |

The final ECR states (Cancelled, Disapproved, Released) are used to classify ECRs in the system and in reports.

The ECR reports are:

- ECR Log Report is a list of all the ECRs. For details, refer to section [13.1.2.10 ECR Log Report](#).
- ECR Report is a printable form of an ECR. For details, refer to section [13.1.2.11 ECR Report](#).

### 13.1.2.2 Submitting an ECR

An ECR is a complex document that is filled in by more than one person. An ECR creator is not expected to fill in the ECR completely. This section defines the information an ECR must have to be submitted.

Use the following procedure to create and submit an Engineering Change Request:

1. Go to **Contents → Change Management → ECRs**.
2. Click **Create New ECR**. The view of a new ECR appears.

Figure 251.

3. Enter the **ECR title** in the **Title** text box, making it as descriptive as possible.
4. Select the appropriate hierarchy:
  - **Physical hierarchy** refers to the Item (Part) itself and any Document associated with it. A bug in software would be classified as a physical hierarchy.
  - **Administrative hierarchy** refers to anything related to the process of running the business. An error in an expense report procedure would be classified as an administrative hierarchy.
5. Specify the ECR intent:
  - **Corrective Action** if the ECR corrects an already occurring problem.
  - **Product Improvement** if the ECR prevents a possible problem or improves the product.
6. Enter the name of the person who requested the change in the **Requested By** text box.
7. Specify the source of the ECR by selecting one of the following from the **Source** list: **Internal**, **Customer**, or **Supplier**.
8. Describe the solution proposed by the ECR Creator in the **Proposed Solution** text box.
9. Go to the **Affected Items** accordion tab to specify which Items are affected by the ECR.
10. If the ECR should request adding Items, use the procedure described in section [13.1.2.3 Adding an Item with an ECR](#).
11. If the ECR should request changing Items, use the procedure described in section [13.1.2.4 Changing an Item with an ECR](#).
12. If the ECR should request deleting Items, use the procedure described in section [13.1.2.5 Deleting an Item with an ECR](#).
13. Go to the **Files** accordion tab to attach all the files concerning this ECR.

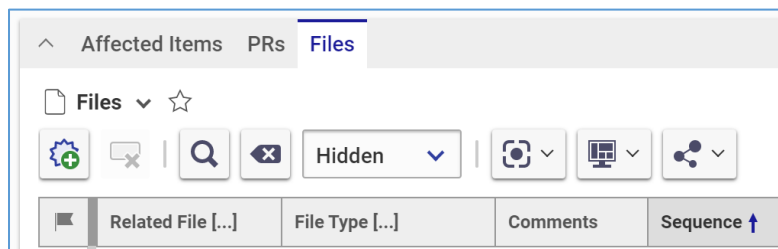


Figure 252.

14. Click **New File** on the Files accordion toolbar. The standard operating system dialog for file browsing and selection appears.
15. Search for and select a file. A new row appears in the **Files** grid with the file attached.

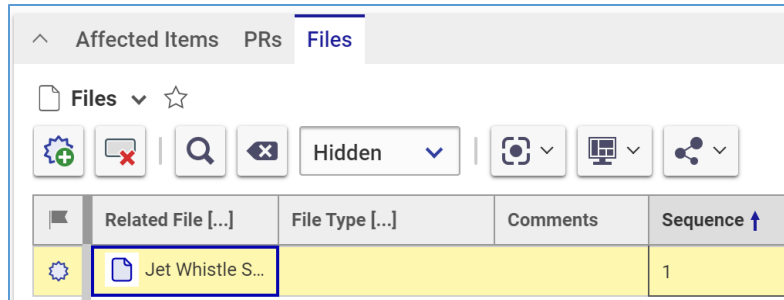


Figure 253.

**Note:** Aras Innovator will automatically assign a file type depending on File's extension when you save the ECR.

16. If it is necessary to provide additional information about the File, enter it in the **Comments** cell.
17. Repeat steps 14-16 for other files related to this ECR.
18. Click **Done** on the **ECR** toolbar. The ECR is created. The automatic ECR properties get their values.
19. Go to **Contents → My Innovator → My InBasket → Search My InBasket**.
20. Go to the **Workflow Activity Completion** dialog for the ECR.

**Workflow Activity Completion**

Workflow: ECR-100001  
Activity: Submit ECR

| Sequence | Required                 | Description                                | Complete                            |
|----------|--------------------------|--------------------------------------------|-------------------------------------|
| 1        | <input type="checkbox"/> | Attach known Affected Items                | <input checked="" type="checkbox"/> |
| 2        | <input type="checkbox"/> | Ensure the ECR is ready for initial review | <input checked="" type="checkbox"/> |

**Vote**

Vote: Submit Delegate Delegate to:

Comments:

**Authentication**

Password:  E-Signature:

**Complete** **Save Changes** **Cancel**

Figure 254.

21. Mark the tasks you completed in the **Tasks** field.
22. Select **Submit** in the **Vote** list.
23. Click **Complete**.

The ECR enters the **Review ECR** activity and awaits the decision of the CSI.

### 13.1.2.3 Adding an Item with an ECR

To request add and release of an affected Item with an Engineering Change Request:

1. Open the ECR.
2. Click **Edit** on the ECR toolbar.
3. Go to the Affected Items accordion tab.
4. Click **New Affected Item** on the Affected Items accordion toolbar. A new row appears at the bottom of the Affected Items grid.

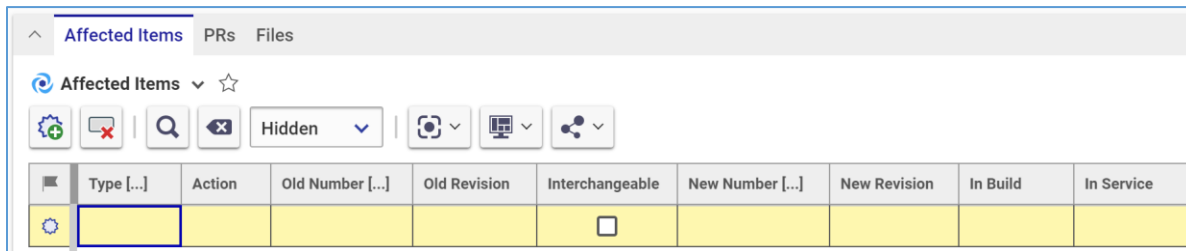


Figure 255.

5. Select **Add** in the **Action** cell.

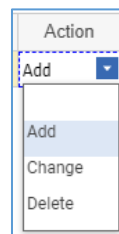


Figure 256.

6. Click the **ellipsis** button in the **New Number** cell. The **Search dialog – Change Controlled Item** appears.
7. Search for and select the new affected Item. The affected Item populates the **New Number** cell.

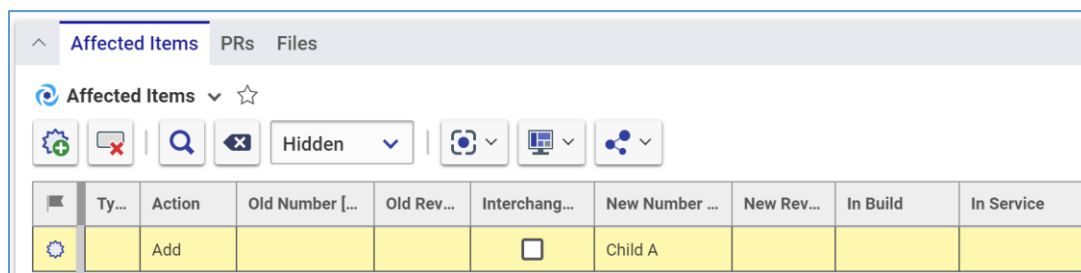


Figure 257.

8. Click **Save** or **Done** on the **ECR** toolbar. The ECR now has an affected Item that needs to be added and released.

### 13.1.2.4 Changing an Item with an ECR

To request changing an existing affected item with an Engineering Change Request:

1. Open the ECR.
2. Click **Edit** on the ECR toolbar.



3. Go to the Affected Items accordion tab.
4. Click **New Affected Item** on the Affected Items accordion toolbar. A new row appears at the bottom of the Affected Items grid.
5. Select **Change** in the **Action** cell.

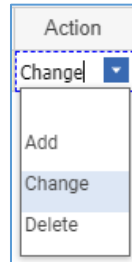


Figure 258.

6. Click the **ellipsis** button in the **Old Number** cell. The **Search dialog – Change Controlled Item** dialog box appears.
7. Using the standard search procedure, search for and select the affected Item to be changed. The selected Item populates the **Old Number** cell.

|  | T.. | Action | Old Number [...] | Old Rev... | Interchang...            | New Number ... | New Rev... | In Build | In Service |
|--|-----|--------|------------------|------------|--------------------------|----------------|------------|----------|------------|
|  |     | Change | Child B          |            | <input type="checkbox"/> |                |            |          |            |

Figure 259.

8. Select the **Interchangeable** cell check box if the old and new Items are interchangeable e.g., if there is no Fit-Form-Function change. When selected, the same Item number is retained. Otherwise, a new Item number will need to be assigned.
9. If the **Interchangeable** cell is not selected, e.g., the old and new Items are not interchangeable due to a Fit-Form-Function change, select the replacing new Item:
  - a. Click the **ellipsis** button in the **New Number** cell. The **Search dialog – Change Controlled Item** appears.
  - b. Search for and select the new affected Item. The new affected Item populates the **New Number** cell.

|  | T..  | Action | Old Number [...] | Old Rev... | Interchang...            | New Number ... | New Rev... | In Build | In Service |
|--|------|--------|------------------|------------|--------------------------|----------------|------------|----------|------------|
|  | <... | Change | Child B          |            | <input type="checkbox"/> | Child A        |            |          |            |

Figure 260.

10. Click **Save** two times on the **ECR** toolbar.
11. Select the action to take for the Items that are being built in the **In Build** cell: **Use Existing**, **Rework**, or **Scrap**.
12. Select the action to take for the Items that are currently in service in the **In Service** cell: **Use Existing**, **Rework**, or **Scrap**.

|  | T..  | Action | Old Number [...] | Old Rev... | Interchang...            | New Number ... | New Rev... | In Build | In Service   |
|--|------|--------|------------------|------------|--------------------------|----------------|------------|----------|--------------|
|  | <... | Change | Child B          | A          | <input type="checkbox"/> | Child A        | A          | Rework   | Use Existing |

Figure 261.

- Click **Save** or **Done** on the ECR toolbar. The ECR has an affected Item requested for changing.

| Affected Items PRs Files                                                                                                                                                                                                                        |      |        |                         |   |                          |                         |   |        |              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|-------------------------|---|--------------------------|-------------------------|---|--------|--------------|
| Affected Items <span>▼</span> <span>☆</span><br>[Icons: Gear, Comment, Search, Close, Hidden, Filter, View, Share]<br>[Icons: Flag, T.., Action, Old Number [...], Old Rev..., Interchang..., New Number ..., New Rev..., In Build, In Service] |      |        |                         |   |                          |                         |   |        |              |
|                                                                                                                                                                                                                                                 | <... | Change | <a href="#">Child B</a> | A | <input type="checkbox"/> | <a href="#">Child A</a> | A | Rework | Use Existing |

Figure 262.

### 13.1.2.5 Deleting an Item with an ECR

To request superseding of an existing affected Item with an Engineering Change Request:

- Open the ECR.
- Click **Edit** on the ECR toolbar.
- Go to the Affected Items accordion tab.
- Click **New Affected Item** on the Affected Items accordion toolbar. A new row appears at the bottom of the Affected Items grid.
- Select **Delete** in the **Action** cell.

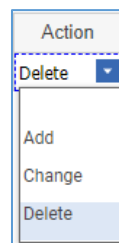


Figure 263.

- Click the ellipsis button in the Old Number cell. The **Search dialog – Change Controlled Item** appears.
- Search for and select the affected Item. The old Affected Item populates the **Old Number** cell.

|  | T.. | Action | Old Number [...] | Old Rev... | Interchang...            | New Number ... | New Rev... | In Build | In Service |
|--|-----|--------|------------------|------------|--------------------------|----------------|------------|----------|------------|
|  |     | Delete | Child B          |            | <input type="checkbox"/> |                |            |          |            |

Figure 264.

- Click **Save** two times on the **ECR** toolbar.
- Select the action to take for the Items that are being built in the **In Build** cell: **Use Existing**, **Rework**, or **Scrap**.

10. Select the action to take for the Items that are currently in service in the **In Service** cell: **Use Existing**, **Rework**, or **Scrap**.

|  | T..  | Action | Old Number [...] | Old Rev... | Interchang...            | New Number ... | New Rev... | In Build | In Service   |
|--|------|--------|------------------|------------|--------------------------|----------------|------------|----------|--------------|
|  | <... | Delete | Child B          | A          | <input type="checkbox"/> |                |            | Rework   | Use Existing |

Figure 265.

11. Click **Save** or **Done** on the ECR toolbar. The ECR has an affected Item assigned for deletion.

| T..  | Action | Old Number [...] | Old Rev... | Interchang...            | New Number ... | New Rev... | In Build | In Service   |
|------|--------|------------------|------------|--------------------------|----------------|------------|----------|--------------|
| <... | Delete | Child B          | A          | <input type="checkbox"/> |                |            | Rework   | Use Existing |

Figure 266.

### 13.1.2.6 Removing an Affected Item from an ECR

Use the following procedure:

1. Follow steps 1-3 of the Adding an Item within an ECR procedure.
2. Click the **Affected Item** and then click **Delete Row** on the **Affected Items** accordion toolbar. The **Affected Item** becomes unavailable.

| T..  | Action | Old Number [...] | Old Rev... | Interchang...            | New Number ... | New Rev... | In Build | In Service   |
|------|--------|------------------|------------|--------------------------|----------------|------------|----------|--------------|
| <... | Delete | Child B          | A          | <input type="checkbox"/> |                |            | Rework   | Use Existing |

Figure 267.

3. Click **Save** or **Done** on the **ECR** toolbar. The **Affected Item** disappears from the grid.

**Note:** The **Affected Item** disappears immediately if the ECR was not saved after it was added.

### 13.1.2.7 Reviewing an ECR

A CSI receives a **Submitted** ECR to decide whether to reject it or to submit it for technical review. This section defines the information the CSI must provide when the decision is made to submit the PR.

To review an Engineering Change Request and submit it for technical review:

1. Open the **ECR** view.
2. Click **Edit** on the **ECR** toolbar.
3. Specify the tech review priority number in the **Priority For Tech Review** field. The higher the priority of the tech review, the lower the number selected.
4. Specify the ECR Owner in the **Assigned Creator** text box.

5. Verify that the information entered by a Creator is correct.
6. Verify that the **Affected Items** accordion tab contains all the Items affected by the ECR.
7. Verify that the **Files** accordion tab has all the necessary files attached.
8. If this ECR is based on a PR:
  - a. Go to the **PRs** accordion tab.

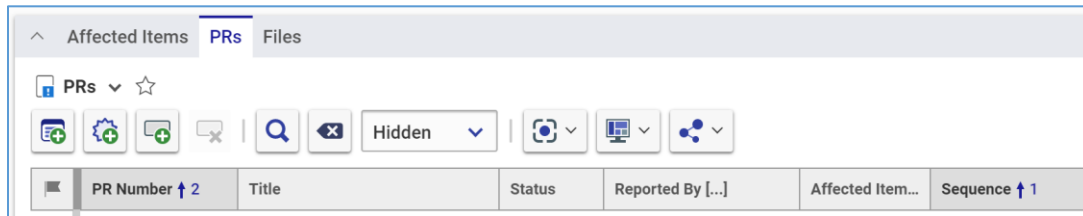


Figure 268.

- b. Click **Select Items**. The **Search dialog – PR** appears.
- c. Search for and select the initiating PR. The selected PR appears as a bottom row in the PRs grid.

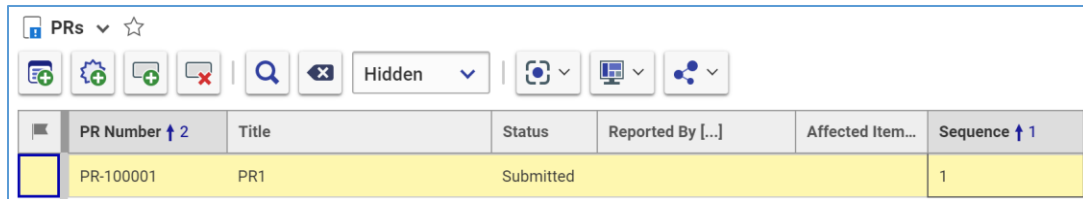
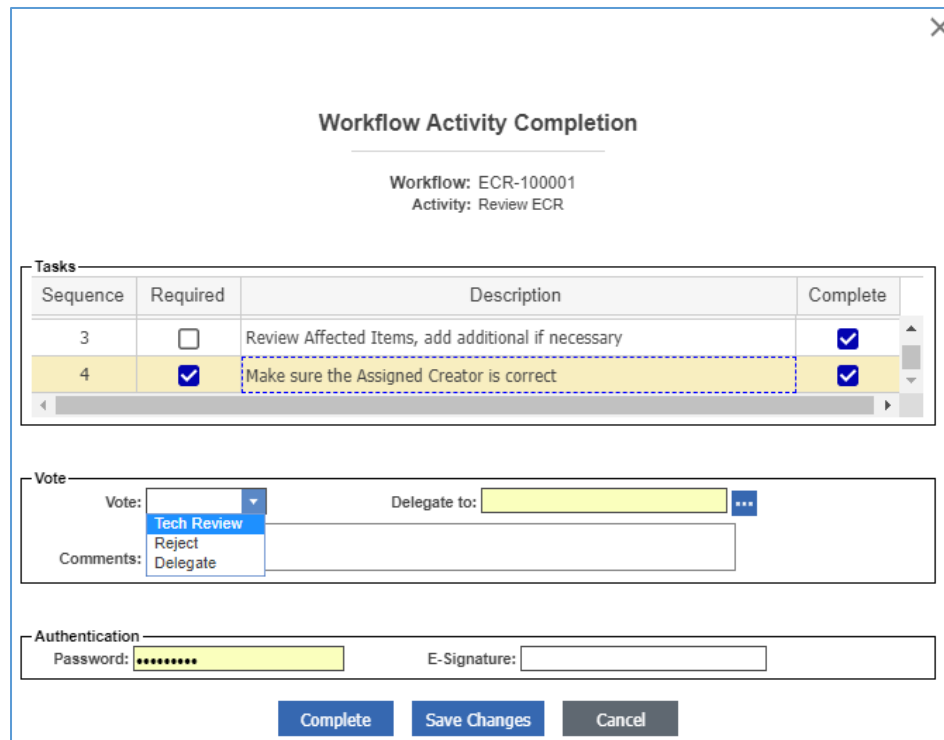


Figure 269.

9. Click **Done** on the ECR toolbar.
10. Go to **Contents → My Innovator → My InBasket → Search My InBasket**.
11. Go to the **Workflow Activity Completion** dialog for the ECR.



**Workflow Activity Completion**

Workflow: ECR-100001  
Activity: Review ECR

| Sequence | Required                            | Description                                        | Complete                            |
|----------|-------------------------------------|----------------------------------------------------|-------------------------------------|
| 3        | <input type="checkbox"/>            | Review Affected Items, add additional if necessary | <input checked="" type="checkbox"/> |
| 4        | <input checked="" type="checkbox"/> | Make sure the Assigned Creator is correct          | <input checked="" type="checkbox"/> |

**Vote**

Vote:

Delegate to:

Comments:

**Authentication**

Password:  E-Signature:

Figure 270.

12. Mark the tasks you completed in the Tasks field.
13. Select **Tech Review** in the Vote list.
14. Click **Complete**.

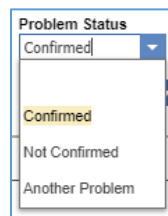
The ECR enters the **Technical Review** activity and awaits technical review by the ECR Owner.

#### 13.1.2.8 Technical Review of an ECR

An ECR Owner reviews an ECR and provides data, comments, and recommendations. The Owner does this job according to the company's policies and procedures. This section defines the information an ECR Owner must provide to comply with the CMII process requirements.

Use the following procedure to conduct a technical review of an Engineering Change Request:

1. Open the ECR.
2. Click **Edit** on the ECR toolbar.
3. Verify that the information already entered in the ECR is correct.
4. Select the proper value in the **Problem Status** list: **Confirmed**, **Not Confirmed**, or **Another Problem**.



**Problem Status**

Confirmed

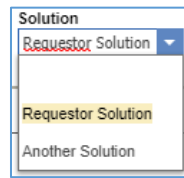
Confirmed

Not Confirmed

Another Problem

Figure 271.

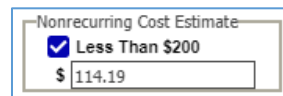
5. Select the proper value in the **Solution** list: **Requestor Solution** or **Another Solution**.



A screenshot of a dropdown menu titled 'Solution'. The menu is open, showing two options: 'Requestor Solution' (which is highlighted in yellow) and 'Another Solution'.

Figure 272.

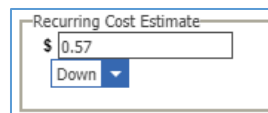
6. Enter the **Nonrecurring Cost Estimate** value in the accompanying text box. If the estimate is less than \$200, select the check box. This is a one-time cost estimation of the ECR solution implementation.



A screenshot of a form section titled 'Nonrecurring Cost Estimate'. It contains a checkbox labeled 'Less Than \$200' which is checked. Below the checkbox is a text box containing the value '\$ 114.19'.

Figure 273.

7. Enter the recurring cost estimation in the **Recurring Cost Estimate** text box. This is the difference in the unit cost of the Item as a result of the change. Select **Up** or **Down** in the accompanying list to indicate whether the unit cost went up or down as a result of the ECR.



A screenshot of a form section titled 'Recurring Cost Estimate'. It contains a text box with the value '\$ 0.57' and a dropdown menu below it with 'Down' selected.

Figure 274.

8. Enter the key timing factors for the implementation of the ECR in the **Key Implementation Timing Factors** text box.
9. Enter additional information in the **Comments** text box.
10. Click **Done** on the **ECR** toolbar.
11. Go to **Contents → My Innovator → My InBasket → Search My InBasket**.
12. Go to the **Workflow Activity Completion** dialog for the ECR.

**Workflow Activity Completion**

Workflow: ECR-100001  
Activity: Technical Review

| Sequence | Required                 | Description                    | Complete                            |
|----------|--------------------------|--------------------------------|-------------------------------------|
| 1        | <input type="checkbox"/> | Assess change impact           | <input checked="" type="checkbox"/> |
| 2        | <input type="checkbox"/> | Get input from other creators  | <input checked="" type="checkbox"/> |
| 3        | <input type="checkbox"/> | Make technical recommendations | <input checked="" type="checkbox"/> |

**Vote**

Vote: Complete Delegate Delegate to:  ...

Comments:

**Authentication**

Password:  E-Signature:

Figure 275.

13. Mark the tasks you completed in the **Tasks** field.
14. Select **Complete** in the **Vote** list.
15. Click **Complete**.

The ECR enters the **Route ECR** activity and awaits a decision of the CSI.

### 13.1.2.9 Routing an ECR

A CSI receives an ECR to decide how to route the ECR: either **Fast Track Approve** for a fast-track ECR or **CRB Approve** for full Change Review Board (CRB) involvement. The CSI makes this decision according to the company's policies and procedures. This section defines which pieces of information the CSI must provide to comply with the CMII process requirements when routing the ECR.

To route an Engineering Change Request:

1. Open the ECR.
2. Click **Edit** on the ECR toolbar.
3. Verify that the information already entered in the ECR is correct.
4. Double-check the **Key Implementation Timing Factors** text box to make sure it contains the correct data.
5. Specify the implementation priority number in the **Priority For Tech Review** field. The higher the priority, the lower the number selected.

Priority For Tech Review

☐ 1
☐ 2
☐ 3
☐ 4

Figure 276.

6. Click **Done** on the **ECR** toolbar.
7. Go to **Contents → My Innovator → My InBasket → Search My InBasket**.
8. Go to the **Workflow Activity Completion** dialog for the ECR.

Workflow Activity Completion

Workflow: ECR-100001
Activity: Route ECR

Tasks

| Sequence | Required                 | Description                      | Complete                            |
|----------|--------------------------|----------------------------------|-------------------------------------|
| 1        | <input type="checkbox"/> | Review technical recommendations | <input checked="" type="checkbox"/> |
| 2        | <input type="checkbox"/> | Prepare Cost Estimate            | <input checked="" type="checkbox"/> |
| 3        | <input type="checkbox"/> | Route the ECR                    | <input checked="" type="checkbox"/> |

Vote

Vote: 

CRB Approve

Fast Track Approve

Delegate

Delegate to:

...

Comments:

Authentication

Password:

.....

E-Signature:

Complete

Save Changes

Cancel

Figure 277.

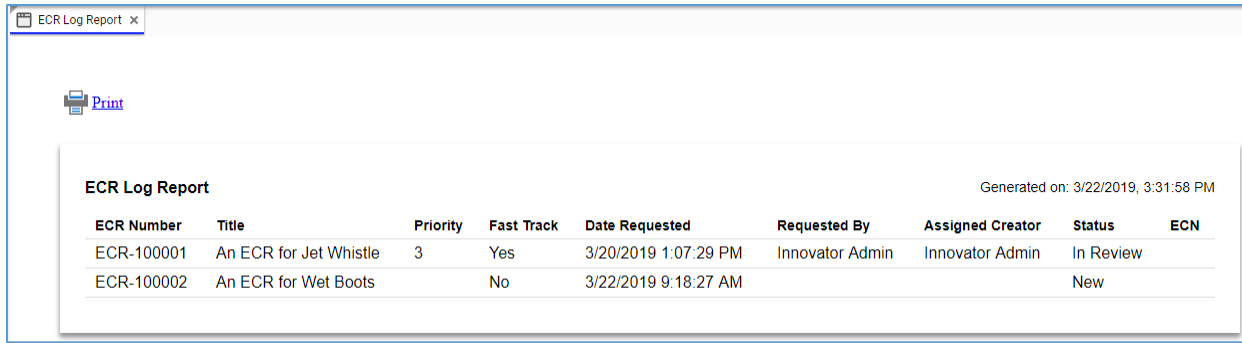
9. Mark the tasks you completed in the **Tasks** field.
10. Select either **CRB Approve** or **Fast Track Approve** in the **Vote** list depending on your decision.
11. Click **Complete**.

The ECR now awaits the final decision to be either **ECR Approved** or **ECR Disapproved**.

#### 13.1.2.10 ECR Log Report

You can get quick information about all the ECRs submitted to Aras Innovator with the ECR Log Report.





ECR Log Report Generated on: 3/22/2019, 3:31:58 PM

| ECR Number | Title                  | Priority | Fast Track | Date Requested       | Requested By    | Assigned Creator | Status    | ECN |
|------------|------------------------|----------|------------|----------------------|-----------------|------------------|-----------|-----|
| ECR-100001 | An ECR for Jet Whistle | 3        | Yes        | 3/20/2019 1:07:29 PM | Innovator Admin | Innovator Admin  | In Review |     |
| ECR-100002 | An ECR for Wet Boots   |          | No         | 3/22/2019 9:18:27 AM |                 |                  | New       |     |

Figure 278.

There are several ways to access the ECR Log Report:

- Go to **Contents** → **Change Management** → **ECRs**, right-click any **ECR**, and then click **Reports** → **ECR Log Report**.

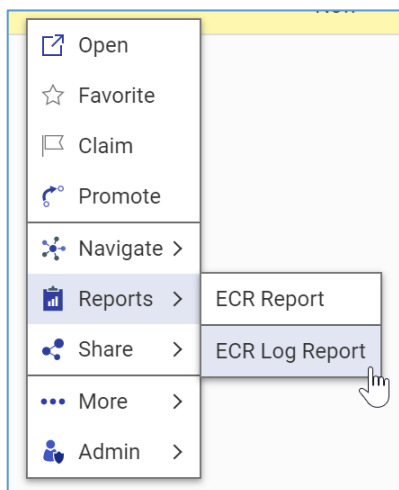


Figure 279.

- Open the view of any ECR. On the **ECR** toolbar, click **Reports**, and then click **ECR Log Report**.

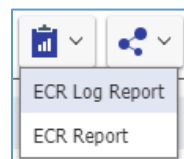


Figure 280.

To quit a Report, close its view in Aras Innovator.

At the top left corner of the ECR Log Report view, there is a **Print** button, which uses the standard browser print function. Depending on the browser and your specific configuration, various print choices, such as printing to a printer or saving to PDF, are available.

### 13.1.2.11 ECR Report

You can print or share an ECR using the ECR Report.

ECR Report x

Print

ECR Report Generated on: 3/22/2019, 3:35:26 PM

ECR NUMBER: ECR-100001 TITLE: An ECR for Jet Whistle STATUS: In Review

BASIS: ☐ Administrative Hierarchy ☒ Physical Hierarchy TYPE: ☐ Product Improvement ☒ Corrective Action REQUESTED BY: Innovator Admin SOURCE: Internal

PROPOSED SOLUTION: Turbo Fan

FAST TRACK: Yes PRIORITY FOR TECH REVIEW: ☐ 1 ☐ 2 ☒ 3 ☐ 4 CREATOR: Innovator Admin PROBLEM STATUS: Confirmed

TECHNICAL REVIEW: SOLUTION: Requestor Solution

NONRECURRING COST: \$ KEY IMPLEMENTATION TIMING FACTORS: COMMENTS:

RECURRING COST: \$

IMPLEMENTATION PRIORITY: ☒ 1 ☐ 2 ☐ 3 ☐ 4

**AFFECTED ITEMS**

| TYPE | NUMBER                   | REVISION | ACTION |
|------|--------------------------|----------|--------|
| Part | Child Part A Alternate 2 | A        | Change |

Figure 281.

There are several ways to access the ECR Report:

- Go to **Contents** → **Change Management** → **ECRs**, right-click an **ECR** to be reported, and then click **Reports** → **ECR Report**.

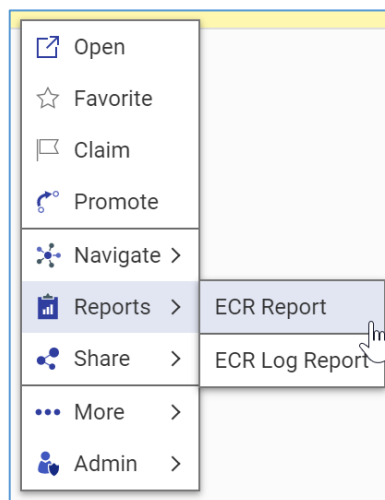


Figure 282.

- Open the view of an ECR. On the **ECR** toolbar, click **Reports**, and then click **ECR Report**.

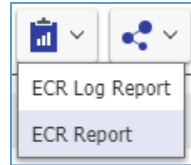


Figure 283.

To quit a Report, close its view in Aras Innovator.

At the top left corner of the ECR Report view, there is a **Print** button, which uses the standard browser print function. Depending on the browser and your specific configuration, various print choices, such as printing to a printer or saving to PDF, are available.

### 13.1.3 Engineering Change Notice (ECN)

An Engineering Change Notice (ECN) is a process for implementing changes within an organization. There are three types of change Actions for an affected Item in the ECN process:

- **Add** — A new affected Item becomes **Released**.
- **Change** — An affected existing Item is modified. The Item becomes **Superseded**, **Obsolete**, or incremented to a **new revision** level to allow modification.
- **Delete** — An existing affected Item is obsolete. The Item becomes **Superseded** or **Obsolete**.

#### 13.1.3.1 Understanding the ECN Process

The ECN workflow has the following roles:

- An ECN Creator is the user who initially created the ECN in the system. This may or may not be the same person as other roles.
- An ECN Owner (**Assigned Creator**), is the engineer or technical person responsible for the technical implications of the change and its ramifications.
- A Change Specialist II (CSII) prepares ECNs for implementing approved ECRs as well as the ECN impact information for the change review board to develop a detailed implementation plan.
- A Change Specialist III (CSIII) audits the ECN implementation using ECNs and detailed implementation plans as checklists.
- The Configuration Management (CM) group.

An ECN traverses between various states during its life cycle depicted in the following figure.



Figure 284.

The ECN workflow together with the ECN life cycle produces the following scenario:



Figure 285.

1. A member of the CM group creates and submits the ECN.
2. The ECN enters the **ECN Planning** activity **assigned to the CSII**. According to the company's rules and policies, the CSII reviews the ECN, assigns an Owner to it, and makes decisions and recommendations about the propagation of the Item change. For example, the CSII can decide whether a part should have a new revision or an entirely new part number. Also, the CSII checks whether a changed part leads to a change of revision or part number of assemblies containing this part. Once the CSII votes **Complete**, the ECN moves further.
3. The ECN enters the **Update Documents** activity **assigned to the Owner**. The Owner is responsible for updating the Item documentation affected by this change, including BOMs, specifications, drawings, and any other documents. The affected Items requiring the change of documentation may include only the items being changed directly but also hierarchies containing these Items. Once the Owner votes **Complete**, the ECN moves further.
4. The ECN enters the **Review Documents** activity **assigned to the CSII**. The CSII decides to either:
  - a. **Reject** — The ECN goes back to the **Update Documents** activity.
  - b. **Approve** — The ECN goes further.
5. The ECN enters the **ECN Audit** activity **assigned to the CSIII**. Once the CSIII votes **Complete**, the ECR goes further.
6. The ECN becomes **Released**. The life cycle states of the affected Items are automatically updated.

The following table shows the ECN lifecycle states and workflow activities in terms of Roles and Permissions.

**Table 5:** The ECN lifecycle and workflow activities with corresponding Roles and Permissions

| Life Cycle State | Workflow Activity | Can Add | Get                              | Update            | Delete | Change access | Can be promoted   | Update responsible |
|------------------|-------------------|---------|----------------------------------|-------------------|--------|---------------|-------------------|--------------------|
| New              | Submit ECN        | CM      | All employees, Creator, Owner CM | Creator, Owner CM | CM     | CM            | Yes               | Creator            |
| In Planning      | ECN Planning      |         | All employees, Creator, Owner CM | Creator, Owner CM | CM     | CM            | Yes               | CSII               |
| In Work          | Update Documents  |         | All employees                    |                   |        |               | Yes               | Owner              |
| In Review        | Review Documents  |         | All employees                    |                   |        |               | Yes               | CSII               |
| In Review        | ECN Audit         |         |                                  |                   |        |               | Yes               | CSIII              |
| Released         | Released          |         | All employees                    |                   |        |               | No, a final state |                    |

There is the ECN report, which is a printable form of an ECN. For details, refer to section [13.1.3.9 The ECN Report](#).

### 13.1.3.2 Creating an ECN

Because an ECN can be used to add, change, or delete affected Items, the ECN creation process can be conventionally divided into general and specific procedures. This section defines the general steps a member of the CM group should use for creating an ECN. The specific procedures are outlined in sections [13.1.3.3 Adding an Item with an ECN](#), [13.1.3.4 Changing Items with an ECN](#), and [13.1.3.5 Deleting an Item with an ECN](#).

To create an Engineering Change Notice:

1. Go to **Contents → Change Management → ECNs**.
2. Click **Create New ECN**. The view of a new ECN appears.

The screenshot shows the 'ECN 1' form. At the top, there are buttons for 'Save', 'Done', and 'Delete'. Below these are tabs for 'ECN', 'Affected Items', 'ECRs', and 'Files'. The 'ECN' tab is active, showing fields for 'ECN Number' (with a dropdown for 'Server Assigned'), 'Title', 'Status', 'Basis' (radio buttons for 'Administrative Hierarchy' and 'Physical Hierarchy'), 'Priority' (radio buttons 1-4), 'Assigned Creator' (with a dropdown), 'Effectivity Date' (with a calendar icon), 'Description', and 'Special Instructions'. There is also a checkbox for 'Customer Approval Required'. At the bottom, there is a table with columns: 'Ty...', 'Action', 'Old Number [...]', 'Old Rev...', 'Interchang...', 'New Number ...', 'New Rev...', 'In Build', and 'In Service'.

Figure 286.

3. Enter the **ECN title** in the **Title** text box, making it as descriptive as possible.
4. Select the appropriate hierarchy for **Basis**:
  - **Administrative hierarchy** refers to anything related to the process of running the business. An error in an expense report procedure would be classified as an administrative hierarchy.
  - **Physical hierarchy** refers to the Item (Part) itself and any Document associated with it. A bug in software would be classified as a physical hierarchy.
5. Specify the change priority number in the **Priority** field. The higher the priority of the change, the lower the number selected.
6. Specify the ECN Owner in the **Assigned Creator** text box.
7. Specify the date on which the ECN goes into effect in the **Effective Date** box. If left blank, it is automatically set to the Item release date.
8. If the change requires customer approval, select the **Customer Approval Required** check box.

9. Describe the change in the **Description** text box.
10. If there are any special instructions for the change, enter them in the **Special Instructions** text box.
11. Go to the **ECRs** accordion tab to attach all the ECRs included in this ECN for tracking and reporting.

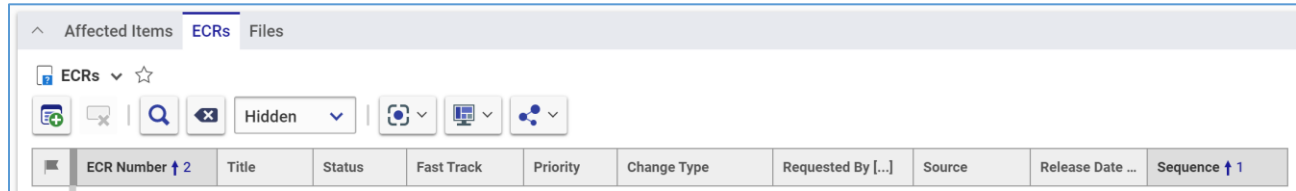


Figure 287.

12. Click **Add ECRs** on the ECRs accordion toolbar. The **Search dialog – ECR** appears.
13. Search for and select the ECRs combined or included in this ECN. The selected ECRs appear as new rows at the bottom of the ECRs grid.

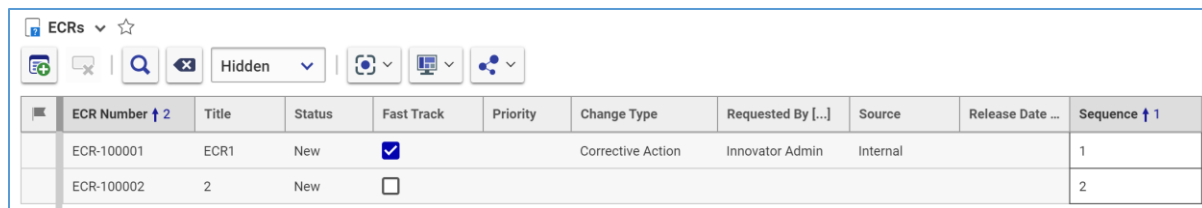


Figure 288.

14. Go to the **Files** accordion tab to attach all the files concerning this ECN.

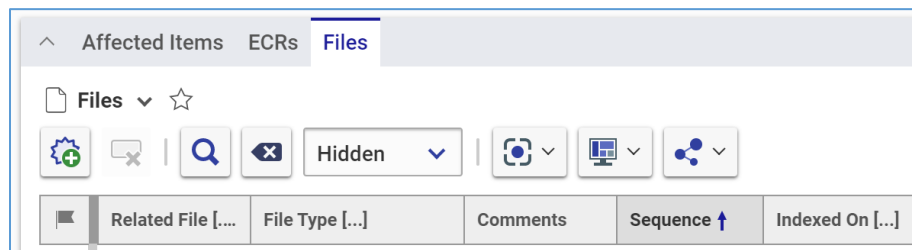


Figure 289.

15. Click **New File** on the Files accordion toolbar. The standard operating system dialog for file browsing and selection appears.
16. Search for and select a file. A new row appears in the Files grid with the file attached.

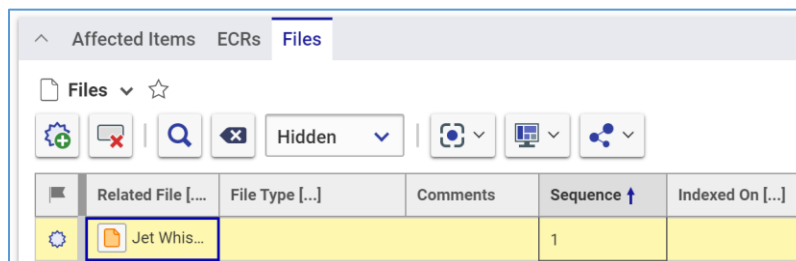


Figure 290.

**Note:** Aras Innovator will automatically assign a file type depending on File's extension when you save the ECN.

17. If it is necessary to provide additional information about the File, enter it in the **Comments** cell.
18. Repeat steps 15-17 for other files concerning this ECN.
19. If you need to add Items with the ECN, use the procedure described in section [13.1.3.3 Adding an Item with an ECN](#).
20. If you need to change Items with the ECN, use the procedure described in section [13.1.3.4 Changing Items with an ECN](#).
21. If you need to delete Items with the ECN, use the procedure described in section [13.1.3.5 Deleting an Item with an ECN](#).
22. Click **Done** on the **ECN** toolbar. The ECN is created and waits to be submitted as defined in the section [13.1.3.7 Submitting an ECN](#).

Figure 291.

### 13.1.3.3 Adding an Item with an ECN

Because an ECN can be used to add, change, or delete affected Items, the ECN creation process can be conventionally divided into general and specific procedures. This section defines the steps a member of the CM group should use for adding affected Items when creating the ECN. Section [13.1.3.2 Creating an ECN](#) outlines the general procedure. The other specific procedures are outlined in sections [13.1.3.4 Changing Items within an ECN](#) and [13.1.3.5 Deleting an Item within an ECN](#).

A new Item has the **Preliminary** State. An ECN promotes the Item to the **Released** State at the end of the ECN process when it is released.

To assign a new affected Item to release with an Engineering Change Notice:

1. Open the ECN.

- Click **Edit** on the **ECN** toolbar.
- Go to the **Affected Items** accordion tab.
- Click **Create Item** on the **Affected Items** accordion toolbar. A new row appears at the bottom of the **Affected Items** grid.

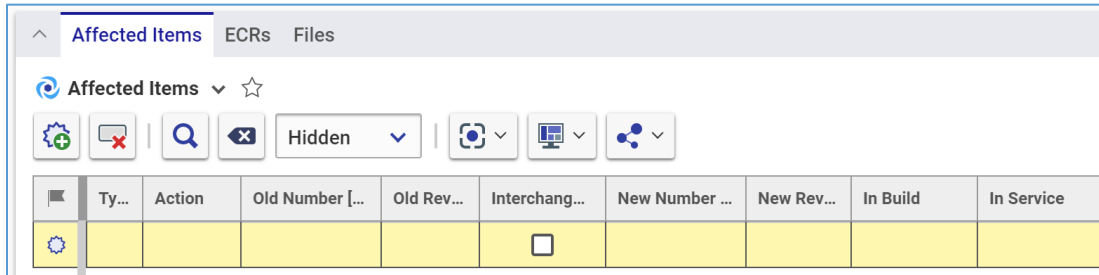


Figure 292.

- Select **Add** in the **Action** cell.

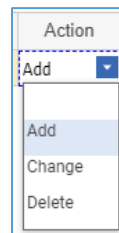


Figure 293.

- Click the **ellipsis** button in the **New Number** cell. The **Search dialog – Change Controlled Item** appears.
- Search for and select the new affected Item. The affected Item populates the **New Number** cell.

|  | Ty... | Action | Old Number [...] | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service |
|--|-------|--------|------------------|--------------|--------------------------|------------------|--------------|----------|------------|
|  |       | Add    |                  |              | <input type="checkbox"/> | Child B          |              |          |            |

Figure 294.

- Click **Save** or **Done** on the **ECN** toolbar. The ECN has an affected Item assigned for adding.

#### 13.1.3.4 Changing Items with an ECN

Because an ECN can be used to add, change, or delete affected Items, the ECN creation process can be conventionally divided into general and specific procedures. This section defines the steps a member of the CM group should use for changing affected Items when creating the ECN. Section [13.1.3.2 Creating an ECN](#) outlines the general procedure. The other specific procedures are outlined in sections [13.1.3.3 Adding an Item with an ECN](#) and [13.1.3.5 Deleting an Item with an ECN](#).

A change for an already Released Item can be either:

- Interchangeable**, the Released Item is automatically revised during the ECN process to create a new Preliminary Item version to keep changes. When the ECN is released at the end of the ECN process, this new Item version is Released.
- Non-Interchangeable**, During the ECN process, any necessary changes are made to a new Preliminary Item. When the ECN is released at the end of the ECN process, the new Item is Released, and the old affected Item is Superseded.



To assign an existing affected Item to be changed when creating an Engineering Change Notice:

1. Follow steps 1-4 of the Adding an Item within an ECN procedure.
2. Select **Change** in the **Action** cell.

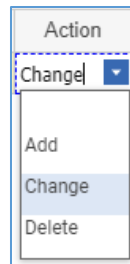


Figure 295.

3. Click the **ellipsis** button in the **Old Number** cell. The **Search dialog – Change Controlled Item** appears.
4. Search for and select the affected Item to be changed. The selected affected Item populates the **Old Number** cell.

|  | Ty... | Action | Old Number [...] | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service |
|--|-------|--------|------------------|--------------|--------------------------|------------------|--------------|----------|------------|
|  |       | Change | Child A          |              | <input type="checkbox"/> |                  |              |          |            |

Figure 296.

5. Select the **Interchangeable** cell check box if the old and new Items are interchangeable, e.g., if there is no Fit-Form-Function change. The new Item version can be replaced or modified without causing any change in functionality to the surrounding Items. When selected, the same Item number will be retained.
6. If the **Interchangeable** cell is not selected, e.g., the old and new Items are not interchangeable due to a Fit-Form-Function change, select the replacement new Item:
  - a. Click the **ellipsis** button in the **New Number** cell. The **Search dialog – Change Controlled Item** appears.
  - b. Using the standard search procedure, search for and select the new affected Item. The new affected Item populates the **New Number** cell. This new item must be in the Preliminary state to pass readiness checks during the ECN process.

|  | Ty... | Action | Old Number [...] | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service |
|--|-------|--------|------------------|--------------|--------------------------|------------------|--------------|----------|------------|
|  | <i... | Change | Child A          |              | <input type="checkbox"/> | Child B          |              |          |            |

Figure 297.

7. Click **Save** two times on the ECN toolbar.
8. Select the action to take on the Items that are being built in the **In Build** cell: **Use Existing**, **Rework**, or **Scrap**.
9. Select the action to take on the Items that are currently in service in the **In Service** cell: **Use Existing**, **Rework**, or **Scrap**.

| Ty... | Action | Old Number [...] | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service   |
|-------|--------|------------------|--------------|--------------------------|------------------|--------------|----------|--------------|
| <i... | Change | Child A          | A            | <input type="checkbox"/> | Child B          | A            | Rework   | Use Existing |

Figure 298.

10. Click **Save** or **Done** on the **ECN** toolbar. The ECN has an affected Item assigned for changing.

| Affected Items ECRs Files                                                                         |        |                         |              |                          |                         |              |          |              |
|---------------------------------------------------------------------------------------------------|--------|-------------------------|--------------|--------------------------|-------------------------|--------------|----------|--------------|
| Affected Items <span>▼</span> ☆<br>[Icons: Gear, Eraser, Search, X, Hidden, View, Monitor, Share] |        |                         |              |                          |                         |              |          |              |
| Ty...                                                                                             | Action | Old Number [...]        | Old Revision | Interchangeable          | New Number [...]        | New Revision | In Build | In Service   |
| <i...                                                                                             | Change | <a href="#">Child A</a> | A            | <input type="checkbox"/> | <a href="#">Child B</a> | A            | Rework   | Use Existing |

Figure 299.

### 13.1.3.5 Deleting an Item with an ECN

Because an ECN can be used to add, change, or delete affected Items, the ECN creating process can be conventionally divided into general and specific procedures. This section defines the steps a member of the CM group should use for superseding affected Items with the ECN. Section [13.1.3.2 Creating an ECN](#) outlines the general procedure. The other specific procedures are outlined in sections [13.1.3.3 Adding an Item with an ECN](#) and [13.1.3.4 Changing Items with an ECN](#).

An already Released item gets Superseded at the end of the ECN process when it is released.

Use the following procedure to assign an existing affected Item to be superseded with an Engineering Change Notice:

1. Follow steps 1-4 of the Adding an Item within an ECN procedure.
2. Select **Delete** in the **Action** cell.

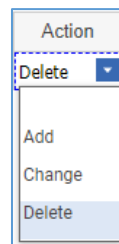


Figure 300.

3. Click the **ellipsis** button in the **Old Number** cell. The **Search dialog – Change Controlled Item** appears.
4. Search for and select the affected Item to be deleted. The selected affected Item populates the **Old Number** cell.

| Ty... | Action | Old Number [...] | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service |
|-------|--------|------------------|--------------|--------------------------|------------------|--------------|----------|------------|
| <i... | Delete | Child A          | A            | <input type="checkbox"/> |                  |              |          |            |

Figure 301.

5. Click **Save** two times on the ECN toolbar.

6. Select the action to take on the Items that are being built in the **In Build** cell: **Use Existing**, **Rework**, or **Scrap**.
7. Select the action to take on the Items that are currently in service in the **In Service** cell: **Use Existing**, **Rework**, or **Scrap**.

|  | Ty... | Action | Old Number [...] | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service   |
|--|-------|--------|------------------|--------------|--------------------------|------------------|--------------|----------|--------------|
|  | <i... | Delete | Child A          | A            | <input type="checkbox"/> |                  |              | Rework   | Use Existing |

Figure 302.

8. Click **Save** or **Done** on the ECN toolbar. The ECN has an affected Item assigned for superseding.

| Affected Items ECRs Files                                                                                 |      |        |                         |              |                          |                  |              |          |              |
|-----------------------------------------------------------------------------------------------------------|------|--------|-------------------------|--------------|--------------------------|------------------|--------------|----------|--------------|
| Affected Items <span>▼</span> <span>☆</span><br>[Icons: Gear, Chat, Search, Hidden, View, Monitor, Share] |      |        |                         |              |                          |                  |              |          |              |
|                                                                                                           | Type | Action | Old Number [...]        | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service   |
|                                                                                                           |      | Delete | Child Part A Substitute | A            | <input type="checkbox"/> |                  |              | Rework   | Use Existing |

Figure 303.

### 13.1.3.6 Removing an Affected Item from an ECN

Use the following procedure:

1. Follow steps 1-3 of the Adding an Item within an ECN procedure.
2. Click the **Affected Item** and then click **Delete Row** on the **Affected Items** accordion toolbar. The **Affected Item** becomes unavailable.

| Affected Items ECRs Files                                                                                 |       |        |                  |              |                          |                  |              |          |              |
|-----------------------------------------------------------------------------------------------------------|-------|--------|------------------|--------------|--------------------------|------------------|--------------|----------|--------------|
| Affected Items <span>▼</span> <span>☆</span><br>[Icons: Gear, Chat, Search, Hidden, View, Monitor, Share] |       |        |                  |              |                          |                  |              |          |              |
|                                                                                                           | Ty... | Action | Old Number [...] | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service   |
|                                                                                                           | <i... | Delete | Child A          | A            | <input type="checkbox"/> |                  |              | Rework   | Use Existing |

Figure 304.

3. Click **Save** or **Done** on the ECN toolbar. The **Affected Item** disappears from the grid.

**Note:** The **Affected Item** disappears immediately if the ECN was not saved after it was added.

### 13.1.3.7 Submitting an ECN

Use the following procedure:

1. Go to **Contents** → **My Innovator** --> **My InBasket** → **Search My InBasket**.
2. Double-click on the Workflow Task to open the **Workflow Activity Completion** dialog for the ECN.

**Workflow Activity Completion**

Workflow: ECN-100003  
Activity: Submit ECN

| Sequence | Required                 | Description                                | Complete                            |
|----------|--------------------------|--------------------------------------------|-------------------------------------|
| 1        | <input type="checkbox"/> | Check the ECN form for completeness        | <input checked="" type="checkbox"/> |
| 2        | <input type="checkbox"/> | Ensure all Affected Items are attached     | <input checked="" type="checkbox"/> |
| 3        | <input type="checkbox"/> | Submit the ECN for implementation planning | <input checked="" type="checkbox"/> |

**Vote**

Vote: Submit Delegate Delegate to:  ...

Comments:

**Authentication**

Password:  E-Signature:

Complete
Save Changes
Cancel

Figure 305.

3. Mark the tasks you completed in the **Tasks** field.
4. Select **Submit** in the **Vote** list.
5. Click **Complete**.

The ECR enters the **ECN Planning** activity assigned to the CSII.

#### 13.1.3.8 ECN Planning

For the **ECN Planning** activity, a CSII makes sure that the planning is accurate and complete by providing all necessary Affected Items information, such as:

- There are new part numbers for the non-interchangeable parts changes.
- The ECN has all other affected items in the hierarchy.
- Affected Items up the hierarchy tree from the actual Item are changed.
- Affected Documents of the parent assemblies (such as BOMs) are changed.

Since each company has its own rules for new part numbers or new revision generation, the CSII is responsible for the accuracy of this information.

#### 13.1.3.9 The ECN Report

You can print or share an ECN using the ECN Report.

ECN Report x

Print

ECN Report Generated on: 3/25/2019, 7:47:55 PM

ECN NUMBER: ECN-100003 TITLE: Changing the Jet Whistle STATUS: In Planning

BASIS: ☐ Administrative Hierarchy ☒ Physical Hierarchy

PRIORITY: ☒ 1 ☐ 2 ☐ 3 ☐ 4

CREATOR: Innovator Admin

CUSTOMER APPROVAL: Not Required RELEASE DATE: EFFECTIVITY DATE: 4/19/2019 12:00:00 AM

DESCRIPTION: Changing the Jet Whistle

SPECIAL INSTRUCTIONS: Changing the Jet Whistle

**AFFECTED ITEMS**

| TYPE | SUPERSEDED              |          | ACTION | SUPERSEDING |          | IN WORK | IN SERVICE   |
|------|-------------------------|----------|--------|-------------|----------|---------|--------------|
|      | NUMBER                  | REVISION |        | NUMBER      | REVISION |         |              |
| Part | Child Part A Substitute | A        | Delete |             |          | Rework  | Use Existing |

Figure 306.

There are several ways to access the ECN Report:

- Go to **Contents** → **Change Management** → **ECNs** → **Search ECNs**, right-click an **ECN** to be reported, and then click **ECN Report**.

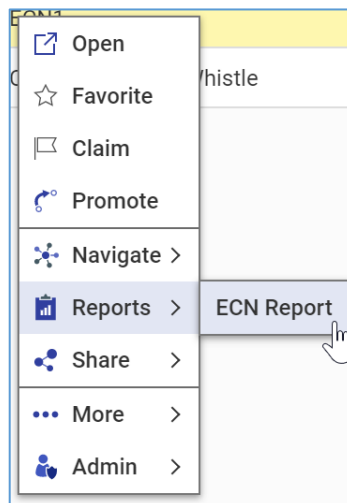


Figure 307.

- Open the view of an ECN to be reported. On the **ECN** toolbar, click **Reports**, and then click **ECN Report**.

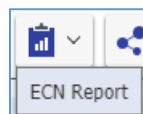


Figure 308.

To quit a Report, close its view in Aras Innovator.

At the top left corner of the ECN Report view, there is a **Print** button, which uses the standard browser print function. Depending on the browser and your specific configuration, various print choices, such as printing to a printer or saving to PDF, are available.

## 13.2 Simplified Change Management

The Simplified change process uses two different kinds of items to manage changes:

- The **Simple Engineering Change Order (ECO)** is a simplified approach for releasing, modifying, or deleting Items. This process is useful when the full CMII process is not required to add or modify an Item.
- The **Manufacturing Change Order (MCO)** allows for modification of a released Part to add a Manufactured Part to the Part's AML. The AML contains a list of Manufactured Parts created by approved Manufacturers and sold by approved Vendors. By using the MCO, the alteration of an AML on a released Part does not require the full CMII process.

Both Items are processed according to the same Simplified Change Management workflow depicted in the following figure.

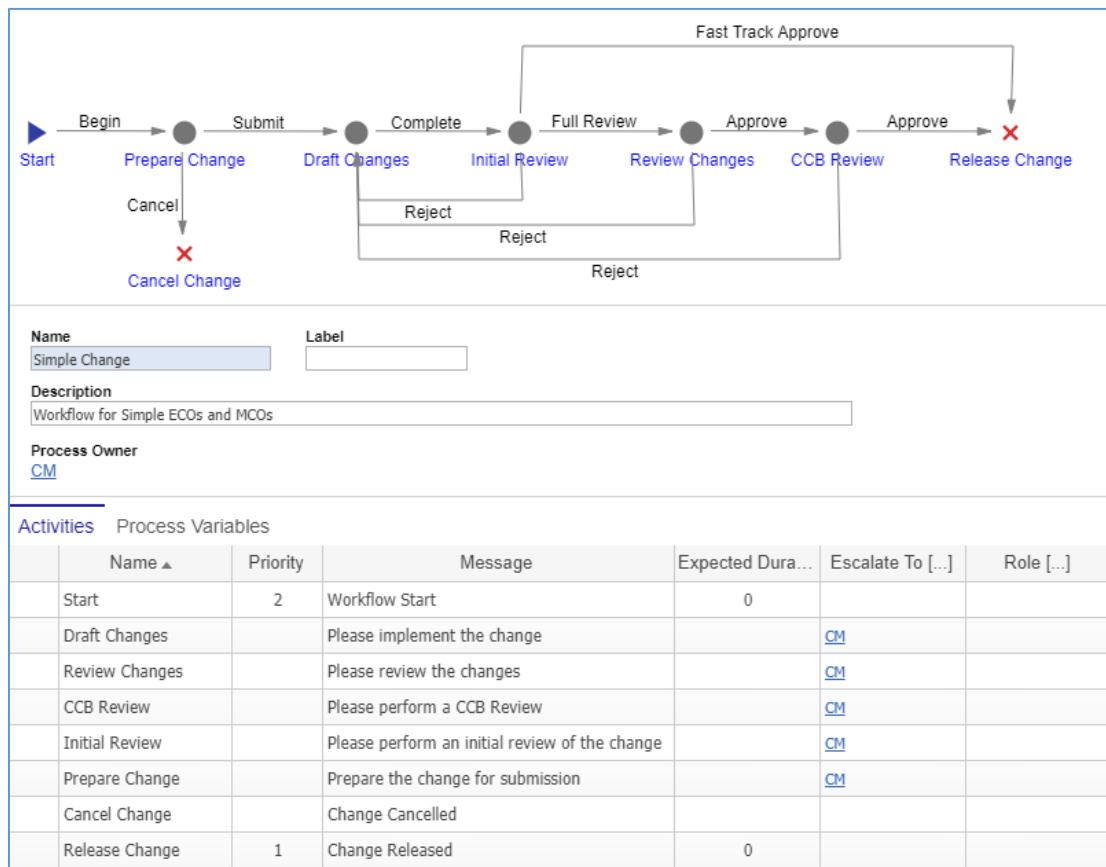


Figure 309.

### 13.2.1 Simple Engineering Change Order (ECO)

A Simple Engineering Change Order (ECO) is a simplified process of implementing changes within an organization.

There are three types of change Actions for an affected Item in the Simple ECO process:

- **Add** — A new affected Item becomes **Released**.
- **Change** — An affected existing Item is modified. The Item becomes **Superseded**, **Obsolete**, or incremented to a new revision level to allow modification.
- **Delete** — An existing affected Item is obsolete. The Item becomes **Superseded** or **Obsolete**.

#### 13.2.1.1 Understanding the Simple ECO Process

The Simple ECO workflow has the following roles:

- A Simple ECO Creator (**Originator**), which is a user who initially created the Simple ECO in the system. This may or may not be the same person as other roles.
- A Simple ECO Owner (**Assigned Creator**), which is an engineer or technical person responsible for technical implications and ramifications of the change to each affected Item.
- A Change Specialist I (CSI) coordinates and prepares Simple ECOs through the workflow.
- Change Analyst (CA).
- The Configuration Management (CM) group.
- The Change Control Board (CCB), which is responsible for review and final approval of an MCO.

A Simple ECO traverses between various states during its life cycle as depicted in the following figure.

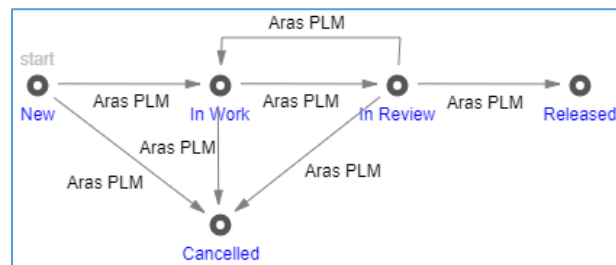


Figure 310.

The Simple Change Management workflow and Simple ECO life cycle together produce the following scenario:

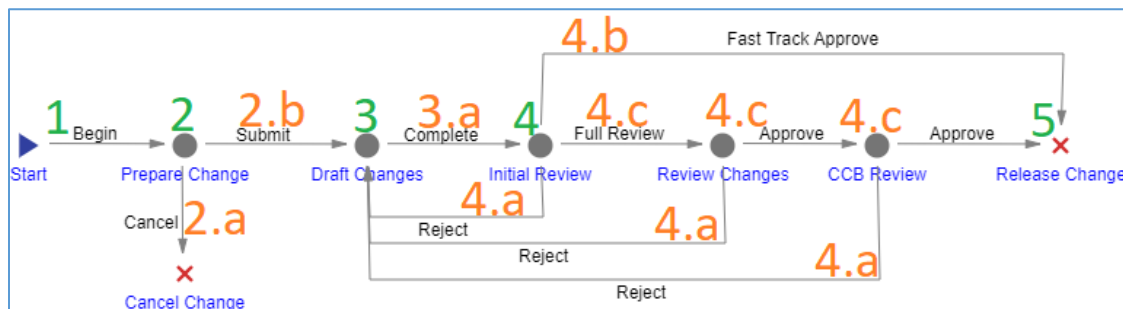


Figure 311.

1. An Aras Innovator user initiates a Simple ECO: creates and saves the Simple ECO. When the Creator saves the Simple ECO, it enters the **In Work** state, and a workflow is automatically started to assign tasks and track the changes to the Simple ECO.
2. The Simple ECO enters the **Prepare Change** activity **assigned to the CSI**. According to the company's rules and policies, the CSI reviews the Simple ECO, assigns an Owner to each Affected Item in the Simple ECO, and makes decisions and recommendations about the propagation of these Item changes. For example, the CSI can decide whether a part should have a new revision or an entirely new part number. Also, the CSI checks whether a changed part leads to a change of revision or part number of assemblies containing this part.
  - a. The CSI can **Cancel** the Simple ECO.
  - b. Once the CSI votes **Submit**, the Simple ECO goes further.
3. The Simple ECO enters the **Draft Changes** activity **assigned to each Owner**. The Owner is responsible for updating the Item documentation affected by this change, like BOMs, specifications, drawings, any other documents. The affected Items requiring the change of documentation may be not only the items being changed directly but also hierarchies containing these Items.
  - Once all the Owners vote **Complete**, the Simple ECO goes further.
4. The Simple ECO enters the **Initial Review** activity **assigned to the CA**. The CA reviews the changes and decides either to:
  - a. **Reject** — The Simple ECO goes back to the **Draft Changes** activity. The Owners can modify and resubmit the Simple ECO for the **Initial Review** activity.
  - b. **Fast Track Approve** — The Simple ECO is **Approved** and **Released**.
  - c. **Full Review** — The Simple ECO enters the **Review Changes** activity **assigned to other team members**. The Simple ECO goes under in-depth formal/informal reviews. Once these reviews are completed, The Simple ECO enters the **CCB Review** activity. Once the CCB audit is completed, the Simple ECO is **Approved** and **Released**. The Simple ECO may also be rejected (4.a) at these activities.
5. Once the Simple ECO is **Released**, the life cycle states of the affected Items are updated.

The following table shows the Simple ECO lifecycle and workflow in terms of Roles and Permissions.

**Table 6:** The Simple ECO lifecycle and workflow activities with corresponding Roles and Permissions

| Life Cycle State | Workflow Activity | Can Add | Get                                          | Update                                | Delete                                | Change access                  | Can be promoted | Update responsible                                  |
|------------------|-------------------|---------|----------------------------------------------|---------------------------------------|---------------------------------------|--------------------------------|-----------------|-----------------------------------------------------|
| New              | Prepare Change    | World   | Aras PLM, CM, Creator, Manager, Owner, World | Aras PLM, CM, Creator, Manager, Owner | Aras PLM, CM, Creator, Manager, Owner | Aras PLM, CM, Creator, Manager | Yes             | CSI                                                 |
| In Work          | Draft Changes     |         | Aras PLM, CM, Manager, Owner, World          | Aras PLM, CM, Manager, Owner          | Aras PLM, CM                          | Aras PLM, CM                   | Yes             | Each Owner (Assigned Creator) of all Affected Items |



| Life Cycle State | Workflow Activity                          | Can Add | Get             | Update   | Delete   | Change access | Can be promoted | Update responsible |
|------------------|--------------------------------------------|---------|-----------------|----------|----------|---------------|-----------------|--------------------|
| In Review        | Initial Review, Review Changes, CCB Review |         | Aras PLM, World | Aras PLM | Aras PLM | Aras PLM      | Yes             | CA                 |
| Released         | Released Change                            |         | Aras PLM, World | Aras PLM |          | Aras PLM      | No              |                    |
| Cancelled        | Cancelled Change                           |         | Aras PLM, World | Aras PLM |          | Aras PLM      | Yes             |                    |

### 13.2.1.2 Creating a Simple ECO

Because a Simple ECO can be used to add, change, or delete affected Items, the Simple ECO creation process can be conventionally divided into general and specific procedures. This section defines the general steps you should use for creating a Simple ECO. The specific procedures are outlined in sections [13.2.1.3 Adding an Item within a Simple ECO](#), [13.2.1.4 Changing an Item within a Simple ECO](#), and [13.2.1.5 Deleting an Item within a Simple ECO](#).

To create a Simple Engineering Change Order:

1. Go to **Contents** → **Change Management** → **ECOs** (one with the green icon).

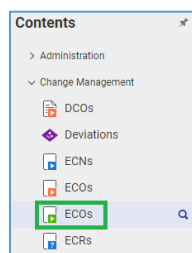


Figure 312.

2. Click **Create New ECO**. The view of a new Simple ECO appears.

Figure 313.

3. Keep the default **Phase In** value in the **Change Category** list.
4. Describe the change in the **Change Description** text box.
5. Describe the reason for the change in the **Reason For Change** text box.
6. Keep the **default** value in the **Originator** box. This is the Simple ECO Creator.
7. Keep the default **Change Specialist I** value in the **Change Analyst** box. This is a person or group assigned to act as a CA in the Simple ECO workflow process.
8. Keep the **Date Originated** box blank. This box is automatically filled with the date on which the Simple ECO is saved.
9. Go to the **Attachments** accordion tab to attach all the files concerning this Simple ECO.

Figure 314.

10. Click **New File** on the Attachments accordion toolbar. The standard operating system dialog for file browsing and selection appears.
11. Search for and select a file. A new row appears in the **Files** grid with the file attached.

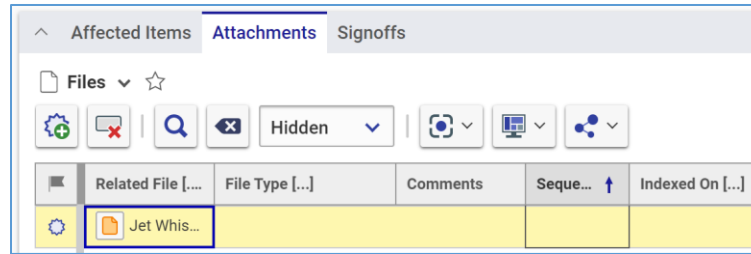


Figure 315.

**Note:** Aras Innovator will automatically assign a file type depending on File's extension when you save the ECO.

12. If it is necessary to provide additional information about the File, enter it in the **Comments** cell.
13. Repeat steps 10-12 for other files associated with this Simple ECO.
14. If you need to add Items with the Simple ECO, use the procedure described in section [13.2.1.3 Adding an Item within a Simple ECO](#).
15. If you need to change Items with the Simple ECO, use the procedure described in section [13.2.1.4 Changing an Item within a Simple ECO](#).
16. If you need to delete Items with the Simple ECO, use the procedure described in section [13.2.1.5 Deleting an Item within a Simple ECO](#).
17. Click **Done** on the **Simple ECO** toolbar. The Simple ECO is created and awaits to be submitted as defined in the section [13.2.1.7 Submitting a Simple ECO](#).

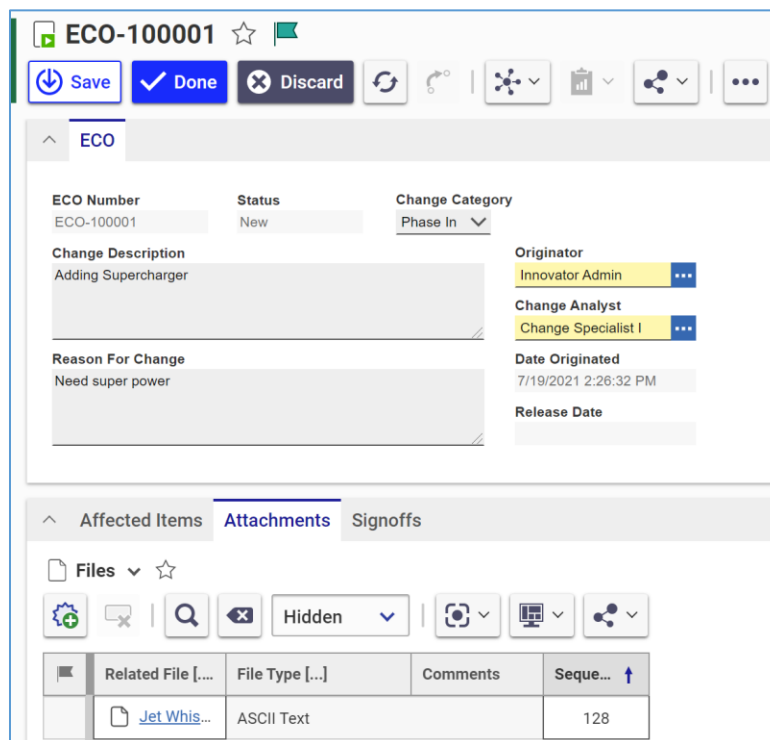


Figure 316.

### 13.2.1.3 Adding an Item within a Simple ECO

Because a Simple ECO can be used to add, change, or delete affected Items, the Simple ECO creating process can be conventionally divided into general and specific procedures. This section defines the steps you should use for adding affected Items when creating a Simple ECO. Section [13.2.1.2 Creating a Simple ECO](#) outlines the general procedure. The other specific procedures are outlined in sections [13.2.1.4 Changing an Item within a Simple ECO](#) and [13.2.1.5 Deleting an Item within a Simple ECO](#).

A new Item has the **Preliminary** State. A Simple ECO promotes the Item to the **Released** State at the end of the Simple ECO process when it is released.

To assign a new affected Item for adding when creating a Simple Engineering Change Order:

1. Open the Simple ECO.
2. Click **Edit** on the Simple ECO toolbar.
3. Go to the **Affected Items** accordion tab.
4. Click **New Affected Item** on the **Affected Items** accordion toolbar. A new row appears at the bottom of the **Affected Items** grid.

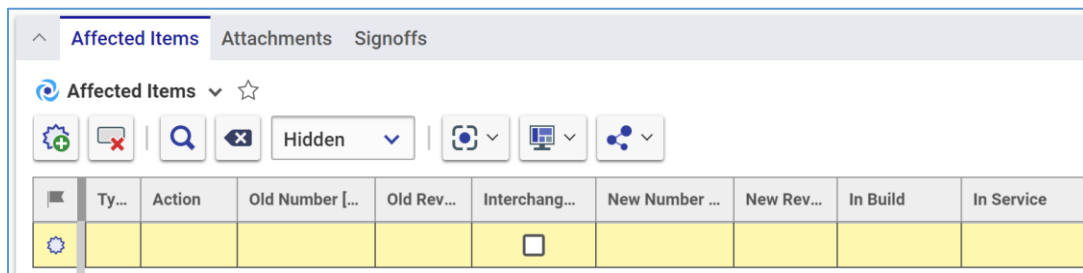


Figure 317.

5. Select **Add** in the **Action** cell.

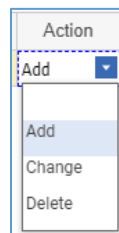


Figure 318.

6. Click the **ellipsis** button in the **New Number** cell. The **Search dialog – Change Controlled Item** appears.
7. Search for and select the new affected Item. The affected Item populates the **New Number** cell.

|  | Ty... | Action | Old Number [...] | Old Rev... | Interchangeable          | New Number [...] | New Revision | In Build | In Service |
|--|-------|--------|------------------|------------|--------------------------|------------------|--------------|----------|------------|
|  |       | Add    |                  |            | <input type="checkbox"/> | Child A          |              |          |            |

Figure 319.

8. Click **Save** or **Done** on the Simple ECO toolbar. The Simple ECO has an affected Item assigned for adding.

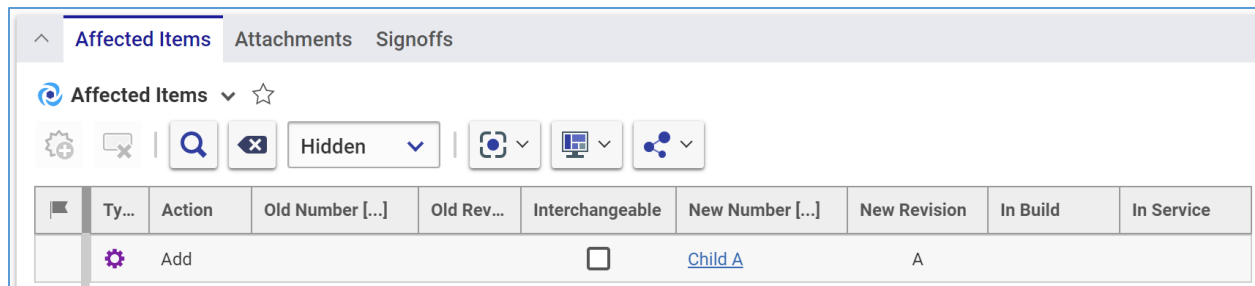


Figure 320.

#### 13.2.1.4 Changing an Item within a Simple ECO

Because a Simple ECO can be used to add, change, or delete affected Items, the Simple ECO creating process can be conventionally divided into general and specific procedures. This section defines the steps you should use for changing affected Items when creating a Simple ECO. Section [13.2.1.2 Creating a Simple ECO](#) outlines the general procedure. The other specific procedures are outlined in sections [13.2.1.3 Adding an Item within a Simple ECO](#) and [13.2.1.5 Deleting an Item within a Simple ECO](#).

A change for an already Released Item can be either:

- **Interchangeable**, the Released Item is automatically revised during the ECN process to create a new Preliminary Item version to make changes. When the ECN is released at the end of the ECN process, this new Item version is Released.
- **Non-Interchangeable**, During the ECN process, any necessary changes are made to a new Preliminary Item. When the ECN is released at the of the ECN process, the new Item is Released, and the old affected Item is Superseded.

To assign an existing affected Item to be changed when creating a Simple Engineering Change Order:

1. Follow steps 1-4 of the Adding an Item within a Simple ECO procedure.
2. Select **Change** in the **Action** cell.

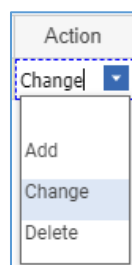


Figure 321.

3. Click the **ellipsis** button in the **Old Number** cell. The **Search dialog – Change Controlled Item** appears.
4. Using the standard search procedure, search for and select the affected Item to be changed. The selected affected Item populates the **Old Number** cell.

|  | Ty... | Action | Old Number [...] | Old Rev... | Interchangeable          | New Number [...] | New Revision | In Build | In Service |
|--|-------|--------|------------------|------------|--------------------------|------------------|--------------|----------|------------|
|  |       | Change | Child A          |            | <input type="checkbox"/> |                  |              |          |            |

Figure 322.

5. Select the **Interchangeable** cell check box if the old and new Items are interchangeable, e.g., if there is no Fit-Form-Function change. The new Item version can be replaced or modified without causing any change in functionality to the surrounding Items. When selected, the same Item number will be retained.
6. If the **Interchangeable** cell is not selected, e.g., the old and new Items are not interchangeable due to a Fit-Form-Function change, select the replacing new Item:
  - a. Click the **ellipsis** button in the **New Number** cell. The **Search dialog – Change Controlled Item** appears.
  - b. Using the standard search procedure, search for and select the new affected Item. The new affected Item populates the **New Number** cell. This new item must be in the Preliminary state to pass readiness checks during the Simple ECO process.

|  | Ty... | Action | Old Number [...] | Old Rev... | Interchangeable          | New Number [...] | New Revision | In Build | In Service |
|--|-------|--------|------------------|------------|--------------------------|------------------|--------------|----------|------------|
|  |       | Change | Child A          |            | <input type="checkbox"/> | Child B          |              |          |            |

Figure 323.

7. Click **Save** two times on the **Simple ECO** toolbar.
8. Select the action to take on the Items that are being built in the **In Build** cell: **Use Existing**, **Rework**, or **Scrap**.
9. Select the action to take on the Items that are currently in service in the **In Service** cell: **Use Existing**, **Rework**, or **Scrap**.

|  | Ty... | Action | Old Number [...] | Old Rev... | Interchangeable          | New Number [...] | New Revision | In Build | In Service   |
|--|-------|--------|------------------|------------|--------------------------|------------------|--------------|----------|--------------|
|  |       | Change | Child A          | A          | <input type="checkbox"/> | Child B          | A            | Rework   | Use Existing |

Figure 324.

10. Click **Save** or **Done** on the Simple ECO toolbar. The Simple ECO has an affected Item assigned for changing.

| Affected Items Attachments Signoffs |       |        |                         |            |                          |                         |              |          |              |
|-------------------------------------|-------|--------|-------------------------|------------|--------------------------|-------------------------|--------------|----------|--------------|
| Affected Items                      |       |        |                         |            |                          |                         |              |          |              |
| Hidden                              |       |        |                         |            |                          |                         |              |          |              |
|                                     | Ty... | Action | Old Number [...]        | Old Rev... | Interchangeable          | New Number [...]        | New Revision | In Build | In Service   |
|                                     |       | Change | <a href="#">Child A</a> | A          | <input type="checkbox"/> | <a href="#">Child B</a> | A            | Rework   | Use Existing |

Figure 325.

### 13.2.1.5 Deleting an Item within a Simple ECO

Because a Simple ECO can be used to add, change, or delete affected Items, the Simple ECO creating process can be conventionally divided into general and specific procedures. This section defines the steps you should use for deleting affected Items when creating a Simple ECO. Section [13.2.1.2 Creating a Simple ECO](#) outlines the general procedure. The other specific procedures are outlined in sections [13.2.1.3 Adding an Item within a Simple ECO](#) and [13.2.1.4 Changing an Item within a Simple ECO](#).

An already Released item gets Superseded at the end of the Simple ECO process when it is released. Use the following procedure to assign an existing affected Item to be superseded with a Simple Engineering Change Order:

1. Follow steps 1-4 of the Adding an Item within a Simple ECO procedure.
2. Select **Delete** in the **Action** cell.

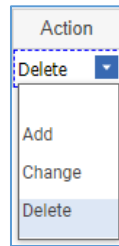


Figure 326.

3. Click the **ellipsis** button in the **Old Number** cell. The **Search dialog – Change Controlled Item** appears.
4. Using the standard search procedure, search for and select the affected Item to be deleted. The selected affected Item populates the **Old Number** cell.

|  | Ty... | Action | Old Number [...] | Old Rev... | Interchangeable          | New Number [...] | New Revision | In Build | In Service |
|--|-------|--------|------------------|------------|--------------------------|------------------|--------------|----------|------------|
|  |       | Delete | Child A          | A          | <input type="checkbox"/> |                  |              |          |            |

Figure 327.

5. Click **Save** two times on the **Simple ECO** toolbar.
6. Select the action to take on the Items that are being built in the **In Build** cell: **Use Existing**, **Rework**, or **Scrap**.
7. Select the action to take on the Items that are currently in service in the **In Service** cell: **Use Existing**, **Rework**, or **Scrap**.

|  | Ty... | Action | Old Number [...] | Old Rev... | Interchangeable          | New Number [...] | New Revision | In Build | In Service   |
|--|-------|--------|------------------|------------|--------------------------|------------------|--------------|----------|--------------|
|  |       | Delete | Child A          | A          | <input type="checkbox"/> |                  |              | Rework   | Use Existing |

Figure 328.

8. Click **Save** or **Done** on the Simple ECO toolbar. The Simple ECO has an affected Item assigned for deleting.

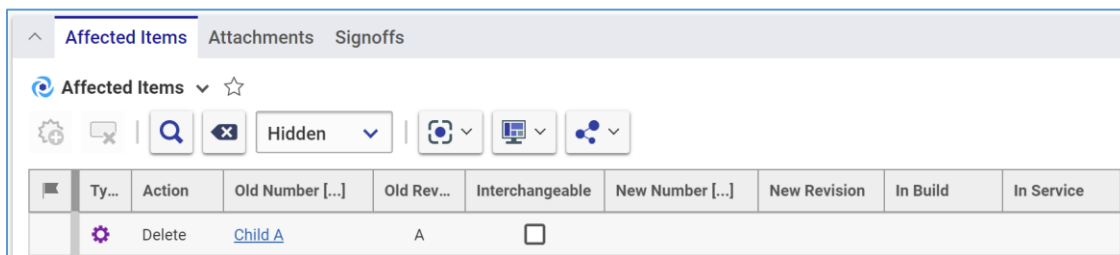


Figure 329.

### 13.2.1.6 Removing an Affected Item from a Simple ECO

Use the following procedure:

1. Follow steps 1-3 of Adding an Item within a Simple ECO procedure.
2. Click the **Affected Item** and then click **Delete Row** on the **Affected Items** accordion toolbar. The **Affected Item** becomes unavailable.

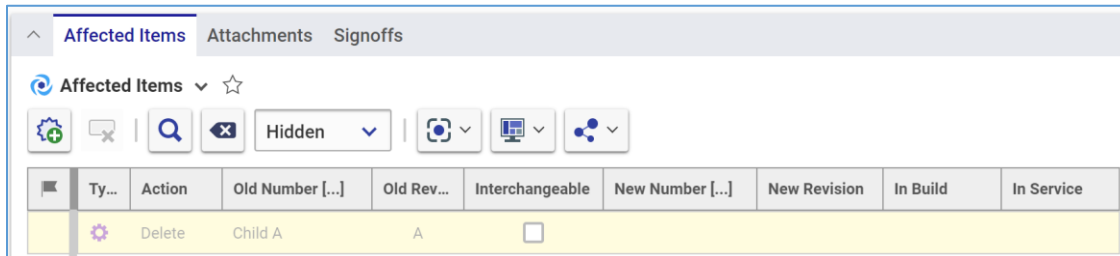


Figure 330.

3. Click **Save** or **Done** on the Simple ECO toolbar. The **Affected Item** disappears from the grid.

**Note:** The **Affected Item** disappears immediately if the Simple ECO was not saved after it was added.

### 13.2.1.7 Submitting a Simple ECO

Use the following procedure:

1. Go to **Contents** → **My Innovator** → **My InBasket** → **Search My InBasket**.
2. Double-click on the Workflow Task to open the **Workflow Activity Completion** dialog for the Simple ECO.

Figure 331.

3. Mark the tasks you completed in the **Tasks** field.
4. Select **Submit** in the **Vote** list.
5. Click **Complete**.

The Simple ECO enters the **Draft Changes** activity assigned to the Assigned Creator.



### 13.2.1.8 Processing a Simple ECO

Once a Simple ECO is submitted, each Affected Item goes to its Assigned Creator for the **Draft Changes** activity. The Assigned Creator checks and may provide the following information:

- There are new part numbers for the noninterchangeable parts changes.
- The ECN has all other affected items in the hierarchy.
- Affected Items up the hierarchy tree from the actual Item are changed.
- Affected Documents of the parent assemblies (such as BOMs) are changed.
- Etc.

Since each company has its own rules for new part numbers or new revision generation, the Assigned Creator is responsible for entering this information manually.

The only automatic update that Aras Innovator does for the Simplified CM process is a new revision for the interchangeable Items changes. This automatic update takes place once the Simple ECO is released.

## 13.2.2 Simple Manufacturing Change Order (MCO)

The Simple Manufacturing Change Order (MCO) is a simplified process used for changing an Approved Manufacturer List (AML) for a released Part. The AML modification does not require the full CMII process because this change does not change the Part itself. The MCO process is suitable for this type of modification.

The MCO process provides a specialized user interface to enable editing the AML without the ability to edit Part properties or relationships.

### 13.2.2.1 Understanding the MCO Process

The MCO workflow has the following roles:

- An MCO Creator (**Originator**), is the user who initially created the MCO in the system. This may or may not be the same person as other roles.
- An MCO Owner (**Assigned Creator**), is the person responsible for the AML change.
- A Change Specialist I (CSI) coordinates and prepares MCOs through the workflow.
- Change Analyst (CA).
- The Configuration Management (CM) group.
- The Change Control Board (CCB), is responsible for review and final approval of an MCO.

An MCO moves between various states during its life cycle as shown in the following figure.

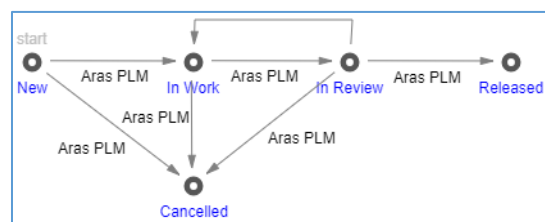


Figure 332.

The Simple Change Management workflow and MCO life cycle produce together the following scenario:

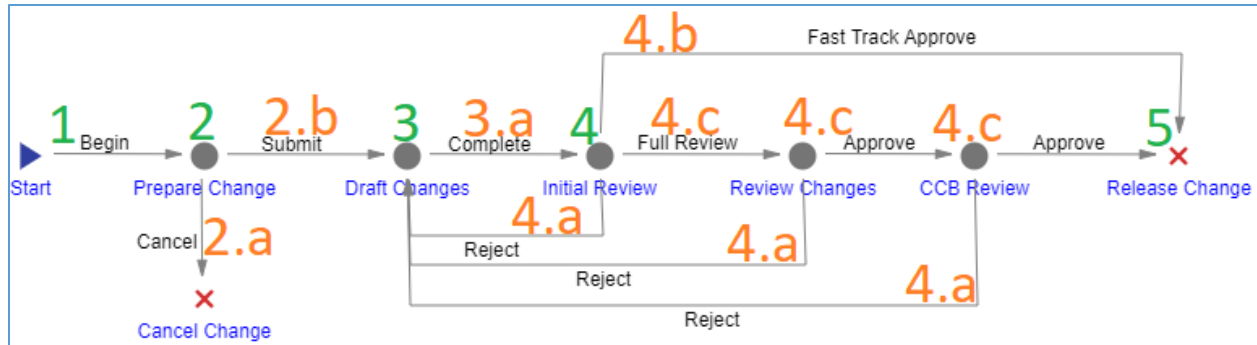


Figure 333.

1. An Aras Innovator user creates and saves the MCO. The creator should assign the Affected Parts to the MCO. When the Creator saves the MCO, it enters the **In Work** state, and a workflow is automatically started to assign tasks and track the changes to the MCO.
2. The MCO enters the **Prepare Change** activity **assigned to the CSI**. According to the company's rules and policies, the CSI reviews the MCO and decides whether to:
  - a. **Cancel** the MCO. It cannot be restored in the future.
  - b. **Submit** the MCO. It goes further.
3. The MCO enters the **Draft Changes** activity **assigned to an Owner**. The Owner is responsible for updating the AML using the AML editor.
  - a. Once the Owner votes **Complete**, the ECR goes further.
4. The MCO enters the **Initial Review** activity **assigned to the CA**. The CA reviews the AML changes and decides either to:
  - a. **Reject** — The MCO goes back to the **Draft Changes** activity. The Owner can modify and resubmit the MCO for the **Initial Review** activity.
  - b. **Fast Track Approve** — The MCO is **Approved** and **Released**.
  - c. **Full Review** — The MCO enters the **Review Changes** activity **assigned to other team members**. The MCO goes under in-depth formal/informal reviews. Once these reviews are completed, The MCO enters the **CCB Review** activity. Once the CCB audit is completed, the MCO is **Released**. The MCO may also be **Rejected** (4.a) at these activities.
5. Once the MCO is released, the affected Part is updated.

The following table shows the Simple ECO lifecycle states and workflow activities in terms of Roles and Permissions.

**Table 7:** The MCO lifecycle and workflow activities with corresponding Roles and Permissions

| Life Cycle State | Workflow Activity | Can Add | Get                                          | Update                                | Delete                                | Change access                  | Can be promoted | Update responsible |
|------------------|-------------------|---------|----------------------------------------------|---------------------------------------|---------------------------------------|--------------------------------|-----------------|--------------------|
| New              | Prepare Change    | World   | Aras PLM, CM, Creator, Manager, Owner, World | Aras PLM, CM, Creator, Manager, Owner | Aras PLM, CM, Creator, Manager, Owner | Aras PLM, CM, Creator, Manager | Yes             | CSI                |

| Life Cycle State | Workflow Activity                          | Can Add | Get                                 | Update                       | Delete       | Change access | Can be promoted | Update responsible |
|------------------|--------------------------------------------|---------|-------------------------------------|------------------------------|--------------|---------------|-----------------|--------------------|
| In Work          | Draft Changes                              |         | Aras PLM, CM, Manager, Owner, World | Aras PLM, CM, Manager, Owner | Aras PLM, CM | Aras PLM, CM  | Yes             | Owner              |
| In Review        | Initial Review, Review Changes, CCB Review |         | Aras PLM, World                     | Aras PLM                     | Aras PLM     | Aras PLM      | Yes             | CA                 |
| Released         | Released Change                            |         | Aras PLM, World                     | Aras PLM                     |              | Aras PLM      | No              |                    |
| Cancelled        | Cancelled Change                           |         | Aras PLM, World                     | Aras PLM                     |              | Aras PLM      | Yes             |                    |

### 13.2.2.2 Creating an MCO

To create a Manufacturing Change Order:

1. Go to **Contents → Change Management → MCOs**.
2. Click **Create New MCO**. The view of a new MCO appears.

Figure 334.

3. Keep the default **Phase In** value in the **Change Category** list.

4. Describe the change in the **Change Description** text box.
5. Describe the reason for the change in the **Reason For Change** text box.
6. Keep the **default** value in the **Originator** box. This is the MCO Creator.
7. Keep the default **Change Specialist I** value in the **Change Analyst** box. This is a person or group assigned to act as a CA in the MCO workflow process.
8. Keep the **Date Originated** box blank. This box is automatically filled with the date on which the MCO is saved.
9. Go to the **Attachments** accordion tab to attach all the files concerning this MCO.

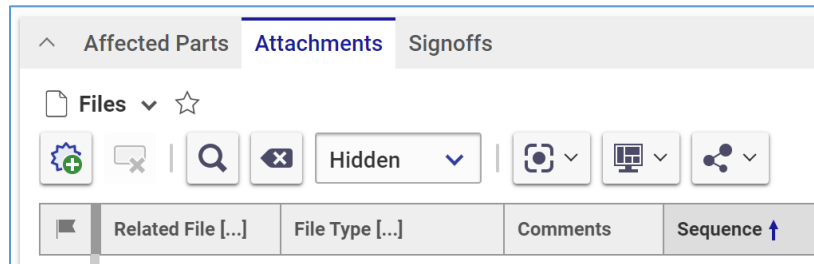


Figure 335.

10. Click **New File** on the Attachments accordion toolbar. The standard operating system dialog for file browsing and selection appears.
11. Search for and select a file. A new row appears in the **Files** grid with the file attached.

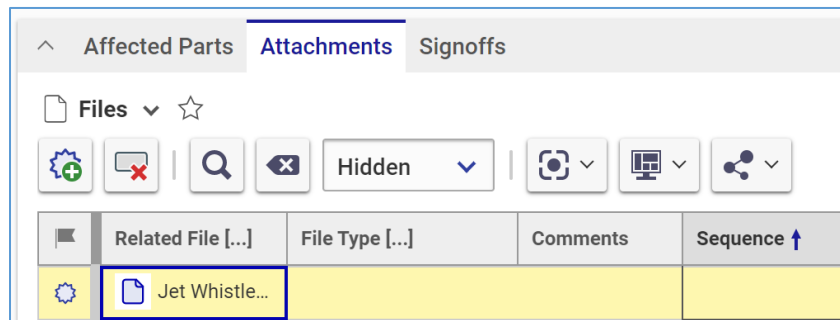


Figure 336.

**Note:** Aras Innovator will automatically assign a file type depending on File's extension when you save the ECN.

12. If it is necessary to provide additional information about the File, enter it in the **Comments** cell.
13. Repeat steps 10-12 for other files concerning this MCO.
14. Go to the **Affected Parts** accordion tab to assign Parts with AMLs this MCO should change.

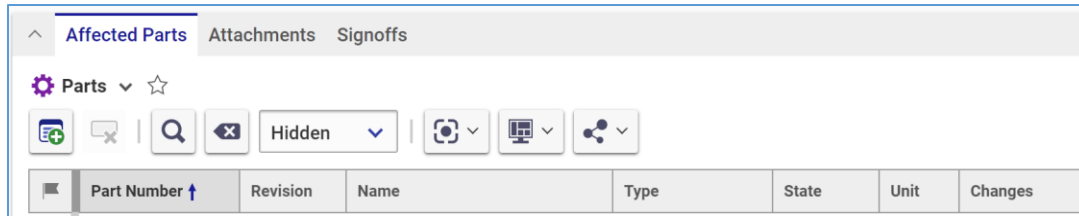


Figure 337.

15. Click **Add Parts** on the Affected Parts accordion toolbar. The **Search dialog – Part** appears.
16. Search and select Parts with AMLs this MCO should change. The selected Parts appear under **Affected Parts**.

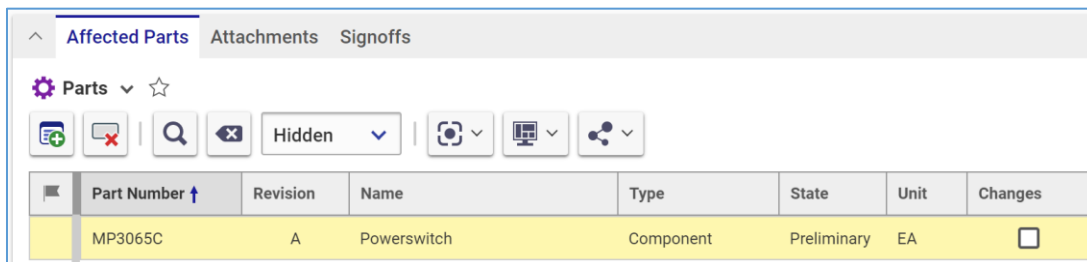


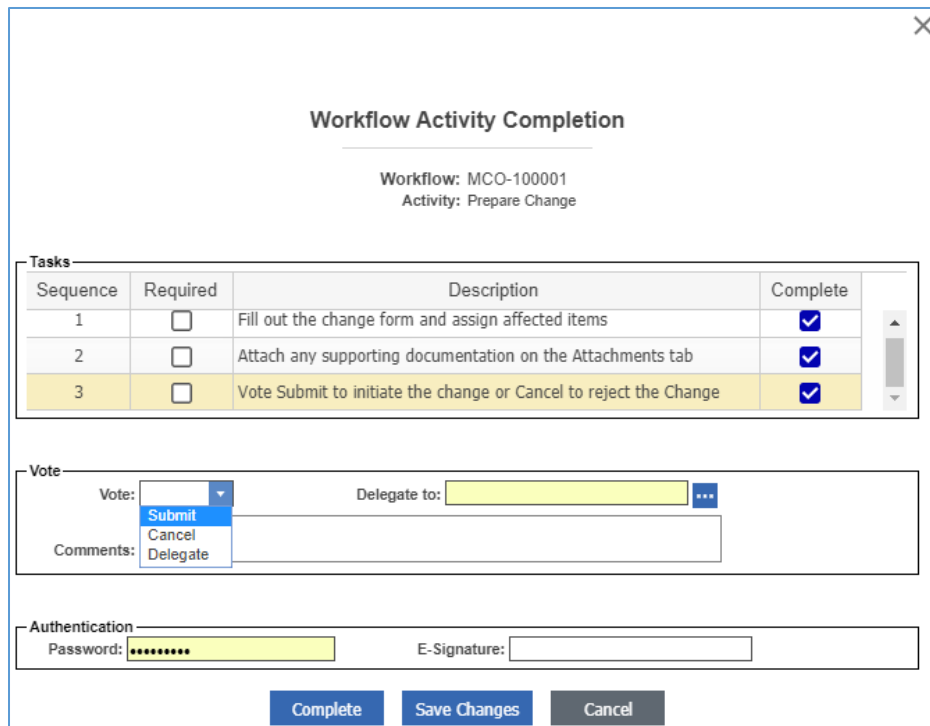
Figure 338.

17. Click **Done** on the MCO toolbar. The MCO is created and waits to be submitted as defined in the section [13.2.2.2 Submitting an MCO](#).

### 13.2.2.3 Submitting an MCO

To submit a Manufacturing Change Order:

1. Go to **Contents → My Innovator → My InBasket → Search My InBasket**.
2. Double-click on the Workflow Task to open the **Workflow Activity Completion** dialog for the MCO.



**Workflow Activity Completion**

Workflow: MCO-100001  
Activity: Prepare Change

| Sequence | Required                 | Description                                                       | Complete                            |
|----------|--------------------------|-------------------------------------------------------------------|-------------------------------------|
| 1        | <input type="checkbox"/> | Fill out the change form and assign affected items                | <input checked="" type="checkbox"/> |
| 2        | <input type="checkbox"/> | Attach any supporting documentation on the Attachments tab        | <input checked="" type="checkbox"/> |
| 3        | <input type="checkbox"/> | Vote Submit to initiate the change or Cancel to reject the Change | <input checked="" type="checkbox"/> |

**Vote**

Vote:

Delegate to:

Comments:

**Authentication**

Password:  E-Signature:

Figure 339.

3. Mark the tasks you completed in the **Tasks** field.
4. Select **Submit** in the **Vote** list.
5. Click **Complete**.

The MCO enters the **Draft Changes** activity assigned to the Owner.

#### 13.2.2.4 Drafting Changes within an MCO

Use the following procedure:

1. Open the **MCO** view.
2. Go to the **Affected Parts** accordion tab.
3. Right-click a **Part** in the **Affected Parts** grid and select **Actions** → **Launch AML Editor**.

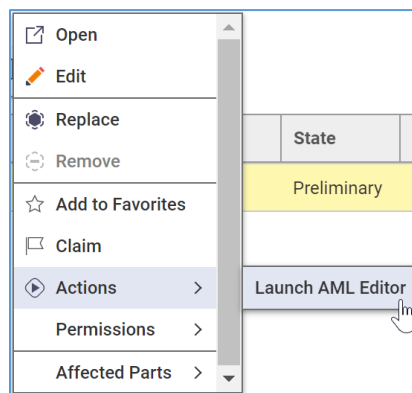


Figure 340.

The AML Editor appears with the AML of the given Part in a separate dialog box.

Figure 341.

4. Edit the AML as described in section [7.7 Approved Manufacturer List \(AML\)](#).
5. Click **Save Changes**. The AML Editor disappears. The given Affected Part gets the **Manual Change State**.

| Part Number | Revision | Name        | Type      | State         | Unit | Changes                  |
|-------------|----------|-------------|-----------|---------------|------|--------------------------|
| MP3065C     | A        | Powerswitch | Component | Manual Change | EA   | <input type="checkbox"/> |

Figure 342.

6. Go to **Contents** → **My Innovator** → **My InBasket** → **Search My InBasket**.
7. Double-click on the Workflow Task to open the **Workflow Activity Completion** dialog for the MCO.

| Sequence | Required                 | Description                                                  | Complete                            |
|----------|--------------------------|--------------------------------------------------------------|-------------------------------------|
| 1        | <input type="checkbox"/> | Review the change form for details of the changes to be made | <input checked="" type="checkbox"/> |
| 2        | <input type="checkbox"/> | Update your affected items to reflect the desired change     | <input checked="" type="checkbox"/> |
| 3        | <input type="checkbox"/> | Vote Complete to indicate that your work is done             | <input checked="" type="checkbox"/> |

Vote: Complete Delegate to:  ...

Comments:

Authentication: Password:  E-Signature:

Complete Save Changes Cancel

Figure 343.

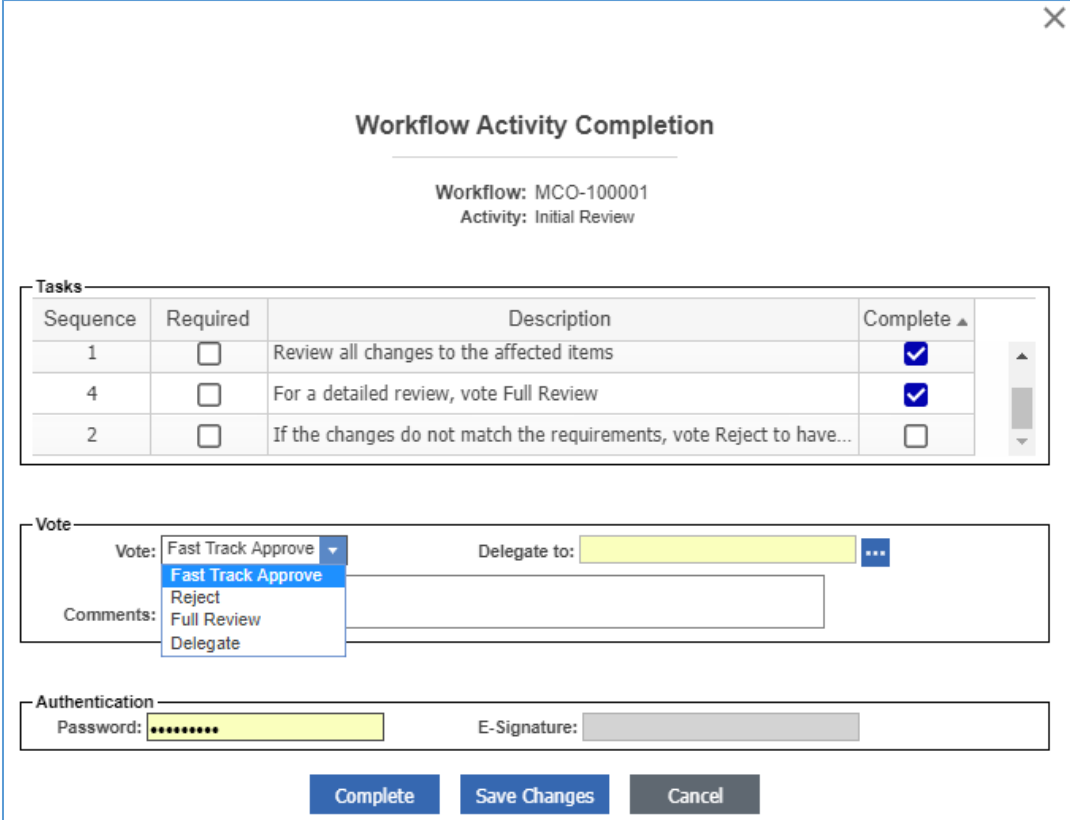
8. Mark the tasks you completed in the **Tasks** field.
9. Select **Complete** in the **Vote** list.
10. Click **Complete**.

The MCO enters the **Initial Review** activity assigned to the CA.

### 13.2.2.5 Initial Reviewing of an MCO

Use the following procedure:

1. Review all changes to the Affected Parts.
2. Ensure each Affected Part assigned to the MCO contains the same information as in the MCO.
3. Go to **Contents** → **My Innovator** → **My InBasket** → **Search My InBasket**.
4. Go to the **Workflow Activity Completion** dialog for the MCO.



**Workflow Activity Completion**

Workflow: MCO-100001  
Activity: Initial Review

| Sequence | Required                 | Description                                                          | Complete                            |
|----------|--------------------------|----------------------------------------------------------------------|-------------------------------------|
| 1        | <input type="checkbox"/> | Review all changes to the affected items                             | <input checked="" type="checkbox"/> |
| 4        | <input type="checkbox"/> | For a detailed review, vote Full Review                              | <input checked="" type="checkbox"/> |
| 2        | <input type="checkbox"/> | If the changes do not match the requirements, vote Reject to have... | <input type="checkbox"/>            |

**Vote**

Vote: Fast Track Approve Fast Track Approve Reject Full Review Delegate

Delegate to:

Comments:

**Authentication**

Password: \*\*\*\*\* E-Signature:

Complete
Save Changes
Cancel

Figure 344.

5. Mark the tasks you completed in the **Tasks** field.
6. Select **the option** appropriate for your decision in the **Vote** list: **Fast Track Approve**, **Reject**, **Full Review**, or **Delegate**.
7. Click **Complete**.

The MCO enters the activity according to your decision.



## 13.3 Express Change Management

Express Change Management (CM) complements the CMII and Simple CM as a streamlined approach for Change Management. It is ideal for organizations that may not require all the rigorous steps of the CMII. An Express CM process can work in parallel with CMII and Simple CM processes.

The Express CM provides two types of items to manage changes:

- The **Express Engineering Change Order (ECO)** is a streamlined authorization and execution of actual Part modification. You can use it for adding, changing, deleting, releasing, and revising Parts and BOMs. The Express ECO uses the Impact Matrix that enables you to visualize and review in a quick way how the modification of a Part will affect other Items in a hierarchy.
- The **Express Document Change Order (DCO)** is a process specifically defined for managing the modification of Documents and CAD Documents.

A Team is a defined logical group of users in Aras Innovator. For the Express ECO and DCO, the PE solution comes out-of-the-box with one default **Product** Team, which has the following roles:

- Team
- Team Manager
- Team Member
- Team Guest

Engagement of each role depends on the Express Change item.

### 13.3.1 Express Engineering Change Order (ECO)

An Express Engineering Change Order (ECO) is a quick process of implementing Part and Part BOM relationship changes using the Impact Matrix.

#### 13.3.1.1 Understanding the Express ECO Process

The **Product** Team roles are engaged in the Express ECO workflow as follows:

- **Team Manager** is assigned to the **Planning** activity and responsible for the Impact Analysis of Affected Parts and setting an effective date.
- **Team Member** is assigned to the **Plan Review** and **Draft Changes** activities for reviewing Impact Matrix and making changes to the Affected Parts and BOMs.

In addition to the **Product** Team, the Express ECO workflow has the following roles:

- An Express ECO Creator is a user who initially created and submitted the Express ECO in the system. This may or may not be the same person as other roles.
- An Express ECO **Change Coordinator** (Owner) is a person responsible for managing the ECO from start to finish.
- The Configuration Management (CM) group.
- The Change Control Board (CCB) is responsible for review and final approval of an Express ECO.

An Express ECO also traverses between various states during its life cycle depicted in the following figure.

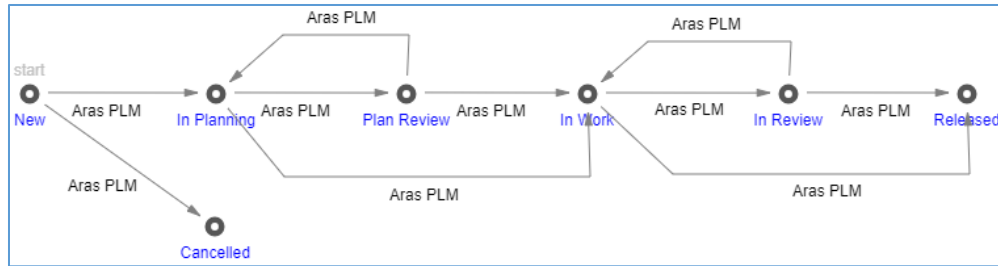


Figure 345.

The Express ECO workflow together with the Express ECO life cycle produces the following scenario:

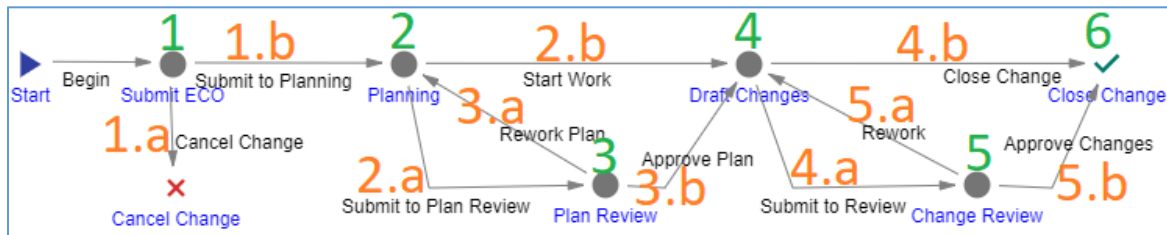


Figure 346.

1. An Aras Innovator user initiates an Express ECO process by creating a new Express ECO. The Creator must review the Express ECO to either:
  - a. **Submit** it. A workflow is automatically started to assign tasks and track the changes to the Express ECO.
  - b. **Cancel** it. The Express ECO cannot be restored in the future.
2. The Express ECO enters the **Planning** activity assigned to the **Team Manager**. The Team Manager performs an impact analysis and ensures that each Affected Part has an Action set in the Impact Matrix. The Express ECO cannot go to the next activity if its Impact Matrix has any Part with the **Item Action** cell blank or **Review**. The Team Manager votes either:
  - a. **Submit to Plan Review** if the Express ECO needs a further detailed review by a Team Member. The Express ECO goes to the **Plan Review** activity.
  - b. **Start Work** if the Express ECO is ready for work. The Express ECO goes to the **Draft Changes** activity.
3. The Express ECO enters the **Plan Review** activity assigned to the **Team Member**. This Team Member must review the Express ECO to vote either:
  - a. **Rework Plan** if the Express ECO needs its plan reworked by the Team Manager. The Express ECO goes back to the **Planning** activity.
  - b. **Approve Plan** if the Express ECO is ready for work. The Express ECO goes to the **Draft Changes** activity.
4. The Express ECO enters the **Draft Changes** activity assigned to the **Team Member**. This Team Member makes changes to the new revisions of Parts and BOMs according to the Express ECO, and, when completed, votes either:
  - a. **Submit to Review** if the Express ECO requires the final review by the CCB. The Express ECO goes to the **Change Review** activity.
  - b. **Close Change** if the Express ECO process is completed and does not require the final review by the CCB. The Express ECO is **Released**.

5. The Express ECO enters the **Change Review** activity **assigned to the CCB**. All the CCB members must conduct the final review of the Express ECO and vote the same decision of either:
  - a. **Rework** if the Express ECO needs any further reworking. The Express ECO goes back to the **Draft Changes** activity.
  - b. **Approve Changes** if the Express ECO process is completed and does not require any further reworking. The Express is ECO **Released**.
6. Once the Express ECO enters the **Close Change** activity, it becomes **Released**. Each Part is automatically updated according to the specified Action.

The following table shows the Express ECO lifecycle states and workflow activities in terms of Roles and Permissions.

**Table 8:** The Express ECO lifecycle and workflow activities with corresponding Roles and Permissions

| Life Cycle State | Workflow Activity | Can Add | Get                                                                      | Update                                                          | Delete                         | Change access  | Can be promoted   | Update responsible |
|------------------|-------------------|---------|--------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------|----------------|-------------------|--------------------|
| New              | Start, Submit ECO | World   | Team Manager, Team Member, Administrators, Owner, All Employees, Creator | Team Manager, Team Member, Administrators, Owner, Creator       | Administrators, Owner, Creator | Administrators | Yes               | Creator            |
| Cancelled        | Cancel Change     |         | Team Manager, Team Member, Administrators, Owner, All Employees, Creator | Administrators                                                  |                                | Administrators | No, a final state |                    |
| In Planning      | Planning          |         | Team Manager, Team Member, Administrators, Owner, All Employees, Creator | Team Manager, Team Member, Administrators, Owner, All Employees | Administrators                 | Administrators | Yes               | Team Manager       |
| Plan Review      | Plan Review       |         | Team Manager, Team Member, Administrators, Owner, All Employees, Creator | Team Manager, Team Member, Administrators, Owner, All Employees | Administrators                 | Administrators | Yes               | Team Member        |
| In Work          | Draft Changes     |         | Team Manager, Team Member, Administrators, Owner, All                    | Team Manager, Team Member, Administrators, Owner,               | Administrators                 | Administrators | Yes               | Team Member        |

| Life Cycle State | Workflow Activity | Can Add | Get                                                                      | Update                                                          | Delete         | Change access  | Can be promoted   | Update responsible |
|------------------|-------------------|---------|--------------------------------------------------------------------------|-----------------------------------------------------------------|----------------|----------------|-------------------|--------------------|
|                  |                   |         | Employees, Creator                                                       | All Employees                                                   |                |                |                   |                    |
| In Review        | Change Review     |         | Team Manager, Team Member, Administrators, Owner, All Employees, Creator | Team Manager, Team Member, Administrators, Owner, All Employees | Administrators | Administrators | Yes               | CCB                |
| Released         | Closed Change     |         | Team Manager, Team Member, Administrators, Owner, All Employees, Creator | Administrators                                                  |                | Administrators | No, a final state |                    |

### 13.3.1.2 Creating an Express ECO

Use the following procedure:

1. Go to **Contents** → **Change Management** → **ECOs** (one with the orange icon).

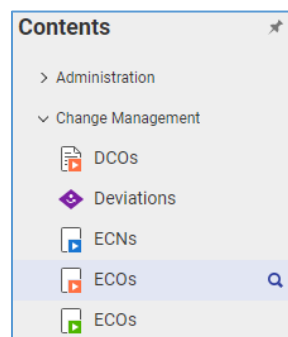


Figure 347.

2. Click **Create New ECO**. The view of a new Express ECO appears.

The screenshot shows the 'ECO 1' form in the Aras Product Engineering 14 User Guide. The form has a header with 'ECO 1' and buttons for 'Save', 'Done', and 'Delete'. Below the header, there are several sections: 'Number' and 'Title' (with a yellow highlight), 'Server Assigned', 'Change Reason', 'Change Description', 'Change Coordinator' (with a dropdown menu), 'Team' (with a dropdown menu), 'Priority' (with radio buttons for '1 - High', '2 - Normal', and '3 - Low'), 'Release Date', and 'Effective Date'. At the bottom, there is a table with columns 'Item Number', 'Name', 'Revision', 'State', and 'Sequence'.

Figure 348.

3. Enter the Express ECO title in the **Title** text box.
4. Describe the reason for the change in the **Change Reason** text box.
5. Describe the change in the **Change Description** text box.
6. Specify change priority using the **Priority** radio button: **1 – High**, **2 – Normal**, or **3 – Low**.
7. If it is necessary to set an Express ECO Owner other than you:
  - a. Click the **ellipses** button in the **Change Coordinator** box. The **Search dialog – Identity** appears.
  - b. Using the standard search procedure, search for and select a Team. The selected Team replaces the default identity.
8. If it is necessary to set a Team other than the default:
  - a. Click the **ellipses** button in the **Team** box. The **Search dialog – Team** appears.
  - b. Using the standard search procedure, search for and select a Team. The selected Team replaces the default identity.
9. Keep the **Release Date** box blank. This box is automatically filled with the date on which the Express ECO is approved and closed.
10. Specify the date change becomes effective in the **Effective Date** box. It is automatically set to the released date if left blank.

**Note:** The effective date is often when items are put into use based on decisions in the organization.

11. Go to the **Attachments** accordion tab to attach all the files concerning this Express ECO.



Figure 349.

12. Click **Create Item** on the **Attachments** accordion toolbar. The standard operating system dialog for file browsing and selection appears.
13. Search for and select a file. A new row appears in the **Files** grid with the file attached.



Figure 350.

**Note:** Aras Innovator will automatically provide a file type to the **File Type** cell depending on the File's extension when you save the Express ECO.

14. If it is necessary to provide additional information about the File, enter it in the **Comments** cell.
15. Attach to the Express ECO Affected Parts, using the procedure described in the section [13.3.1.4 Managing the Impact Matrix](#).
16. Click **Done** on the Express ECO toolbar. The Express ECO is created and waits to be either submitted or cancelled as defined in the section [13.3.1.3 Voting on an Express ECO](#).

**ECO-00001001** ☆

**Edit** [Refresh] [Undo] [Share] [Print] [More]

**ECO**

|                                                                                                                        |                                                                |
|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Number</b><br>ECO-00001001                                                                                          | <b>Title</b><br>Jet Whistle Turbo Thrust Enhancement           |
| <b>Change Reason</b><br>The Jet Whistle should be heard on Mars.                                                       | <b>Change Description</b><br>Adding more power to Turbo Thrust |
| <b>Change Coordinator</b><br><a href="#">Innovator Admin</a>                                                           | <b>Team</b><br><a href="#">Product Team</a>                    |
| <b>Release Date</b><br>                                                                                                | <b>Effective Date</b><br>                                      |
| <b>Priority</b><br><input type="radio"/> 1 - High<br><input type="radio"/> 2 - Normal<br><input type="radio"/> 3 - Low |                                                                |

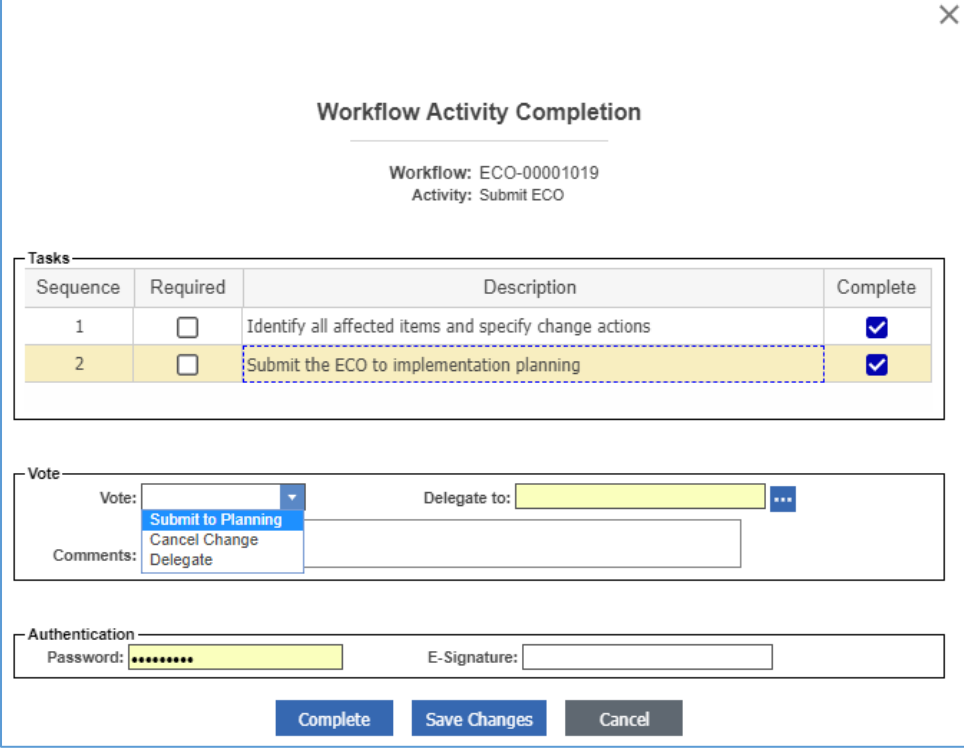
**New**

Figure 351.

### 13.3.1.3 Voting on an Express ECO

To vote on an Express ECO:

1. Go to **Contents** → **My Innovator** → **My InBasket** → **Search My InBasket**.
2. Double-click on the Workflow Task to open the **Workflow Activity Completion** dialog for the Express ECO.



The dialog box is titled "Workflow Activity Completion" and includes a close button (X) in the top right corner. Below the title, it displays "Workflow: ECO-00001019" and "Activity: Submit ECO".

The **Tasks** section contains a table with the following data:

| Sequence | Required                 | Description                                            | Complete                            |
|----------|--------------------------|--------------------------------------------------------|-------------------------------------|
| 1        | <input type="checkbox"/> | Identify all affected items and specify change actions | <input checked="" type="checkbox"/> |
| 2        | <input type="checkbox"/> | Submit the ECO to implementation planning              | <input checked="" type="checkbox"/> |

The **Vote** section includes a "Vote:" dropdown menu with options: "Submit to Planning", "Cancel Change", and "Delegate". A "Delegate to:" field with a search icon is also present. A "Comments:" text area is located below the dropdown.

The **Authentication** section contains a "Password:" field with masked characters and an "E-Signature:" field.

At the bottom, there are three buttons: "Complete", "Save Changes", and "Cancel".

Figure 352.

3. Mark the completed tasks in the **Tasks** field.
4. Select the option per your decision in the **Vote** list.
5. Click **Complete**.

The Express ECO enters the next activity as defined in the Express ECO workflow.

### 13.3.1.4 Managing the Impact Matrix

The Impact Matrix is a tool for Express Change Management of Parts in Express ECOs. For your review, it can also visualize how the modification of an Affected Part will Impact other Items in a hierarchy.

You can manage the change in a quick way using the **Impact Matrix** accordion toolbar and context menu.

The **Impact Matrix** accordion toolbar displays the following buttons:

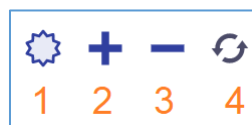


Figure 353.

1. **Add Item** launches the Search dialog – Change Controlled Item to add Affected Parts to the Impact Matrix.
2. **Expand All** unfolds all the assembly trees.
3. **Collapse All** folds back all the assembly trees.
4. **Refresh** shows the latest updates.



The Impact Matrix context menu features:

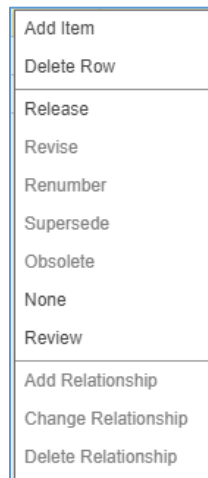


Figure 354.

- Actions for a Part:
  - **Add Item** launches the **Search dialog – Change Controlled Item** to add Affected Parts to the Impact Matrix.
  - **Delete Row** removes the Part from the Impact Matrix.
  - **Release** performs the initial release of a Part, promoting the Part from the **Preliminary** state to the **Released** state.
  - **Revise** creates a new revision of an existing released Part. The Part will be superseded by a new revision of the Part.
  - **Renumber** replaces an existing Part with its copy. The existing Part is copied as a new Part, which is assigned a new Part number, and released.
  - **Supersede** provides a new part number for the original part. The original Part is superseded.
  - **Obsolete** promotes an existing released or superseded Part to the **Obsolete** state.
  - **Review** puts the Part under **Review**. Action needs to be selected on the Part before the ECO can be processed. The ECO cannot be completed if any Part is under **Review**.
  - **None** states that the Part is explicitly not affected or impacted by the change. No action will be taken on the Part. Under certain circumstances, this action may not be allowed if an Affected Item requires this Part to be modified.
- Actions for a BOM relationship of a previously released Part (each selected action will appear in the **Relationship Action** column of the Impact matrix once the ECO is processed):
  - **Add Relationship** adds another released Part to the BOM of the given released Part.
  - **Modify Relationship** modifies the existing BOM of the given released Part. You can specify a new quantity and adjust the sequence number of the BOM.
  - **Delete Relationship** removes the existing BOM from the given affected released Part.

The following procedure explains the main Impact Matrix management features:

1. Open the **Express ECO**.
2. Click **Edit** on the **Express ECO** toolbar.

- Go to the **Impact Matrix** accordion tab.
- Click **Add Item** on the Impact Matrix accordion toolbar. The **Search dialog – Change Controlled Item** appears.
- Search for and select an Affected Part. The selected Affected Part and Parts Impacted by this Part appear in the Impact Matrix tree.

| Impact Matrix |             |          |             |          |          |      |  |
|---------------|-------------|----------|-------------|----------|----------|------|--|
| Item Number   | Name        | Revision | State       | Sequence | Quantity | Unit |  |
| Parent Part   | Parent Part | A        | Preliminary |          |          | EA   |  |

Figure 355.

- Click **Expand All** on the **Impact Matrix** toolbar. You can see what Parts in assemblies are Impacted when the given Part is changed. The given Part is always violet-highlighted. The given Part's Parents are above the Part. The given Part's Children are below the Part.

| Impact Matrix  |                |          |             |          |          |      |             |
|----------------|----------------|----------|-------------|----------|----------|------|-------------|
| Item Number    | Name           | Revision | State       | Sequence | Quantity | Unit | Item Action |
| Parent Part    | Parent Part    | A        | Preliminary |          |          | EA   |             |
| Child Part B   | Child Part B   | D        | Preliminary | 2        | 1        | EA   | Review      |
| Child Part B.1 | Child Part B.1 | A        | Preliminary | 1        | 1        | EA   |             |
| Child Part B.2 | Child Part B.2 | A        | Preliminary | 2        | 1        | EA   |             |

Figure 356.

- Right-click the **given Part** and then click **Delete Row**. The given Part and its children disappear from the Impact Matrix.

| Child Part B |                |
|--------------|----------------|
| Child        | Add Item       |
| Child        | Delete Row     |
| Child        | Child Part B.1 |
| Child        | Child Part B.2 |

Figure 357.

- Click **Save** or **Done** on the **Express ECO** toolbar to save the changes in the Express ECO.

### 13.3.1.5 Reviewing an Express ECO

When reviewing an Express ECO, you must analyze each Affected and Impacted Part to set Action to it. An Express ECO cannot go any further if any Affected or Impacted Part has no Action set. Some Actions may invalidate the Express ECO.

The following procedure explains the process of reviewing and setting Actions:

- Open the **Express ECO**.
- Click **Edit** on the Express ECO toolbar.
- Go to the **Impact Matrix** accordion tab.
- Check whether each Part has an Action.
- Check set Actions for validity.
- Right-click a **Part** and then click the proper **Action**. For Actions details, refer to [13.3.1.4 Managing the Impact Matrix](#).

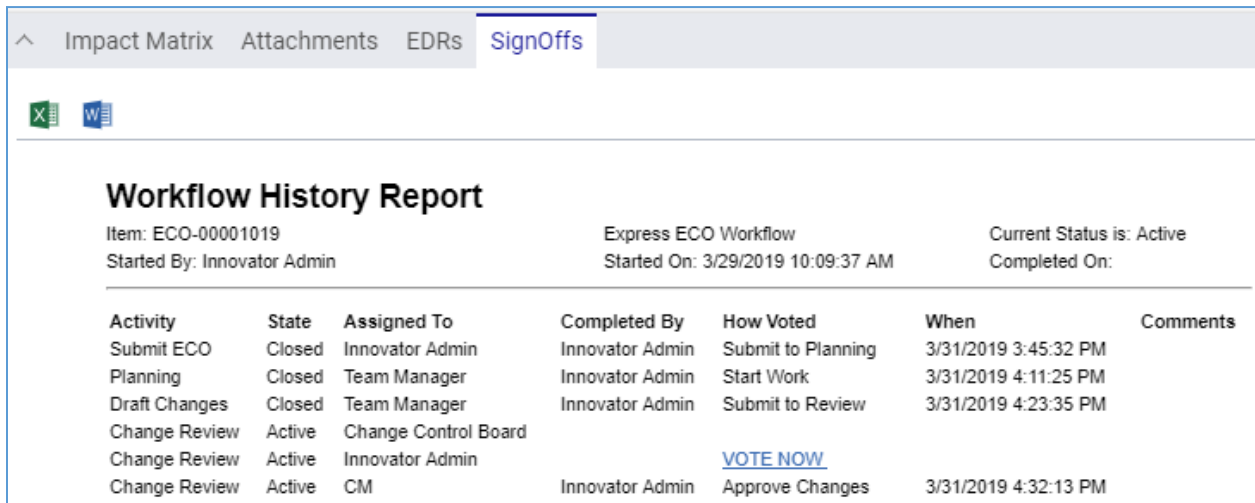
- Click **Save** or **Done** on the **Express ECO** toolbar to save the changes in the Express ECO.

### 13.3.1.6 Viewing and Processing an Express ECO Signoff Activities

From the Express ECO view, you can review the current and previous Activities assigned for an Express ECO. You can also vote within an Activity if you have rights.

Use the following procedure:

- Open the **Express ECO** view.
- Go to the **SignOffs** accordion tab.



The screenshot shows the 'SignOffs' tab selected in the top navigation bar. Below the navigation bar, there are icons for Excel and Word. The main content area is titled 'Workflow History Report'. It displays metadata for an Express ECO: Item: ECO-00001019, Started By: Innovator Admin, Express ECO Workflow Started On: 3/29/2019 10:09:37 AM, and Current Status is: Active. Below this, a table lists the workflow activities.

| Activity      | State  | Assigned To          | Completed By    | How Voted                | When                 | Comments |
|---------------|--------|----------------------|-----------------|--------------------------|----------------------|----------|
| Submit ECO    | Closed | Innovator Admin      | Innovator Admin | Submit to Planning       | 3/31/2019 3:45:32 PM |          |
| Planning      | Closed | Team Manager         | Innovator Admin | Start Work               | 3/31/2019 4:11:25 PM |          |
| Draft Changes | Closed | Team Manager         | Innovator Admin | Submit to Review         | 3/31/2019 4:23:35 PM |          |
| Change Review | Active | Change Control Board |                 |                          |                      |          |
| Change Review | Active | Innovator Admin      |                 | <a href="#">VOTE NOW</a> |                      |          |
| Change Review | Active | CM                   | Innovator Admin | Approve Changes          | 3/31/2019 4:32:13 PM |          |

Figure 358.

- Click the **VOTE NOW** link. The **Workflow Activity Completion** dialog for the Express ECO Activity appears.
- Use steps 3-5 of the [Voting on an Express ECO](#) procedure.

### 13.3.1.7 Checking an Express ECO for Completeness

At any point in an Express ECO process, you can use the **Completeness Check Results** dialog to ensure that the Express ECO has all the required information provided and that each Part has an Action assigned. Using this dialog early avoids resolving errors later during the workflow process when the cause may be more difficult to determine.

There are several ways to access the **Completeness Check Results** dialog for an Express ECO:

- Go to **Contents** → **Change Management** → **ECOs**, right-click the **Express ECO** to be checked, and then click **Check for Completeness**.

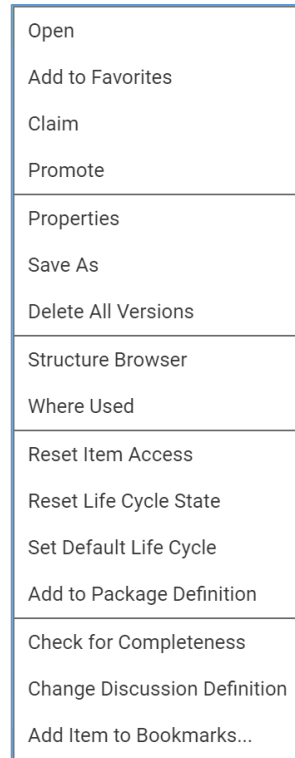


Figure 359.

- Open the view of an Express ECO to be checked, on the **Express ECO** toolbar, click **More**, and then click **Check for Completeness**.

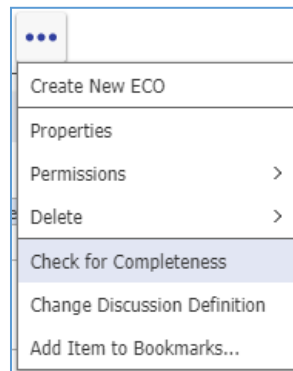


Figure 360.

When accessed, the **Completeness Check Results** dialog shows you checking results:

- If an Express ECO is ready to process, there is the message: "Completeness Check succeeded. Express ECO is ready to move forward."

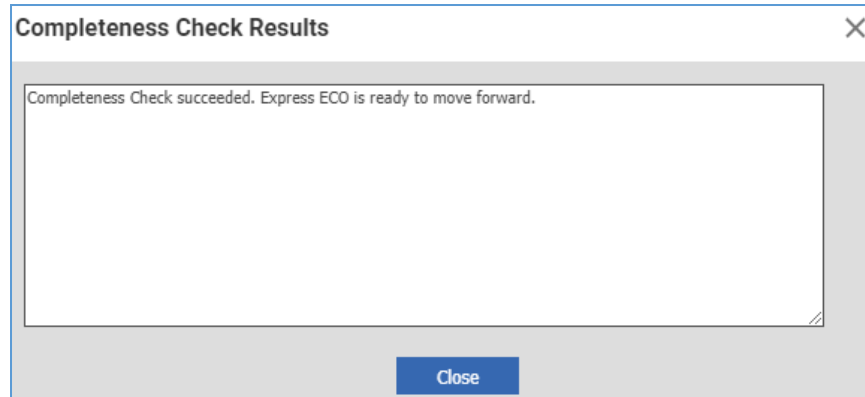


Figure 361.

- If an Express ECO is not ready to process, there are issues preventing the Express ECO from completeness and their sources. Make sure to correct each issue.

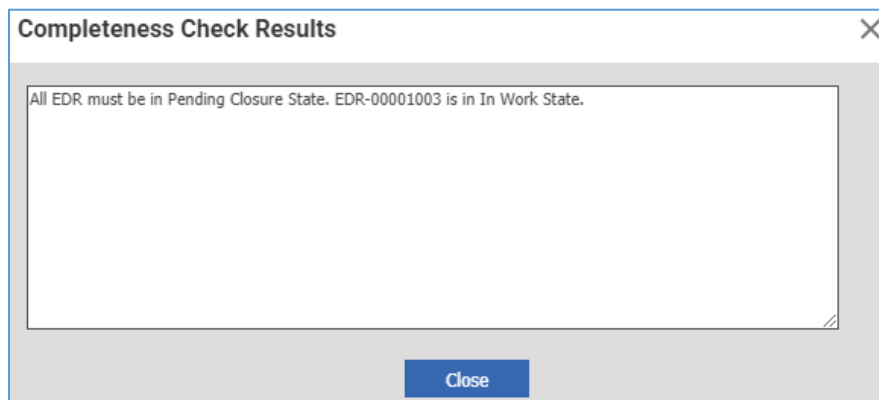


Figure 362.

### 13.3.1.8 Viewing and Revising Parts Released within an Express ECO

Once the Express ECO workflow is completed, you can search for the released Parts and check that they all are **Released**.

When a released Part must be changed to resolve a problem or provide improvement, an Express ECO is created to apply the change, revise the Part, and review the impact of other Affected Parts in an assembly.

The following procedure explains the Part revising process:

1. Open the **Express ECO**.
2. Click **Edit** on the Express ECO toolbar.
3. Go to the **Impact Matrix** accordion tab.
4. Click **Add Item** on the **Impact Matrix** accordion toolbar. The **Search dialog – Change Controlled Item** appears.
5. Search for and select an Affected Part(s). The selected Affected Part appears in the Impact Matrix tree.

| Impact Matrix Attachments EDRs SignOffs |               |          |          |          |          |      |             |              |   |
|-----------------------------------------|---------------|----------|----------|----------|----------|------|-------------|--------------|---|
|                                         |               |          |          |          |          |      |             |              |   |
| Item Number                             | Name          | Revision | State    | Sequence | Quantity | Unit | Item Action | EDR Grouping | N |
| MP2936                                  | Acrylic Cover | A        | Released |          |          | EA   |             |              |   |
| MP2951                                  | Cover         | A        | Released | 15       | 1        | EA   | Review      |              |   |

Figure 363.

- Right-click each **Part** and click **Review** to see what Parts in assemblies are Impacted when the Part clicked is changed.

| Item Number | Name                      | Revision | State       | Sequence | Quantity | Unit | Item Action |  |
|-------------|---------------------------|----------|-------------|----------|----------|------|-------------|--|
| MP0103      | MakerBot Replicator wi... | A        | Preliminary |          |          | EA   |             |  |
| MP2936      | Acrylic Cover             | A        | Released    | 50       | 1        | EA   | Review      |  |
| MP2955      | Front Window Panel        | B        | Released    | 10       | 1        | EA   | Review      |  |
| MP2951      | Cover                     | A        | Released    | 15       | 1        | EA   | Review      |  |
| MP2948      | Cover Hardware            | A        | Released    | 5        | 1        | EA   | Review      |  |
| MP2948      | Cover Hardware            | A        | Preliminary | 10       |          |      |             |  |
| MP2295      | M3 Square Nut             | A        | Released    | 10       | 15       | EA   |             |  |
| MP0370      | M3x16 Socket Cap Bolt     | A        | Released    | 5        | 15       | EA   |             |  |
| MP2936      | Acrylic Cover             | A        | Preliminary | 10       |          |      |             |  |
| MP0102      | Cover Option for Maker... | A        | Preliminary |          |          | EA   |             |  |
| MP2936      | Acrylic Cover             | A        | Released    | 5        | 1        | EA   | Review      |  |
| MP2955      | Front Window Panel        | B        | Released    | 10       | 1        | EA   | Review      |  |
| MP2951      | Cover                     | A        | Released    | 15       | 1        | EA   | Review      |  |
| MP2948      | Cover Hardware            | A        | Released    | 5        | 1        | EA   | Review      |  |
| MP2936      | Acrylic Cover             | A        | Preliminary | 10       |          |      |             |  |

Figure 364.

- Right-click each Affected and Impacted **Part** and then click **Revise**.

| Item Number | Name                      | Revision | State       | Sequence | Quantity | Unit | Item Action |  |
|-------------|---------------------------|----------|-------------|----------|----------|------|-------------|--|
| MP0103      | MakerBot Replicator wi... | A        | Preliminary |          |          | EA   |             |  |
| MP2936      | Acrylic Cover             | A        | Released    | 50       | 1        | EA   | Revise      |  |
| MP2955      | Front Window Panel        | B        | Released    | 10       | 1        | EA   | Revise      |  |
| MP2951      | Cover                     | A        | Released    | 15       | 1        | EA   | Revise      |  |
| MP2948      | Cover Hardware            | A        | Released    | 5        | 1        | EA   | Revise      |  |
| MP2948      | Cover Hardware            | A        | Preliminary | 10       |          |      |             |  |
| MP2295      | M3 Square Nut             | A        | Released    | 10       | 15       | EA   |             |  |
| MP0370      | M3x16 Socket Cap Bolt     | A        | Released    | 5        | 15       | EA   |             |  |
| MP2936      | Acrylic Cover             | A        | Preliminary | 10       |          |      |             |  |
| MP0102      | Cover Option for Maker... | A        | Preliminary |          |          | EA   |             |  |
| MP2936      | Acrylic Cover             | A        | Released    | 5        | 1        | EA   | Revise      |  |
| MP2955      | Front Window Panel        | B        | Released    | 10       | 1        | EA   | Revise      |  |
| MP2951      | Cover                     | A        | Released    | 15       | 1        | EA   | Revise      |  |
| MP2948      | Cover Hardware            | A        | Released    | 5        | 1        | EA   | Revise      |  |
| MP2936      | Acrylic Cover             | A        | Preliminary | 10       |          |      |             |  |

Figure 365.

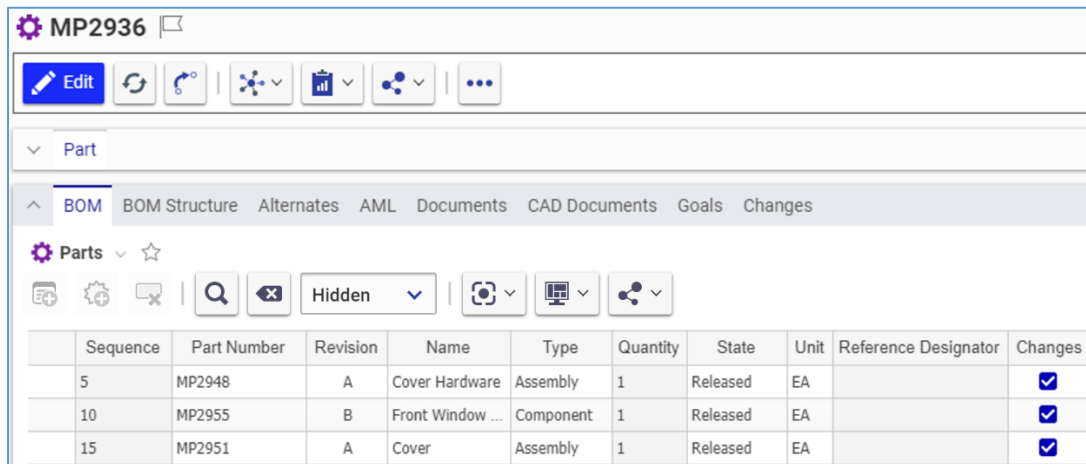
- Click **Done** on the Express ECO toolbar. The Express ECO is ready for revising Parts.

### 13.3.1.9 Processing Parts within an Express ECO

This section describes the revision of released Parts as an example of processing Parts within an Express ECO.

To process Parts, follow the Express ECO workflow as explained in section [13.3.1.1 Understanding the Express ECO Process](#) and notice how the state of Parts is changing:

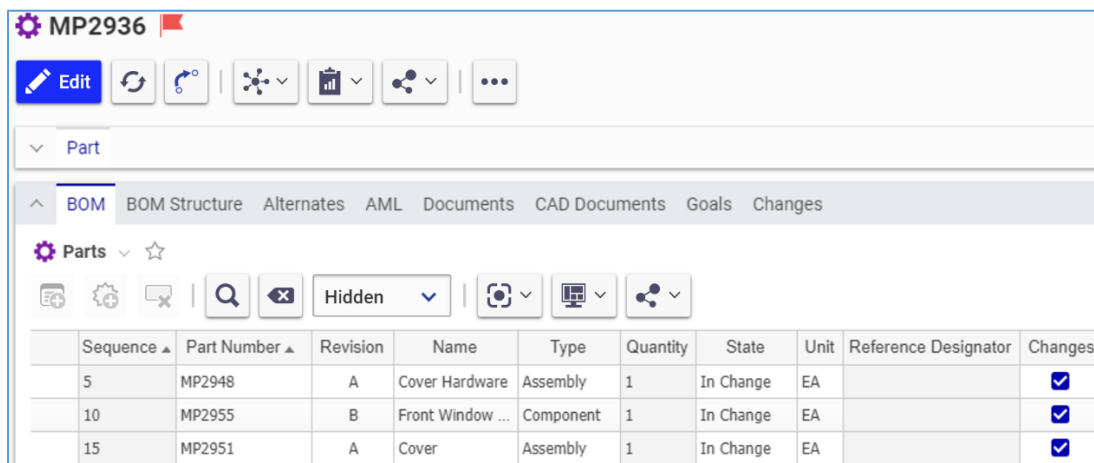
- Submit the Express ECO.** The **Changes** check mark is automatically set on each Affected Part to indicate that it is under a change process.



| Sequence | Part Number | Revision | Name             | Type      | Quantity | State    | Unit | Reference Designator | Changes |
|----------|-------------|----------|------------------|-----------|----------|----------|------|----------------------|---------|
| 5        | MP2948      | A        | Cover Hardware   | Assembly  | 1        | Released | EA   |                      | ✓       |
| 10       | MP2955      | B        | Front Window ... | Component | 1        | Released | EA   |                      | ✓       |
| 15       | MP2951      | A        | Cover            | Assembly  | 1        | Released | EA   |                      | ✓       |

Figure 366.

- Planning.** After submission, the Affected Parts are set to the **In Change** state and are not allowed to be edited, note the **red claiming flag**.



| Sequence | Part Number | Revision | Name             | Type      | Quantity | State     | Unit | Reference Designator | Changes |
|----------|-------------|----------|------------------|-----------|----------|-----------|------|----------------------|---------|
| 5        | MP2948      | A        | Cover Hardware   | Assembly  | 1        | In Change | EA   |                      | ✓       |
| 10       | MP2955      | B        | Front Window ... | Component | 1        | In Change | EA   |                      | ✓       |
| 15       | MP2951      | A        | Cover            | Assembly  | 1        | In Change | EA   |                      | ✓       |

Figure 367.

- Change Review.** During the review process, the Affected Parts are promoted to the **In Review** state and a new major revision. No edits are allowed on the original revision.

| Sequence | Part Number | Revision | Name             | Type      | Quantity | State     | Unit | Reference Designator | Changes                             |
|----------|-------------|----------|------------------|-----------|----------|-----------|------|----------------------|-------------------------------------|
| 5        | MP2948      | B        | Cover Hardware   | Assembly  | 1        | In Review | EA   |                      | <input checked="" type="checkbox"/> |
| 10       | MP2955      | C        | Front Window ... | Component | 1        | In Review | EA   |                      | <input checked="" type="checkbox"/> |
| 15       | MP2951      | B        | Cover            | Assembly  | 1        | In Review | EA   |                      | <input checked="" type="checkbox"/> |

Figure 368.

4. **Close Change.** The Express ECO is **Released**.

| Item Number | Name          | Revision | State       | Sequence | Quantity | Unit | Item Action | EDR Grouping | New Num... | State    | Revision |
|-------------|---------------|----------|-------------|----------|----------|------|-------------|--------------|------------|----------|----------|
| MP2936      | Acrylic Cover | A        | Superseded  |          |          | EA   | Revise      |              | MP2936     | Released | B        |
| MP2936      | Acrylic Cover | A        | Preliminary | 10       |          |      |             |              |            |          |          |
| MP2948      | Cover Hard... | A        | Superseded  | 5        | 1        | EA   | Revise      |              | MP2948     | Released | B        |
| MP0370-01   | M3x16 Sock... | A        | Released    | 5        | 15       | EA   |             |              |            |          |          |

Figure 369.

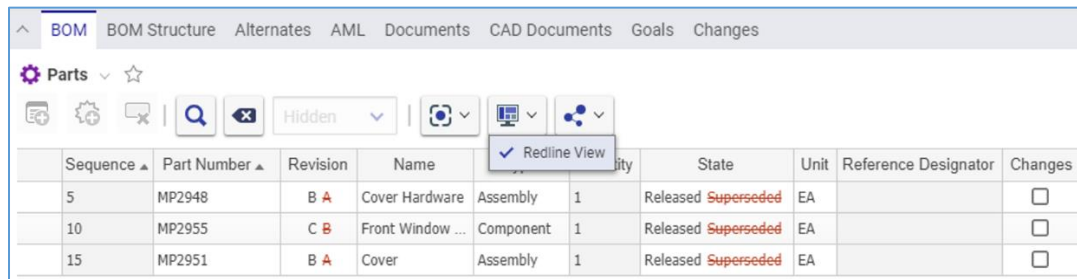
5. The newly revised Parts are **Released** at the new major Revision.

| Sequence | Part Number | Revision | Name             | Type      | Quantity | State    | Unit | Reference Designator | Changes                  |
|----------|-------------|----------|------------------|-----------|----------|----------|------|----------------------|--------------------------|
| 5        | MP2948      | B        | Cover Hardware   | Assembly  | 1        | Released | EA   |                      | <input type="checkbox"/> |
| 10       | MP2955      | C        | Front Window ... | Component | 1        | Released | EA   |                      | <input type="checkbox"/> |
| 15       | MP2951      | B        | Cover            | Assembly  | 1        | Released | EA   |                      | <input type="checkbox"/> |

Figure 370.



You can use **Redline View** for displaying the current Part changes in a Parent BOM.



| Sequence | Part Number | Revision | Name             | Assembly  | Quantity | State                      | Unit | Reference Designator | Changes                  |
|----------|-------------|----------|------------------|-----------|----------|----------------------------|------|----------------------|--------------------------|
| 5        | MP2948      | B A      | Cover Hardware   | Assembly  | 1        | Released <b>Superseded</b> | EA   |                      | <input type="checkbox"/> |
| 10       | MP2955      | C B      | Front Window ... | Component | 1        | Released <b>Superseded</b> | EA   |                      | <input type="checkbox"/> |
| 15       | MP2951      | B A      | Cover            | Assembly  | 1        | Released <b>Superseded</b> | EA   |                      | <input type="checkbox"/> |

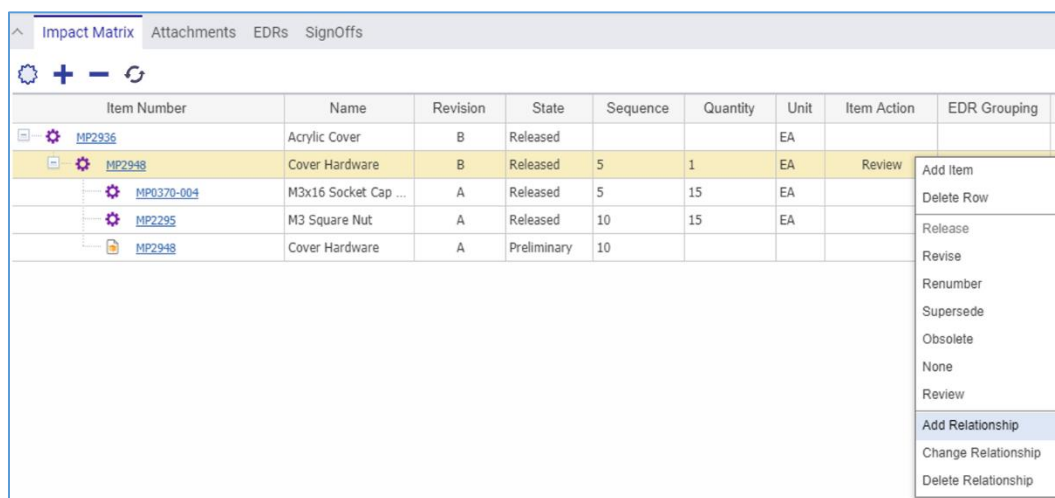
Figure 371.

### 13.3.1.10 Modifying Part BOM Relationships within an Express ECO

Using specialized Relationship Actions on the Impact Matrix, which are discussed in section [13.3.1.4 Managing the Impact Matrix](#), you can add, change, and remove Parts from a BOM, as well as modify information on the BOM Entry itself (e.g., Quantity and Sequence Number).

To modify a BOM, follow the Express ECO workflow as explained in section [13.3.1.1 Understanding the Express ECO Process](#) and notice how the BOM is changing:

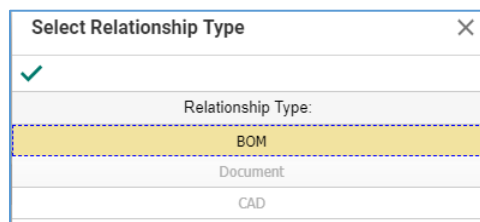
1. Create an Express ECO with a Relationship Action.



| Item Number | Name                 | Revision | State       | Sequence | Quantity | Unit | Item Action | EDR Grouping |
|-------------|----------------------|----------|-------------|----------|----------|------|-------------|--------------|
| MP2936      | Acrylic Cover        | B        | Released    |          |          | EA   |             |              |
| MP2948      | Cover Hardware       | B        | Released    | 5        | 1        | EA   | Review      |              |
| MP0370-004  | M3x16 Socket Cap ... | A        | Released    | 5        | 15       | EA   |             |              |
| MP2295      | M3 Square Nut        | A        | Released    | 10       | 15       | EA   |             |              |
| MP2948      | Cover Hardware       | A        | Preliminary | 10       |          |      |             |              |

Figure 372.

2. Select BOM in the **Select Relationship Type** dialog.



Select Relationship Type

Relationship Type:

- BOM**
- Document
- CAD

Figure 373.

3. When adding or replacing a Part, select a new Part in the **Search Dialog – Part**. The Affected Parent should have **Revise** action set as its BOM change will create a new Revision. Notice the changes in the Impact Matrix.

| Impact Matrix Attachments EDRs SignOffs |                       |          |             |          |          |      |             |              |            |       |          |                     |          |          |                  |  |
|-----------------------------------------|-----------------------|----------|-------------|----------|----------|------|-------------|--------------|------------|-------|----------|---------------------|----------|----------|------------------|--|
| Item Number                             | Name                  | Revision | State       | Sequence | Quantity | Unit | Item Action | EDR Grouping | New Number | State | Revision | Relationship Action | Sequence | Quantity | New Relationship |  |
| MP2936                                  | Acrylic Cover         | B        | Released    |          |          | EA   |             |              |            |       |          |                     |          |          |                  |  |
| MP2948                                  | Cover Hardware        | B        | Released    | 5        | 1        | EA   | Revise      |              |            |       |          |                     |          |          |                  |  |
| MP0370-004                              | M3x16 Socket Cap Bolt | A        | Released    | 5        | 15       | EA   |             |              |            |       |          |                     |          |          |                  |  |
| MP2295                                  | M3 Square Nut         | A        | Released    | 10       | 15       | EA   |             |              |            |       |          |                     |          |          |                  |  |
| MP2948                                  | Cover Hardware        | A        | Preliminary | 10       |          |      |             |              |            |       |          |                     |          |          |                  |  |
|                                         |                       |          |             |          |          |      |             |              |            |       |          |                     | Attach   |          | MP2690           |  |

Figure 374.

- Complete the Express ECO creation. Check the BOM under changes. Note the **red claiming flag**.

| MP2948                                                                                                                                                |             |          |                       |           |          |          |      |                      |                          |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|-----------------------|-----------|----------|----------|------|----------------------|--------------------------|--|
| <div> <div>Edit</div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> </div>                                                  |             |          |                       |           |          |          |      |                      |                          |  |
| Part                                                                                                                                                  |             |          |                       |           |          |          |      |                      |                          |  |
| BOM BOM Structure Alternates AML Documents CAD Documents Goals Changes                                                                                |             |          |                       |           |          |          |      |                      |                          |  |
| <div> <div>Parts</div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> </div> |             |          |                       |           |          |          |      |                      |                          |  |
| Sequence                                                                                                                                              | Part Number | Revision | Name                  | Type      | Quantity | State    | Unit | Reference Designator | Changes                  |  |
| 5                                                                                                                                                     | MP0370-004  | A        | M3x16 Socket Cap Bolt | Component | 15       | Released | EA   |                      | <input type="checkbox"/> |  |
| 10                                                                                                                                                    | MP2295      | A        | M3 Square Nut         | Component | 15       | Released | EA   |                      | <input type="checkbox"/> |  |

Figure 375.

- Submit the Express ECO.

| MP2948                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |          |                  |           |          |          |      |                      |                          |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|------------------|-----------|----------|----------|------|----------------------|--------------------------|--|
| <div> <div>Edit</div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |          |                  |           |          |          |      |                      |                          |  |
| Part                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |          |                  |           |          |          |      |                      |                          |  |
| <div> <div>Part Number</div> <div>MP2948</div> <div>Revision</div> <div>B</div> <div>State</div> <div>Released</div> <div>Assigned Creator</div> <div></div> <div>Name</div> <div>Cover Hardware</div> <div>Designated User</div> <div></div> <div>Type</div> <div>Assembly</div> <div>Unit</div> <div>EA</div> <div>Make / Buy</div> <div>Make</div> <div>Cost (Calculated)</div> <div>1.4250</div> <div>Effective Date</div> <div>3/31/2019 4:35:16 PM</div> <div>Long Description</div> <div></div> <div>Changes Pending</div> <div><input checked="" type="checkbox"/></div> </div> |             |          |                  |           |          |          |      |                      |                          |  |
| BOM BOM Structure Alternates AML Documents CAD Documents Goals Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |          |                  |           |          |          |      |                      |                          |  |
| <div> <div>Parts</div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |          |                  |           |          |          |      |                      |                          |  |
| Sequence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Part Number | Revision | Name             | Type      | Quantity | State    | Unit | Reference Designator | Changes                  |  |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MP0370-004  | A        | M3x16 Socket ... | Component | 15       | Released | EA   |                      | <input type="checkbox"/> |  |
| 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | MP2295      | A        | M3 Square Nut    | Component | 15       | Released | EA   |                      | <input type="checkbox"/> |  |

Figure 376.

6. **Planning.** After submission, the Affected Parts are set to the **In Change** state and not allowed to be edited.

**MP2948**

**Part**

Part Number: MP2948, Revision: B, State: In Change, Assigned Creator: [User], Designated User: [User]

Name: Cover Hardware

Type: Assembly, Unit: EA, Make / Buy: Make, Cost (Calculated): 1.4250, Effective Date: 3/31/2019 4:35:16 PM

Long Description: [Text Area]

☒ Changes Pending

**BOM** | BOM Structure | Alternates | AML | Documents | CAD Documents | Goals | Changes

**Parts**

| Sequence | Part Number | Revision | Name             | Type      | Quantity | State    | Unit | Reference Designator | Changes                  |
|----------|-------------|----------|------------------|-----------|----------|----------|------|----------------------|--------------------------|
| 5        | MP0370-004  | A        | M3x16 Socket ... | Component | 15       | Released | EA   |                      | <input type="checkbox"/> |
| 10       | MP2295      | A        | M3 Square Nut    | Component | 15       | Released | EA   |                      | <input type="checkbox"/> |

Figure 377.

| Item Number | Name            | Revisi... | State       | S... | Quantity | Unit | Item Action | E... | New Number | State       | Revisi... | Relationship Action | Sequ... | Quantity | New Relationship |
|-------------|-----------------|-----------|-------------|------|----------|------|-------------|------|------------|-------------|-----------|---------------------|---------|----------|------------------|
| MP2948      | Cover Hardware  | B         | In Change   |      |          | EA   | Revise      |      | MP2948     | Preliminary | C         |                     |         |          |                  |
| MP0370-004  | M3x16 Socket... | A         | Released    | 5    | 15       | EA   |             |      |            |             |           |                     |         |          |                  |
| MP2295      | M3 Square Nut   | A         | Released    | 10   | 15       | EA   |             |      |            |             |           |                     |         |          |                  |
| MP2948      | Cover Hardware  | A         | Preliminary | 10   |          |      |             |      |            |             |           |                     |         |          |                  |

Figure 378.

7. **Change Review.** During the review process, the Affected Parts are promoted to the **In Review** state and a new major revision, which has the new BOM with changes. No edits are allowed on the original revision.



ECO

Number: ECO-00001020 Title: SD Card fit issues

Change Reason: Change Description:

Released

Priority: ☐ 1 - High ☐ 2 - Normal ☐ 3 - Low

Change Coordinator: Innovator Admin Team: Product Team

Release Date: 4/1/2019 Effective Date:

Impact Matrix Attachments EDRs SignOffs

| Item Number | Name            | Revision | State       | S... | Q... | Unit | Item Action | ED... | New Number | State    | Revision | Relationship Action | S... | Quantity | New Relationship |
|-------------|-----------------|----------|-------------|------|------|------|-------------|-------|------------|----------|----------|---------------------|------|----------|------------------|
| MP2948      | Cover Hardwa... | B        | Superseded  |      |      | EA   | Revise      |       | MP2948     | Released | C        |                     |      |          |                  |
| MP0370-004  | M3x16 Socket... | A        | Released    | 5    | 15   | EA   |             |       |            |          |          |                     |      |          |                  |
| MP2295      | M3 Square Nut   | A        | Released    | 10   | 15   | EA   |             |       |            |          |          |                     |      |          |                  |
| MP2948      | Cover Hardwa... | A        | Preliminary | 10   |      |      |             |       |            |          |          |                     |      |          |                  |
|             |                 |          |             |      |      |      |             |       |            |          |          | Attach              |      |          | MP2690           |

Figure 381.

The newly revised Parts are **Released** at the new major Revision.

Part

Part Number: MP2948 Revision: C State: Released Assigned Creator:

Name: Cover Hardware Designated User:

Type: Assembly Unit: EA Make / Buy: Make Cost (Calculated): 1.4450 Effective Date: 4/1/2019 6:50:31 AM

Long Description:

☐ Changes Pending

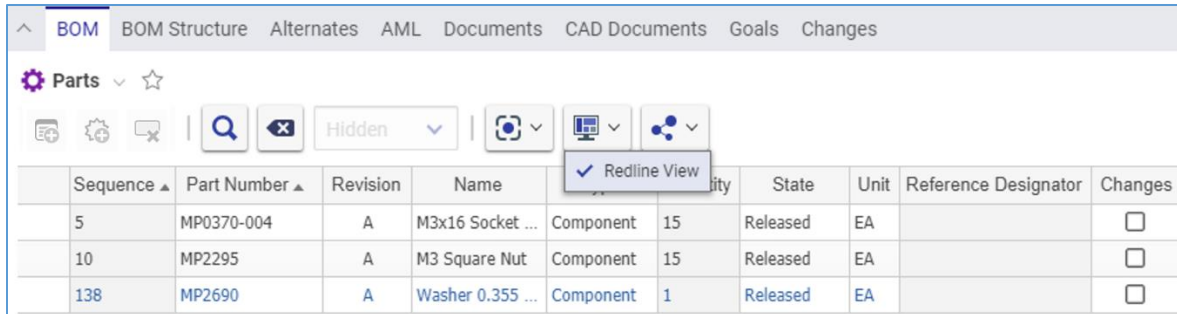
BOM BOM Structure Alternates AML Documents CAD Documents Goals Changes

Parts

| Sequence | Part Number | Revision | Name             | Type      | Quantity | State    | Unit | Reference Designator | Changes                  |
|----------|-------------|----------|------------------|-----------|----------|----------|------|----------------------|--------------------------|
| 5        | MP0370-004  | A        | M3x16 Socket ... | Component | 15       | Released | EA   |                      | <input type="checkbox"/> |
| 10       | MP2295      | A        | M3 Square Nut    | Component | 15       | Released | EA   |                      | <input type="checkbox"/> |
| 138      | MP2690      | A        | Washer 0.355 ... | Component | 1        | Released | EA   |                      | <input type="checkbox"/> |

Figure 382.

You can use **Redline View** for displaying the current Part changes in a Parent BOM.



| Sequence | Part Number | Revision | Name             | Quantity | State    | Unit | Reference Designator | Changes                  |
|----------|-------------|----------|------------------|----------|----------|------|----------------------|--------------------------|
| 5        | MP0370-004  | A        | M3x16 Socket ... | 15       | Released | EA   |                      | <input type="checkbox"/> |
| 10       | MP2295      | A        | M3 Square Nut    | 15       | Released | EA   |                      | <input type="checkbox"/> |
| 138      | MP2690      | A        | Washer 0.355 ... | 1        | Released | EA   |                      | <input type="checkbox"/> |

Figure 383.

### 13.3.2 Express Document Change Order (DCO)

An Express Document Change Order (DCO) is a quick process for implementing Document and CAD Document changes. You can quickly release a new Document, change, revise, or obsolete an existing Document.

In this section, the term Document means an item of both Document and CAD Document ItemTypes.

#### 13.3.2.1 Understanding the Express DCO Process

The **Product** Team roles are engaged in the DCO workflow as the following:

- **Team Manager** is assigned to the **Planning** and **Initial Review** activities for reviewing Affected Documents and routing DCOs.
- **Team Member** is assigned to the **Draft Changes** for making changes to the Affected Documents.

In addition to the **Product** Team, the Express ECO workflow has the following roles:

- A DCO Creator, is a user who initially created and submitted the DCO in the system. This may or may not be the same person as other roles.
- A DCO **Change Coordinator** (Owner), is a person responsible for managing the DCO from start to finish.
- The Configuration Management (CM) group.
- The Change Control Board (CCB), is responsible for review and final approval of an MCO.

A DCO traverses between various states during its life cycle depicted in the following figure.

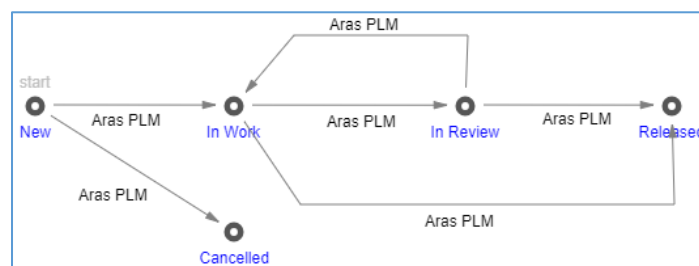


Figure 384.

The DCO workflow together with the DCO life cycle produces the following scenario:

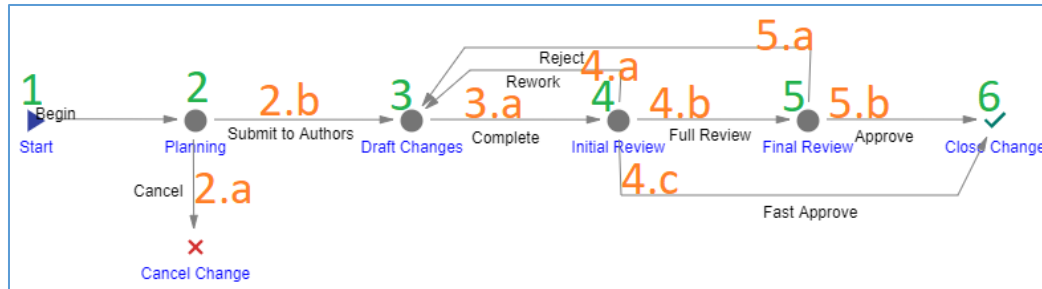


Figure 385.

1. **Start.** An Aras Innovator user initiates a DCO process by creating a new DCO. Once the DCO is saved, a workflow is automatically started to assign tasks and track the changes to it.
2. The Express ECO enters the **Planning** activity assigned to the **Team Manager**. The Team Manager reviews the DCO to vote either:
  - a. **Submit to Authors.** The DCO is ready for work and goes to the **Draft Changes** activity.
  - b. **Cancel.** The DCO is cancelled and cannot be restored in the future.
3. The DCO enters the **Draft Changes** activity assigned to **Team Members**. These Team Members make changes to the new revisions of Documents according to the DCO, and, when completed, vote:
  - a. **Complete** to propagate the DCO to the **Initial Review** activity.
4. The DCO enters the **Initial Review** activity assigned to the **Team Manager**. The Team Manager reviews the DCO and Affected Documents in details to vote either:
  - a. **Full Review** if the DCO requires the final review by the CCB. The DCO goes to the **Final Review** activity.
  - b. **Fast Approve** if the DCO is ready for release. The DCO goes to the **Close Change** activity.
  - c. **Rework** if the DCO or Affected Documents need any further reworking by the Team Manager. The DCO goes back to the **Draft Changes** activity.
5. The Express ECO enters the **Final Review** activity assigned to the **CCB**. All the CCB members must conduct the final review of the DCO and vote the same decision of either:
  - a. **Approve** if the DCO process is completed and does not require any further reworking. The DCO goes to the **Close Change** activity.
  - b. **Reject** if the DCO needs any further reworking. The DCO goes back to the **Draft Changes** activity.
6. Once the DCO enters the **Close Change** activity, it becomes **Released**. Each Document is automatically released and versioned accordingly.

The following table shows the DCO lifecycle and workflow in terms of Roles and Permissions.

**Table 9:** The Express DCO lifecycle and workflow activities with corresponding Roles and Permissions

| Life Cycle State | Workflow Activity            | Can Add | Get                                                                      | Update                                                          | Delete                         | Change access  | Can be promoted   | Update responsible |
|------------------|------------------------------|---------|--------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------|----------------|-------------------|--------------------|
| New              | Start, Planning              | World   | Team Manager, Team Member, Administrators, Owner, All Employees, Creator | Team Manager, Team Member, Administrators, Owner, Creator       | Administrators, Owner, Creator | Administrators | Yes               | Creator            |
| Cancelled        | Cancel Change                |         | Team Manager, Team Member, Administrators, Owner, All Employees, Creator | Administrators                                                  |                                | Administrators | No, a final state |                    |
| In Work          | Draft Changes                |         | Team Manager, Team Member, Administrators, Owner, All Employees, Creator | Team Manager, Team Member, Administrators, Owner, All Employees | Administrators                 | Administrators | Yes               | Team Member        |
| In Review        | Initial Review, Final Review |         | Team Manager, Team Member, Administrators, Owner, All Employees, Creator | Team Manager, Team Member, Administrators, Owner, All Employees | Administrators                 | Administrators | Yes               | CCB                |
| Released         | Closed Change                |         | Team Manager, Team Member, Administrators, Owner, All Employees, Creator | Administrators                                                  |                                | Administrators | No, a final state |                    |

### 13.3.2.2 Creating a DCO

Because a DCO can be used to release, change, revise, or obsolete Affected Documents, the DCO creating process can be conventionally divided into general and specific procedures. This section defines the general steps you should use for creating a DCO. The specific procedures are outlined in sections [13.3.2.3 Releasing a Document within a DCO](#), [13.3.2.4 Replacing a Document within a DCO](#), [13.3.2.5 Revising Documents within a DCO](#), and [13.3.2.6 Obsoleting a Document within a DCO](#).

To create a DCO:

1. Go to **Contents → Change Management → DCOs**.
2. Click **Create New DCO**. The view of a new DCO appears.



**DCO 1**

Save Done Delete

DCO

Number: Server Assigned Title: [Yellow Highlighted Box]

Change Reason: [Text Area]

Priority:
 

- ☐ 1 - High
- ☐ 2 - Normal
- ☐ 3 - Low

Change Coordinator: Innovator Admin Team: Product Team Release Date: [Text Box]

Change Description: [Text Area]

Affected Items Attachments SignOffs

Affected Items

Hidden

| Type [...] | Action | Old Number [...] | Old Revision | Interchangeable | New Number [...] | New Revision | In Build | In Service |
|------------|--------|------------------|--------------|-----------------|------------------|--------------|----------|------------|
|------------|--------|------------------|--------------|-----------------|------------------|--------------|----------|------------|

Figure 386.

- Enter the DCO title in the **Title** text box.
- Describe the reason for the change in the **Change Reason** text box.
- If it is necessary to set the Change Coordinator other than you:
  - Click the **ellipses** button in the **Change Coordinator** box. The **Search dialog – Identity** appears.
  - Search for and select an Identity.
- If it is necessary to set a Team other than the default:
  - Click the **ellipses** button in the **Team** box. The **Search dialog – Team** appears.
  - Using the standard search procedure, search for and select a Team. The selected Team replaces the default identity.
- Describe the change in the **Change Description** text box.
- Specify change priority using the **Priority** radio button: **1 – High**, **2 – Normal**, or **3 – Low**.
- Keep the **Release Date** box blank. This box is automatically filled with the date on which the Express ECO is approved and closed.
- If the DCO should release Documents, use the procedure described in the section [13.3.2.3 Releasing a Document within a DCO](#).
- If the DCO should change Documents, use the procedure described in the section [13.3.2.4 Replacing a Document within a DCO](#).
- If the DCO should revise Documents, use the procedure described in the section [13.3.2.5 Revising Documents within a DCO](#).

13. If the DCO should obsolete Documents, use the procedure described in the section [13.3.2.6 Obsoleting a Document within a DCO](#).
14. Go to the **Attachments** accordion tab to attach all the files concerning this DCO.

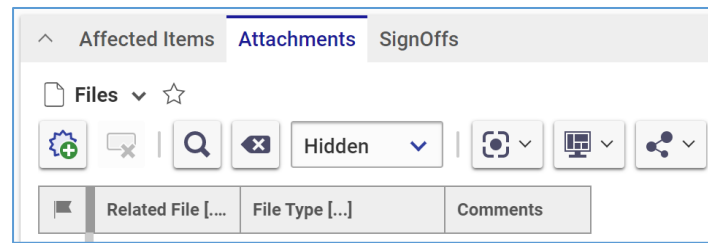


Figure 387.

15. Click **Create Item** on the **Attachments** accordion toolbar. The standard operating system dialog for file browsing and selection appears.
16. Search for and select a file. A new row appears in the **Files** grid with the file attached.

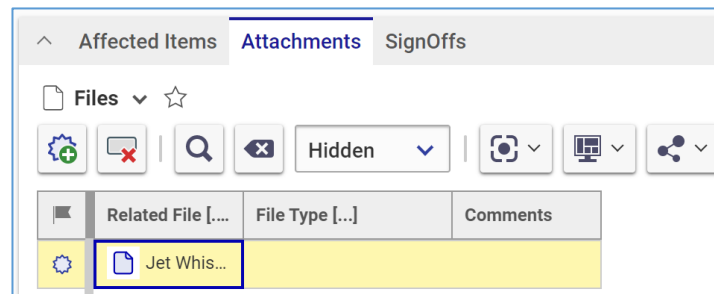


Figure 388.

**Note:** Aras Innovator will automatically provide a file type to the **File Type** cell depending on the File's extension when you save the DCO.

17. If it is necessary to provide additional information on the File, enter it in the **Comments** cell.
18. Click **Done** on the **DCO** toolbar. The DCO is created and awaits to be either submitted or cancelled as defined in the section [12.3.2.8 Voting on a DCO](#).

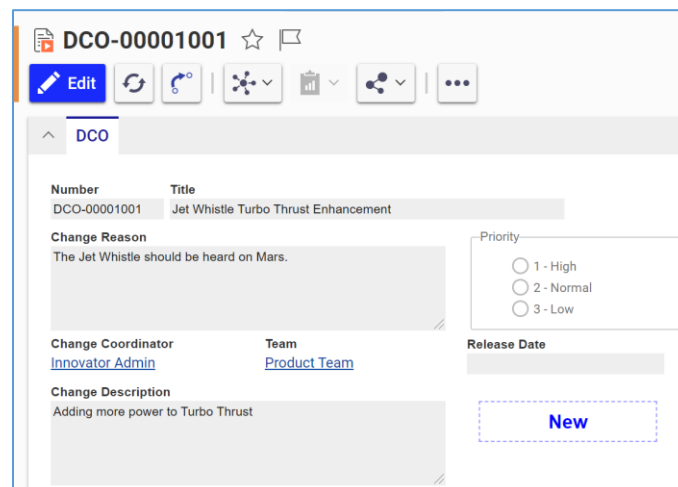


Figure 389.

### 13.3.2.3 Releasing a Document within a DCO

Because a DCO can be used to release, change, or obsolete Affected Documents, the DCO creating process can be conventionally divided into general and specific procedures. This section defines the steps for releasing affected Documents with a DCO.

To assign an affected Document in the **Preliminary** state for release:

1. Open the **DCO**.
2. Click **Edit** on the DCO toolbar.
3. Go to the **Affected Items** accordion tab.
4. Click **Create Item** on the **Affected Items** accordion toolbar. A new row appears at the bottom of the **Affected Items** grid.

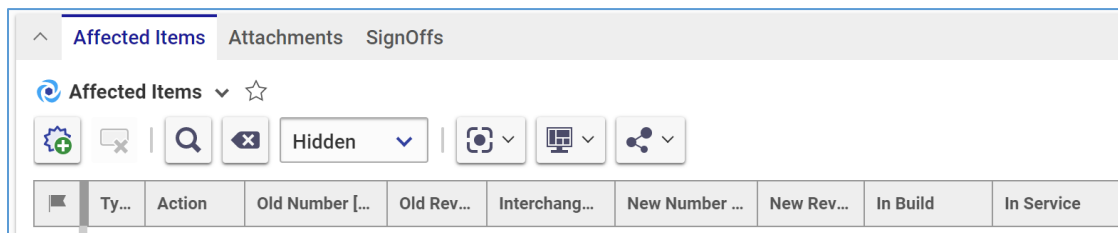


Figure 390.

5. Select **Add** in the **Action** cell.

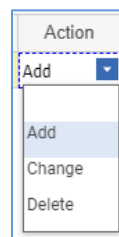


Figure 391.

6. Click the **ellipsis** button in the **New Number** cell. The **Search dialog – Change Controlled Item** appears.
7. Search for and select the new affected Document. The affected Document populates the **New Number** cell.

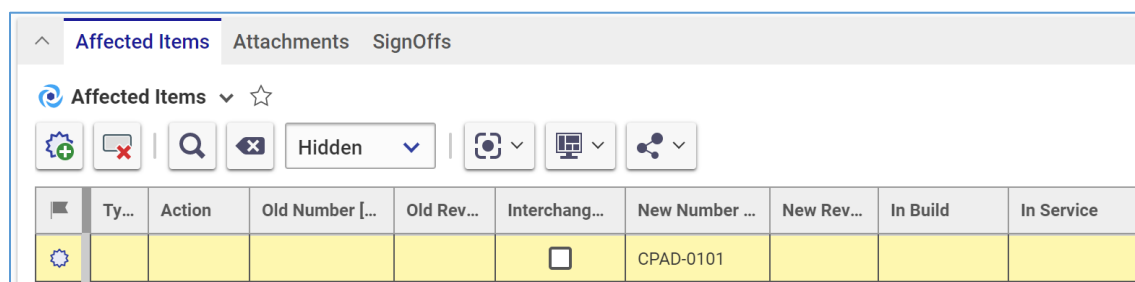


Figure 392.

8. Click **Save** or **Done** on the **DCO** toolbar. The DCO has an Affected Document in the **Preliminary** state assigned for release.

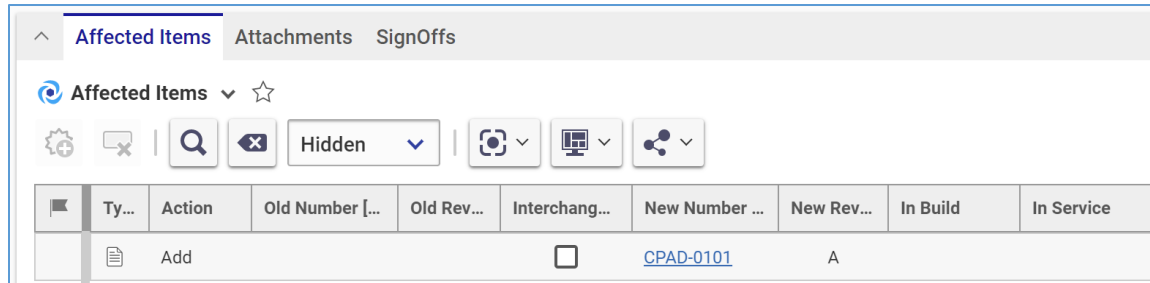


Figure 393.

#### 13.3.2.4 Replacing a Document within a DCO

The DCO **Change** Action for a Document does not modify the contents of the Document or attached files; it informs users that the old, **Superseded**, Document should no longer be used, and a new document is available.

Use the following procedure to assign an existing affected Document to be changed:

1. Follow steps 1-4 of the Adding a Document within a DCO procedure.
2. Select **Change** in the **Action** cell.

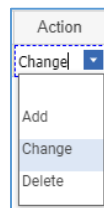


Figure 394.

3. Click the **ellipsis** button in the **Old Number** cell. The **Search dialog – Change Controlled Item** appears.
4. Search for and select the affected Document to be changed. The selected Document populates the **Old Number** cell.

|  | Type | Action | Old Number [...]        | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service |
|--|------|--------|-------------------------|--------------|--------------------------|------------------|--------------|----------|------------|
|  |      | Change | Child Part A Substitute |              | <input type="checkbox"/> |                  |              |          |            |

Figure 395.

5. Leave the **Interchangeable** cell check box blank.
6. Click the **ellipsis** button in the **New Number** cell. The **Search dialog – Change Controlled Item** appears.
7. Search for and select the new Affected Document. The new affected Document populates the **New Number** cell.

|  | Type | Action | Old Number [...]        | Old Revision | Interchangeable          | New Number [...]          | New Revision | In Build | In Service |
|--|------|--------|-------------------------|--------------|--------------------------|---------------------------|--------------|----------|------------|
|  |      | Change | Child Part A Substitute |              | <input type="checkbox"/> | Child Part A Substitute 2 |              |          |            |

Figure 396.

8. Click **Save** two times on the **DCO** toolbar.
9. Select the action to take on the Items that are being built in the **In Build** cell: **Use Existing**, **Rework**, or **Scrap**.

10. Select the action to take on the Items that are currently in service in the **In Service** cell: **Use Existing**, **Rework**, or **Scrap**.

| Type | Action | Old Number [...] | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service   |
|------|--------|------------------|--------------|--------------------------|------------------|--------------|----------|--------------|
|      |        | CPAD-0103        | A            | <input type="checkbox"/> | CPAD-0101        | A            | Rework   | Use Existing |

Figure 397.

11. Click **Save** or **Done** on the **DCO** toolbar. The DCO has an Affected Document assigned for changing.

| Type | Action | Old Number [...] | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service   |
|------|--------|------------------|--------------|--------------------------|------------------|--------------|----------|--------------|
|      |        | CPAD-0103        | A            | <input type="checkbox"/> | CPAD-0101        | A            | Rework   | Use Existing |

Figure 398.

### 13.3.2.5 Revising Documents within a DCO

A released Document cannot be changed. A DCO can be issued to increment the revision to the next level and reset the lifecycle back to the preliminary state, allowing modifications. This controls who and when can make edits to a new revision of a Document.

Use the following procedure to assign an existing affected Document to be revised:

1. Follow steps 1-4 of the Adding a Document within a DCO procedure.
2. Select **Change** in the **Action** cell.

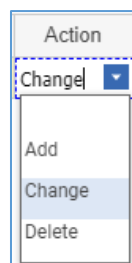
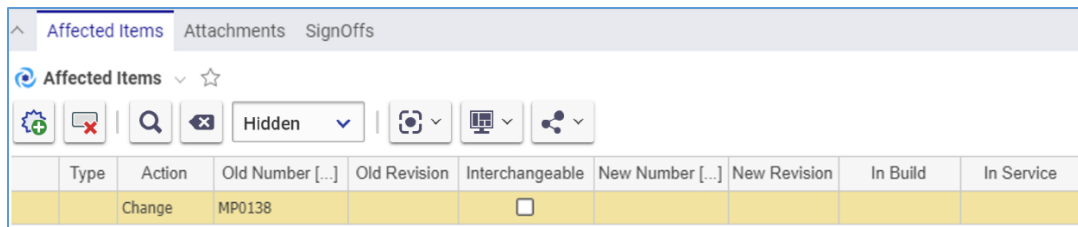


Figure 399.

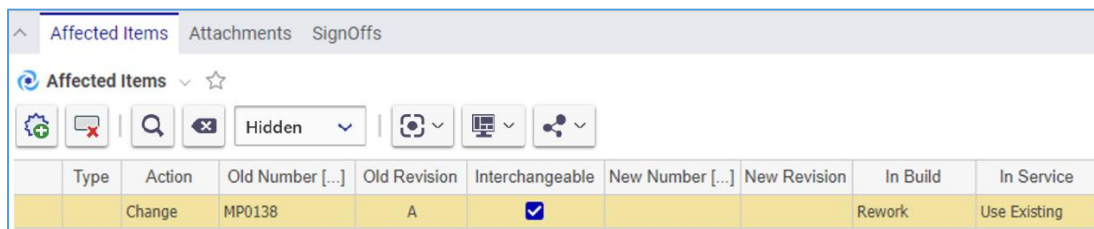
3. Click the **ellipsis** button in the **Old Number** cell. The **Search dialog – Change Controlled Item** appears.
4. Search for and select the affected Document to be revised and updated. The selected Document populates the **Old Number** cell.



| Type | Action | Old Number [...] | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service |
|------|--------|------------------|--------------|--------------------------|------------------|--------------|----------|------------|
|      | Change | MP0138           |              | <input type="checkbox"/> |                  |              |          |            |

Figure 400.

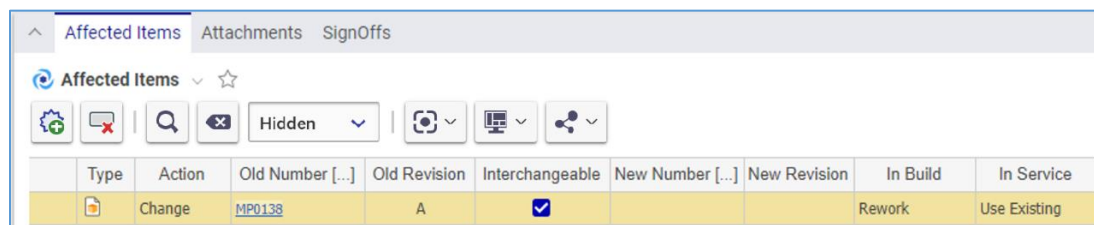
5. Select the **Interchangeable** cell check box.
6. Click **Save** two times on the **DCO** toolbar.
7. Select the action to take on the Items that are being built in the **In Build** cell: **Use Existing**, **Rework**, or **Scrap**.
8. Select the action to take on the Items that are currently in service in the **In Service** field: **Use Existing**, **Rework**, or **Scrap**.



| Type | Action | Old Number [...] | Old Revision | Interchangeable                     | New Number [...] | New Revision | In Build | In Service   |
|------|--------|------------------|--------------|-------------------------------------|------------------|--------------|----------|--------------|
|      | Change | MP0138           | A            | <input checked="" type="checkbox"/> |                  |              | Rework   | Use Existing |

Figure 401.

9. Click **Save** or **Done** on the **DCO** toolbar. The DCO has an Affected Document assigned for revising.



| Type                                                                                | Action | Old Number [...]       | Old Revision | Interchangeable                     | New Number [...] | New Revision | In Build | In Service   |
|-------------------------------------------------------------------------------------|--------|------------------------|--------------|-------------------------------------|------------------|--------------|----------|--------------|
|  | Change | <a href="#">MP0138</a> | A            | <input checked="" type="checkbox"/> |                  |              | Rework   | Use Existing |

Figure 402.

### 13.3.2.6 Obsoleting a Document within a DCO

The DCO **Delete** Action for a Document does not delete the Document or its attached files; it informs users that this Document is **Obsolete** and should no longer be used.

Use the following procedure to assign an existing Affected Document to be:

1. Follow steps 1-4 of the Adding a Document within a DCO procedure.
2. Select **Delete** in the **Action** cell.

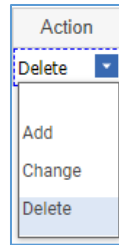


Figure 403.

3. Click the **ellipsis** button in the **Old Number** cell. The **Search dialog – Change Controlled Item** appears.
4. Search for and select the affected Document to be deleted. The selected affected Document populates the **Old Number** cell.

| Affected Items Attachments SignOffs                                                                                                                                               |      |        |                  |              |                          |                  |              |          |            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|------------------|--------------|--------------------------|------------------|--------------|----------|------------|
| Affected Items <span>⌵</span> <span>☆</span><br><span>⚙️</span> <span>🗑️</span> <span>🔍</span> <span>🔒</span> Hidden <span>⌵</span> <span>🔄</span> <span>🖨️</span> <span>🔗</span> |      |        |                  |              |                          |                  |              |          |            |
|                                                                                                                                                                                   | Type | Action | Old Number [...] | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service |
|                                                                                                                                                                                   |      | Delete | CPAD-0102        |              | <input type="checkbox"/> |                  |              |          |            |

Figure 404.

5. Click **Save** two times on the **DCO** toolbar.
6. Select the action to take on the Items that are being built in the **In Build** cell: **Use Existing**, **Rework**, or **Scrap**.
7. Select the action to take on the Items that are currently in service in the **In Service** cell: **Use Existing**, **Rework**, or **Scrap**.

| Affected Items Attachments SignOffs                                                                                                                                               |      |        |                  |              |                          |                  |              |          |              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|------------------|--------------|--------------------------|------------------|--------------|----------|--------------|
| Affected Items <span>⌵</span> <span>☆</span><br><span>⚙️</span> <span>🗑️</span> <span>🔍</span> <span>🔒</span> Hidden <span>⌵</span> <span>🔄</span> <span>🖨️</span> <span>🔗</span> |      |        |                  |              |                          |                  |              |          |              |
|                                                                                                                                                                                   | Type | Action | Old Number [...] | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service   |
|                                                                                                                                                                                   |      | Delete | CPAD-0102        |              | <input type="checkbox"/> |                  |              | Rework   | Use Existing |

Figure 405.

8. Click **Save** or **Done** on the **DCO** toolbar. The DCO has an affected Document assigned for deleting.

| Affected Items Attachments SignOffs                                                                                                                                               |      |        |                           |              |                          |                  |              |          |              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|---------------------------|--------------|--------------------------|------------------|--------------|----------|--------------|
| Affected Items <span>⌵</span> <span>☆</span><br><span>⚙️</span> <span>🗑️</span> <span>🔍</span> <span>🔒</span> Hidden <span>⌵</span> <span>🔄</span> <span>🖨️</span> <span>🔗</span> |      |        |                           |              |                          |                  |              |          |              |
|                                                                                                                                                                                   | Type | Action | Old Number [...]          | Old Revision | Interchangeable          | New Number [...] | New Revision | In Build | In Service   |
|                                                                                                                                                                                   |      | Delete | <a href="#">CPAD-0102</a> | A            | <input type="checkbox"/> |                  |              | Rework   | Use Existing |

Figure 406.

### 13.3.2.7 Removing a Document from a DCO

To remove a Document from a DCO so that it is not processed:

1. Follow steps 1-3 of the Adding a Document within a DCO procedure.
2. Click the **Document** and then click **Delete Row** on the **Affected Items** accordion toolbar. The **Document** becomes unavailable.

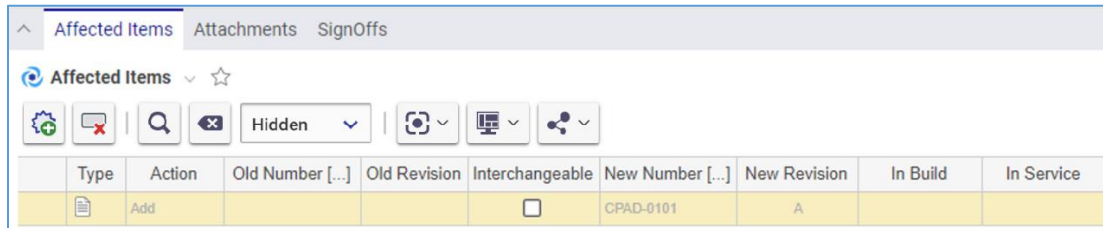


Figure 407.

3. Click **Save** or **Done** on the DCO toolbar. The **Document** disappears from the grid.

**Note:** The **Document** disappears immediately if the DCO was not saved after it was added.

### 13.3.2.8 Voting on a DCO

Use the following procedure:

1. Go to **Contents** → **My Innovator** → **My InBasket** → **Search My InBasket**.
2. Double-click on the Workflow Task to open the **Workflow Activity Completion** dialog for the DCO.

Workflow Activity Completion

Workflow: DCO-00001006  
Activity: Planning

Tasks

| Sequence | Required                 | Description                                                      | Complete                            |
|----------|--------------------------|------------------------------------------------------------------|-------------------------------------|
| 1        | <input type="checkbox"/> | Fill out the change form and assign the affected document        | <input checked="" type="checkbox"/> |
| 2        | <input type="checkbox"/> | Perform impact analysis and include any other affected docume... | <input checked="" type="checkbox"/> |
| 3        | <input type="checkbox"/> | Define change actions                                            | <input checked="" type="checkbox"/> |

Vote

Vote:

Delegate to:

Submit to Authors

Cancel

Delegate

Refuse

Comments:

Authentication

Password:

E-Signature:

Complete

Save Changes

Cancel

Figure 408.

3. Mark the tasks you completed in the **Tasks** field.
4. Select the option per your decision in the **Vote** list.



- Click **Complete**.

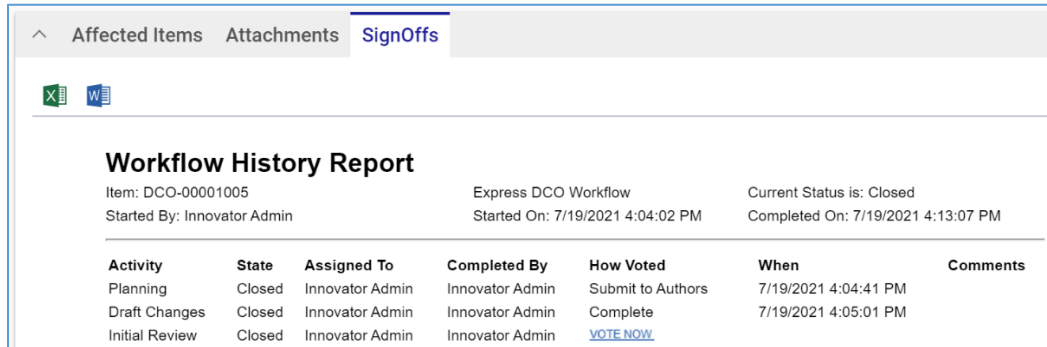
The DCO enters the **next** activity as defined in the DCO workflow.

### 13.3.2.9 Viewing and Processing DCO Signoff Activities

From the DCO view, you can review the current and previous Activities assigned for the DCO. You can also vote within an Activity if you have rights.

Use the following procedure:

- Open the DCO.
- Go to the **SignOffs** accordion tab.



| Workflow History Report     |        |                                  |                 |                                    |                      |          |
|-----------------------------|--------|----------------------------------|-----------------|------------------------------------|----------------------|----------|
| Item: DCO-00001005          |        | Express DCO Workflow             |                 | Current Status is: Closed          |                      |          |
| Started By: Innovator Admin |        | Started On: 7/19/2021 4:04:02 PM |                 | Completed On: 7/19/2021 4:13:07 PM |                      |          |
| Activity                    | State  | Assigned To                      | Completed By    | How Voted                          | When                 | Comments |
| Planning                    | Closed | Innovator Admin                  | Innovator Admin | Submit to Authors                  | 7/19/2021 4:04:41 PM |          |
| Draft Changes               | Closed | Innovator Admin                  | Innovator Admin | Complete                           | 7/19/2021 4:05:01 PM |          |
| Initial Review              | Closed | Innovator Admin                  | Innovator Admin | <a href="#">VOTE NOW</a>           |                      |          |

Figure 409.

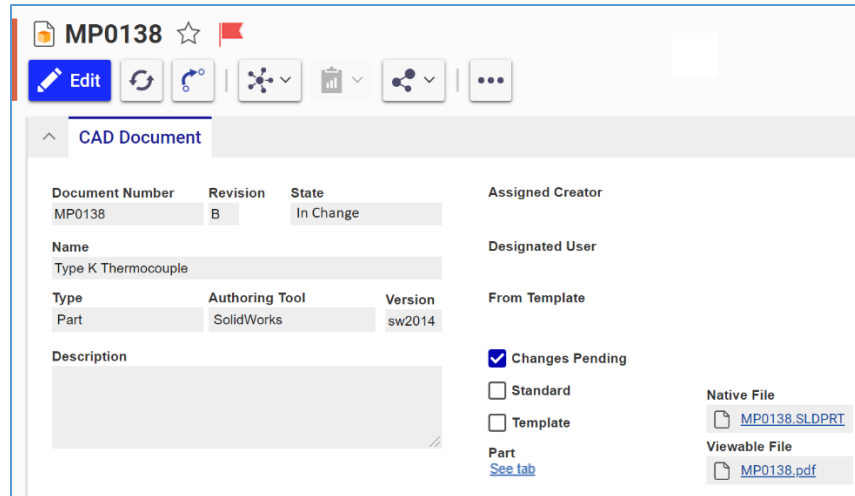
- Click the **VOTE NOW** link. The **Workflow Activity Completion** dialog for the DCO Activity appears.
- Use steps 3-5 of the **Voting on a DCO** procedure.

### 13.3.2.10 Processing Documents within a DCO

This section describes revising a released Document as an example of processing Documents within an Express ECO.

To process Documents, follow the Express DCO workflow as explained in section [13.3.2.1 Understanding the Express DCO Process](#) and notice how the state of Documents is changing:

- Submit the DCO** to the **Draft Changes** activity. The old Document version is in the **In Change** state and has **Changes Pending** check mark set automatically to indicate that this Document is under a change process.



**MP0138** ☆

[Edit](#) [Refresh](#) [Undo](#) [Share](#) [Print](#) [More](#)

**CAD Document**

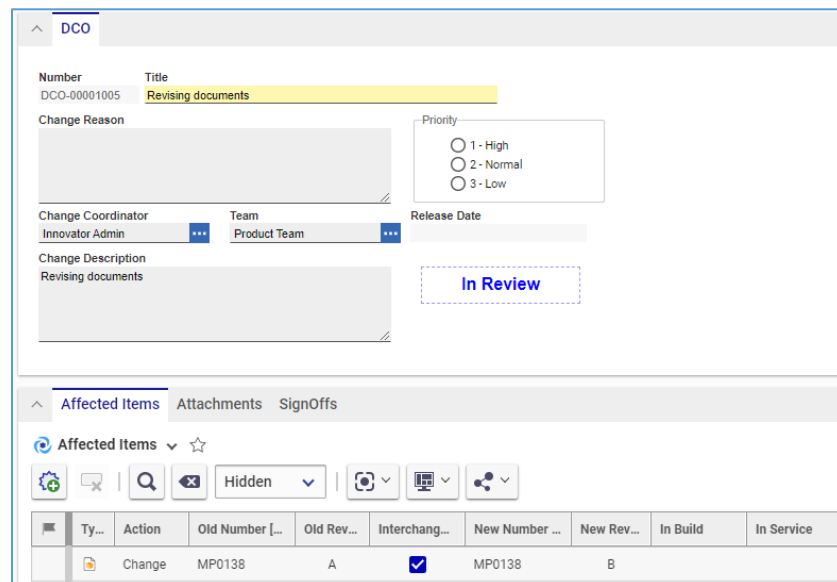
|                                    |                                     |                           |                                                                                                                                                    |
|------------------------------------|-------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Document Number</b><br>MP0138   | <b>Revision</b><br>B                | <b>State</b><br>In Change | <b>Assigned Creator</b>                                                                                                                            |
| <b>Name</b><br>Type K Thermocouple |                                     |                           | <b>Designated User</b>                                                                                                                             |
| <b>Type</b><br>Part                | <b>Authoring Tool</b><br>SolidWorks | <b>Version</b><br>sw2014  | <b>From Template</b>                                                                                                                               |
| <b>Description</b>                 |                                     |                           | <input checked="" type="checkbox"/> <b>Changes Pending</b><br><input type="checkbox"/> <b>Standard</b><br><input type="checkbox"/> <b>Template</b> |

**Native File**  
[MP0138.SLDPR](#)

**Viewable File**  
[MP0138.pdf](#)

Figure 410.

The DCO has a link to the original and new major revisions of the Document.



**DCO**

|                               |                                    |
|-------------------------------|------------------------------------|
| <b>Number</b><br>DCO-00001005 | <b>Title</b><br>Revising documents |
|-------------------------------|------------------------------------|

**Change Reason**

**Priority**  
☐ 1 - High  
☐ 2 - Normal  
☐ 3 - Low

**Change Coordinator**  
Innovator Admin

**Team**  
Product Team

**Release Date**

**Change Description**  
Revising documents

**In Review**

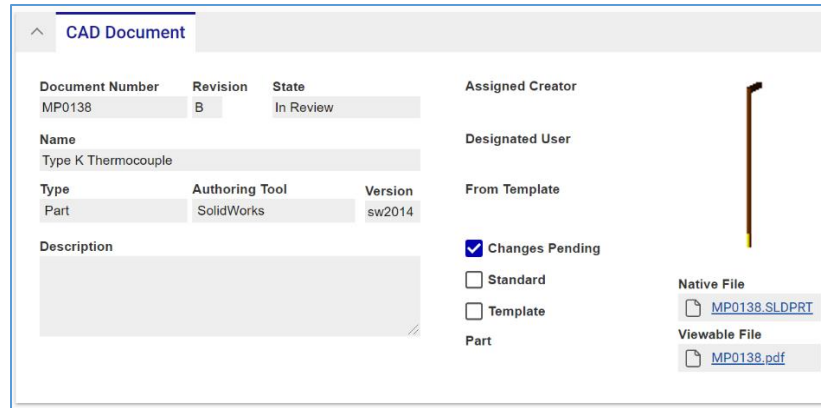
**Affected Items** | Attachments | SignOffs

**Affected Items** ☆

| Ty...  | Action | Old Number [...] | Old Rev... | Interchang...                       | New Number ... | New Rev... | In Build | In Service |
|--------|--------|------------------|------------|-------------------------------------|----------------|------------|----------|------------|
| Change |        | MP0138           | A          | <input checked="" type="checkbox"/> | MP0138         | B          |          |            |

Figure 411.

The new major revision of the Document is in the **Preliminary** state and has **Changes Pending** check mark set automatically to indicate that this Document is under a change process.



**CAD Document**

| Document Number | Revision | State     | Assigned Creator |
|-----------------|----------|-----------|------------------|
| MP0138          | B        | In Review |                  |

Name  
Type K Thermocouple

Type  
Part

Authoring Tool  
SolidWorks

Version  
sw2014

Description

Designated User

From Template

☒ Changes Pending

☐ Standard

☐ Template

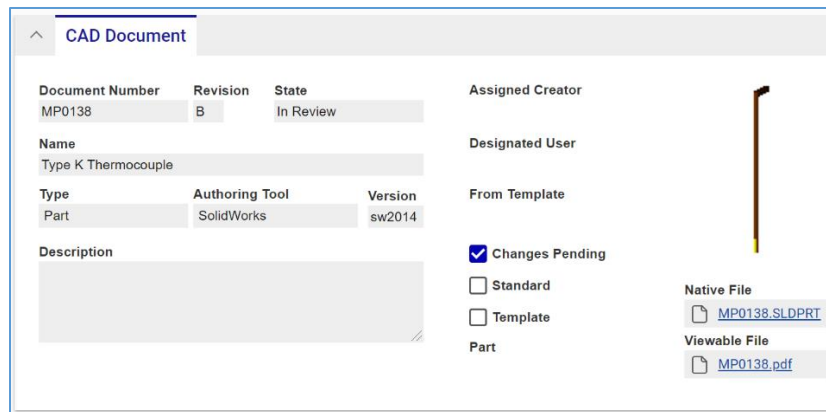
Part

Native File  
[MP0138.SLDPR](#)

Viewable File  
[MP0138.pdf](#)

Figure 412.

- The **Initial Review** activity. The Affected Document is set to the **In Review** state and is not allowed to be edited.



**CAD Document**

| Document Number | Revision | State     | Assigned Creator |
|-----------------|----------|-----------|------------------|
| MP0138          | B        | In Review |                  |

Name  
Type K Thermocouple

Type  
Part

Authoring Tool  
SolidWorks

Version  
sw2014

Description

Designated User

From Template

☒ Changes Pending

☐ Standard

☐ Template

Part

Native File  
[MP0138.SLDPR](#)

Viewable File  
[MP0138.pdf](#)

Figure 413.

- The **Final Review** activity. The Affected Document is still in the **In Review** state and is not allowed to be edited.
- Close Change**. The DCO is **Released**.

DCO-00001005

Edit

DCO

Number: DCO-00001005, Title: Revising documents

Change Reason: [Empty]

Change Coordinator: Innovator Admin, Team: Product Team

Release Date: 7/19/2021

Priority: 1 - High, 2 - Normal, 3 - Low

Change Description: Revising documents

Released

Affected Items

| Ty...  | Action | Old Number [...] | Old Rev... | Interchang...                       | New Number ... | New Rev... | In Build | In Service |
|--------|--------|------------------|------------|-------------------------------------|----------------|------------|----------|------------|
| Change |        | MP0138           | A          | <input checked="" type="checkbox"/> | MP0138         | B          |          |            |

Figure 414.

- The newly revised Document is **Released** at the new major Revision.

MP0138

Edit

CAD Document

Document Number: MP0138, Revision: B, State: Released

Name: Type K Thermocouple

Type: Part, Authoring Tool: SolidWorks, Version: sw2014

Description: [Empty]

Assigned Creator: Designated User

From Template: ☐ Changes Pending, ☐ Standard, ☐ Template

Native File: MP0138.SLDPRIT

Viewable File: MP0138.pdf

Figure 415.